Ira A. Fulton Schools of Engineering

Arizona State University

School of Computing and Augmented Intelligence

CS / CSE Capstone Proposal Catalog

Fall 2026

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162. Northern Watch Services Inc. - Virtual Try ON
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165. Appy.yo - WeVibe: AI-Powered Dating and Compatibility Recommendation App
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Studio XE - Accessibility Conformance and Interface Consistency for an Enterprise Platform

Proposer: David Lutz david.lutz@communityhdp.com

Project Contact: Rohith Reddy Byreddy rohith@snackiebeans.com

Project Description: Studio XE is a software platform used daily by organizations in healthcare, finance, and legal services; a meaningful share of its users rely on keyboard navigation or assistive technology. The application does not currently meet a defensible accessibility standard: of roughly 2,600 interface files, only about 12 percent contain any accessibility markup, and there is no automated accessibility testing of any kind. As a consequence the organization cannot produce a Voluntary Product Accessibility Template (VPAT) — the conformance document that enterprise, healthcare, and government purchasers request by name — which closes off an entire class of customer. A related problem compounds it: the platform's 14 functional areas each carry their own conventions for navigation, empty screens, loading behavior, error handling, and filtering — inconsistency that disproportionately harms users navigating by keyboard or screen reader. The student team measures conformance against WCAG 2.2 Level AA, automates what can be automated (axe-core and accessibility linting in continuous integration), remediates the core interface to zero critical and serious automated violations on the 20 highest-traffic screens, establishes full keyboard operability of the shell, command palette, and data grid, authors the interaction specification the platform lacks, brings all 14 areas into conformance or logged exception, and publishes a VPAT 2.5 traceable to test evidence rather than assertion. All work occurs against a synthetic dataset — no production systems, customer data, or protected health information is accessible at any point.

Student Learning Goals: Students learn accessibility engineering as it is practiced commercially: measuring against a published standard (WCAG 2.2 AA), automating conformance checking in CI, remediating a large production codebase, performing documented screen-reader evaluations (NVDA, VoiceOver) across real user workflows, and producing the compliance artifact (VPAT 2.5) that enterprise procurement actually requests. Roughly half the work is standards interpretation, measurement, and technical writing; half is implementation — an unusual and valuable combination.
The team also authors an interaction specification governing empty, loading, and error states, filter behavior, and data-table defaults across 14 functional areas — experience in writing a standard other engineers must follow, then driving a large surface into conformance against it. Acceptance criteria are numeric and fixed at term start; work merges into a platform in active production use under employee-standard code review.

Deliverables: • Automated accessibility testing (axe-core, eslint-plugin-jsx-a11y) integrated into continuous integration.

- Zero critical or serious automated violations across the 20 highest-traffic screens.
- Full keyboard operability of shell, command palette, and data grid, with a

documented, re-runnable traversal test.

- A screen-reader evaluation log across five primary workflows, with findings and resolutions.
- A written interaction specification accepted by the sponsor; all 14 functional areas conforming or carrying a logged, justified exception.
- A published VPAT 2.5 traceable to test evidence, plus a final presentation with measured before-and-after conformance data.

Desired Background:

- Front-End: working React and TypeScript.
- Standards: one detail-oriented member willing to work from standards documents (WCAG, VPAT) is essential.
- Tooling: axe-core, eslint-plugin-jsx-a11y, screen readers (NVDA, VoiceOver) — sponsor provides the environment.
- Suits software engineering mixed with IT or cybersecurity students; strong ethics-integration framing.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

ASU Ira A. Fulton Schools of Engineering, School of Electrical, Computer and Energy Engineering (Dr. Jeff Zhang's lab) - Adaptive Compute Scaling for RL-Based LLM Post-Training

Proposer: Dr. Jeff (Jun) Zhang jeffzhang@asu.edu

Project Description: Today's best reasoning-focused AI models (the kind behind "thinking" modes in tools like ChatGPT) are built in two stages: first the model learns general language ability, then it goes through a second stage called reinforcement learning post-training, where it gets noticeably better at reasoning. This second stage works in a loop: the model generates several possible answers to a problem, a reward system scores those answers, and the model updates itself based on those scores. This loop repeats thousands of times and is one of the most computationally expensive parts of building modern AI because every step runs at full computational effort regardless of whether that much effort is actually needed. This project explores hardware-software co-design as a way to cut that cost: building software that senses, in real time, how much effort a given step actually requires and pairing that software with the hardware best suited to act on it with GPUs for the bulk of the work, and potentially a custom FPGA circuit for specific repetitive sub-tasks where a purpose-built piece of hardware can outperform a general-purpose one. The goal is to show that scaling effort up and down intelligently, and matching it to the right hardware, can meaningfully cut training cost without hurting how well the model learns.

Student Learning Goals: Students will learn the mechanics of reinforcement learning post-training and reward modeling which are the same techniques behind today's reasoning-focused AI models and get hands-on experience with GPU cluster computing and PyTorch. Beyond the ML side, this project is a genuine introduction to hardware-software co-design: understanding not just how to write correct code, but where computational cost actually lives in a system and how software decisions and hardware capabilities have to be designed together to realize real speed gains rather than theoretical ones. Students interested in the hardware side will have the option to prototype a small accelerator kernel on an FPGA (a reconfigurable chip that can be custom-built for one specific task); students more interested in ML or systems can focus on the adaptive-effort logic or performance benchmarking instead.

Deliverables: A working prototype that adjusts computational effort during RL post-training based on real-time signals from the reward system and the model's own generation process. A benchmark suite comparing this adaptive approach against the standard fixed-effort baseline, measuring both training time and final model quality.

Desired Background: Comfortable with Python with some exposure to PyTorch, TensorFlow, or a similar ML framework; coursework in machine learning is helpful. For students interested in the hardware sub-track, digital logic, or any FPGA/HLS exposure is a plus but not required (hardware piece is optional and can be scoped based on team interest). Comfortable working on Linux/command line.

Delivery Mode: In-person

Additional Materials: N/A

N/A - Agentic AI Capstone & Campus Event Matchmaker

Proposer: Shatrughna Upadhyay supadh18@asu.edu

Project Description: Universities host a massive ecosystem of student organizations and capstone programs that constantly seek funding from industry partners. Connecting campus event organizers with potential corporate sponsors is often an inefficient, manual process. This AI-driven platform acts as a centralized matchmaking system tailored for higher education. It enables university hosts to create event profiles and automatically connects them with targeted industry sponsors based on academic domain, technical requirements, and budget. It utilizes an adaptation of the Gale-Shapley stable matching algorithm alongside LLM-driven autonomous outbound email outreach to acquire and pair sponsors.

Student Learning Goals: Students will gain hands-on experience building a modern dual-sided marketplace and integrating generative AI into production workflows. They will learn to manage complex relational databases, implement algorithmic matchmaking (Gale-Shapley), generate vector embeddings for semantic search, and deploy serverless event-driven architecture using Next.js, Python, and Supabase.

Deliverables: - A deployed web application with secure, dual-role dashboards (University Organizers vs. Industry Sponsors).
- An NLP-powered matching engine that pairs project requirements with sponsor tech stacks.
- An autonomous outbound email pipeline (via the Resend API) that drafts and sends personalized sponsor invitations.
- A complete source code repository with comprehensive technical documentation and a handover manual.

Desired Background: Proficiency in JavaScript/TypeScript (React/Next.js) and Python. A foundational understanding of relational databases (PostgreSQL), RESTful APIs, and basic machine learning concepts (such as working with LLM prompts and vector embeddings).

Delivery Mode: In-person

Expeditise LLC - AI Advertising Content Platform for Expeditise Products

Proposer: Ali Sarabi asarabi1@asu.edu

Project Description: Expeditise LLC is an early-stage startup in the ASU Venture Devil program building a family of AI products, with more launching over the next year. Every launch and every update needs advertising material, and today that work is manual and repeated from scratch for each product. This capstone team will build an advertising content platform that Expeditise uses across all of its products: the operator supplies raw material for whichever product is being promoted, and the platform produces a complete, ready-to-use set of advertising assets for it.

The central workflow is deliberately simple from the outside. Someone picks a product, drops in a set of images and screenshots plus a few lines of text, chooses a goal such as signups or installs and a set of target channels, and the platform returns everything needed to advertise: generated ad media sized and formatted correctly for each platform, captions, ad copy, hashtags, and several variants to choose between. In the first phase the platform stops there. It creates and organizes the material, validates that each piece actually meets the target platform's requirements, and hands it back for review, download, and use. Nothing is published automatically and no campaign runs on its own.

The second phase connects the platform to the products' social accounts so posting no longer means leaving the tool. Instagram, Facebook, X, LinkedIn, TikTok, and YouTube accounts are linked through their official APIs, and the operator browses the generated assets, selects which ones go to which account, and posts them immediately or at a chosen time. Publishing stays an explicit human decision; the engineering work is account authentication and token refresh, validation against each platform's format and policy rules before a post is accepted, and queueing, retries, visible failure states, and rate-limit handling so a selected post reliably lands.

Because the platform serves a product family rather than a single app, each product is configured as its own brand profile: voice and tone, colours, logo, approved claims, destination links, and accounts. Adding the next Expeditise product should be a configuration step, not a new codebase. Alongside this, the team will build the measurement side: tracked links generated for every asset, landing pages generated per product and audience, experiments comparing headlines and creatives, and reporting that ties results back to the material that produced them, so the platform can show what is actually working rather than what looks busy.

Guardrails are part of the engineering work: platform terms-of-service and rate-limit compliance, human approval before anything reaches an account, disclosure of AI-generated content, and privacy compliance for anything involving user data. Students are also strongly encouraged to use agentic AI in their own development workflow, including coding agents, AI-assisted review, and generated assets, while remaining accountable for everything they ship. The goal is a platform in real use for the next Expeditise product launch, with measured results the team can point to, not only a dashboard.

Student Learning Goals: Students will gain experience in growth engineering, the software layer behind modern marketing, combining API integration, LLM pipelines, media processing, data engineering, experimentation, and product analytics on real products with real audiences.

• Integrating and authenticating against third-party social and marketing APIs, including OAuth, tokens, webhooks, pagination, rate limits, and sandbox versus production environments
• Building agentic LLM pipelines that generate, critique, and revise advertising copy within brand, factual, and safety constraints
• Automated media generation and processing: composing graphics from templates, cropping and reformatting to each platform's requirements, and assembling short video from stills
• Designing queueing, retry, and idempotency logic for a service that publishes to outside platforms on a person's behalf
• Multi-tenant design, so one platform can serve several products with separate brands, accounts, and campaigns
• Event tracking, data pipelines, and attribution modelling from first impression through signup, install, and activation
• Designing and analysing A/B tests, including significance testing and avoiding false positives
• Building internal dashboards and automated reporting that a non-technical founder can act on directly
• Engineering under real-world constraints: platform policy and format rules, API rate limits, privacy law, and the reputational cost of automation that misbehaves
• Using agentic AI as a working development tool while staying fully accountable for the code that ships

Deliverables: 1. Multi-Product Foundation and Tracking

• Brand profiles per product covering voice, colours, logo, approved claims, destination links, and target audiences, so a new product is added by configuration rather than by new code
• Tracked links generated for every asset and channel, plus event tracking across product sites with a documented schema, so results can be attributed even when a post is placed by hand

2. Advertising Content Generation from Images and Text (Phase 1)

• An intake workflow where an operator selects a product, uploads images and screenshots, adds a few lines of text, selects a goal and target channels, and receives a complete draft set of advertising assets
• Automated media generation: template-based graphics with logo, headline, and device mockups, cropping and resizing to each platform's feed, story, reel, and carousel formats, short video assembled from stills with captions, and generated alt text
• Language-model generation and ranking of captions, ad copy, hashtags, and variants

under brand, factual, and safety constraints
• Per-platform validation of every generated piece against aspect ratio, resolution, file size, duration, character limit, and hashtag rules, so nothing is delivered that the target platform would reject

3. Asset Library, Review, and Export
• A searchable library where generated campaigns are stored and versioned per product, channel, and goal, with review states such as draft, approved, and used
• Export in ready-to-use bundles: per-platform folders, downloadable archives, copy in a form that pastes cleanly, and a suggested posting calendar the operator can follow
• A feedback loop where reviewer edits and rejections are recorded and used to improve later generations

4. Social Account Connection and Assisted Posting (Phase 2)
• Account linking for Instagram and Facebook (Meta Graph), X, LinkedIn, TikTok, and YouTube through their official APIs, with OAuth, token refresh, and per-product account permissions
• A posting screen where the operator selects an asset and target accounts and publishes immediately or at a chosen time, with publishing always an explicit human action
• Pre-publish policy and format checks, queueing, retries, visible failure states, rate-limit handling, and a kill switch

5. Landing Pages, Experiments, and Reporting
• Programmatic generation of landing pages per product, use case, keyword, and audience, with SEO metadata, structured data, sitemaps, and performance budgets
• Experiments comparing headlines, creatives, and calls to action through tracked links and landing page variants, with significance testing
• A reporting view and weekly summary agent that ties results back to the assets that produced them and proposes the next experiments

6. Testing, Deployment, and Documentation
• Automated tests for the generation pipeline and for API integrations against mocks and sandbox accounts, plus monitoring and alerting for the live services
• Deployment, secrets management, and media storage infrastructure for the production platform
• Documentation of the architecture, event schema, API integrations, and runbooks for adding a new product and a new channel

Desired Background: Students should have experience or interest in the following areas:
• Full-stack development with JavaScript/TypeScript or Python, REST APIs, and a modern web framework such as Next.js or React
• Integrating third-party APIs, including OAuth flows, webhooks, pagination, and rate limits
• Backend and service work: job scheduling, queues, databases, and cron or serverless

scheduled functions
• Image and video processing with libraries such as Pillow, sharp, or FFmpeg is a plus
• Data work: event tracking, SQL, dashboards, and enough statistics to run and read an A/B test
• Experience with LLM APIs and agentic AI is highly recommended
• Interest in marketing, growth, SEO, content, or social platforms; no marketing coursework is required, but curiosity about why people click matters
• Interest in startups and in work whose results are publicly measurable

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Extra Sauce Agency - AI Agent for Faster Content Distribution

Proposer: Manny Jassal manny@extrasauceagency.com

Project Contact: Manny Jassal manny@extrasauceagency.com

Project Description: Build an AI-powered "Content Command Center" that helps companies turn one piece of content into dozens of high-quality posts across channels—fast.

This project focuses on designing and developing an AI agent system that can ideate, write, repurpose, and distribute content while maintaining a consistent brand voice. The system will simulate a team of AI agents (researcher, strategist, copywriter, editor, etc.) working together to help users go from idea → published content in minutes.

Students will work on real-world problems like reducing content production time, eliminating "what should I post?" decision fatigue, and enabling founders and marketing teams to scale their content presence across platforms.

This is not just a prototype project—we are actively building this product and exploring GTM, meaning your work has the potential to directly influence a live startup.

Student Learning Goals: Students will gain hands-on experience building a real AI product from the ground up, not just theoretical models.

You will:

Design and implement AI agent workflows (multi-agent systems)
Work with LLMs to solve real product use cases (ideation, rewriting, personalization)
Build features that integrate with APIs (social platforms, content tools, storage systems)
Apply product thinking: UX, onboarding, and "time-to-value" optimization
Learn how to translate user problems into AI-driven solutions
Collaborate in a startup-like environment with fast iteration cycles

By the end of the project, you will have experience across AI engineering, product design, and real-world application building—highly relevant for roles in AI, SaaS, and product development.

Deliverables: Students will be expected to deliver a working prototype (or set of features) that contribute to the overall AI agent platform.

Possible deliverables include:

AI agent workflows (e.g., ideation agent, copywriter agent, repurposing agent)
A conversational interface for interacting with the system

Content repurposing engine (long-form → short-form content)
Brand voice personalization system (memory + tone matching)
Integration prototypes (e.g., LinkedIn posting, Google Drive/Notion ingestion)
UX flows for onboarding and content creation
Documentation + demo of the product

Bonus (for standout teams):

Performance analysis system for content optimization
Real-time content grading or feedback system
Trend scraping or inspiration feed

Desired Background: We are looking for motivated students who are excited to build at the intersection of AI, product, and content.

Ideal backgrounds include:

Computer Science, Software Engineering, Data Science, or related fields
Experience with Python, JavaScript, or full-stack development
Familiarity with APIs and/or working with LLMs (OpenAI, etc.)
Interest in AI agents, automation, or SaaS products
Curiosity about content, marketing, or creator tools (not required, but a plus)

Most importantly:

Strong problem-solving mindset
Willingness to build, test, and iterate quickly
Interest in working on something that could become a real product

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

Honeywell Aerospace - AI Agent for Quotation

Proposer: Naseeruddin Mohammed Naseeruddin.Mohammed@honeywellaerospace.com

Project Contact: Charles Dairman Charles.Dairman@HoneywellAerospace.com

Project Description: Every engineering project/program needs to have a well-defined plan with costs, timeline, and resources carefully defined. Doing this effectively and accurately is laborious task for the project engineers and program managers. This project team will work to understand the nuances of project planning, the data streams a project engineer has available to them to build sound engineering project plans and embrace the power of AI to arm project engineers and program managers with agentic tools to expedite various tasks associated with the early states of project planning.

Student Learning Goals: Analyze engineering project planning processes by identifying the inputs, workflows, and decision points used to develop engineering estimates, Statements of Work (SOWs), Work Breakdown Structures (WBSs), schedules, and resource plans.
Design an AI-assisted project planning system that supports engineering project managers through automated generation of planning artifacts while maintaining human oversight and decision-making.
Develop secure AI workflows capable of processing proprietary engineering data while ensuring compliance with company security, confidentiality, and data governance requirements.
Integrate heterogeneous enterprise data sources by designing methods to consume and interpret information from systems such as SAP, Accolade, AET, SharePoint repositories, historical SOWs, Integrated Master Schedules (IMSs), and other engineering documentation.
Implement retrieval and reasoning capabilities that allow AI agents to extract relevant historical project information and generate context-aware recommendations for new engineering projects.
Develop conversational AI interfaces that guide engineers through the project planning process by asking clarifying questions when information is incomplete or ambiguous.
Generate engineering planning documents including preliminary cost estimates, Statements of Work (SOWs), Work Breakdown Structures (WBSs), resource plans, and schedule recommendations based on historical project data and user input.
Apply responsible AI engineering practices by evaluating model performance, identifying potential inaccuracies or hallucinations, and implementing validation mechanisms to ensure trustworthy engineering recommendations.
Design maintainable AI software architectures that support future enhancements, additional enterprise data sources, and evolving engineering planning processes.
Validate AI-generated outputs by comparing generated planning artifacts against historical project data and sponsor feedback to measure accuracy, completeness, and usability.
Apply software engineering best practices including modular architecture, version control, documentation, testing, and deployment strategies throughout the project lifecycle.
Communicate technical and business solutions effectively through technical reports, design

reviews, demonstrations, and presentations that explain the AI system's capabilities, limitations, and engineering value.
Collaborate as a multidisciplinary engineering team by integrating expertise in software engineering, artificial intelligence, systems engineering, and project management to deliver a production-ready prototype.

Deliverables: Develop and AI agent for engineering to reduce time spent on generating estimates, creating statements of work (SOWs), work breakdown structures (WBS), etc.
Utilize AI to digest a wide varied of historical data (AET, accolade, SAP, sharepoints, SOWs, IMSs, etc.)
As this is Honeywell proprietary information, the data must not be provided to any 3rd party. (see note)
Historical data should be consumed without requiring significant massaging of the data (agent should prompted for details if needed).
Ideally the input for requesting a new estimate is to provide the SOW and perhaps ask a series or questions prompted by the AI.
Response should of course be the estimate, but also WBS Structure (with compliance matrix), BOEs, even the IMS.

Desired Background: Python, Java, JavaScript, or C#
Experience with Large Language Models (LLMs) and AI frameworks
Prompt engineering and AI agent development
Retrieval-Augmented Generation (RAG)
Database design and SQL
REST APIs and system integration
Git/GitHub version control
Object-oriented software development
Cloud computing (Azure, AWS, or Google Cloud)
Data processing and ETL pipelines
Software testing and debugging
Preferred Experience
Machine learning or generative AI coursework
AI agent development (LangChain, LangGraph, CrewAI, AutoGen, Semantic Kernel, etc.)
Enterprise software development
Document processing and natural language processing (NLP)
Working with structured and unstructured data
Agile/Scrum software development
Technical documentation

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

- US Citizenship requirement

Kids At Risk Action - AI Assisted Podcast Conversion For Child Advocacy

Proposer: Mike E. Tikkanen mike@invisiblechildren.org

Project Contact: Mike E. Tikkanen mike@invisiblechildren.org

Project Description: Kids At Risk Action (KARA) aims to enhance the accessibility and dissemination of its extensive library of child abuse research and articles by converting them into AI-assisted podcasts. This project addresses the challenge of reaching a broader audience through engaging audio content that can be easily shared on social media platforms. By leveraging AI technology, the project seeks to transform written content into compelling podcasts, making critical information more accessible to the public and raising awareness about child abuse issues. The goal is to create a series of podcasts that maintain the integrity and depth of the original research while being engaging and informative for listeners. This project provides learners with the opportunity to apply their skills in AI, audio production, and content adaptation, ensuring the final product is both informative and impactful.

Student Learning Goals: Learners on this project gain hands-on experience translating complex, research-driven content into accessible audio stories for a broad public audience. They deepen their understanding of child abuse and child protection issues while practicing ethical, trauma-informed communication. Students also build practical skills in AI-assisted script generation, audio editing, and podcast production workflows, from drafting and voicing episodes to quality control. Through iterative feedback with KARA, they learn how to preserve the integrity and nuance of original research while adapting tone, structure, and length to keep listeners engaged and informed.

Deliverables: The project deliverables include a series of AI-assisted podcasts based on KARA's existing research and articles. Podcasts should be cover key topics from the original content. The team will also distribute podcasts on various social media platforms. Additionally, a brief report outlining the process, challenges, and recommendations for future podcast conversions will be submitted.

Desired Background: For this KARA AI-podcast project, the ideal student background would combine communication, tech, and issue awareness:

Strong writing and summarization skills, especially turning research or policy material into clear, engaging scripts for general audiences.

Basic familiarity with audio production tools (e.g., Audacity, GarageBand, or similar) and a willingness to learn recording, editing, and level-balancing for podcast-quality sound.

Interest or coursework in AI, digital media, or human-computer interaction, including comfort experimenting with AI text and voice tools and evaluating outputs critically and

ethically.

Background or strong interest in social work, psychology, child welfare, or related fields so they can handle child-abuse content sensitively and maintain accuracy and trauma-informed framing.

Students don't need to be experts in all four areas, but the project is best suited to learners who bring at least two of these strengths and are motivated to grow in the others.

Delivery Mode: In-person

Additional Materials: https://invisiblechildren.org/blog/, https://invisiblechildren.org/karas-critical-conversation-podcasts/

AP Technologies - AI engine tool

Proposer: Lovi Rastogi lovi.rastogi@apmac.ca

Project Contact: Navneet Garg navneet.garg@apmac.ca

Project Description: AP-Tech aims to assist local small and medium businesses (SMBs) in understanding their current challenges and future aspirations. This project involves creating a SaaS product that has AI engine to calculate the productivity factor in $ value and optimize the decision making of organizations that intends to hire their workforce. An example would be hiring a sales team and this tool will help determine the estimated value of each employee that they bring revenue on weekly basis with 95% confidence limits, subject to the strength of the model. This project will provide learners with an opportunity to apply their knowledge of business analysis, s/w development, s/w engineering, s/w architecture, stakeholder engagement, and strategic planning in a real-world context.

Key tasks include:

- Creating an architecture of the SaaS model
- Translate the requirements into BRD.
- Decompose the requirements into individual working modules.
- Create AI models that predict the behavior of future workforce.
- Test the models with 95% confidence limits.
- Create low level documentation and make the system low code development.

Student Learning Goals: Domain expertise and knowledge
Providing specialized knowledge in the project subject area, with industry context.

Skills, knowledge and expertise
Sharing knowledge in specific technical skills, techniques, methodologies required for the project.

Hands-on support
Direct involvement in project tasks, offering guidance, and demonstrating techniques.

Tools and/or resources
Providing access to necessary tools, software, and resources required for project completion.

Regular meetings
Scheduled check-ins to discuss progress, address challenges, and provide feedback.

Deliverables: The deliverables for this project include:

- An architecture (IT stack - technology, security, data, integration, business) of the deployment model
- A report for overall features implemented.
- A set of documentation for future permanent workforce to take over.

These deliverables will provide AP Tech with valuable insights to launch the product with MVP that can help organizations in making informed decisions. There is a high potential of absorbing the right candidate as permanent role with the company on successful launch of the product.

Desired Background: Artificial intelligence
predictive analytics
sas business intelligence (bi)
software as a service (saas)
new product development
product roadmap development
agile project management
organizational leadership

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Arizona State University - AI for Business: Developing Smart Business Negotiation Bots

Proposer: Yalin Wang ylwang@asu.edu

Project Contact: Yalin Wang ylwang@asu.edu

Project Description: This project focuses on building intelligent AI-powered business negotiation bots using large language models (LLMs) and conversational AI technologies. The project continues the success of our previous capstone project (https://showcase.asucapstone.com/survey/10632) and expands it into more advanced AI-driven negotiation systems for real-world business applications.

Students will develop smart negotiation agents capable of understanding business contexts, communicating naturally with users, and generating strategic negotiation responses. The project will explore modern AI technologies such as:

Large language models (LLMs)
Prompt engineering and AI agents
Reinforcemnet learning
Retrieval-augmented generation (RAG)
Multi-agent negotiation systems
Business analytics and decision support

Potential application areas include:

Sales and customer negotiation
Procurement and supplier interactions
Contract negotiation assistance
Business training simulators
AI-powered customer service systems

This project is ideal for students interested in artificial intelligence, natural language processing, business technologies, and human-AI interaction.

Student Learning Goals: Students will gain hands-on experience in:

Building AI-powered conversational systems
Working with large language models and generative AI
Designing negotiation strategies and dialogue systems
Prompt engineering and AI agent development
Full-stack AI application development
Evaluating AI system performance and user interaction

Students will learn how modern AI technologies are transforming business operations and

customer engagement. The project also provides opportunities to work on interdisciplinary problems involving both AI and business strategy.

Participants will strengthen their technical, research, and communication skills while developing portfolio-quality AI applications relevant to today's rapidly growing AI industry.

Deliverables: Expected project outcomes include:

AI-powered business negotiation chatbot prototypes
Conversational workflows and negotiation strategies
Fine-tuned or prompt-engineered LLM systems
User testing and performance evaluation reports
Web-based or API-driven demo applications
Final technical report and project presentation
Well-documented and reproducible code repositories

Advanced teams may further explore:

Multi-agent negotiation simulations
Personalized negotiation strategies
Integration with real business datasets
Research publications or startup-oriented prototypes

Desired Background: Well motivated. Expected but not required experience in machine learning, large language models, and natural language processing.

Delivery Mode: In-person

AP Technologies - AI Product Development

Proposer: Lovi Rastogi lovi.rastogi@apmac.ca

Project Contact: Navneet Garg navneet.garg@apmac.ca

Project Description: The project aims to develop a comprehensive plan for an innovative AI system designed to transform decision-making processes across various industries. The goal is to create a framework that leverages advanced machine learning algorithms to analyze data and provide actionable insights, enabling businesses to make more informed and calculated decisions. This project will focus on identifying key industry challenges and opportunities where AI can be applied to enhance decision-making efficiency and accuracy.

The team will create a product map, project plan, implementation timelines, Business Requirement Documents that outlines the necessary steps for implementation. By applying classroom knowledge of project management, product management, AI and data analysis, learners will gain practical experience in crafting a strategic plan that addresses real-world business needs.

Student Learning Goals: Mentorship

Domain expertise and knowledge
Providing specialized knowledge in the project subject area, with industry context.

Skills, knowledge and expertise
Sharing knowledge in specific technical skills, techniques, methodologies required for the project.

Hands-on support
Direct involvement in project tasks, offering guidance, and demonstrating techniques.

Regular meetings
Scheduled check-ins to discuss progress, address challenges, and provide feedback.

Deliverables: The project deliverables include a detailed development plan that outlines the following deliverables.

Product features and a roadmap for the next 2-3 years.
Create an MVP for this product to be launched.
Craft a project plan for developing the product. Identify all teams.
Develop a repository of Business Requirement Document that will outline the functional and non-functional requirements.

The deliverables will be presented in a professional document and a presentation format, summarizing the findings and recommendations for AP Tech.

Desired Background: Artificial intelligence
predictive analytics
sas business intelligence (bi)
software as a service (saas)
new product development
product roadmap development
agile project management
organizational leadership

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: www.ap-tech.ai

Expeditise LLC - AI Reply Assistant for iOS - Screenshot Sharing, Dynamic Island Access, and Shortcuts

Proposer: Ali Sarabi asarabi1@asu.edu

Project Description: Expeditise LLC, an early-stage startup in the ASU Venture Devil program, builds AI tools for everyday communication. Its AI reply assistant reads the conversation the user is looking at and generates several suggested replies in a chosen tone such as professional, casual, funny, or formal. The Android version does this with a floating button that captures the screen on demand and reads it with OCR.
iOS reaches other apps through Apple's extension points rather than a floating overlay, so the iOS version uses a different interaction model, already decided and set out in the next section. The team will deliver an iOS product built on three surfaces: a share sheet extension that accepts a screenshot of any conversation, a Live Activity that puts the assistant in the Dynamic Island so the user can watch it work and tap through to the replies, and App Intents and Shortcuts for one-tap invocation. The chosen reply goes to the clipboard for a one-tap paste, so the product reaches every app without replacing the keyboard the user already has.

An existing iOS codebase and a working Android version are in place, so the team extends a real product rather than starting from an empty repository. The engineering interest is in the constraints: an app extension runs briefly under a hard memory budget, so OCR, model calls, and history have to be placed carefully across the extension, the main app, the App Group they share, and the backend, and the whole path must stay fast enough that suggestions arrive before the user gives up. Students are strongly encouraged to use agentic AI in their own development workflow while remaining accountable for everything they ship.

iOS Interaction Model
The interaction model below is the design the team is asked to build.
• Screenshot and share. The primary surface. The user captures any conversation and shares it to the assistant through a share or action extension, which works in every app without special access to any of them.
• On-device OCR. The shared screenshot is read with the Vision framework on the phone itself, which keeps message content off the network and removes a round trip from the critical path.
• Dynamic Island quick access. A Live Activity places the assistant in the Dynamic Island once a screenshot is shared, showing progress and giving the user a compact, always-visible place to tap through to the suggested replies without switching apps.
• Clipboard hand-off. The chosen reply is placed on the clipboard for a one-tap paste, so the assistant reaches every app without replacing the keyboard the user already has.
• Shortcuts and App Intents. Actions exposed through App Intents allow invocation from Siri, Spotlight, the Action button, and user automations without opening the app first.
• Extension budget. App extensions get little memory and a short life, so model calls,

history, and anything heavy belong in the main app or the backend, shared through an App Group, with the extension kept thin.
• App Store review. Apps using Live Activities draw review attention, so privacy labels, clear data disclosure, and a Live Activity tied to genuine ongoing work are part of building it correctly.

Student Learning Goals: Students will gain hands-on experience shipping a real iOS product built from multiple app extensions under strict platform, memory, and privacy constraints, and will finish with a TestFlight build they can show.
• Native iOS development in Swift with SwiftUI and UIKit, including app lifecycle, storage, and settings
• Building and debugging iOS app extensions: share and action extensions, and the App Group boundary between them and the main app
• Working within hard platform limits, including extension memory budgets, short execution windows, and background execution restrictions
• On-device computer vision using the Vision framework for OCR, including parsing a screenshot into who said what
• Integrating LLM APIs with streaming, timeouts, retries, and cost control, and designing a prompt system for different response tones
• App Intents, Shortcuts, and ActivityKit Live Activities, including interactive presentations in the Dynamic Island
• Performance engineering with Instruments: latency budgets, memory profiling, and measuring time from tap to first suggestion on real hardware
• Privacy engineering and App Store review: permission flows, privacy labels, data minimization, and preparing a submission that passes
• Using agentic AI as a working development tool while staying fully accountable for the code that ships

Deliverables: 1. iOS Foundation and Main Application
• Bring the existing iOS codebase up to date so it builds and runs on current Xcode and current devices, with the project structure, dependencies, and build steps documented
• Main app covering onboarding, account and authentication, response-style settings, and recent suggestion history, with an App Group as the shared store between app and extensions
• Privacy handling: minimize what leaves the device, state plainly what is sent and why, and avoid retaining message content beyond the active session

2. Screenshot and Share Sheet Path (Primary Surface)
• A share or action extension that accepts a screenshot of any conversation and returns suggested replies, kept within the extension's memory and execution limits
• On-device OCR using the Vision framework, with layout heuristics that separate the conversation into messages and infer who said what

• Response tones (professional, casual, funny, formal) plus rewrite of a draft the user supplies, consistent with the Android version
• One-tap copy of the chosen reply to the clipboard, so the user pastes it straight into the conversation

3. Dynamic Island Access and Quick Invocation
• A Live Activity that appears when a screenshot is shared, shows progress in the Dynamic Island, and lets the user tap through to the suggested replies without leaving the app they are in
• Interactive elements in the expanded presentation where the platform allows them, so common actions such as copying a reply or switching tone are reachable directly
• App Intents and Shortcuts support for invocation from Siri, Spotlight, the Action button, or an automation, plus correct fallback presentation on devices without a Dynamic Island

4. AI Integration, Backend, and Performance
• LLM integration with streaming responses, timeouts, retries, and cost controls, sharing the prompt and response-style system with the Android product where practical
• A stated latency budget with measurements on real devices from tap to first visible suggestion, plus caching and prefetching where they help
• An optional on-device model path for supported hardware, with a cloud fallback and a documented comparison of quality, speed, and cost

5. Testing, App Review, and Release
• Unit and UI tests, a device matrix that includes phones without a Dynamic Island, and memory-pressure tests for the share extension
• A TestFlight build, privacy nutrition labels, clear disclosure of what leaves the device, and a review-risk checklist covering the Live Activity
• Documentation of the architecture, the extension boundaries, the prompt system, and the platform limits that shaped the design

Desired Background: Students should have experience or interest in the following areas:
• Swift and iOS development with SwiftUI or UIKit; prior experience with app extensions is a strong plus but can be learned during the project
• Access to a Mac capable of running current Xcode and to a physical iPhone for testing, which this project genuinely requires
• REST APIs, JSON, authentication, and basic backend concepts
• Experience with LLM APIs, and interest in prompt design for different tones and contexts
• Interest in computer vision and OCR, image handling, and text parsing
• Comfort with debugging and profiling tools, and patience for platform restrictions and App Store review
• Knowledge of agentic AI is highly recommended
• Interest in startups and in shipping a real app that users install

An Android version of this product already exists and an iOS codebase is in place, so the team joins a live product rather than a blank page.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: https://expressreply.com

MentaMorph Inc - AI Scenarios & Personalization Engine

Proposer: Susan Meredith susan@mentamorph.com

Project Contact: Susan Meredith susan@mentamorph.com

Project Description: This project develops the adaptive intelligence layer of MentaMorph, enabling AI-assisted scenario generation and personalized learning pathways. It integrates gameplay data and user behavior to dynamically adjust scenarios and difficulty.

Student Learning Goals: Students will explore applied AI in a real product context, learning how to design systems that adapt to users and balance automation with educational goals.

Deliverables: • Scenario generation framework
- Multiple playable scenarios
- Personalization/adaptive logic
- Integration with data and gameplay systems

Desired Background: • Python or similar programming
- Familiarity with AI/ML concepts
- Strong problem-solving skills

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

GenLedge - AI Software developer - Autonomous Finance

Proposer: Kulwant Yadav Kulwant@genledge.ai

Project Description: Imagine a finance department that never sleeps. MagicMode is building the world's first autonomous finance organization — persistent AI agents that independently process invoices, categorize transactions, and manage month-end close with less human intervention. As an intern, you will work directly with the founder to build real agent workflows using the Anthropic Claude API, FastAPI, and live accounting integrations with QuickBooks and Xero. Your work goes into a real product serving real businesses.

Student Learning Goals: Agentic AI development, LLM tool-use, OAuth 2.0 accounting integrations, full-stack Python and TypeScript, and startup product development.

Deliverables: Functional AI Agent Prototype

MCP Server Integration - Develop or extend a Model Context Protocol (MCP) server that connects the agent to a live accounting data source e.g. QuickBooks Online or Xero

FastAPI Backend Endpoint(s) - Implement backend routes that support agent task orchestration

Technical Documentation - Deliver a concise developer handoff doc covering architecture decisions, setup instructions,integration process and known limitations, so the codebase is ready for the next contributor.

Demo Presentation - Deliver a live walkthrough demonstrating the agent's capabilities, the accounting workflows it automates, and its measurable impact on finance team efficiency.

Desired Background: Computer Science & Computer Engineering

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

MyEdMaster LLC - AI-based marketing software

Proposer: John Leddo john@myedmaster.com

Project Contact: John Leddo john@myedmaster.com

Project Description: The goal is to create an AI-based marketing program that accepts a product description of what the company wants to sell and a description of the target market. It then does web scraping to identify potential customers and their contact information. Once these are identified, it sends out marketing emails to the customers using the company's email address as the sender.

Student Learning Goals: Students will learn to create a practical tool for use in a business that employs AI and was created with the assistance of AI code generation tools.

Deliverables: A prototype of the tool

Desired Background: large language models, web scraping, Python

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

ATOMS - AI-Enabled Robot Control: From Specifications to Autonomous Manipulation

Proposer: shiv patel shiv.patel@gmail.com

Project Description: This project uses systems engineering principles to control a robot arm. Students write a formal specification for a ball-sorting task — requirements, architecture diagrams, test cases — then use that specification to drive three different AI approaches to making the robot perform the task autonomously.

The emphasis is on specification quality, not code volume. A well-written spec enables AI tools to generate code, defines how to evaluate LLM-driven control attempts, and sets the discipline for collecting training data. Some coding is required to get things working, but the core skill being developed is writing specifications precise enough that AI can act on them.

Three approaches, same spec, same acceptance criteria — students compare the results and learn which AI tools fit which problems.
Hardware and tutorials are provided. No prior robotics experience required.

Lerbot and Nvidia Jetson are provided

Student Learning Goals: How to write specifications that are precise enough for AI systems to act on. Students will learn requirements engineering, architecture decomposition, traceability, and formal test case design — then see firsthand how specification quality directly determines whether an AI-generated system works or fails. Along the way, students get hands-on experience with a physical robot arm, NVIDIA Jetson edge computing, computer vision model training, LLM tool integration, and imitation learning. But the through-line is always the spec — it's the document that ties everything together and the skill that transfers regardless of which AI tools come next.

Deliverables: System specification document — requirements, architecture, test cases, and traceability matrix for a manipulation task
Working autonomous manipulation — robot arm performs the specified task using a trained imitation learning policy, building on prior team's progress
Demonstrated spec-to-behavior loop — modify the specification (e.g., change target object, sorting criteria, or placement location), retrain, and show the robot's behavior changes accordingly
Trained models and demonstration datasets published to Hugging Face Hub
Test results evaluated against each specification revision's acceptance criteria
Demo video showing how specification changes drive different robot behaviors

Desired Background: Systems Engineering, Computer Science, Computer Engineering, Embeded Systems

Python

Comfortable with linux command line (required - Jetson runs Ubuntu)
Machine Learnings basics (nice to have)

Most important is a genuine desire to learn and excitement for AI and robotics. This is an intense, hands-on project — you'll work with physical hardware, train real models, and push into technologies that are months old, not years old. You will learn a lot. But this is the wrong project to pick if you're looking to coast for a grade.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://github.com/huggingface/lerobot

MyEdMaster LLC - AI-generated course

Proposer: John Leddo john@myedmaster.com

Project Contact: John Leddo john@myedmaster.com

Project Description: Students will use AI to create a program that accepts a description of a desired course and any certification requirements or tests the user will have to take upon completion of the course. The program will conduct an assessment of what the student already knows about the subject matter the course is intended to cover and will create, in real time, a personalized version of the course that matches what the user still needs to learn and the user's learning style while insuring at the end, the course will have covered all material necessary so that the user can pass what ever certification test s/he needs to. We have some existing software that can be used in the current project that assesses a user's current level of knowledge.

Student Learning Goals: Students will learn how to use AI code generation tools as well as create software that customizes content to user's as opposed to providing each one with the same content.

Deliverables: a prototype of the software

Desired Background: Students will be familiar with code generation tools, large language models and AI/machine learning

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Being Love Incorporated - AI-Powered Book Creation and Publishing System

Proposer: Kimberlee Klein eelrebmik@gmail.com

Project Contact: Kimberlee Klein eelrebmik@gmail.com

Project Description: A previous student team successfully transformed a 20-year digital archive containing thousands of blog posts into a searchable knowledge base. The next phase is to build an AI-powered publishing system capable of transforming that archive into books.

Students will design and develop a system that can analyze the existing archive, identify related themes and content, distinguish the author's original commentary from quoted or referenced material, organize relevant content into book and chapter structures, and use AI to assist in creating coherent manuscripts while preserving the author's voice and ideas.

The goal is not to create a single book manually. The goal is to build a repeatable software pipeline that can be used to generate multiple books from the archive over time, with appropriate human review and editorial controls.

Student Learning Goals: Students will gain hands-on experience designing and developing a real-world AI-assisted software system using a large existing dataset. They will explore practical applications of artificial intelligence, natural language processing, data organization, search and retrieval, prompt design, and workflow automation.

Students will also gain experience evaluating AI-generated outputs, addressing challenges such as source attribution and preservation of author voice, designing human-review checkpoints, and integrating multiple components into a functional end-to-end system.

The project provides an opportunity to move beyond individual AI prompts and experiment with how AI can be incorporated into a scalable software pipeline that solves a real-world information management and publishing problem.

Deliverables: The primary deliverable will be a functional prototype of an AI-powered book creation system that works with the existing searchable archive.

Expected deliverables include:

* A system for identifying and grouping related content from the archive by topic or theme
* AI-assisted identification and separation of original author commentary from quoted or referenced material
* Generation of proposed book concepts and chapter structures based on archive content

* An AI-assisted workflow for transforming selected content into coherent manuscript drafts while preserving the author's voice
* Human review and approval checkpoints throughout the book-creation process
* A functional end-to-end prototype demonstrating the creation of at least one book or substantial manuscript from the archive
* Source code and technical documentation
* Documentation of the AI models, prompts, architecture, and workflows used
* Recommendations and documentation allowing future teams to improve and scale the system to produce additional books

Desired Background: Students with a background or interest in computer science, software engineering, artificial intelligence, machine learning, natural language processing, or full-stack development would be well suited for this project.

Experience with APIs, databases, search/retrieval systems, AI/LLM integration, and workflow automation would be beneficial. Students do not need publishing experience, but an interest in solving complex problems involving large amounts of text-based data would be valuable.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: https://globalwarming-arclein.blogspot.com/

Genovation AI - AI-Powered PLC & Industrial Robot Code Generation Platform

Proposer: Faisal Ahmed faisal.a@genovation.ai

Project Contact: Jamal Hossain contact@genovation.ai

Project Description: Genovation AI is building an AI assistant that helps industrial engineers generate, review, and export production-ready PLC ladder-logic code from plain English. We're hiring one team of 4 to 5 ASU CS/CSE seniors to build it over two semesters, with a paid Anthropic Claude API key, a private GitHub org, cloud infrastructure, a 15+ year industrial-robotics advisor, and a real automotive pilot partner waiting to use what you ship.

This is not a toy project. The software you write is intended to run in real factories. We've written this posting so you know exactly what you're getting into before you apply, and so the right people apply.

We want a Software/AI Developer with PLC or Industrial Automation experience, preferably with Python or C++, LLM integration, IEC 61131-3 languages, and robotics or control-system knowledge.

Student Learning Goals: Industrial system integrators spend 150 to 200 engineering hours per project hand-writing PLC code in ladder logic, the code that runs conveyors, robotic welders, assembly lines, and safety systems. Skilled programmers bill $75 to $100/hour, and a single mid-size integrator can spend $300K to $400K/year on this labor alone. No purpose-built AI tool exists for this workflow.

The opening: Rockwell Studio 5000, used by ~90% of North American industrial automation, imports and exports programs as L5X, which is plain XML. That means ladder logic is structured text an AI can parse, learn from, generate, and validate. That's the technical foundation that makes this feasible, and what your team will build on.

- You'll have: a real industry pilot partner (a Tier 1 automotive integrator) keeping the work grounded;
- You'll have: a domain expert advisor with 15+ years of FANUC and ABB robotics experience reviewing your output weekly;
- You'll leave with: a portfolio piece in three of the highest-demand fields in North America: AI engineering, industrial software, and robotics.

Deliverables: The two-semester roadmap

Semester 1 (Months 1 to 4): Foundations

- An L5X parser and generator that reads real files, structures them, and regenerates valid L5X (verified by Studio 5000 import)
- A baseline AI pipeline: natural language to L5X using the Claude API with structured prompting
- Vector database setup: ingest sample ladder-logic programs, embed, and retrieve relevant examples
- A basic web interface (text input, API call, output display)
- An accuracy baseline report: 30 test prompts scored by the domain expert

Semester 2 (Months 5 to 8): The real product
- A full context-aware RAG pipeline using the client program library
- A visual ladder-logic renderer (real rungs, contacts, coils, branches)
- A validation rules engine (declared tags? valid schema? E-stop interlocks present?)
- An approval workflow with full audit logging
- Clean L5X export, a metrics dashboard, and pilot-deployment documentation

Desired Background: You'll join a team of 4 to 5. Each role below maps to a distinct skill set; depending on final team size, some people will wear two hats. Read the role that fits you and be honest about the "experience you need" list. It's what we'll talk through in your interview.

Role 1: AI / LLM Integration Lead

1 seat · CS · the AI heart of the product

You own the part of the system that turns plain-English requests into valid industrial code. If LLMs and retrieval pipelines are your thing, this is the seat where your work is most visible.

What you'll own
- Anthropic Claude API integration and prompt engineering
- Design and implementation of the RAG (Retrieval-Augmented Generation) pipeline
- Framework setup with LangChain or LlamaIndex
- Accuracy measurement, evaluation harnesses, and iteration

Experience you need
- You have built and shipped at least one real LLM-powered application, not just a tutorial
- Hands-on with LangChain, LlamaIndex, or direct API prompt chaining
- Strong Python: async code, API error handling, parsing JSON/XML responses
- You can explain what RAG is, and when to use zero-shot vs. few-shot vs. retrieval, without notes
- You have worked with embeddings and semantic search in some form

Bonus experience (not required)
- Prompt-optimization or fine-tuning workflows
- Familiarity with engineering or industrial domains
- You've built a chatbot or code assistant of any kind

What you'll walk away knowing

Production-grade LLM integration, RAG architecture, and how to measure and improve generation quality on a task where correctness is non-negotiable.

Role 2: Backend Engineer
1 seat · CS · solid Python required
You build the spine of the platform and own its single most technically critical component: the L5X parser and generator that reads and writes real ladder-logic files.
What you'll own
- FastAPI backend architecture and API endpoints
- PostgreSQL database design
- The L5X XML parser and generator, the most critical piece of the system
- Authentication and session management

Experience you need
- You have built and deployed a REST API somewhere real, not just localhost
- Strong Python: OOP, error handling, working with external libraries
- Experience parsing and generating XML or JSON in Python
- Basic database design: you can write SQL and understand normalization
- Git proficiency: clean commit history, PR workflow

Bonus experience (not required)
- FastAPI specifically (not just Flask/Django)
- lxml, ElementTree, or similar XML libraries
- Any exposure to industrial or binary file formats
- Docker

What you'll walk away knowing
How to design a backend around a strict, real-world file format, validate generated output rigorously, and build APIs that fail gracefully when an upstream AI service goes down.

Role 3: Frontend Engineer
1 seat · CS or CSE · React required, SVG non-negotiable
You build the interface engineers actually trust. The centerpiece is a visual ladder-logic renderer that draws contacts, coils, and branches as real diagrams, not raw XML.
What you'll own
- The React web application
- SVG-based ladder-logic renderer (rungs, contacts, coils, branches)
- Approval workflow UI: approve / edit / reject, all logged
- Metrics dashboard and L5X export / download

Experience you need
- You have shipped a React app with real complexity, not a to-do list
- You can build SVG graphics programmatically in React: this is non-negotiable for the renderer
- Comfortable with state management (useState, useReducer, or Redux)
- You can integrate REST APIs with fetch or axios
- Clean, readable component architecture

Bonus experience (not required)

- D3.js or similar data-visualization experience
- Rendering technical diagrams (flowcharts, circuits, anything similar)
- TypeScript
- UI/UX sensibility: this tool must feel trustworthy to AI-skeptical engineers

What you'll walk away knowing

How to render complex technical diagrams in SVG, design state for review/approval flows, and build interfaces that earn trust from expert users.

Role 4: Robotics / Embedded Systems Specialist

1 seat · CSE strongly preferred · robotics exposure required

You extend the platform from PLCs into industrial robots, generating FANUC TP and ABB RAPID code. You'll reverse-engineer proprietary formats and make sure generated motion is safe.

What you'll own

- FANUC TP and ABB RAPID code-generation module (Semester 2)
- Research and documentation of robot programming file formats
- Integration of robot code generation into the main pipeline
- Hardware/software interface docs for the pilot client; validation of robot path outputs

Experience you need

- Coursework or project experience in robotics, embedded systems, or industrial automation
- Familiarity with at least one robot environment (FANUC, ABB, ROS, or equivalent)
- You can read and understand structured robot code, even if not expert-level
- Python proficiency
- Ability to research undocumented formats and reverse-engineer their structure

Bonus experience (not required)

- You've programmed a FANUC or ABB robot, even in simulation
- ROS experience
- Motion planning, kinematics, or simulation tools (RoboGuide, RobotStudio)
- Manufacturing or automotive exposure

What you'll walk away knowing

How AI-generated motion gets validated for real-world safety, how proprietary robot formats are structured, and how a software pipeline reaches onto a factory floor.

Role 5: Full-Stack Generalist / DevOps & Testing Lead

1 seat · CS · versatile required

You're the glue. You make four separate services run as one system, keep quality high with real tests and CI, and own the path from a developer's laptop to a client's private server.

What you'll own

- CI/CD pipeline (GitHub Actions)
- Vector database setup and management (Chroma local to Pinecone production)

- System integration: making all components talk to each other
- Automated testing (pytest backend; Playwright or Cypress frontend)
- Deployment and infrastructure (Docker, cloud hosting); technical docs and README

Experience you need

- Experience setting up and running Docker containers
- You've written real automated tests, not just 'I know what unit tests are'
- Functional across both frontend and backend (not expert in either)
- Git workflow discipline: branching, PR reviews, release tagging
- You've deployed a full-stack app to a cloud provider (AWS, GCP, Azure, Heroku, Railway, or anything)

Bonus experience (not required)

- GitHub Actions or similar CI/CD
- Vector databases specifically
- Worked in a team of 3+ with structured workflow
- Secrets / API-key management awareness

What you'll walk away knowing

How to integrate a multi-service system, enforce quality through automated testing and CI, and deploy software into a restricted, real-world client environment.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

General Dynamics Mission Systems - AI-Powered Visual Inspection System for Rescue 21 Remote Fixed Facilities Preventive Maintenance.

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Daren Cole daren.cole@gd-ms.com

Project Description: General Dynamics Mission Systems (GDMS), as the primary lifecycle sustainment contractor for the U.S. Coast Guard's Rescue 21 system, would like to create an AI-powered Visual Inspection and Damage Assessment (VIDA) platform to modernize the annual preventative maintenance process at Remote Fixed Facilities (RFFs). During each annual preventative maintenance visit, field technicians capture an average of 340 photographs documenting the condition of site infrastructure — including communication towers, VHF/UHF and direction-finding antennas, electronics shelters, power systems, generators, solar panels, and structural support components. Given the remote and often harsh coastal and maritime environments in which these sites operate, exposure to salt air, extreme weather, and the elements make corrosion, structural fatigue, and environmental damage a persistent and costly challenge. The VIDA platform will ingest these photo sets and apply computer vision and deep learning models trained specifically on Rescue 21 infrastructure to automatically detect and classify degradation across all site components.

The AI system will leverage multimodal large language model (LLM) capabilities combined with fine-tuned image recognition models to analyze each photograph for a range of defects including surface corrosion, rust formation, paint deterioration, structural cracking, antenna deformation, loose or missing hardware, water intrusion, vegetation encroachment, and damage to power systems and cabling. Solution should be based on open-source software. Each detected issue will be tagged, geolocated to a specific site asset, and assigned a severity rating — ranging from Informational to Critical — based on the degree of degradation observed and its potential impact on site operability. The system will also compare current-year photos against historical baselines from prior maintenance cycles, enabling trend analysis to identify components that are progressively deteriorating and may require proactive intervention before the next scheduled visit, reducing the risk of unexpected outages that could compromise maritime search and rescue communications.

Upon completing its analysis of all 340 photos per site, the VIDA platform will automatically generate a structured, prioritized Site Condition Report (SCR) that is routed to the appropriate GDMS engineering, quality, and field service personnel. The report will highlight all flagged items with annotated photo evidence and recommended corrective actions. By replacing what was previously a time-intensive manual photo review process with an AI-driven workflow, GDMS anticipates a significant reduction in the time from site visit to corrective action initiation, improved consistency and accuracy in defect identification across all RFF locations nationwide, and better overall mission readiness of the Rescue 21 network — directly supporting the Coast Guard's life-saving maritime

distress response mission.

Student Learning Goals: Deep dive into AI and application to real life problems.

Deliverables: Project deliverables are phased through the two-semester sequence and include:

- First semester would be to generate requirements/analysis and to develop the ability to load, search, and identify issues found in site photos, focusing on corrosion, damage, misalignment, or missing items.
- Second semester develop the tool further to enhance the search capability and to generate a Site Condition Report (SCR).

Desired Background: Anyone wanting to learn more about AI tools and how to apply them to real world applications

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Qualcomm Inc. - AI100 powered Prompt-to-Plan Travel Planner with Visual Itineraries and Personalized Recommendations

Proposer: Emma Lacey elacey@qti.qualcomm.com

Project Contact: Rohit Mann rohimann@qti.qualcomm.com

Project Description: Build a travel planning system where a user enters a natural-language prompt (e.g., "I want a beach vacation for 5 days, relaxed pace, great food"), and the system generates:
a structured itinerary (day-by-day),
a visual itinerary board (images per day/activity) using an image model, and
personalized recommendations using embeddings + retrieval based on a user's past trips/preferences.
The capstone emphasizes a practical, end-to-end pipeline: preference capture → retrieval/ranking → itinerary generation → visual presentation → iteration/refinement.

Student Learning Goals: End-to-end applied GenAI system building: retrieval + ranking + generation + UI.
Practical experience with embeddings, vector search, evaluation, and safety constraints.

Deliverables: Final system and deliverables published on Github

Desired Background: Skills Needed: Python, building REST APIs (FastAPI/Flask or Node), Basic ML concepts (embeddings, similarity search, ranking), Web app development, Data ingestion + normalization
Nice to have: LLM orchestration (prompting, tool/function calling, evaluation), Vector DB experience, UI/UX for itinerary visualization

Delivery Mode: In-person

Kids At Risk Action - AI-assisted search and triage over KARA's content

Proposer: Mike E. Tikkanen mike@invisiblechildren.org

Project Contact: Mike E. Tikkanen mike@invisiblechildren.org

Project Description: Goal: Build an AI-powered interface where users describe a situation and receive curated KARA resources (articles, podcasts, external links) with clear, safety-first disclaimers.

Why Fulton students fit: Leverages student skills in AI agents, NLP, retrieval systems, and system integration that EPICS and other ASU programs emphasize.

Multi-semester growth:

Phase 1: Semantic search over KARA content with basic filters.

Phase 2: AI-generated "reading paths" and summaries for different audiences.

Phase 3: Lightweight personalization, analytics on usage gaps, and governance tools.

Student Learning Goals: Students on the KARA interactive storytelling tool would gain a long-term, real-client design and development experience that blends technical skills, human-centered design, and civic learning.

Authentic, multi-semester project work
Work in a team to define, design, build, test, and iteratively improve a real system for a community partner, rather than a short, closed-ended class assignment.

Experience joining a project midstream (or handing it off), learning how to work with legacy code, documentation, and evolving requirements over multiple semesters.

Technical and design growth
Apply and deepen skills in software development, data modeling, and interactive/UX design while solving an ill-defined, real-world problem instead of a textbook exercise.

Practice human-centered design: talking with stakeholders, incorporating user feedback, and making design decisions that balance clarity, ethics, and emotional impact in a sensitive domain.

Professional, team, and civic learning
Build teamwork and project-management skills on a multidisciplinary team, including communication with a "client" (KARA) and coordination of roles, milestones, and deliverables.

Reflect on the social impact of their work, developing empathy, civic responsibility, and an appreciation of how engineering skills can directly support vulnerable children and families.

Deliverables: Expected deliverables center on a working storytelling tool plus the artifacts needed for ethical use and multi-semester continuation.

Product and content deliverables
Functional interactive web experience that walks users through one or more anonymized CPS/foster-care pathways with clickable choices, timelines, and explanations.

Story/data schema and a small library of KARA-vetted narratives, showing how events, decisions, and outcomes are structured and stored for future expansion.

Design, ethics, and testing deliverables
UX package with wireframes, user flows, screen mockups, and rationale for accessibility and trauma-informed design choices (language, warnings, exits).

Short testing report summarizing at least one round of user or stakeholder feedback, key usability/impact findings, and prioritized improvements.

Technical and handoff deliverables
Code repository with readme, architecture diagram, setup/deployment instructions, and clear guidance on how to add or modify storylines.

Handoff brief (slides or short report) outlining current status, known limitations, and recommended next steps for future EPICS or capstone teams working with KARA.

Desired Background: For this KARA interactive storytelling tool, the ideal is a multidisciplinary team with strong software, UX, and social-issue awareness.

Technical background
Students in computer science, software engineering, informatics, or related computing majors to build the web app, design story data structures, and handle integrations/analytics.

Comfort with basic web frameworks, version control, and working in a team-based project environment similar to other EPICS and eProjects offerings.

UX and human-centered background
Students with user-experience or design focus (e.g., Graphic Information Technology – UX, Human Systems Engineering – UX) to lead research, information architecture, visual design, and accessibility.

Experience or interest in human-centered design methods: interviews, journey mapping, prototyping, and iterative usability testing with a community partner.

Social issue and communication background
Work with KARA and if possible, at least one team member with interest or coursework in social work, psychology, criminology, family studies, or related social sciences to help interpret CPS/foster-care processes and keep narratives respectful.

Strong written/oral communication skills across the team to work with KARA as a client, document decisions, and write clear, trauma-informed narrative content and disclaimers.

Delivery Mode: In-person

AloWish - AloWish: Accessible Travel Planning and Risk Assessment Platform

Proposer: Hossein Shahid hossein@alowish.com

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: AloWish is developing a digital travel platform designed to help travelers with disabilities, health conditions, and accessibility needs make safer and more personalized travel decisions.
The platform will combine traveler information with destination, accommodation, transportation, and service-provider data to assess potential travel risks and recommend suitable options. Core functionality will include traveler profiles, personalized risk assessment, destination insights, search and filtering, and matching travelers with appropriate travel services.
Students will design and develop the AloWish application from the ground up, including the user interface, backend, database, and core recommendation functionality. The project may also incorporate external APIs and data sources to support destination and travel-service information.
The final solution will demonstrate a functional end-to-end travel-planning experience in which users can enter their individual needs and preferences and receive relevant destination information, risk insights, and personalized travel recommendations.

Student Learning Goals: - Design and develop a real-world software product from initial requirements through implementation.
- Develop a full-stack web application including front-end, backend, and database components.
- Design data structures for traveler profiles, destinations, accessibility information, and travel services.
- Develop recommendation, matching, and risk-assessment functionality.
- Integrate external APIs and third-party data sources.
- Apply accessibility and inclusive-design principles throughout the application.
- Gain experience with authentication, privacy, secure data handling, testing, and debugging.
- Collaborate directly with an industry startup while following professional software engineering and product-development practices.

Deliverables: - Define the technical architecture and application requirements for the AloWish platform.
- Design and develop the AloWish front-end user experience.
- Develop the backend services and database architecture.
- Implement traveler profiles capturing relevant health, accessibility, and travel preferences.
- Develop destination and service-provider search, filtering, and recommendation functionality.

- Implement a risk-assessment and matching system that considers traveler needs and destination information.
- Integrate approved external travel, destination, transportation, or accessibility data sources where appropriate.
- Apply accessibility, authentication, privacy, and secure data-handling practices.
- Perform functional, usability, integration, and system testing.
- Deliver a functional end-to-end prototype or deployable application.
- Provide source code, database documentation, API documentation, setup instructions, testing results, and technical documentation.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:
- Computer Science
- Software Engineering
- Full-Stack Web Development
- Front-End Development
- Back-End Development
- Database Design
- Artificial Intelligence / Machine Learning
- Recommendation Systems
- Python
- JavaScript / TypeScript
- React or similar front-end frameworks
- REST APIs and third-party API integration
- Cloud Computing
- Authentication and Application Security
- UI/UX Design
- Web Accessibility
- Software Testing and Debugging

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

Aptamer Therapeutics - Aptamer Therapeutics: Connected Biosensor Platform for Personalized Medication Management

Proposer: Ehsan Daneshgar ehsan@launchpath.ca

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: Aptamer Therapeutics is developing a smart healthcare platform that combines a mobile application, embedded biosensor technology, and artificial intelligence to support personalized medication dosage recommendations based on patient health information and physiological data. The system is intended for use with all sorts of biomarkers (i.e., Glucose, caffeine, hemoglobin).

An existing software platform has already been developed, including user authentication, patient profiles, medical history management, dosage assessment workflows, Bluetooth communication, and backend services. Students will begin by evaluating the current platform through functional, technical, and usability testing to identify gaps, areas for improvement, and opportunities to enhance the overall user experience and system performance. Based on their findings, students will propose enhancements and implement approved improvements alongside the project's primary development objectives.

The project will then focus on advancing the hardware integration, enabling real-time glucose monitoring, improving communication between the biosensor and mobile application, and enhancing the intelligence of the dosage recommendation system using both sensor data and patient health history. Students will work across mobile development, backend services, embedded systems, Bluetooth communication, AI integration, and data visualization to further develop the platform into a more complete, reliable, and scalable healthcare solution.

Student Learning Goals: Students participating in this project will have the opportunity to:

-Extend and enhance an existing digital health platform used for personalized medication management.
-Develop and maintain mobile application features using React Native and modern software development practices.
-Design and enhance backend APIs, database functionality, and cloud-based services.
-Integrate embedded hardware, biosensors, and Bluetooth Low Energy (BLE) communication with a mobile application.
-Gain hands-on experience working with microcontrollers, sensor data acquisition, and hardware/software integration.
-Explore AI-assisted decision support and health data analysis within a real-world healthcare application.
-Collaborate with industry sponsors while following agile development and software engineering best practices.

-Participate in system testing, debugging, validation, technical documentation, and deployment activities.

Deliverables: -Evaluate the existing platform and document opportunities for improvement.
-Recommend and implement approved enhancements to functionality, usability, and system performance.
-Integrate the glucose monitoring hardware with the existing mobile application.
-Develop and validate reliable Bluetooth Low Energy (BLE) communication between the biosensor and the application.
-Enhance embedded firmware to support real-time physiological data collection.
-Improve the AI-assisted dosage recommendation workflow using sensor and historical patient data.
-Expand health analytics, visualization, and patient history features.
-Improve backend APIs, database functionality, and overall system reliability.
-Conduct end-to-end testing and validation of the integrated hardware and software platform.
-Deliver updated technical documentation, deployment guides, and user documentation.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:

-Mobile application development (React Native preferred)
-JavaScript / TypeScript
-Backend development (Node.js, REST APIs)
-Database systems (MongoDB or similar)
-Embedded systems or microcontroller programming
-Bluetooth Low Energy (BLE) communication
-Artificial intelligence or machine learning fundamentals
-UI/UX development
-Software testing and debugging

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

ASU - Capstone - ASU Capstone Judging App

Proposer: Satya parupudi parupudi@asu.edu

Project Description: Capstone judging app

Student Learning Goals: Php, iOS, android

Deliverables: Full stack app, mobile apps

Desired Background: Any

Delivery Mode: In-person

Required Agreements:

- ["None"]

SCAI - ASU Capstone Project Manager

Proposer: Dr. Satya Parupudi parupudi@asu.edu

Project Description: Website designed to keep capstone teams in sync with a shared digital calendar, monitor and oversee job progress with time and activity updates. Allows sponsors to view real-time what tasks teams are working on.

Student Learning Goals: Web development, database configuration, process management

Deliverables: Build out phase one of the project

Desired Background: Frontend and backend software development is highly preferred

Delivery Mode: In-person

ASU School of Computing and Augmented Intelligence (SCAI) - ASU SCAI Capstone Sponsor Integration Platform

Proposer: Steven Osborn sdosborn@asu.edu

Project Contact: Steven Osborn sdosborn@asu.edu

Project Description: The ASU SCAI Capstone Sponsor Integration Platform is an existing web application designed to centralize and automate the management of the SCAI Capstone program. The platform currently supports sponsor project proposals, student project preferences, project assignments, seminar submissions, judge applications, project catalogs, email communication, editable website content, and administrative workflows.

The Fall 2026 team will continue development of the existing platform alongside the current Capstone 2 team. Students will work within an established production codebase to improve existing features, complete new functionality, address technical debt, strengthen security, and further automate administrative processes.

The incoming team will also help provide coverage for areas currently owned by individual team members and contribute to areas where additional development expertise is needed.

Student Learning Goals: Students will gain experience developing and maintaining a real production web application used to support the ASU SCAI Capstone program. They will work with an existing codebase, database, APIs, authentication and security systems, automated email workflows, and administrative tools.

Students will gain practical experience with software development practices including Git/GitHub collaboration, requirements gathering, sprint planning, code reviews, testing, debugging, database design, UI/UX development, security, and deployment. They will also collaborate directly with the existing Capstone 2 development team and project stakeholders.

Deliverables: Continued development and maintenance of the Capstone Sponsor Integration Platform
Improvements to sponsor, student, seminar, and judge submission workflows
Enhancements to project assignment and matching functionality
Improvements to administrative dashboards and reporting
Additional automation of repetitive Capstone administration processes
Security, reliability, and performance improvements
UI/UX and mobile usability improvements
Testing and documentation for implemented features
Production-ready code integrated into the existing application

Desired Background: We are looking for students who can complement the existing development team and take ownership of different areas of the platform.

Desired experience includes:

Full-stack web development
JavaScript/TypeScript and React/Next.js
SQL and relational databases
REST APIs and third-party integrations
UI/UX and responsive frontend development
Authentication and web application security
Automated testing and debugging
Git/GitHub collaborative development

Students do not need experience in every area. A team with a mix of frontend, backend, database, security, testing, and UI/UX experience would be ideal.

Delivery Mode: In-person

Additional Materials: N/A

SCAI - ASU SCIA - ASU SCIA - ABET Fullstack Development

Proposer: Steven Osburn sdosburn@asu.edu

Project Contact: Steven Osburn sdosburn@asu.edu

Project Description: Develop the core data infrastructure that extracts, transforms, and secures information from Canvas and ABET sources, providing a stable backend for reporting and analysis.

Student Learning Goals: Real-World Data Engineering Practice
Students learn how to design and implement a full backend data infrastructure, from data extraction to transformation to secure delivery.
They experience real engineering constraints such as API rate limits, data inconsistencies, schema evolution, and reliability requirements.
2. Working with Production APIs (Canvas)
Students gain hands-on experience authenticating to and consuming a large-scale REST API. They learn how to retrieve structured data (courses, assignments, rubrics, outcomes) and handle pagination, errors, retries, and efficient querying.
3. Web Scraping and Data Normalization
Students learn to build a robust scraping pipeline for ABET criteria updates. They gain experience with HTML parsing, change-tracking, normalization of unstructured text, and scheduling recurring jobs.
4. Database Design and Data Warehousing
Students learn to design a normalized MySQL database, including: Entity-relationship modeling, Object-Relational Mapping (ORM), and maintaining data integrity/access control
They experience how backend schemas support analytics and reporting requirements.
5. ETL (Extract, Transform, Load) Pipelines
Students learn professional ETL practices such as: Automated nightly updates, validating data quality, schema mapping (e.g., Canvas outcomes → ABET outcomes), logging, monitoring, and handling failed jobs
6. Secure Backend API Development
Students gain experience designing and implementing RESTful APIs that provide controlled access to processed data for another team. They learn about endpoint design, authentication, authorization, and secure data delivery.
7. DevOps, CI/CD, and Deployment
Students practice:
Maintaining existing CI/CD pipelines for automated testing and deployment
Using Docker for containerized environments
Monitoring system uptime and performance
They learn how professional software teams automate and maintain reliable backend services.
8. System Security and Compliance
Students develop understanding of:
Encryption (in transit and at rest)

Role-based access control
Secure handling of tokens and credentials
Compliance considerations related to education data (FERPA-adjacent thinking)
9. Agile Project Management & Team Collaboration
Students experience real-world team workflows including:
Sprint planning and task breakdown
Technical documentation writing
Cross-team API coordination (sharing data across teams)
They learn to communicate effectively within specialized roles (API dev, DBA, DevOps, etc.).

Deliverables: Tasks
- Frontend: You create web routes with Symfony's defined MVC Architecture
- Web App: You manage a PHP web application that communicate with FastAPI endpoints defined by teammates
- API Backend: You manage a python FastAPI project that fulfills jobs for the sake of the Web Application
- Refactoring: You refactor a legacy PHP application into a more refined application using the Symfony framework
- Quality Assurance: You create functional testing to ensure the functionality of the web application prior to deployment
- Sysadmin: You have the skills to log into the remote cPanel server to ensure application functionality
- Documentation: You document the code you create to ensure that it's maintainable for future developers
- Security: you ensure that the application meets FERPA-Compliant security standard

Desired Background: To be successful, students should ideally have completed or be enrolled in:
Database Systems or equivalent
Software Engineering / Systems Analysis
Information Assurance (Only required for the Sysadmin task)
Nice-to-have courses:
Web Development
DevOps / Cloud Computing
Data Engineering or ETL Systems
Software QA
Also a strong preference for those with one or all of:
Frontend development experience
Server setup and integration
Tools that we use:
Docker / Docker compose
PHP 8.3 with the Symfony Framework
Python FastAPI

Web Host Managers (cPanel)
Unix Command line

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: N/A

Winterton Brothers - Audit, Improve & Expand an AI-Powered E-Commerce Product Intake, Data & Media Pipeline

Proposer: Paul Winterton p.winterton@wintertonbrothers.com

Project Contact: Paul Winterton p.winterton@wintertonbrothers.com

Project Description: Winterton Brothers has an existing AI-powered e-commerce product intake and enrichment workflow developed through a previous student project. The core workflow is expected to already perform a substantial portion of the required processing. This project will inherit that system, audit how it works, identify weaknesses and incomplete areas, improve its accuracy, reliability, speed and cost efficiency, and expand it with additional AI-driven capabilities.

The workflow processes product photographs and other intake evidence to identify products, create or match reusable Product Master records, extract listing-specific condition and completeness information, research and validate product data, and generate structured output.

Students will improve the existing architecture through areas such as tiered AI processing, automated validation gates, retry and escalation logic, source corroboration, improved exception handling, and more efficient use of AI models and APIs.

The project will also expand the Product Master workflow with conditional AI-generated media. Depending on the product and available evidence, this may include improved hero imagery, enhanced product-content and lifestyle graphics, informational graphics, and up to one short product video when appropriate.

The approaches described above are current working hypotheses rather than fixed implementation requirements. Students will be encouraged to audit the existing solution, research alternatives, challenge assumptions, and propose approaches that improve accuracy, robustness, maintainability, processing speed, or cost.

Student Learning Goals: Students will work with and improve an existing real-world AI automation system rather than starting from a blank project.

Semester 1 will focus on understanding and auditing the inherited workflow, identifying technical weaknesses and incomplete areas, improving the core product intake and processing pipeline, and strengthening validation, retry, escalation, and cost-control strategies. Students will also research AI models, APIs, computer-vision tools, media-generation services, and validation approaches that may be useful for the more advanced capabilities planned for Semester 2.

Semester 2 will focus on implementing and integrating selected advanced capabilities, including conditional AI-generated product media, automated media validation, retry and

escalation handling, and continued cost and performance optimization.

Students will gain practical experience with multimodal AI, computer vision, OCR, generative AI, REST APIs, workflow orchestration, structured data, automated validation, model selection, cost and latency optimization, debugging, and integration with Microsoft Power Automate and SharePoint.

Students will also have an opportunity to independently research problems, evaluate competing technical approaches, and recommend improvements rather than simply implementing a predetermined solution.

Deliverables: Semester 1:
- Audit and technical assessment of the inherited workflow.
- Improvements to the existing product intake, validation, identification, and enrichment pipeline.
- Tiered AI processing with appropriate validation, retry, escalation, and exception-handling logic.
- Improved separation and coordination between reusable Product Master processing and listing-specific product processing.
- Research and recommendations for AI models, APIs, media-generation tools, AI media validation, and other advanced Semester 2 capabilities.
- A proposed technical plan for the Semester 2 expansion based on research and findings from the inherited system.

Semester 2:
- Integrated conditional Product Master media-generation capabilities.
- Reusable hero-image generation and processing.
- Conditional generation of 0–5 enhanced product-content, lifestyle, feature, or informational graphics based on product need.
- Up to one short product video when video adds meaningful value.
- Automated AI validation of generated media with retry and escalation handling.
- Continued optimization of processing cost, speed, reliability, and model/API selection.
- Final integrated workflow, technical documentation, testing results, and handoff materials.

Desired Background: Students should have experience or strong interest in software development, API integration, and AI/ML systems.

Useful skills include Python, JavaScript, or similar programming languages; REST APIs and JSON; computer vision or multimodal AI; generative AI; workflow automation; debugging; data validation; and Git/version control.

Experience with Microsoft Power Automate, SharePoint, cloud APIs, image-processing services, or e-commerce systems would be helpful but is not required. Students should be

comfortable researching unfamiliar technologies, comparing technical approaches, and working with an existing codebase and workflow architecture.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://wintertonbrothers-my.sharepoint.com/:f:/p/p_winterton/IgCi4g_b9AmFQ7oV6AASl55cAWYV1jAcRpMJI8czjvwQfb4?e=FfJUX7

DigiClips - AWS AMAZON LIGHTSAIL INSTRUCTIONS PROJECT

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: AWS AMAZON LIGHTSAIL INSTRUCTIONS
Instructions how to put up on AWS Amazon Lightsail
https://docs.google.com/document/d/1-RY3blWKGOMFNnASyrxRSU0lxB3_ivrD-N-KReNlJao/edit?usp=sharing
https://www.digiclipsinc.com/digiprojects/AWS_Final_report_DigiClips.pdf
https://www.digiclipsinc.com/digiprojects/AWS%20Instructions%20and%20Recoding.mp4

https://bitnami.com/

Putting code up on github and being able to move it to AWS Amazon Lightsail
https://github.com/Olafaloofian/React-Frontend-Lightsail-Deployment

Connecting to the TV and Radio processed data
SSH3 virtual directory set up to make sure you are connecting to the media processed data.

SEARCH ENGINE DEPLOYMENT: AWS_ASU_TEAM2_FA25_SE_DOC

To make the projects frontend presentable in the cloud up on AWS Amazon Lightsail with https://44.231.94.92/ and new domain name www.digiclipsproject.com and UPGRADING TO ANGULAR 19 or maybe angular 20 first would have to merge it digiclips henry's github master repository branch then it would have to go up on digi-frontend local computer for some more testing then a production build and put up on AWS Amazon Lightsail for final presentation.

DIGICLIPS NEW VERSION PROJECT explains the frontend needed with presenting
general public subscribers information administration
With
MAIN MENU
SUBMENU LOGIN
SUBMENU GENERAL PUBLIC
SUBMENU SUBSCRIBER
SUBMENU INFORMATION
SUBMENU ADMINISTRATION

This explains what is needed and to connect to the frontend projects.
The processed media needs to be able to process on our local computers and be able to generate media results reports for both the general public and subscribers to present and demonstrate the projects.

The information should be able to for now show off the project worked on and there needs some more testing and put the new version for presentation and for further testing up on www.omnis.com to replace the old version www.digiclipsinc.com

The administration should just need to present this project, not be the administrators.
==========
CAPSTONE TEAMS WORKING ON AWS AMAZON LIGHTSAIL PROJECT: Oregon Fall 2025 ASU_2_Fall 2025 TEAM OSU_6_Spring 2026 Team
==========
https://docs.github.com/en/subscriptions-and-notifications/get-started/configuring-notifications

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: To help other capstone students and put up their projects up on AWS Amazon Lightsail on www.digiclipsproject.com

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials:
https://docs.google.com/document/d/1zyJZf8pPqlvNhIskuoYasB8oibuJhE9MZD-O7gkrH0g/edit?usp=sharing

Feminine Intelligence Agency - Blab Labs: A Real-Time Platform for Structured, Facilitated Group Conversations

Proposer: Lauren Nignon lauren.nignon@gmail.com

Project Contact: Lauren Nignon lauren.nignon@gmail.com

Project Description: Blab Labs (a product of the Feminine Intelligence Agency) is a web-based platform for structured, facilitated group conversations. It runs live video sessions in which participants move through a fixed sequence of conversation phases and take timed speaking turns, creating a predictable, equitable format where people who are very different from one another can talk openly and authentically. It is explicitly not a therapy or mental-health tool; it is a space for genuine conversation governed by clear rules and structure.

A prior capstone team built a working prototype: a Node.js/Express and Socket.IO server, a LiveKit-powered live video layer, and a vanilla-JavaScript front end handling host controls, session phases, timed speaking rounds, and intro/closing song playback. Next year's team will extend this existing codebase into a more complete product. The project begins with a formal, sponsor-supported review of the inherited work — standing up the development environment, walking through every existing feature, and reviewing the code, deployment, and documentation — so the team knows exactly what exists and what is problematic before building anything new.

Core work includes participant accounts and session scheduling; an in-session "conversational menu" that lets each speaker signal the kind of response they want (for example Just Listening, Vent Mode, Reflective Questions, Practical Advice, Heavy Share, or Joy); host and moderation tools (mute, remove, rename, transfer host, lock session, manage a waitlist, and report/flag); reliability improvements with automated unit and end-to-end tests; and a data-privacy and safety implementation. Stretch goals include rules-based participant matching (by time zone, then language, then cohort), a sponsor-facing analytics view, broader accessibility and mobile support, and optional live captions.

Because sessions involve live video among adults who may not know one another, privacy and safety are core requirements: sessions are live and not recorded by default, participants consent to camera and microphone at join, roles and permissions separate hosts from participants, and only limited data (name, email, and session history) is stored, with deletion on request. Participation is limited to adults (18+). FIA provides day-one resources including repository access, funded LiveKit credentials, cloud hosting and budget, existing documentation, weekly meetings with the founder, and recruited pilot participants from existing conversation groups. The front end is currently vanilla JavaScript, and the team is welcome to modernize specific modules (for example with React) where it improves maintainability.

Student Learning Goals: Students will gain hands-on, end-to-end full-stack web development experience while extending a real, deployed codebase rather than starting from scratch. They will learn to read, run, and audit an inherited production codebase

before extending it; build real-time, event-driven systems using WebSockets (Socket.IO) and WebRTC video (LiveKit); design and implement authentication, data modeling, and responsible handling of user data (consent, access control, retention, and deletion); write automated unit and end-to-end tests for a real-time application; gather requirements and deliver iteratively alongside a founder/sponsor; deploy and operate the app (cloud hosting, environment configuration, CI); design for a sensitive, human-centered use case with attention to accessibility, trust, and safety; and collaborate professionally using git/GitHub, code review, and agile workflows.

Deliverables: Core:
A "state of the codebase" onboarding audit report (feature inventory, known issues and technical debt, and keep/refactor/rebuild recommendation), co-reviewed with the sponsor, plus a working development environment for every team member
Participant accounts / authentication and session scheduling and registration
In-session conversational menu feature
Host and moderation controls (mute, remove, rename, transfer host, lock session, waitlist, report/flag)
Reliability improvements to the real-time session and video layer, with automated unit and end-to-end tests (e.g. Playwright/Cypress) and defined success criteria (session-join success above 99%, video connection within ~3 seconds, no more than one unexpected dropout per 30 participants)
A data privacy and safety implementation (consent at join, roles/permissions, reporting/blocking, data retention and deletion) plus a pre-pilot security and data-flow review
Updated technical documentation and a working, deployed instance
Final demo, presentation, and handoff documentation
Stretch:
Rules-based participant matching/grouping (time zone within ±2 hours, then language, then cohort)
Sponsor-facing analytics view (sessions held, attendance, turns taken vs. passed, no-show rate)
Broader accessibility (WCAG-oriented) and mobile-responsive support
Optional live captions / transcription

Desired Background: Proficiency in JavaScript and core web fundamentals (HTML/CSS)
Experience with Node.js/Express and comfort with both REST and real-time (WebSocket) patterns is a plus
Front-end experience (vanilla JS or a framework such as React); willingness to work in an existing vanilla-JS / ES-modules codebase
Some exposure to authentication and databases (for accounts, scheduling, and stored session data)
Familiarity with git/GitHub and collaborative, code-reviewed development
Bonus skills: WebRTC/LiveKit or video streaming, automated testing (Playwright/Cypress),

cloud deployment (e.g. Render), UI/UX design, or interest in accessibility and human-centered design
Comfort working directly with a startup founder and iterating on evolving requirements

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: www.feminineintelligence.agency

Blue Almond - Blue Almond: Smart Retail Experience and Analytics Platform

Proposer: Ehsan Daneshgar ehsan@launchpath.ca

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: Blue Almond is developing an intelligent retail platform that combines connected smart shelf hardware, cloud services, and web-based analytics to help businesses measure customer engagement with in-store product displays. Motion detection, interactive surveys, campaign management tools, and connected retail technologies work together to collect meaningful customer interaction data that can be used to evaluate marketing performance across multiple retail locations.

A functional platform has already been developed, including hardware connectivity, campaign and shelf management, cloud-based backend services, analytics dashboards, customer survey workflows, and administrative tools. This project focuses on advancing the platform into its next stage of development.

Students will begin by evaluating the existing hardware and software platform through functional, technical, and usability testing to understand its current capabilities, identify opportunities for improvement, and recommend enhancements that strengthen the overall product. Once approved, students will implement those improvements while expanding the platform with new functionality aligned with the product roadmap.

As part of the next phase of development, students will also explore opportunities to enhance the platform through AI-powered retail analytics and intelligent customer engagement. This may include evaluating how multisensory experiences—such as sound (including ASMR-inspired experiences), lighting, visual content, and interactive media—can improve shopper engagement while generating meaningful retail insights. Students will prototype and validate innovative features that support Blue Almond's long-term vision of creating intelligent, data-driven retail experiences.

Project work may include enhancing hardware reliability, improving communication between connected shelf devices and cloud services, expanding campaign management features, refining customer survey workflows, strengthening analytics and reporting, supporting multi-device deployments, improving scalability and performance, implementing AI-powered retail insights, and enhancing the overall user experience.

Student Learning Goals: Students participating in this project will have the opportunity to:

Work on a real-world IoT platform that combines embedded hardware, cloud services, and web applications.
Evaluate an existing platform to identify technical, functional, and usability improvements.

Design, propose, and implement meaningful product enhancements.
Develop full-stack software using modern frontend, backend, and cloud technologies.
Improve communication between connected hardware devices and cloud services.
Enhance analytics, campaign management, and customer engagement workflows.
Explore AI-powered retail analytics and shopper behavior insights using real-world interaction data.
Prototype innovative customer engagement features, including multisensory retail experiences and intelligent campaign optimization.
Gain experience designing scalable systems that support multiple connected devices.
Participate in software testing, hardware validation, deployment, and technical documentation while following agile development practices.

Deliverables: Review and evaluate the existing hardware and software platform to identify opportunities for improvement.
Recommend and implement approved enhancements to existing features, workflows, and overall system performance.
Improve communication and reliability between connected shelf devices and backend services.
Expand campaign management, survey workflows, analytics, and administrative functionality.
Prototype innovative retail engagement features that support Blue Almond's long-term product roadmap.
Develop AI-driven analytics, reporting, or recommendation capabilities using customer interaction data.
Improve scalability to support multiple connected devices and retail deployments.
Conduct functional, usability, hardware, and system testing to validate improvements.
Deliver updated technical documentation, deployment guides, and source code.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:

Full-stack software development
React, TypeScript, or modern frontend frameworks
Backend development (REST APIs, cloud services)
Database systems
Embedded systems or IoT development
Raspberry Pi or similar edge devices
Artificial Intelligence or Machine Learning
Data visualization and analytics
UI/UX design

Software testing and system optimization

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

Viking Roots - Build a Next-Generation AI Social Platform Around Photos & Identity

Proposer: Diana Bristow diana.bristow@icloud.com

Project Contact: Diana Bristow diana.bristow@icloud.com

Project Description: What if social media actually understood your photos—who's in them, how people are connected, and the stories behind them?

Viking Roots (KinSnap) is an AI-powered social platform built around photos, identity, and connection. Users can upload and tag images, recognize people across photos, and explore how they're connected, while adding stories in a more conversational way.

You can think of it loosely as Instagram with smarter tagging, Facebook with more meaningful connections, and a layer of AI that helps organize and bring context to it all.

The platform already exists in a working form, developed through a 2025/26 Capstone project. It includes user accounts, photo upload and tagging, a structured data model for people and relationships, search, admin tools, and an initial interface.

The focus now is on taking that foundation and turning it into a product people genuinely enjoy using.

From early user feedback, a few things are clear. People want better privacy controls, tagging that is faster and less frustrating, and clearer ways to connect and navigate relationships. This project is about turning those needs into features that actually work well and feel natural.

Depending on the team's strengths and interests, this could include improving tagging and identity matching, adding AI-assisted features, making the interface more interactive, strengthening performance, or exploring how a mobile version fits alongside the platform.

Student Learning Goals: You'll get hands-on experience working on a modern, real-world platform, not a theoretical project.

That includes building features for a social product, working with image-heavy systems (uploads, tagging, search), and designing how people and relationships are structured and connected behind the scenes.

You'll also be translating actual user feedback into product decisions, improving things like privacy, usability, and overall flow—so it's not just about writing code, but about building something people will actually use.

On the technical side, you'll work with APIs, authentication, and platform integration, and

get exposure to tools like AWS, Stripe Identity Verification, and Figma as part of the development process.

You'll be working in a real team environment using Git and shared workflows, with an emphasis on clean handoff and collaboration.

You'll also get a look at how a startup is actually built, from product decisions to getting something ready for real users. This includes exposure to early-stage planning, feature prioritization, and how products are positioned and funded.

Based on how the project has run so far, students can expect to take part in real business meetings, help define deliverables and timelines, divide work across the team, and contribute to both the technical build and the overall direction of the product.

Deliverables: This is a full academic year project running from September to April, giving you the opportunity to work through product testing, user feedback, and contribute to a live platform as it moves through active use and public launch.

By the end of the project, the team will have delivered a functional, startup-ready social media platform centred around photo sharing, tagging, and connections between people, with heritage elements built naturally into the experience. The platform will include stronger privacy controls, improved and simplified tagging, and better ways for users to connect and navigate relationships.

A working version of the platform will already be live, and your focus will be on shaping the core user experience—refining how it feels, how it performs, and how people actually use it. This will include developing a fully workable admin panel, advancing photo tagging with AI-assisted features to recognize the same people across images within groups or locations, coordinating with a companion mobile app direction, and creating a clear, practical user guide.

All development should be production-ready, clearly structured, and supported by practical documentation so the platform can continue to grow beyond the Capstone. The goal is to leave behind something that is not just working, but actively used and ready to scale as a real product.

Desired Background: This project is best suited for students in computer science, software engineering, full-stack development, UI/UX design, information systems, or related programs, particularly those interested in building real-world applications and working on production-level systems.

Experience with modern web development is helpful, especially front-end frameworks such as React, along with backend development in Node, Python, or similar environments. Familiarity with APIs, database design, and version control will make it easier to contribute

early on.

The current stack and workflow include tools such as React, Python, AWS, Stripe Identity, Figma for design, and Discord for team collaboration, along with structured data formats such as GEDCOM. Students don't need to know everything in advance—strong problem-solving ability, curiosity, and a willingness to learn are just as important. Students who enjoy coding challenges or competitions will likely feel right at home.

An interest in building social platforms, working with image-based features, or exploring AI-driven functionality is a strong asset. Experience with accessibility or user experience design is also valuable.

Students who are comfortable working in a team, open to feedback, and interested in building something real that will continue beyond the course will be a good fit.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Winterton Brothers - Build a Production-Ready eBay and Shopify Commerce Integration Platform

Proposer: Paul Winterton p.winterton@wintertonbrothers.com

Project Contact: Paul Winterton p.winterton@wintertonbrothers.com

Project Description: Winterton Brothers is developing a centralized internal commerce system for managing inventory and marketplace activity across multiple sales channels. This project will create the integration foundation connecting eBay and Shopify to a centralized Microsoft SharePoint and Power Automate environment.

Students will design and build separate eBay and Shopify event routers using each platform's current APIs, webhooks or notification services. Incoming events will be validated, checked for duplicates, enriched through platform API calls when required, normalized and routed into shared child workflows.

The system will fully process inventory-related events, including new orders, payment confirmation, inventory reservations, cancellations, inventory releases and placement of paid orders into a fulfillment queue. Returns, refunds, disputes and supported customer communications will also be routed and recorded in structured SharePoint lists, creating the foundation for future operational automation.

The team will design and create the required SharePoint lists outside the company's existing product-reference lists. Channel listing IDs, variant IDs and marketplace SKUs must be mapped to Winterton Brothers' internal product and SKU identifiers.

Separate nightly reconciliation workflows must compare eBay and Shopify records against the centralized SharePoint system, identify missing or outdated records and safely process corrections through the same shared workflows.

Development will occur entirely in sandbox environments using a dedicated Power Platform environment, eBay sandbox accounts, a Shopify development store and a controlled SharePoint site. The architecture must support later transition to production credentials and future expansion to additional channels such as Amazon.

The project excludes shipping-carrier integrations, Power Apps development, marketplace listing creation, accounting, sales analytics and production deployment.

Student Learning Goals: Students will gain practical experience designing a real event-driven commerce integration system using:

REST APIs, OAuth authentication, webhooks and platform notifications
Microsoft Power Automate solutions, parent flows and reusable child flows
Microsoft SharePoint data architecture and relational record design

Event normalization across two different commerce platforms
Internal-to-channel SKU and identifier mapping
Idempotency and duplicate-event prevention
Error handling, retries, controlled replay and exception management
Scheduled reconciliation and incremental synchronization
Secure configuration using environment variables and connection references
Automated testing, technical documentation and deployment planning

Students will work directly with the project sponsor to research requirements, propose the data architecture, develop the system through iterative sprints and demonstrate a functioning end-to-end sandbox solution.

Deliverables: Approved system architecture and data model
eBay and Shopify event maps
Shared normalized event structure
SharePoint list architecture, relationships, identifiers and statuses
Flow and data-movement diagrams
eBay integration
REST notification intake and validation
Event router
Topic-specific child flows
API enrichment and internal SKU resolution
Shopify integration
Webhook intake and validation
Event router
Topic-specific child flows
API enrichment and internal SKU resolution
Central SharePoint workflow system
Orders and order lines
Channel SKU and listing mappings
Inventory reservations
Paid-order fulfillment queue
Returns, refunds, disputes and supported customer communications
Processing errors and integration exceptions
Reconciliation and synchronization history
Operational inventory workflows
Create and update orders
Reserve inventory
Release inventory following cancellation
Route paid orders into the fulfillment queue
Prevent inventory changes when a SKU cannot be mapped
Reliability framework
Duplicate-event rejection
Standardized error-handling scopes

Correlation and flow-run tracking
Retry and controlled replay processes
Authentication and connection-failure handling
Nightly reconciliation
Separate eBay and Shopify reconciliation flows
Incremental retrieval from the previous successful checkpoint
Identification and repair of missing or outdated records
Reconciliation summaries and failure reporting
Testing and handoff
Documented test scenarios and results
Power Platform Solution export
Source files and configuration documentation
User and developer documentation
Sandbox-to-production deployment guide
Final demonstration and presentation

Desired Background: Desired Background

Students should have experience or interest in:

Software engineering and systems integration
REST APIs, JSON, OAuth and webhook-based architecture
Microsoft Power Automate or comparable workflow platforms
Database or structured data design
Error handling, testing and debugging
Git or other source-control practices
E-commerce systems or marketplace integrations

Experience with Microsoft SharePoint, Power Platform, Shopify APIs or eBay APIs is helpful but not required. Students should be comfortable researching unfamiliar APIs and translating business requirements into a reliable technical solution.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

MentaMorph Incorporated - Business Scenario Design & Simulation

Proposer: Susan Meredith susan@mentamorph.com

Project Description: This project develops a structured series of business-focused scenarios to create sustainable cash flow. The team will design scenario logic, economic models, and decision pathways that build in complexity. The first scenario is an Entrepreneurship scenario with a focus on sales & marketing then Service adding employees, Product which includes Inventory and Margins, and Scale adding processes and sustainable operations.The project will use AI logic and other elements from earlier projects.

Student Learning Goals: Students will work at the intersection of systems thinking, business modeling, and simulation design. They will learn how to translate real-world business dynamics—such as revenue generation, cost structures, and scaling—into interactive, decision-based scenarios.

Deliverables: • Scenario framework (rules, variables, progression)
- 4 designed scenarios aligned to the Money in Motion sequence
- Cash flow and decision models for each scenario
- Scenario documentation and integration specifications
- Prototype or partially implemented scenarios

Desired Background: • Systems thinking and problem-solving
- Interest in business or entrepreneurship
- Experience with modeling or logic design
- Bonus: game design or finance

Delivery Mode: In-person

Required Agreements:

- ["Non-disclosure agreement"
- "Intellectual property agreement"]

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

Honeywell Aerospace - Cable Design Tool

Proposer: Kyle Tappendorf Kyle.Tappendorf@HoneywellAerospace.us

Project Contact: Charles Dairman Charles.Dairman@HoneywellAerospace.com

Project Description: On the surface, it might seem that making a cable for flight hardware or testing would be simple. Just grab some wire and a few connectors and the job is done, right? Not at all! The reality is that cable design is a labor intensive task. The correct wire type and gauge must be chosen; the correct size and material connector must be chosen; the matching or corresponding backshell must be chosen; correct pinout must be done to keep crosstalk and other signal quality issues at bay; components must be chosen that can handle the environmental conditions required (e.g. high temperature or high humidity); not to mention putting all that information into a drawing that can be easily version-controlled and "made official" or published for use. It is easy for an engineer to make mistakes, especially with material mismatches or electrical current ratings. One possible solution to reduce engineering development costs is to create an automated "Cable Design Tool." The engineer designing the cable would input information into the tool, such as connector and wiring information, and the tool would do the rest, quickly generating a .dwg or .vsdx file that could be released as a formal drawing.

The software tool will need to meet the following requirements:
The tool must include a database of all approved Honeywell Part Numbers for connectors, backshells, and wires
The tool must include a method for requesting the addition of new part numbers not already present in the tool.
The tool must support a graphical user interface (GUI)
The user must be able to select the desired parts, such as connectors and wires
The tool must show only parts that are compatible with each other
The user must be able to input the number of connectors and "branches" present in the cable
The user must be able to add net names for each wire in the cable design
Based on user input through the GUI, the tool must generate a source file for the drawing. Acceptable formats are .creo, .dwg, and .vsdx
The tool must contain version control capability
The tool can be developed using AI agents such as Github Copilot.
The tool must be easily maintained by a single engineer

Student Learning Goals: By the completion of this project, students will be able to:

Analyze engineering requirements by translating customer needs and aerospace cable design standards into functional and technical software requirements.
Design a scalable software architecture that integrates a graphical user interface, parts database, compatibility validation engine, and drawing generation module.

Develop a database-driven engineering application capable of storing, retrieving, and managing approved cable components, including connectors, backshells, and wiring information.
Implement engineering validation logic that automatically verifies component compatibility, electrical constraints, and allowable design configurations to reduce human error.
Develop an intuitive graphical user interface (GUI) that enables engineers to efficiently create cable assemblies through guided user input.
Generate engineering documentation automatically by producing industry-standard cable drawings or source files (.DWG, .VSDX, or Creo) from user-defined cable configurations.
Apply software engineering best practices, including version control, modular design, testing, documentation, and maintainable code to support long-term sustainability by a single engineer.
Leverage AI-assisted software development tools (e.g., GitHub Copilot or similar AI agents) responsibly to improve development efficiency while maintaining code quality and engineering integrity.
Design for maintainability and extensibility by creating software that allows future engineers to easily update part databases, add new components, and modify system functionality.
Validate software functionality through comprehensive testing, demonstrating that generated cable designs satisfy project requirements and produce accurate engineering outputs.
Communicate engineering solutions effectively through technical documentation, design reviews, demonstrations, and presentations to industry sponsors.
Collaborate within an engineering team by applying project management methodologies, issue tracking, code reviews, and professional communication throughout the software development lifecycle.

Deliverables: Students should produce:
Requirements specification document
Software architecture and system design documentation
Component database containing approved Honeywell part numbers
GUI-based Cable Design Tool
Compatibility validation engine
Automated drawing generation module (.DWG, .VSDX, or Creo)
Version-controlled source code repository
User documentation and maintenance guide
Test plan and validation report
Final demonstration to Honeywell engineers
Final technical report and project presentation

Desired Background: The software tool will need to meet the following requirements:
The tool must include a database of all approved Honeywell Part Numbers for connectors, backshells, and wires

The tool must include a method for requesting the addition of new part numbers not already present in the tool.
The tool must support a graphical user interface (GUI)
The user must be able to select the desired parts, such as connectors and wires
The tool must show only parts that are compatible with each other
The user must be able to input the number of connectors and “branches” present in the cable
The user must be able to add net names for each wire in the cable design
Based on user input through the GUI, the tool must generate a source file for the drawing. Acceptable formats are .creo, .dwg, and .vsdx
The tool must contain version control capability
The tool can be developed using AI agents such as Github Copilot.
The tool must be easily maintained by a single engineer.

Delivery Mode: In-person

Required Agreements:

- ["Intellectual property agreement"
- "US Citizenship requirement"]

N/A - CacheMatrix: Context-Enabled Semantic Caching Proxy for Cost Reduction

Proposer: Shatrughna Upadhyay supadh18@asu.edu

Project Description: As enterprise organizations scale LLM applications, API token costs and inference latency become primary bottlenecks. Naive architectures route every query directly to the LLM, but telemetry shows users frequently ask conceptually identical questions using slightly different phrasing. CacheMatrix is a Semantic Caching Gateway that intercepts these queries. It converts incoming text into high-dimensional vector embeddings and searches for conceptually similar queries previously processed. If a semantic match is identified, the system bypasses the LLM entirely and returns the cached response in milliseconds, drastically improving user experience and reducing API billing.

Student Learning Goals: Students will bridge the gap between academic theory and enterprise-grade AI deployment. They will gain hands-on experience with advanced backend architecture, asynchronous Python programming, cryptographic hashing, and vector database management. Furthermore, they will learn to transition from procedural coding to building high-performance, cost-optimized AI system infrastructure.

Deliverables: By the end of the capstone sequence, the team must deliver: 1. A containerized FastAPI proxy server that intercepts client requests. 2. O(1) exact-match caching using SHA-256 hashing. 3. Local embedding generation using open-source Sentence Transformers. 4. Vector similarity search integration utilizing an open-source database (Valkey or PostgreSQL with pgvector). 5. Automated Time-To-Live (TTL) expiration logic. 6. A load-testing script demonstrating the empirical latency reduction and theoretical cost savings

Desired Background: Proficiency in Python. Fundamental understanding of RESTful APIs, data structures, and algorithmic complexity. Basic familiarity with Docker and database concepts is highly preferred.

Delivery Mode: In-person

Feminine Intelligence Agency - ChatBoy: Modernizing and Extending a Manipulation-Awareness Web App with AI-Graded Assessment and Partner Integrations

Proposer: Lauren Nignon lauren.nignon@gmail.com

Project Contact: Lauren Nignon lauren.nignon@gmail.com

Project Description: ChatBoy is a web application from the Feminine Intelligence Agency (FIA) that teaches people to recognize manipulation, coercion, social red flags, and microaggressions through interactive scenarios and assessments. A prior capstone team built a working platform on a PHP/MySQL backend (Dockerized, version-controlled on GitHub, and deployed to demo.chatboy.app) with account security and email 2FA, a categorized quiz engine, a scenario library, badges and an anonymized leaderboard, progress tracking, translation, and an admin dashboard. It works, but it is still a prototype, and this project is the chance to turn a real, deployed application into a polished product.
The work has genuine range. Students begin by tracking down and closing the security issue behind a live host suspension, then design and ship contemporary, demonstrable features: an AI-graded skills assessment, a modern React/TypeScript assessment and results experience, and system-to-system API integrations. It is real product ownership on software that is actually going live, not a throwaway class exercise.
The team's first task is to test the existing software systematically. FIA will provide a database of test scenarios so the team can exercise the app methodically, establish a clear baseline of what works and what is broken, and produce a prioritized roadmap for the two semesters. The core work then has two parts. The first is professionalization: bounded bug fixing, hardening the application against cybersecurity vulnerabilities, and establishing reliable automated deployment, so ChatBoy behaves like a real product. The second is building three capabilities FIA needs to operate ChatBoy in the real world:
A before-and-after skills assessment. The app presents a written scenario (authored by FIA), the user submits a written response, and an AI grades that response against a scoring rubric that FIA develops and the students implement. Running this before and after the learning experience lets FIA measure each user's improvement. The team owns the product and interaction design of the assessment experience, results visualization, and feedback, building them in a modern front-end.
Shareable links to specific sections. FIA can hand out links that open specific sections or modules for interested parties and partners, including free tiers for partner organizations, with control over expiration and revocation.
Badge integration with FIA's platform. When a user earns a badge, the system pushes the event to FIA's main web platform through its API, matching the user to the right account, so their profile on that platform updates automatically.
The proven PHP/MySQL backend is retained and stabilized rather than rewritten, which keeps the project grounded and puts the focus on quality, security, reliability, and integration, while the new user-facing features are built in React/TypeScript against the PHP backend so students also gain current front-end experience. The AI grading uses an

existing LLM API (such as OpenAI, Google Gemini, or Anthropic Claude) with structured outputs, applied against FIA's rubric; students do not train a model. Because the assessment grades responses to FIA-authored hypothetical scenarios rather than users' real personal disclosures, its scope is bounded; responses carry a clear disclaimer that ChatBoy is educational and not professional advice, and the API is configured to disable vendor use of submitted content for model training. If time remains after the core work, the team improves the app's feedback and gamification mechanics further.

Access to the existing GitHub repository (feminineintelligenceagency/ChatBoyApp) and the database (schema and seed data).
The test scenarios for systematic testing, and the written assessment scenarios and scoring rubric for the before-and-after assessment (FIA designs these; the students implement them).
A labeled validation set of roughly 30–50 example responses with rubric scores, so the team can tune and verify the AI grader.
Approved privacy and data-retention requirements for assessment responses and scores, provided before any user data is collected: responses and scores are retained no longer than 12 months, are deletable on user request, are viewable only by designated FIA reviewers with access logged, and no identifying user information is sent to the LLM vendor.
API access and specification for FIA's main web platform, including how ChatBoy users map to platform accounts through a secure server-to-server identifier, for the badge integration.
Requirements for partner access and free tiers.
Funded AI API access within a small monthly budget for the grading feature; operating within it through caching, request throttling, and efficient model choices is a stated requirement.
A sponsor-run pilot: FIA recruits 8–12 adult participants to complete the pre/post assessment during the project, targeting at least 8 completed pre/post assessments and one feedback session; the team implements at least three prioritized changes from the findings.
Access to the hosting/cPanel account and the live URL (demo.chatboy.app).
A named sponsor contact (FIA founder Lauren, with developer Manoj Kumar) available for weekly check-ins.
Written onboarding materials: the prior team's handoff document, a detailed how-to guide, and the database dump.
Openness to migrating to a low-cost modern host if the current environment proves incompatible with the CI/CD and security goals, subject to sponsor approval.
Repository access and participation governed by a sponsor agreement aligned with the university's capstone IP policy.

Student Learning Goals: This is real-world software quality, security, and integration engineering on a live, inherited codebase, with visible product ownership. Students gain hands-on experience with systematic software testing and test automation; debugging and fixing defects in an existing PHP/MySQL application; application security and hardening (authentication, authorization and access control, input validation, and remediating real

vulnerabilities); CI/CD and reliable deployment with GitHub Actions; modern front-end development (React/TypeScript) for the assessment, results, and feedback experience, including usability testing; integrating an existing LLM API for rubric-based grading with structured outputs; building API integrations between systems (pushing events to an external platform with retries, account matching, and API contract testing); and implementing access and entitlement features such as shareable section links and partner tiers. They also design against sponsor-approved privacy requirements, observe real participants using what they built during the pilot, and practice working within an inherited codebase and communicating with a client sponsor.

Deliverables: Core (required):
Systematic test pass: using FIA's scenarios, exercise the existing app end to end, document a baseline of current behavior, and deliver a prioritized roadmap. Acceptance: a documented test report and roadmap reviewed with the sponsor.
Bounded bug fixing: resolve all critical and high-severity defects plus at least 10 medium-severity defects, formally deferring the remaining backlog. Acceptance: severity classification agreed with the sponsor, and each fixed defect has an automated test that reproduces the original problem and passes after the fix.
Security hardening and host-suspension resolution: proper authorization/access control on all privileged admin endpoints, input validation, a documented security review (relevant OWASP Top 10 items), and investigation and resolution of the prior host suspension (the handoff documents a known admin-authorization gap that may be the cause). Acceptance: an authorization test for every privileged endpoint, a completed review checklist, and a documented root-cause finding and fix for the suspension.
Reliable deployment: a GitHub Actions CI/CD pipeline with automated tests and a reproducible rollback on the chosen host. Acceptance: a demonstrated deploy-and-rollback, with CI running tests on each change.
Before-and-after skills assessment (with front-end): implement the pre/post assessment in which the app presents an FIA-authored scenario, collects the user's written response, and grades it with an existing LLM API against FIA's rubric; and design and ship the assessment experience, a results and progress visualization, and one polished feedback screen in React/TypeScript, with room for the team's own product and interaction design decisions. Acceptance: on FIA's validation set, the grader achieves 100% structured-output validity, at least 80% agreement with human rubric scores, the same score band on at least 90% of repeated runs, and cost under $0.05 per assessment; before/after results are stored and visualized.
Shareable section links and partner access: generate links that grant access to specific sections or modules, with configurable expiration, revocation, and authorization, plus a mechanism to provision free-tier access for partner organizations. Acceptance: a link can be created that opens a specific section for an authorized partner, access can be granted and revoked, and links expire as configured.
Badge API integration: when a user earns a badge, push the event to FIA's main web platform through its API, matching the ChatBoy user to the correct platform account via a secure server-to-server identifier, with retries and a queue so events are not lost when the

platform is temporarily unavailable. Acceptance: earning a badge triggers a verified API call that updates the account, with graceful handling when the platform is down, and the event-driven design is documented in the handoff.
Privacy controls: implement FIA's approved consent, retention (no more than 12 months), deletion, access (designated FIA reviewers only), and logging controls for assessment responses and pre/post scores.
Documentation and handoff: updated technical documentation and a written handoff guide for the following team.
Stretch (if time allows):
Improved feedback across more assessment types (identify the pattern, explain why it is concerning, suggest a better response).
Expanded gamification: enhanced streaks, achievement milestones, and progress display.
Broader front-end modernization of additional UI areas.

Phase 1 (Semester 1): systematically test the existing app; document a baseline and a prioritized roadmap; investigate and resolve the prior host-suspension issue; complete the security review and begin bounded bug fixing; stand up the GitHub Actions CI/CD pipeline with automated tests and rollback; decide (with the sponsor) by the end of Phase 1 whether to remain on cPanel or migrate hosts, so the choice is settled before Phase 2; and prototype the AI-graded assessment against FIA's rubric.
Phase 2 (Semester 2): complete the bounded bug fixing and security hardening; design and ship the assessment experience, results and progress visualization, and feedback screen in React/TypeScript; implement the shareable section links and partner free tiers and the badge-to-platform API integration; run the sponsor pilot and act on its findings; professionalize the software (reliability, documentation, and handoff). Stretch work (further feedback and gamification) is scheduled into remaining capacity.
Suggested team split: roughly two students on backend and security, two on the React/TypeScript front-end, and one on AI grading and API integrations, with responsibilities adjusted after the Phase 1 audit.
Contingency: if the initial audit uncovers major architectural or security defects, the protected core is deployment reliability, security remediation, the AI-graded assessment, and one integration; partner tiers or the badge integration may move to stretch.

Desired Background: Comfort with full-stack web development on PHP and MySQL (or willingness to ramp up quickly), and a solid grasp of databases and SQL. Front-end skills in JavaScript, ideally React/TypeScript, for the assessment and results experience. Experience or interest in software testing and test automation, debugging existing code, and web security fundamentals (authentication, access control, input validation). Experience consuming and building REST APIs, since the project integrates with FIA's platform and an LLM grading API. Familiarity with Git, CI/CD, and Docker is helpful. Genuine interest in the

project's mission of helping people recognize manipulation and protect themselves is valued.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: www.feminineintelligence.agency

Circle.ooo - Circle.ooo

Proposer: Jo Clark jo@circle.ooo

Project Contact: Jo Clark jo@circle.ooo

Project Description: Agentic AI - development of coordinated agents to work together for event planning

Student Learning Goals: Using AI tools and automation for real world tasks

Deliverables: App, Web App, MCP App and/or OpenAI App plus backend (API / integrations)

Desired Background: Computer engineering with a focus on AI tools, agentic and experimenting with AI tools for business outcomes

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Jo Clark - Circle.ooo

Proposer: Jo Clark jo@circle.ooo

Project Contact: Jo Clark jo@circle.ooo

Project Description: Deep Events Vertical AI Native

Student Learning Goals: Extensive use of AI tools for an AI native project

Deliverables: Key workflows for every aspect of Events using AI Agents

Desired Background: AI, machine learning, AI tools, AI workflows, voice AI, unique UI/UX

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://www.cir-cle.ai

Clear Impression - Clear Impression: Next-Gen Operational OS

Proposer: Niels Hansen b.niels.hansen@gmail.com

Project Contact: Niels Hansen b.niels.hansen@gmail.com

Project Description: Develop a high-performance, SEO-optimized business website and operational OS for Clear Impression Window Cleaning. The system includes a 'One-Tap' client portal for scheduling and education, a field-agent app for geolocation-based sales tracking (KPIs: door knocks, backyard access, sales conversion), a computer vision estimator for instant quoting, an AI persona avatar for FAQ, and a secure backend for pricing management and automated marketing. The system is designed for high-concurrency, multi-city scalability, and local SEO dominance.

Student Learning Goals: Students will gain full-stack web development experience, focusing on high-performance infrastructure, secure database design, AI integration, mobile-first UI/UX, and real-time geolocation analytics. They will solve complex problems in system architecture, scaling, and automation, directly impacting business revenue and operational efficiency.

Deliverables: 1. A secure, SEO-optimized, multi-tenant website. 2. A 'One-Tap' mobile client web app with automated booking, SMS marketing, and AI avatar integration. This should include a Content Management System (CMS) backend specifically for the "What to Expect" content.3. A geolocation-enabled field sales app for real-time KPI tracking. 4.AI Avatar integration into website 5. "Live-Status" Client Dashboard
The Idea: Similar to a ride-sharing app, give the client a link that shows exactly when you are en route, when the job starts, and a "Before & After" photo upload feature where they get a notification the moment the window is clean.
6. Multi-Tenant Scalability & Franchise Readiness
The Idea: Structure the codebase so that if you decide to hire a crew or open a new service territory, you can "spin up" a new instance of your backend with its own city-specific SEO pages and pricing logic in minutes.

Desired Background: Full-stack development, database security, UI/UX design, geolocation API integration, and AI/ML model implementation.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Orchard Mortgage LLC - Company Website Integration

Proposer: Frank Frayer frank.frayer@orchardmortgage.com

Project Contact: Frank Frayer / Chrisstian Volkner frank.frayer@orchardmortgage.com

Project Description: This capstone project examines opportunities to enhance the digital customer experience at Orchard Mortgage by modernizing its website and online resources. The project will evaluate improvements to website navigation, educational content, self-service tools, customer communication, and access to mortgage products and Loan Officer information. Through industry research, benchmarking, and stakeholder feedback, the project will develop recommendations to improve customer engagement, increase transparency, and simplify the home financing process. The goal is to create a more user-centered digital experience that strengthens borrower confidence, supports business growth, and positions Orchard Mortgage as a technology-driven mortgage lender.

Student Learning Goals: Through this capstone project, the student will gain practical experience applying organizational leadership principles to a real-world business challenge. The student will learn how to assess an organization's digital customer experience, identify opportunities for improvement, and develop strategic recommendations that align with organizational goals and customer needs.

The student will strengthen skills in qualitative research, industry benchmarking, stakeholder engagement, project management, and evidence-based decision-making. Additionally, the project will provide experience in evaluating digital transformation initiatives, analyzing customer-centered design practices, and understanding how technology, communication, and innovation influence organizational performance. By completing this project, the student will enhance their ability to lead change, solve complex business problems, and develop solutions that improve customer satisfaction while supporting organizational growth and competitive advantage.

Deliverables: The following deliverables will be produced as part of this capstone project:

Website Assessment Report – An evaluation of the current Orchard Mortgage website, identifying strengths, weaknesses, and opportunities for improvement.
Digital Customer Experience Recommendations – A strategic plan outlining enhancements to website navigation, accessibility, self-service features, and customer communication.
Website Wireframes and Sitemap – Conceptual layouts illustrating an improved website structure, navigation, and customer journey.
Interactive Tool Recommendations – Specifications for mortgage calculators, educational resources, AI chatbot functionality, and online appointment scheduling.
Comprehensive Mortgage Program Library Framework – A proposed structure for centralized product pages, including eligibility, benefits, FAQs, educational content, and Loan Officer contact information.
Equity Advance Resource Center Design – A proposed resource hub containing program

information, eligibility guidelines, educational materials, partner information, and customer success stories.
Implementation Roadmap – A phased plan outlining recommended priorities, resource requirements, and future implementation strategies.
Final Capstone Report and Presentation – A comprehensive report summarizing research findings, recommendations, and expected organizational benefits, accompanied by a professional presentation of the project's results.

Desired Background: Students undertaking this project should have a foundational understanding of organizational leadership, business strategy, and project management. Coursework or experience in research methods, change management, and organizational improvement will support the successful completion of the project.

Students should also possess basic knowledge of digital customer experience, website usability, and customer-centered design principles. While technical web development or programming experience is not required, familiarity with website functionality, digital communication tools, and online customer engagement strategies is beneficial.

Experience in the mortgage, financial services, or banking industry is helpful but not required. Students should be able to conduct qualitative research, analyze organizational processes, evaluate industry best practices, and develop evidence-based recommendations that align with organizational goals and customer needs.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials:
https://orchard.com/intake/address?utm_source=bing&utm_medium=cpc&utm_campaign=677622214&utm_content=1170981828696495&utm_term=orchard&location=78531&adlabel=branded2&locationinterest=&market=PHX&bn_adid=73186712827134&bn_dev=c&bn_target=kwd-73187121462019&bn_type=o&bn_matchtype=p&msclkid=256c58466bc6156ea7f6320544fa0f5f

Appy.yo - Cookful: AI Product Scanning and Recipe Creation Tool

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: Cookful needs UI/UX improvements and stronger AI functionality, especially around scanning products and creating recipes. Many people waste food because they forget what they have, do not know what to cook, or struggle to use ingredients before they expire. Cookful is designed to help users scan products, track available ingredients, and receive recipe ideas based on what they already have. The current focus is to improve the user experience and implement AI-powered scanning and recipe creation.

Student Learning Goals: UI/UX design, AI implementation, image recognition, barcode/product scanning, recipe generation, front-end development, back-end development, database design, JavaScript, React, Python, API integrations.

Deliverables: A fully functioning software solution with improved UI/UX, AI product scanning, and AI recipe creation. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.

Desired Background: UI/UX Designer 1
AI/Product Scanning Developer 1–2
Front-end Developer 1
Back-end Developer 1
Recipe Logic/Product Research Support 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

General Dynamics Mission Systems - Correlated Operational Root-cause and Troubleshooting Expert System (CORTEX)

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Felisa Hicks Felisa.Hicks@gd-ms.com

Project Description: Remote system support technicians are often required to troubleshoot complex site, system, and equipment alarms under time pressure. Existing troubleshooting resources, such as Pcause (Probable Cause) files for alarms in DX NetOps Spectrum, tickets in the Customer Relationship Management (CRM) database, shared technician notes, and engineering tribal knowledge, contain valuable information, but they are often distributed across multiple systems and are difficult to correlate quickly during live operations. This capstone project challenges students to design and build a proof-of-concept AI tool that ingests historical CRM, Spectrum, and systems analysis data to identify recurring alarm patterns, site-specific trends, and likely root causes. The system will recommend smarter troubleshooting steps to technicians by learning from historical failures, corrective actions, alarm sequences, affected sites, equipment types, and prior ticket resolutions. Conceptually, the tool acts as an advanced, data-driven version of existing PCause troubleshooting files.

The proposed system will combine structured data analytics, natural language processing, vector search, and potentially graph-based machine learning to reason over both tabular event data and unstructured ticket narratives. Students will investigate whether a vector database, a graph database, a Graph Neural Network (GNN), or a hybrid architecture is best suited for representing relationships among sites, alarms, equipment, failure modes, corrective actions, and ticket outcomes.

At a high level, the tool will:

- Ingest and normalize historical CRM records, Spectrum alarm data, and CRM tickets
- Identify recurring alarm sequences, site-specific issues, equipment correlations, and failure trends
- Convert historical troubleshooting knowledge into searchable embeddings
- Optionally model site/alarm/equipment relationships
- Provide technicians with recommended troubleshooting steps, probable causes, related historical cases, and confidence/explanation information
- Present results through a simple operator-facing dashboard or query interface
- Enable maintainers to review, validate, and improve AI-generated recommendations over time

The final proof-of-concept does not need to connect directly to live operational systems. Instead, students may work from exported, sanitized, or representative datasets provided by the project sponsor. The project should emphasize safe, auditable AI behavior: recommendations must be traceable to source data, historical tickets, known corrective

actions, or approved troubleshooting procedures. The tool should not autonomously execute actions against operational systems; it should assist technicians by providing relevant information and suggesting next-best troubleshooting steps.

A successful project will demonstrate that historical maintenance and alarm data can be transformed into an intelligent, traceable troubleshooting assistant. Performance will be evaluated using a held-out set of historical CRM incidents that were not used to build or tune the system. For each incident, the system's predicted probable cause and recommended troubleshooting steps will be compared with the documented root cause and corrective action. Evaluation measures will include top-1 and top-3 root-cause accuracy, precision/recall or F1 score for root-cause categories where sufficient labels exist, retrieval accuracy for whether relevant historical cases are returned, and source-grounding accuracy for whether recommendations are supported by cited tickets or approved procedures. CCC subject-matter experts will review a representative sample of recommendations for technical correctness, usefulness, and traceability. The system will also be compared with a baseline keyword-search or existing workflow. The proof-of-concept will be considered successful if it produces auditable recommendations, retrieves relevant historical cases, and demonstrates measurable improvement over the baseline according to the agreed evaluation criteria.

Student Learning Goals: Students will gain hands-on, portfolio-quality experience across several high-demand technical areas.

o Data Engineering and System Integration: Students will learn how to ingest, clean, normalize, and merge data from multiple enterprise-style sources. This may include working with CSV exports, database extracts, JSON records, REST APIs, or mock interfaces representing CRM, Spectrum, and systems data.

- Key topics include:
  - Data cleaning and normalization
  - Schema design for alarm, site, ticket, and equipment records
  - Entity resolution across different databases
  - ETL/ELT pipeline development
  - Handling missing, inconsistent, or noisy operational data
  - Building reproducible data processing workflows

o Artificial Intelligence and Machine Learning: Students will apply modern AI/ML methods to a real-world operational troubleshooting problem. The project may include both traditional analytics and advanced AI techniques.

- Potential AI/ML work areas include:
  - Natural language processing of CRM ticket narratives
  - Embedding generation for historical tickets and troubleshooting text
  - Semantic search using vector databases
  - Clustering of similar alarms, sites, symptoms, or failure modes
  - Classification of likely root cause categories
  - Recommendation algorithms for next troubleshooting steps

- Confidence scoring and explanation generation
- Evaluation of AI recommendation accuracy against historical outcomes

o Graph Analytics and Knowledge Representation: A central technical challenge is representing relationships among sites, alarms, equipment, failure modes, tickets, corrective actions, and outcomes. Students will study whether graph-based modeling improves troubleshooting recommendations.

- Potential graph-related work includes:
- Designing a knowledge graph schema
- Building a graph database of sites, alarms, equipment, tickets, and causes
- Running graph queries to identify related failures or recurring patterns
- Evaluating centrality, community detection, and path-based reasoning
- Investigating Graph Neural Networks for predictive root cause analysis
- Comparing graph-based retrieval against vector-based retrieval

o AI Agent and Retrieval-Augmented Generation Design: The system may use retrieval-augmented generation architecture to produce grounded troubleshooting recommendations. Students will learn how to build AI assistants that use enterprise data responsibly.

- Topics include:
- Retrieval-Augmented Generation, or RAG, architecture
- Prompt engineering for troubleshooting workflows
- Structured AI outputs using JSON schemas
- Guardrails to prevent unsupported recommendations
- Source citation and traceability back to historical tickets
- Human-in-the-loop review and feedback mechanisms
- Deterministic-first workflows with AI-assisted summarization

o Full-Stack Software Development: Students will build a user-facing interface for technician troubleshooting support. The interface should allow operators to search by site, alarm, equipment, or symptoms and receive ranked recommendations.

- Potential dashboard capabilities include:
- Alarm trend visualization
- Site-level issue history
- Search interface for historical tickets
- Recommended probable causes
- Suggested troubleshooting steps
- Similar historical incidents
- Links or references to source tickets
- Confidence and explanation display
- Feedback buttons for operator validation

- Possible implementation technologies include FastAPI, Flask, Streamlit, Dash, React, PostgreSQL, Neo4j, Qdrant, Chroma, FAISS, Elasticsearch, or similar open-source tools.

o Software Engineering and MLOps: Students will also gain experience developing a maintainable, testable software system.

- Topics include:

- Git-based collaborative development
- Containerization using Docker
- Database deployment and management
- API design and documentation
- Unit and integration testing
- Model evaluation and benchmarking
- Data versioning and repeatable experiments
- Technical documentation and user manual creation

Deliverables: Project deliverables are phased through the two-semester sequence and include:

o Weekly verbal or written progress reports on work completed and problems encountered

o Stakeholder interviews or workflow analysis summary for technician troubleshooting use cases

o Trade study report with technology recommendations, including comparison of vector database, graph database, and possible GNN approaches

o Data schema and ingestion design for CRM, Spectrum, and systems data sources

o Sanitized or representative dataset preparation and documentation

o Preliminary Design Review package

o Prototype ETL/data processing pipeline

o Prototype vector search and/or graph knowledge base implementation

o AI recommendation engine for likely root cause and next troubleshooting steps

o Operator-facing dashboard or query interface

o Testing and validation report using automated evaluation against held-out historical incidents, supplemented by SME review of a representative stratified sample of recommendations; includes accuracy, retrieval, traceability, baseline-comparison, and data-quality results

o Final implementation of the proof-of-concept system

o Comprehensive project documentation and user manual

o Project presentation and live demonstration

• Desired Final Demonstration: The final project demonstration should show an end-to-end troubleshooting workflow.

o A representative demonstration scenario may include:

- The user selects or enters a site and active alarm condition.
- The system retrieves related Spectrum alarm history, CRM tickets.
- The system identifies similar historical incidents and recurring alarm patterns.
- The system recommends likely root causes and ranked troubleshooting steps.
- The dashboard displays supporting evidence, including similar tickets, prior corrective actions, affected equipment, and trend visualizations.
- The user can provide feedback on whether the recommendation was useful.

o The demonstration should emphasize traceability, explainability, and operator trust. The interface will not present all matching historical records at once; instead, it will show a

ranked, concise evidence summary for each recommendation, including the top three to five supporting incidents or approved procedures, key metadata, relevance scores, and links to the full source records. Operators may filter or expand the evidence set by site, alarm, equipment, date, or failure mode as needed.

Desired Background: Background information required to successfully execute the project includes the following. Not all students are expected to possess all skills; the team should collectively cover these areas, with the faculty advisor and industry mentor providing supplemental guidance.

- Programming
  - Strong Python programming skills are highly desirable. Most data processing, AI/ML, backend API development, and automation will likely be implemented in Python. Familiarity with Pandas, NumPy, scikit-learn, SQLAlchemy, and Jupyter notebooks is helpful.
- Databases and Data Engineering
  - Students should have, or be willing to learn, basic database concepts including relational schemas, SQL queries, indexing, joins, and data normalization. Experience with PostgreSQL, SQLite, or similar databases is helpful. Experience with ETL pipelines or data cleaning is a plus.
- Machine Learning and Natural Language Processing
  - Prior coursework or project experience in machine learning is desirable. Useful topics include classification, clustering, embeddings, semantic search, model evaluation, and natural language processing. Experience with PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LlamaIndex, or similar tools is helpful but not required.
- Graphs, Networks, and Knowledge Representation
  - Familiarity with graph data structures is helpful. Students may work with graph databases such as Neo4j or graph libraries such as NetworkX. For more advanced implementations, students may investigate Graph Neural Networks using PyTorch Geometric or Deep Graph Library.
- Web Development
  - Basic web application development experience is useful for building the operator dashboard. Experience with Streamlit, Dash, Flask, FastAPI, React, JavaScript, or REST APIs is desirable.
- Cybersecurity and Data Handling Awareness
  - Because the project may involve operationally relevant data, students must be comfortable following sponsor guidance for data protection, sanitization, export control, and access restrictions. The proof-of-concept should avoid transmitting sensitive data to unauthorized cloud services. Local or approved on-premises model inference is preferred unless otherwise authorized.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: The project can likely be completed using sponsor-provided sanitized datasets and open-source software. Hardware needs are modest unless the team chooses to run larger AI models locally. • Estimated Total: o Approximately $150–$300 if existing compute resources are used. o Approximately $1,150–$2,800 if dedicated local AI/GPU hardware is required. • Software Note o The core software stack can be implemented using open-source tools such as Python, Pandas, scikit-learn, PyTorch, Hugging Face Transformers, FastAPI, Streamlit, PostgreSQL, Neo4j Community Edition, NetworkX, Qdrant, Chroma, FAISS, or similar technologies. o The project should prioritize local or approved on-premises processing for any sensitive or proprietary datasets.

The Boeing Company - Create lifecycle artifacts for open source linux software

Proposer: Andrew Augenstein andrew.e.augenstein@boeing.com

Project Contact: Andrew Augenstein andrew.e.augenstein@boeing.com

Project Description: Linux open source software is mature from a code development perspective, but often lacks software development lifecycle artifacts such as requirements, design, and test. With improved AI models, it should be possible to create these artifacts from a mature code base as a starting point for software developers/engineers to review and validate.

Student Learning Goals: Understanding of open source, AI usage for generating software development lifecycle artifacts, industry software development processes.

Deliverables: Documentation on prompting and process necessary to generate artifacts, and the actual requirements, design, and test artifacts.

Desired Background: Embedded / Operating System software development, AI usage for software development.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Parents Challenge - CRM Program System

Proposer: Deborah Hendrix dhendrix@parentschallenge.org

Project Contact: Deborah Hendrix dhendrix@parentschallenge.org

Project Description: Parents Challenge currently utilizes a CRM system to manage and track parent and scholar participation within our program. While the platform has been customized to support our operations, it still requires significant manual data entry, workarounds, and ongoing manipulation to generate the information needed for effective program management. Key functions such as tracking participation, processing changes in requests, updating parent profiles, managing session registrations, and maintaining accurate records are not as streamlined or automated as needed.
We are seeking to develop a more robust, fully customized system designed around the unique complexities of our program model. The new solution would centralize data management, automate workflows, improve reporting capabilities, and enhance the user experience for both staff and families. In addition, the system would integrate with the two mobile applications currently in development, which are intended to support parent navigation of resources, access to information, funding requests, and overall engagement with Parents Challenge services.

Student Learning Goals: Computer Science majors would gain substantial applied, real-world learning experiences across software development, systems design, data management, and user-centered technology. This would be an excellent experiential learning project because it combines technical complexity with community impact. This is the kind of project that gives students portfolio-quality experience while helping a mission-driven organization modernize its operations. It is stronger than a typical class assignment because it includes real users, real constraints, and measurable impact.

Deliverables: Over two semesters, Computer Science students would be expected to complete the planning, design, development, and testing of a customized parent and scholar management system for Parents Challenge. Deliverables would include a full needs assessment, system architecture, database design, user interface prototypes, and a functional web-based CRM platform. Students would also develop integrations with the two existing mobile applications to allow seamless sharing of user profiles, registrations, funding requests, and communication tools. Additional deliverables would include reporting dashboards, workflow automation features, security protocols, testing documentation, staff training materials, and a final implementation roadmap for future expansion.

Desired Background: Students participating in this two-semester project should have foundational knowledge in computer science, software development, and database systems. Preferred experience includes programming in web development languages, database design and SQL, API integration, and user interface development. Familiarity with mobile app connectivity, cloud platforms, cybersecurity best practices, and agile project management is highly desirable. Students should also have strong problem-solving skills,

the ability to work collaboratively in teams, and an interest in developing technology solutions that create community impact.

Delivery Mode: In-person

Additional Materials: Donorview is the current CRM system we utilize

General Dynamics Mission Systems - Daffodil: DFDL Schema Development and Embedded Deployment for Binary Data Parsing

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Mike Hoffman Mike.Hoffman@gd-ms.com

Project Description: Apache Daffodil is an open-source implementation of the Data Format Description Language (DFDL), an OGF/ISO standard for describing binary and text data formats using XML Schema annotations. Rather than writing custom parsers, developers define a DFDL schema and Daffodil handles parsing raw binary data into a structured XML info set automatically.

This two-phase project applies Daffodil to an embedded systems context. In Phase 1, students define MVP requirements, author DFDL schemas for a selected embedded binary protocol, and validate correct binary-to-XML parsing on a desktop environment with a reusable test harness. In Phase 2, students use Daffodil's C code generation backend — which compiles DFDL schemas into portable C, eliminating the JVM dependency — to deploy the parser on the AMD VEK385 or similiar (Versal AI Edge Gen 2, ARM Cortex-A78AE + Cortex-R52 + FPGA). Students then validate correctness against Phase 1 output and characterize performance on the embedded target.

Student Learning Goals: Students will develop proficiency in: Data Format Description Language (DFDL) and data interoperability standards; Apache Daffodil framework (schema authoring, CLI, test tooling); XML Schema definition and annotation; embedded C development targeting ARM Cortex processors; cross-compilation and toolchain setup for embedded Linux; requirements definition and MVP scoping for a software tool; parser validation and round-trip (parse/unparse) testing; performance profiling on resource-constrained hardware; and technical documentation and presentation of engineering results.

Deliverables: Project deliverables are phased through the two-semester sequence and include:

- Semester 1: MVP requirements document defining scope and success criteria; DFDL schema(s) for the selected embedded binary protocol; validated desktop parsing pipeline (binary input to XML output); test data set and automated test harness with documented pass/fail results.
- Semester 2: C code generated from DFDL schema via Daffodil C backend; embedded application compiled and running on AMD VEK385; performance characterization report (latency, memory footprint, throughput); final demonstration to project sponsor and written project report.

Desired Background: Background information required to successfully execute the project includes:

- C/C++ programming (required); familiarity with XML or structured data formats (preferred); basic embedded systems concepts (preferred); software engineering fundamentals such as version control and unit testing (required). DFDL expertise is not required -- students will develop this on the project. Java or Scala experience is a plus for Phase 1 desktop work.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: AMD VEK385 Versal AI Edge Series evaluation board (to be provided by GD-MS, approx. $2,000). Apache Daffodil is open-source and free. Development workstations provided by students/ASU. No additional purchases anticipated.

MentaMorph Inc - Data & Analytics Engine

Proposer: Susan Meredith susan@mentamorph.com

Project Contact: Susan Meredith susan@mentamorph.com

Project Description: This project builds the data infrastructure that captures, stores, and analyzes gameplay, learning assessments, and user behavior over time. It also incorporates licensing and usage data from the platform to evaluate engagement relative to access (e.g., seats used vs. purchased). The system will provide structured data and analytics outputs to support dashboards and AI-driven features.

Student Learning Goals: Students will gain experience designing scalable data systems and translating raw data into actionable insights. They will learn how analytics supports both educational outcomes and business intelligence.

Deliverables: • Data models and storage systems

- Data ingestion pipelines
- Analytics engine for user and group metrics
- Reporting APIs
- Sample reports (progress, engagement, license utilization)

Desired Background: • Databases (SQL/NoSQL)

- Backend programming
- Basic data analysis or statistics

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

DPAS - Dataset Generator testing and bug fixes

Proposer: PATRICK DIXON patjdixon@dpas-inc.com

Project Contact: PATRICK DIXON patjdixon@dpas-inc.com

Project Description: Fully test, identify bugs, and fix the application

Student Learning Goals: Experience in project management and software testing

Deliverables: Fully functional application that is "bug free" and well performing

Desired Background: JAVA programming, software testing, project management

Delivery Mode: In-person

Additional Materials: https://github.com/DPAS-INC/dataset-generator

Studio XE - Design System Consolidation for a Large-Scale Enterprise Web Application

Proposer: David Lutz david.lutz@communityhdp.com

Project Contact: Rohith Reddy Byreddy rohith@snackiebeans.com

Project Description: Studio XE is a software platform that organizations in healthcare, finance, and legal services use to run daily operations. Its web application has grown to roughly 4,300 interface components spanning 14 functional areas, built incrementally by different contributors over several years. That growth produced a specific, measurable problem: the application contains seven competing sets of visual design definitions — color, spacing, typography — that evolved independently, alongside roughly 730 distinct color values written directly into individual files. Any change to branding, theming, dark mode, or customer-specific appearance is effectively blocked, because there is no single place to make it.

The student team consolidates the seven systems into one: a single design-token module covering color, spacing, radius, elevation, and type scale in light and dark appearance, with a documented public API. The team migrates the application onto it, adds automated lint enforcement so the condition cannot recur, then builds a component catalog (Storybook) and automated visual regression testing so a change to any shared component displays its effect across the entire application before it is approved. This is a consolidation and engineering project, not a visual redesign: students unify an existing look and make it enforceable. All work occurs against a synthetic dataset — no production systems, customer data, or protected health information is accessible at any point.

Student Learning Goals: Across 30 weeks, students perform a large-scale legacy migration under active development — the class of work professional front-end engineers spend much of their careers on and coursework rarely reaches. They design a public token API other engineers must live with, execute a codemod across ~4,300 components in migration slices merged weekly, and build the CI enforcement (lint rules, visual regression gates) that keeps an engineering standard true after they leave.

Students practice production discipline throughout: pull-request-only workflow into a sponsor-controlled repository, code review held to the sponsor's employee standard, numeric acceptance criteria fixed at term start (distinct hex values 730 → ≤30; catalog ≥900 entries; 100% coverage of shared primitives and dashboard components), and biweekly working demos — running software, not slides. They finish with merged, production-shipped work on a platform with real customers, attributable in a public portfolio.

Deliverables:

- A single design-token module (color, spacing, radius, elevation, type scale; light and dark) with published API documentation; the other six modules removed or reduced to deprecated re-exports.
- Migration complete: distinct hard-coded color values reduced from ~730 to ≤30, each survivor documented as a deliberate exception.
- An automated lint rule and CI gate that fails the build on new unreferenced values,

preventing recurrence.

- Verified light and dark appearance across the 20 highest-traffic screens, with before-and-after evidence.
- Component catalog: 100% of shared primitives and dashboard components, ≥900 entries against an inventory agreed at term start.
- Automated visual regression testing integrated into the approval workflow, plus a written contribution guide sufficient for a new engineer to add a conforming component without assistance.

Desired Background:

- Front-End: strong TypeScript and React; Tailwind CSS.
- Tooling: Storybook, ESLint, GitHub Actions (sponsor provides the full environment preconfigured).
- One member with design sensibility valuable but not required.
- Out of scope: server/database work, new product features, visual redesign.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Arizona State University - Developing Vision-Language Models for Surgical Video Analysis

Proposer: Yalin Wang ylwang@asu.edu

Project Contact: Yalin Wang ylwang@asu.edu

Project Description: Vision-Language Models (VLMs) are a new generation of AI systems that jointly understand visual information and human language. Recent models such as GPT-4V and LLaVA have demonstrated remarkable abilities in image understanding, reasoning, and multimodal interaction. This project explores how these technologies can be applied to neurosurgery video analysis.

Students will develop AI systems capable of analyzing surgical videos, recognizing surgical workflows and instruments, and generating meaningful textual descriptions or summaries. The project combines computer vision, natural language processing, and medical AI, offering students hands-on experience with state-of-the-art multimodal learning techniques.

This interdisciplinary project is ideal for students interested in AI research, healthcare technologies, computer vision, and large-scale deep learning systems.

Student Learning Goals: Students will gain practical experience in:

Vision-language models (VLMs) and multimodal AI
Deep learning and transformer architectures
Video understanding and computer vision
Medical AI applications and healthcare data
Research methods, experimentation, and model evaluation

Participants will work with real surgical video data, read recent AI research papers, implement modern deep learning models, and develop end-to-end AI pipelines using frameworks such as PyTorch and Hugging Face.

The project provides valuable preparation for graduate research, AI industry roles, and interdisciplinary healthcare AI applications.

Deliverables: Expected project outcomes include:

Baseline and fine-tuned vision-language models
Experimental evaluations and performance analysis
Prototype systems for surgical video captioning or question answering

Advanced teams may also contribute to research publications or open-source medical AI tools.

Desired Background: Solid background in machine learning, image analysis, and large language models.

Delivery Mode: In-person

DigiClips - DIGICLIPS GENERAL PUBLIC PROJECT

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: Menu For GENERAL PUBLIC
SUBMENU GENERAL PUBLIC PROJECT
SUBMENU GENERAL PUBLIC as part of DIGICLIPS NEW VERSION PROJECT to present on AWS Amazon Lightsail
new domain name www.digiclpsproject.com/generalpublic
==========
DigiClips ®

Search Language Translation Subscriber User Profile Help
Contact Us Opinion Chat Sign Out

General Public Media Search Engine
==========
CAPSTONE TEAMS WORKING ON GENERAL PUBLIC PROJECT: ASU_5_Fall 2025 TEAM CSUS_1 Fall 2025 IOWA STATE TEAM 5
==========
FOR PRESENTATION DEVELOPMENT MEDIA SEARCH ENGINE
IMPORTANT: Make user friendly. First test and fix problems on the digi-frontend computer. Then merge to digiclips henry's github search engines master branch with comments notes linked to readme and version number. Then do a production build and put up on

PRESENTATION: Should be presented in the cloud as part of DIGICLIPS NEW VERSION PROJECT and put up on AWS Amazon Lightsail this domain name www.digiclipsprojects.com
Description of what is needed. Should be coded in Angular 20 https://angular.dev/
IMPORTANT: As they become a new user they need to be approved by ADMINISTRATION
NEED FIXED: https://www.britopian.com/earned/media-monitoring/
IMPORTANT: To make very user friendly and not to miss any media data
IMPORTANT: Please merge code a lot to digiclips henry's github search engine master repository branch with with comments notes linked to readme and with version number and on team name branch

1. GENERAL PUBLIC - Description of what is needed. Should be done in Angular 19 or 20 https://angular.dev/, This would be a non subscribers and backed by Advertising Dollars through Administration.
Take off Option and Few Options and More Options - just put the Options below where to search. GENERAL PUBLIC OPTIONS to generate media results reports, stream segments,

but a way to encourage the users to become a subscriber when they need to. There is a way to click on the reports and stream the data but with no DETAIL BUTTONS and no way of editing the reports, downloading or any of the Media Analytics. I believe they would just use the Search and Option and no Email Alerts page they can use of course Language Translation. Must have the legal disclaimer. Must have a way to encourage them to become a subscriber. From the Media Results Report to click on the segment on the report with a button to COMMENT and CONVERSATION.
ADVERTISING ON WEBSITES This explains how to set up a Use dummy ads for presenting and demonstrating.
Media to be covered to generate media results reports
Free To Air Antenna TV stations - Radio Station - Web streaming TV - Youtube Tv - Podcast - Siriusxm - Newspapers - Magazines - Social Media - Blogs

IMPORTANT: As they become a new user they need to be approved by ADMINISTRATION

1D. Selection of users from AWS Amazon Lightsail the frontend website when a new user wants to get email alerts notification to the administrators who can approve them. Keep track through a dashboard spreadsheet. Remote connect through codentv1a on our local system. There are 2 types of users: General Public - Subscribers.

List of subscriber services like email alerts media analytics buttons
GENERAL PUBLIC AND SUBSCRIBER
https://docs.google.com/document/d/1OF7AnKaMs0PTSaZcIvTtllvDBklyBzzaBftqut_RljE/edit?usp=sharing

GENERAL PUBLIC MEDIA RESULTS REPORTS
https://docs.google.com/document/d/1808jD-tPIid8uvKf2f0zJezAmfYbZFGPm9SnrCaBqN0/edit?usp=sharing

To have this working with the generated media results reports
Comment Reply Like Dislike Notification
DIGI COMMENT REPLY <DETAILED EXPLANATION

GENERAL PUBLIC OPTIONS
https://docs.google.com/document/d/16y_NDrKB6cInYSVClmPGyzYFKejlTydtQd0yo7NrL1M/edit?usp=sharing

Backed by
ADVERTISING ON WEBSITES This will explain about advertising on websites. As far as

presenting digiclips projects, a dummy ads like this can be created to show an example.

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: With 30 days before the end of semester. Present project up on AWS Amazon Lightsail on www.digiclipsproject.com/generalpublic

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials:
https://docs.google.com/document/d/1TvTRTmg8F7UqVDk1stpWOM5l8OpozxyQw3n3ADjoCW4/edit?usp=sharing

DigiClips - DIGICLIPS MEDIA TRANSFER

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: MEDIA TRANSFER

Should be like https://fromsmash.com/

For Subscribers and Administrators users to be able to send their data whoever they need to after they have downloaded the segments to their computer..

Should be for now a part of www.digiclpsproject.com/subscriber/mediatranfer and www.digiclipsproject.com/administration/mediatransfer up on AWS Amazon Lightsail.

IMPORTANT: For now done through
http://router.asus.com/Main_Login.asp

Connecting through AWS Amazon Lightsail at first through www.digiclipsproject.com with the DIGICLIPS NEW VERSION PROJECT

We have a static IP address

Should be very much like www.wetransfer.com or like Smash https://fromsmash.com/
Users should be approved by Administrators

Should have a way to send email to who needs to download the segments, plus
Email the users to confirm the segment was sent. Also set up a time to do the
Media transfer.

Data transferred should be stored on our local system not on AWS Amazon Lightsail and on the Seagate SD Drive connected to the ASUS Router. It should be auto deleted every 30 days adjusted by the Administrator.

EMAIL ALERTS: Can be done through digiclips.mediatransfer@gmail.com

GITHUB WETRANSFER
https://github.com/orgs/WeTransfer/repositories

GITHUB FROMSMASH
https://github.com/search?q=fromsmash&type=repositories

Transferring of any data should have this MEDIA TRANSFER

A button for Subscribers only.
A button for Administration only

Legal Disclaimer
FAIR USE NOTICE. This email may contain links to broadcast media content available in the research database provided by DIgiClips,Inc.. and clipped by the sender of this email. Digiclips is a provider of monitoring and analytics services and research tools only, does not sell, distribute or syndicate content, and does not own the rights in the broadcast media content available in the DigiClips media database. The recipient of this email is required to comply with applicable copyright and other laws and may not sell, copy, modify, re-broadcast, distribute, perform or display broadcast media content in any infringing or illegal manner. This email is intended for the named recipient only. The recipient is responsible for ensuring that all viewing, storage, sharing and other usage of the content accessible through this email constitutes "fair use" or is otherwise non-infringing under applicable law. If the recipient intends to make any usage of the content that may not constitute "fair use" or otherwise be non-infringing, DigiClips strongly recommends that you contact the relevant rights holder to obtain a suitable license in such content.

==========
CAPSTONE TEAMS WORKED ON THIS

OHIO STATE 2024 Team - merged code to the digiclips henry's github master repository branch
ASU TEAM 4 - also have worked on this and ASU_13_Fall 2025 TEAM and OSU_4_Spring 2026 Team

==========
DESCRIPTION NEEDED FOR SUBSCRIBER AND ADMINISTRATION

BUTTONS like MEDIA TRANSFER to SEND DATA
Maximum size like 2.0 GB - can be changed by Administration
EMAIL ALERTS to send data to download also to who also sent the data
Through digiclips.mediatransfer@gmail.com
Auto delete after 30 days controlled by Administration
MYSQL relational database to keep track of who the user is and who is downloading the data.
Can also work for existing clients.
==========
Who would use Media Transfer would be for Subscribers and Administrators
 plus also on digilap01 and digiclap02 laptop computers for further local testing
Need to change this https://hbremers.github.io/dcWeb/requestservice
 maybe this would work for testing https://hbremers.github.io/dcWeb/mediatransfer

==========
PRESENT AND DELIVERABLE

DIGICLIPS NEW VERSION PROJECT

NEW VERSION to put up on AWS Amazon Lightsail
www.digiclipsproject.com/subsriber/mediatransfer and put up on www.digiclipsproject.com/administration/mediatransfer then come up with a domain name connecting to the processed media data on our local computers to generate media results reports.

AWS AMAZON LIGHTSAIL INSTRUCTIONS
https://docs.google.com/document/d/1zyJZf8pPqlvNhIskuoYasB8oibuJhE9MZD-O7gkrH0g/edit?usp=sharing
==========
IMPORTANT: What is also needed is to be able to use MEDIA TRANSFER on my laptops digilap01 and digilap02 so we can send out data from either one of these laptops. This needs to also be done through the ASUS router and through saved on the Seagate drive.

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: To put DigiClips Media Transfer project up on AWS Amazon Lightsail on www.digilclipsproject.com/subscriber and on www.digiclipsproject.com/generalpublic

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials:
https://docs.google.com/document/d/1V8Ay_gJwgJxOnMIIwJdMoD46d_szQuYVHK24nYZ7lQg/edit?usp=sharing

DigiClips - DIGICLIPS NEW VERSION PROJECT

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: DIGICLIPS NEW VERSION PROJECT

NEW VERSION to put up on AWS Amazon Lightsail and put up on up with a domain name like www.digiclipsproject.com connecting to the processed current media data on our local computers to generate media results reports for both subscribers and general public and show off information and administration.

ROUGH DRAFT:
https://www.digiclipsinc.com/digiclipsnewversion/DigiclipsMainMenu.htm

AWS AMAZON LIGHTSAIL INSTRUCTIONS
https://docs.google.com/document/d/1zyJZf8pPqlvNhIskuoYasB8oibuJhE9MZD-O7gkrH0g/edit?usp=sharing put up on www.digiclipsproject.com

IMPORTANT NOTE: Been worked on by other Capstone Teams the code is up on digiclips henry's github repository master branch with notes comments links to readme and with version number for research and to study and to use. Please email bobshapiro40@gmail.com and hbremers@gmail.com your resume with your code experience, your email address, phone number and github branch. Set up google weekly meetings where we can help you set up your WORK ENVIRONMENT

First make a plan then create the plan, merge first to digiclips henry's github master branch. It takes learning about the frontend https://angular.dev/ upgrading to angular20.

Let us know if you have any ideas to get this project better and much more user friendly. It also takes connecting and working with other capstone students to divide up the project MASTER UNIVERSITY SPREADSHEET which can be done through https://discord.com. DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK

As each part of the project it will take clicking on the BUTTON https://www.digiclipsinc.com/underconstruction.htm until it is working. Then put on the local computers for further testing, then once working do a production build and put up on AWS Amazon Lightsail also on https://44.231.94.92/ and REALLY IMPORTANT your final deliverable and presentation have a domain name like may www.digiclipsproject.com .

DIGICLIPS PROBLEM NEED SOLVED
https://docs.google.com/document/d/1OclbPLNrCWEqKASeb8EBUfJCDhCiEBkVj3d6T-PZzoM/edit?usp=sharing

USERS OF THE SOFTWARE WOULD BE 2 TYPES OF USERS

GENERAL PUBLIC
https://docs.google.com/document/d/1TvTRTmg8F7UqVDk1stpWOM5l8OpozxyQw3n3ADjoCW4/edit?usp=sharing
And
SUBSCRIBERS
https://docs.google.com/document/d/1QBJ8ZbZLepvRNEbLhgsaF2Q2g2DGWcDqu__iyJDaY3Y/edit?usp=sharing
==========
DigiClips Menus for DigiClips New Version
MAIN MENU PROJECT
MAIN MENU
SUBMENU SIGN UP
https://docs.google.com/document/d/1WhqIpioNdhQTpQbHtJHQNv7gtDvHaTyVJvkNh3mvN7g/edit?usp=sharing
NOTE: Login For Register Users
NOTE: Sign Up For new users and be approved by administrator □ Accept Terms
□ Accept Terms □ Remember Me □ Forgot Password
NOTE: Sign Out For users to sign out
NOTE: All users LEGAL TERMS OF USE must approve this NON DISCLOSURE AGREEMENT
NOTE: For any Registered Users.
NOTE: Administration needs email alert plus needed dashboard spreadsheet
==========
DIGICLIPS GENERAL PUBLIC PROJECT
SUBMENU LOGIN PROJECT
SUBMENU LOGIN
General Public Subscriber Information Administration Forget Password
==========
DIGICLIPS GENERAL PUBLIC PROJECT SUBMENU GENERAL PUBLIC PROJECT
SUBMENU GENERAL PUBLIC
Search Chat Language Translation Subscriber User Profile Contact Us Opinion Help Sign Out
==========
DIGICLIPS SUBSCRIBER PROJECT
SUBMENU SUBSCRIBER PROJECT
SUBMENU SUBSCRIBER
Search Email Alerts Archive Media Media Contacts Media Transfer Language Translation User Profile Contact Us Opinion Help Chat Sign Out
==========
DIGICLIPS ADMINISTRATION PROJECT

SUBMENU ADMINISTRATION PROJECT
SUBMENU ADMINISTRATION
Administration Chat Media Transfer Help
==========
DIGICLIPS PROJECT INFORMATION AWS AMAZON LIGHTSAIL
SUBMENU INFORMATION PROJECT
SUBMENU INFORMATION
Information

DIGICLIPS ® Request Service Rate Sheet About Us Samples Opinion Customer Review Chat Contact Us Download DigiClips App

==========
==
IMPORTANT TO MAKE PRESENTABLE
https://docs.google.com/document/d/13sjWZxgqHHPcnl320xkEnYYs-pywDYNjNjrCnP35iPA/edit?usp=sharing

TO PRESENT YOUR PROJECT:

BACKEND teams - needs to connect with teams working on frontend to generate media results reports on AWS Amazon Lightsail https://44.231.94.92/ new domain name www.digiclipsproject.com

FRONTEND - needs to connect with backend and In Cloud Team

ADMINISTRATION - does not go up on AWS Amazon Lightsail for presentation goes up on codentv1a

NOTE: Must be be able to through SEARCH OPTIONS EMAIL ALERTS PAGES generate the current media for both GENERAL PUBLIC and SUBSCRIBERS
==========
BACKEND PROCESSING DATA
https://docs.google.com/document/d/1jiaOhnZjKJamTMRmap_-qzXr794HW4XYkXEUT0DfMK8/edit?usp=sharing

NOTE: Needs to process the current media data to be stored on the archived drives and auto delta around every 30 days for now, but can be changed by the Administrator.

NOTE: Current media data needs to be processed and then imported into mysql relational database.

NOTE: Then to generate Media Results reports for GENERAL PUBLIC - SUBSCRIBERS

==========
GENERAL PUBLIC
https://docs.google.com/document/d/1TvTRTmg8F7UqVDk1stpWOM5l8OpozxyQw3n3ADjoCW4/edit?usp=sharing

OPTION IMPORTANT: Take down OPTIONS word
https://docs.google.com/document/d/1z03UTGtF8uiISEinxHFEXUusl03Wa6FnWJ5AEfwYMTY/edit?usp=sharing

NOTE: Through down
https://www.digiclipsinc.com/digiview/DROPDOWN%20BOXES_01.htm
NOTE: For users to choose all or which country market media television web streaming radio newspapers magazines social media podcast blogs siriusxm - also time ranges and times
NOTE: Choose for Television Free To Air Antenna also All Closed Caption Speech to text left and right channels, video text recognition image recognition lip read to text.
NOTE: Choose Web Streaming All Closed Caption Speech to text left and right channels, video text recognition image recognition lip read to text.
NOTE: Choose for Radio podcast siriusxm speech to text in left and right channels.
NOTE: Newspapers Magazines Social Media Blogs - scrape the web for the reports and delete cookies rssreader http://www.rssreader.com/download.htm
https://zapier.com/blog/best-rss-feed-reader-apps/ or Open source for rss-feed-readers: https://opensource.com/article/17/3/rss-feed-readers maybe use www.duckduckgo.com
NOTE: Backed by Advertisements
NOTE: Generate for General Public Media Results Reports with Comment on each segment and detail steam each segment.

Generate Media Results Reports like this for the General Public.
GENERAL PUBLIC MEDIA RESULTS REPORTS
https://docs.google.com/document/d/1808jD-tPIid8uvKf2f0zJezAmfYbZFGPm9SnrCaBqN0/edit?usp=sharing

NOTE: Has Comments to click on a segment.
NOTE: □ Click on for general public users to be able to edit which segment they need from media results reports

Example: GENERAL PUBLIC MEDIA RESULTS REPORTS
Example of segment

DPS (TELEVISION)
TV HITS 3

□ KCNC-DEN (CBS), 29 Mar 2021 05:02 - CBS4 This Morning-5AM

KCNC-DEN (CBS) - Mar 29, 2021 07:02
...wants his son to be remembered. We are following a developing story. Denver school board member tae anderson says he welcomes any investigation after allegations that he is actually assaulted.
Comment Reply Like Dislike Notification

COMMENTS
https://docs.google.com/document/d/1gjlUTh9iZPZKx0aW1V2yPZi4w-4Kgd3kpS5G1qM7nsI/edit?usp=sharing
NOTE: Users can make a comment or reply and others can also reply back to what was said.
NOTE: General public users can click on a segment and stream the segments also with comments.
==========
SUBSCRIBERS
https://docs.google.com/document/d/1QBJ8ZbZLepvRNEbLhgsaF2Q2g2DGWcDqu__iyJDaY3Y/edit?usp=sharing

OPTION IMPORTANT: Take down the word OPTION

NOTE: Through down
https://www.digiclipsinc.com/digiview/DROPDOWN%20BOXES_01.htm
NOTE: For users to choose all or which country market media television web streaming radio newspapers magazines social media podcast blogs siriusxm - also time ranges and times
NOTE: Choose for Television Free To Air Antenna also All Closed Caption Speech to text left and right channels, video text recognition image recognition lip read to text.
NOTE: Choose Web Streaming All Closed Caption Speech to text left and right channels, video text recognition image recognition lip read to text.
Use for project CNBC http://www.freeintertv.com/view/id-2566
NOTE: Choose for Radio podcast siriusxm speech to text in left and right channels.
NOTE: Newspapers Magazines Social Media Blogs - scrape the web for the reports and delete cookies rssreader http://www.rssreader.com/download.htm
https://zapier.com/blog/best-rss-feed-reader-apps/ or Open source for rss-feed-readers: https://opensource.com/article/17/3/rss-feed-readers maybe use www.duckduckgo.com
NOTE: Backed by Subscriber fees
NOTE: Generate for Subscribers Public Media Results Reports and detail stream each segment with Buttons on each segment.

EMAIL ALERTS
https://docs.google.com/document/d/1YjMkvuaIJOQGIbUZez6l6JlFDhmh6QutVK_8KxSra5E/edit?usp=sharing
IMPORTANT: For subscriber

NOTE: Through down
https://www.digiclipsinc.com/digiview/DROPDOWN%20BOXES_01.htm
NOTE: For users to choose all or which country market media television web streaming radio newspapers magazines social media podcast blogs siriusxm - also time ranges and times
NOTE: Choose for Television Free To Air Antenna also All Closed Caption Speech to text left and right channels, video text recognition image recognition lip read to text.
NOTE: Choose Web Streaming All Closed Caption Speech to text left and right channels, video text recognition image recognition lip read to text.
Use for project CNBC http://www.freeintertv.com/view/id-2566
NOTE: Choose for Radio podcast siriusxm speech to text in left and right channels.
NOTE: Newspapers Magazines Social Media Blogs - scrape the web for the reports and delete cookies rssreader http://www.rssreader.com/download.htm
https://zapier.com/blog/best-rss-feed-reader-apps/ or Open source for rss-feed-readers: https://opensource.com/article/17/3/rss-feed-readers maybe use www.duckduckgo.com
NOTE: Backed by Advertisements for general public
NOTE: Backed by Subscription fees for subscribers
NOTE: Generate for Subscribers Public Media Results Reports and detail stream each segment with Buttons on each segment.

Generate Media Results Reports like this for the Subscribers
SUBSCRIBER MEDIA RESULTS REPORT
https://docs.google.com/document/d/1yY0yBFRO63yyu1BMkDTYKLNX-7aAQclXvAZHMazarrU/edit?usp=sharing

NOTE: Subscriber use EMAIL ALERTS keep updated on keywords and subjects that as mentioned in the media will be emailed to the subscriber.
NOTE: □ Click on for subscriber users to be able to edit which segment they need from media results reports.

NOTE: For Subscriber below each segment should be something like this with
Country Market Audience Numbers Media Values Publicity Value Length Sentiment
Done also through Nielson through Administration and into Mysql Relational Database to generate media results reports for subscribers.
NOTE: There is a rough draft for this but this data would come from Nielsen through Administration.
NOTE: This is a segment that should be able to click on to do a detailed streaming of the segment.
For Television:
https://www.digiclipsinc.com/digiview/Sample%20of%20Stream_Television.htm
For Radio: https://www.digiclipsinc.com/digiview/Sample%20of%20Stream_Radio.htm

NOTE: Subscribers should be able to connect with MEDIA AND JOURNALISTS CONTACTS

should be done through Administration and through Mysql Relational Database

DPS (TELEVISION)
TV HITS 3

□ KCNC-DEN (CBS), 29 Mar 2021 05:02 - CBS4 This Morning-5AM
KCNC-DEN (CBS) - Mar 29, 2021 07:02
...wants his son to be remembered. We are following a developing story. Denver school board member tae anderson says he welcomes any investigation after allegations that he is actually assaulted...

Country: USA Market: Denver, Colorado Audience Numbers Media Values Publicity Value Length Sentiment

NOTE: Subscriber Media Results can click on □ edit media results reports with each segment to detail stream each segment with
BUTTONS

DETAILS STREAMING BUTTONS
https://docs.google.com/document/d/1JlXV_sO1A7sXJ41kiJtz46NTblG1CIC1JP2-6hM42UA/edit?usp=sharing

NOTE: BUTTONS

PREVIOUS SEGMENT - NEXT SEGMENT - EDITOR - DOWNLOAD - MEDIA ANALYTICS - MEDIA TRANSFER

NOTES: On Detailed Media Streaming
PREVIOUS SEGMENT Previous segment is clicking on to get the segment before
NEXT SEGMENT Next segment is to get to the segment after the segment
EDITOR To be able to edit the Previous Segment with the segment and the next segment like this https://www.avs4you.com/
DOWNLOAD To download segment and edited Television Radio podcast siriusxm radio segments and Newspapers Magazine Social Media Blogs
MEDIA ANALYTICS with
Country Market Audience Numbers Media Value Publicity Value Length Sentiment
□ Positive □ Neutral □ Negative
MEDIA TRANSFER To transfer media data like www.wetransfer.com or like https://fromsmash.com/
==========
LANGUAGE TRANSLATION
NOTE: For the front end
https://www.digiclipsinc.com/digiview/DigiClips_Language_Translator.htm

NOTE: Users can choose which language for both Media Results Reports and for the frontend of MAIN MENU - SUBMENU LOGIN - SUBMENU SIGN UP - SUBMENU SIGN OUT - SUBMENU HELP SUBMENU SIGN OUT - SUBMENU HELP - SUBMENU GENERAL PUBLIC - SUBMENU SUBSCRIBER - SUBMENU INFORMATION - SUBMENU INFORMATION - SUBMENU ADMINISTRATION - SEARCH - OPTIONS - EMAIL ALERTS - INFORMATION - ADMINISTRATION.

==========

ARCHIVE

https://archive.org/

NOTE: This can be also used generating some media results reports

NOTE: Once the Option page is working on the local system it will be tested on another system.

==========

DIGICLIPS MAIN MENUS FOR NEW VERSION EXPLANATION

MAIN MENU

SUBMENU LOGIN

SUBMENU SIGN UP

SUBMENU SIGN OUT

SUBMENU HELP

SUBMENU GENERAL PUBLIC

SUBMENU SUBSCRIBER

SUBMENU INFORMATION

SUBMENU ADMINISTRATION

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: To work with the other capstone teams and put the new version projects up on AWS Amazon Lightsail on www.digiclipsproject.com

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials: https://docs.google.com/document/d/1lRaMCHg2ik3-zD-wmeHSE_5gnv5t0zRk4xvqwBWmA4s/edit?usp=sharing

DigiClips - DIGICLIPS SUBSCRIBER PROJECT

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: DIGICLIPS ®

Search Email Alerts Archive Media Media Contacts Media Transfer
Language Translation User Profile Contact Us Opinion Help Chat Sign Out

Subscriber Media Search Engine

==========
DIGICLIPS SUBSCRIBER PROJECT
CUSTOMERS FOR SUBSCRIPTIONS
==========
ROUGH DRAFT SUBSCRIBERS

IMPORTANT: This is what should be deliverable and presentable on www.digiclipsprojects.com/subscriber
Explain what is needed for the SUBSCRIBER.
More details of what is needed EMAIL ALERTS
For now just the Denver, Colorado Television and Radio Stations administration can add on many more markets stations and the media.

Work with other university capstone teams MASTER SPREADSHEET CAPSTONE TEAMS
DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK

Please first merge to DigiClips henry's github master repository with comments notes linked to readme and version number. Then put up on digi-frontend computer then put up on AWS Amazon Lightsail www.digiclipsproject.com/subscriber

Description of what is needed to present the SUBSCRIBERS project on www.digiclipsproject.com/subcriber
Should show how to pay to subscribe.

What is needed should be: www.digiclipsproject.com/subscriber

SUBSCRIBERS projects
OPTION Take off Option and Few Option and More Options - just put the Options below where to search

EMAIL ALERTS
www.digiclipsproject/subscriber/emailalerts

CHARTS AND GRAPHS PROJECT

Code develop in Angular 20 https://angular.dev/

APPROVE SIGN IN SUBSCRIBER: On www.digiclpsproject.com/subscriber
==========
Choice of use
□ 1 day □ 2 days □ 3 days □1 week □ 2 weeks □1 month □ 3 months □ 6 Months
□ 9 months □ One year
==========
IMPORTANT NOTES: CHOICE OF USE There should be a way for the subscriber to contact administration for an administrators to be able to help the user generate media results reports or help set them up with the email alerts and help the subscriber with editing the Television and Radio segments and to a transcript plus send the segments via MEDIA TRANSFER
==========
IMPORTANT NOTE: For Subscriber
So a GENERAL PUBLIC user can choose to be a SUBSCRIBER and choose how much time they need to be a SUBSCRIBER. Needs to charge the SUBSCRIBER for each one of the times. Needs a way to pay for their use and a way through the MYSQL relational database to keep track of the user. Once done with their time they should get cut off until they want to SUBSCRIBE and they get back in when they need to subscribe again. Plus they can choose how much they need then be taken off. They can pay on line via credit card like somehow through www.clover.com or somehow do a direct deposit like through www.bill.com
==========
WHAT IS NEEDED FOR SUBSCRIBERS

DETAILS STREAMING BUTTONS FOR SUBSCRIBERS

PREVIOUS SEGMENT - NEXT SEGMENT - EDITOR - DOWNLOAD - MEDIA ANALYTICS - MEDIA TRANSFER

https://www.digiclipsinc.com/underconstruction.htm

SUBSCRIBER EMAIL ALERTS
www.digiclipsproject.com/subscriber/emailalerts

MEDIA AND JOURNALISTS CONTACTS
BUTTONS http://www.digiclipsinc.com/digiview/TV_Image.png
EDITOR http://www.digiclipsinc.com/digiview/Sample%20of%20Stream_Television.htm
==========
BACKEND PROCESSED MEDIA DATA
BACKEND TV RADIO PROCESSED DATA needs to have the current processed media data which is on CODENTV1C - CODENTV2A - DIGI62 local computers. Generate these media results reports.

IMPORTANT: SSH3 virtual directory set up to make sure you are connecting to the media processed data.
==========
What is needed with each button to present the projects.
==========
Search
NOTE: To search all of the media and general media results reports
https://libanswers.snhu.edu/faq/8739
A boolean search is a method of searching that uses operators like AND, OR, and NOT to combine keywords and narrow or broaden search results.
then the ability to click on these buttons

SUBSCRIBER MEDIA RESULTS REPORT
□ KCNC-DEN (CBS), 29 Mar 2021 05:02 - CBS4 This Morning-5AM
KCNC-DEN (CBS) - Mar 29, 2021 07:02
...wants his son to be remembered. We are following a developing story. Denver school board member tae anderson says he welcomes any investigation after allegations that he is actually assaulted...

Market: Denver, Colorado Audience Numbers Publicity Value Length Sentiment

IMPORTANT: MEDIA ANALYSIS
IMPORTANT NOTE: Subscriber can us this □ check off and be able to edit and choose which segments they want.

IMPORTANT NOTE: Users can click on each segment and stream the details some like this with each button
USE THIS UNTIL THE BUTTON IS WORKING:
https://www.digiclipsinc.com/underconstruction.htm

BUTTONS http://www.digiclipsinc.com/digiview/TV_Image.png
EDITOR http://www.digiclipsinc.com/digiview/Sample%20of%20Stream_Television.htm
PREVIOUS SEGMENT
NOTE: https://www.digiclipsinc.com/digiview/TV_Image.png
NEXT SEGMENT
NOTE: https://www.digiclipsinc.com/digiview/TV_Image.png
EDITOR
NOTE: https://www.digiclipsinc.com/digiview/TV_Image.png
DOWNLOAD
NOTE: https://www.digiclipsinc.com/digiview/TV_Image.png
MEDIA ANALYTICS
NOTE: https://www.digiclipsinc.com/digiview/TV_Image.png
MEDIA TRANSFER
NOTE: https://www.digiclipsinc.com/digiview/TV_Image.png

Option
IMPORTANT Take off Option and Few Options and More Options - just put the Options below where to search.

Media Type
□ All □ Television Free To Air Antenna □ Web Streaming Television □ Radio
□ Podcast □ Newspapers □ Magazines □ Social Media □ Blogs
==========
Geographical Location County Markets
NOTE: Right now just for presenting to just cover the Denver, Colorado market, with Administration to add on more markets and media.
==========
To choose Format: □ All □ email □ doc □ pdf □ spread sheet □ web page

FOLLOW THIS ROUGH DRAFT:
https://www.digiclipsinc.com/digiview/DigiClips_Search_Option.htm
==========
Email Alerts
NOTE: Description of what is needed. SUBSCRIBER EMAIL ALERTS

NOTE: To choose which market you need an email alert which media markets and date range. To choose Format: □ ALL □ email □ doc □ pdf □ spread sheet □ web page
ROUGH DRAFT: https://www.digiclipsinc.com/digiview/DigiClips_EmailAlerts.htm

NOTE: What the subscriber report should look like SUBSCRIBER MEDIA RESULTS REPORT

Media Analytics
MEDIA ANALYTICS
NOTE: Explains how the subscribers would use MEDIA ANALYTICS and how different ways to obtain the media analytics to generate the media results reports.

Media Contacts
MEDIA AND JOURNALISTS CONTACTS
NOTE: This explains how to contact the media and journalists
Media Transfer
MEDIA TRANSFER
NOTE: This explains about transferring data like TV segments Radio Transcripts

Language Translation
LANGUAGE TRANSLATION PROJECT
NOTE: To be with the frontend and the back and start with Spanish then Administration should be able to add on more languages

Archive Media
ARCHIVE MEDIA NOTE:
What should be done is capture and archive all of the media and be able to generate media results reports with the buttons. Not right away and only for subscribers not at this point.

User Profile
NOTE: Should put the profile of the user and have the data imported into mysql relational database. A way for the user to change password.

Contact Us
NOTE: A way to contact administrators with email text message and phone number and stored in the mysql relational database.

Chat
NOTE: A way to chat something very much like https://www.omnis.com/chat/client.php This has been worked on and is merged to digiclips henry's github master repository branch. DIGI CHAT

Opinion
NOTE: A way for users to be able to put in their opinion about what they think about the webpage and ideas to make it better.

Help
NOTE: A way to help the user how to use the frontend
HELP SUBSCRIBERS

Sign out
NOTE: A way to sign out

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: We like 30 days before end of semester. We like to present the project on AWS Amazon Lightsail on www.digiclipsproject.com/subscriber

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials:
https://docs.google.com/document/d/1xWl_GvvQBaMji7CxAwVnhNgOK5mxWM_4HzseHx00VeY/edit?usp=sharing

DigiClips - DIGICLIPS SUBSCRIRBER DETAILS BUTTONS

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: Subscribers
SUBSCRIBER MEDIA RESULTS REPORT generated from OPTION and from EMAIL ALERTS
Should be able to have details with these buttons.
SUBSCRIBER BUTTONS

Television Radio Podcast Siriusxm
Editor like this https://www.avs4you.com/avs-video-editor.aspx
With Buttons PREVIOUS SEGMENT - NEXT SEGMENT - EDITOR - DOWNLOAD - MEDIA ANALYTICS - MEDIA TRANSFER

Another button needed for subscribers: MEDIA AND JOURNALISTS CONTACTS

9A. Details Stream Page Television and Web Streaming
For Subscribers
BUTTONS http://www.digiclipsinc.com/digiview/TV_Image.png
EDITOR http://www.digiclipsinc.com/digiview/Sample%20of%20Stream_Television.htm
NOTE: Code was worked on by Colorado University 2022-23 should be up on digiclips henry's repository master branch.

10A. Details Stream Page Radio Podcast Radio Siriusxm Radio
For Subscribers BUTTONS: http://www.digiclipsinc.com/digiview/Radio_Image.png
EDITOR http://www.digiclipsinc.com/digiview/Sample%20of%20Stream_Radio.htm

With Buttons PREVIOUS SEGMENT - NEXT SEGMENT - EDITOR - DOWNLOAD - MEDIA ANALYTICS - MEDIA TRANSFER

NOTES: On Detailed Media Streaming
PREVIOUS SEGMENT Previous segment is clicking on to get the segment before
NEXT SEGMENT Next segment is to get to the segment after the segment
EDITOR To be able to edit the Previous Segment with the segment and the next segment like this https://www.avs4you.com/ to choose the resolution for television segments.
DOWNLOAD To download segment and edited Television Radio segments and Newspapers Magazine Social Media Blogs podcast siriusxm radio
MEDIA ANALYTICS with
Country Market Audience Numbers Media Value Publicity Value Length Sentiment
□ Positive □ Neutral □ Negative
MEDIA TRANSFER To transfer media data like www.wetransfer.com or like https://fromsmash.com/
==========

First have the details streaming buttons work through digi-frontend then do a production build and put it up on AWS Amazon Lightsail www.digiclipsproject.com
==========
DOWNLOAD To download segment and edited Television Radio segments and Newspapers Magazine Social Media Blogs podcast siriusxm radio
https://sourceforge.net/directory/video-downloaders/windows/
https://www.reddit.com/r/ffmpeg/comments/sigacg/downloading_segmented_videos/
==========
CAPSTONE TEAMS WORKING ON DETAILS STREAMING BUTTONS PROJECT

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: To put the project up on AWS Amazon Lightsail on www.digiclipsproject.com/subscriber

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials:
https://docs.google.com/document/d/1JlXV_sO1A7sXJ41kiJtz46NTblG1CIC1JP2-6hM42UA/edit?usp=sharing

Targeted Media - Digital monitoring tool for local economy - Backend 2026

Proposer: Desmond Peartree sales@azjazz.com

Project Contact: Desmond Peartree sales@azjazz.com

Project Description: PROJECT DETAILS
The main goal for the project is to develop a digital monitoring tool to visualize insightful statistics of and for the a local market. This tool will target small businesses and consumers alike, providing them with valuable data and analytics to make informed decisions and strategies for the local economic market.
This will involve several different steps for the students, including:
- Researching the specific statistics and data points relevant to the local market versus the national views.
- Present the statistics in an insightful and easy-to-understand manner.
- Testing the tool for accuracy and functionality.
- Gathering feedback from potential users to make improvements.

Student Learning Goals: A comprehensive understanding of the statistics and data relevant to the local market. Applications of AI and cybersecurity

Deliverables: To continue building on this foundation, the next cohort should focus on:
Expanding Data Visualizations
Build out additional widgets (50+ data points identified)
Standardize visualization formats for consistency
User Testing & Data Validation
Test survey distribution with real users
Validate data quality and refine survey questions
Platform Optimization
Improve integration between the survey system and the website
Enhance usability of dashboards and data displays
Technical Infrastructure Evaluation
Assess long-term scalability of current platform
Explore migration to a more modern, flexible tech stack
From Desmond: “To be included in the ongoing effort, to be able to select and intersect two data results. Later efforts will overlay cities and states.”
The developed digital monitoring tool with all source materials.
Slide deck presentation for showcasing the tool and its capabilities.
A comprehensive “white paper” report outlining the research findings and the functionality of the tool.

Desired Background: Cybersecurity, Software development, front-end development, back-end development, UI/UX design, spreadsheet logic, AI implementation, database design, workflow automation, Python, JavaScript, React, API integrations.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: https://theblkcommunity.com

Targeted Media - Digital monitoring tool for local economy - UI/UX 2026

Proposer: Desmond Peartree sales@azjazz.com

Project Contact: Desmond Peartree sales@azjazz.com

Project Description: PROJECT DETAILS
The main goal for the project is to develop a digital monitoring tool to visualize insightful statistics of and for the a local market. This tool will target small businesses and consumers alike, providing them with valuable data and analytics to make informed decisions and strategies for the local economic market.
This will involve several different steps for the students, including:
- Researching the specific statistics and data points relevant to the local market versus the national views.
- Present the statistics in an insightful and easy-to-understand manner.
- Testing the tool for accuracy and functionality.
- Gathering feedback from potential users to make improvements.

Student Learning Goals: A comprehensive understanding of the statistics and data relevant to the local market.

Deliverables: - A fully functional UI/UX digital monitoring tool with an intuitive user interface.
- Data visualization techniques applied to present the statistics effectively.
- Polling, testing and feedback analysis to showcase improvements made to the tool.
Final deliverables should include:
- The developed digital monitoring tool with all source materials.
- Slide deck presentation for showcasing the tool and its capabilities.
- A comprehensive “white paper” report outlining the research findings and the functionality of the tool.

Desired Background: Python, SQL, AWS Certified, Agile, JavaScript, Angular, HTML5, CSS3 Next.js, MongoDB,

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: https://theblkcommunity.com

Outer Snail Productions, LLC - Digital Video Manipulation - Phase 3

Proposer: Eric Trant eric@outersnail.com

Project Contact: Eric Trant eric@outersnail.com

Project Description: This project is to develop an integral element to a live video editing application, focused on interacting and integrating several live video feeds. An extended goal will be to integrate the beta version of the features into a demonstration UI developed by a previous Capstone team.

Student Learning Goals: Video file formatting and engagement. UI integration.

Deliverables: A beta-version of the live video integration application, demonstrating successful integration of at least three simultaneous feeds. A stretch goal will be integrating the new features into the previous team's front-end.

Desired Background: Familiarity with video file formats. UI development.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Expeditise LLC - Dynamic Interactive Website & Product Website Improvements

Proposer: Ali Sarabi asarabi1@asu.edu

Project Description: Expeditise is building AI solutions for everyday problems through a family of product websites and applications. The current Expeditise website introduces the company and its products, including ExpressReply, ExpressExplain, and ExpCalendar. This project will improve the main Expeditise website and related product websites by transforming them into a dynamic, modern, interactive, and scalable web experience.

The capstone team will redesign the website structure, visual layout, product presentation, and user flows. The new experience should include high-quality product images, product demos, animated sections, clear calls to action, and responsive layouts that work smoothly on desktop, tablet, and mobile devices. Each product page should better explain the problem being solved, the main features, the expected user workflow, and how customers or service providers can connect with Expeditise.

A major part of the project will be connecting the website to service providers and business workflows. This may include provider onboarding forms, request-a-service flows, waitlist/contact forms, product-specific lead capture, a lightweight admin dashboard, and database/API integration so that website activity can be managed and expanded over time. The final result should be a polished public website ecosystem that supports product discovery, credibility, engagement, and future growth.

Student Learning Goals: Students will gain practical experience building a production-ready startup website ecosystem that combines UI/UX design, frontend engineering, backend integration, product storytelling, and AI product presentation.

• Designing dynamic, responsive, and interactive websites using modern web frameworks
• Creating product-focused landing pages that explain features, use cases, pricing/waitlist options, and customer benefits
• Integrating product images, screenshots, demo sections, animations, and visual storytelling into a professional website
• Building secure contact, provider onboarding, waitlist, and lead-capture workflows connected to databases or backend services
• Improving website performance, accessibility, search engine readiness, analytics tracking, and maintainability
• Collaborating in an agile startup-style workflow with weekly progress demos, design reviews, and iterative feedback

Deliverables: 1. Discovery, Audit, and Website Strategy

• Review the current Expeditise website and 3-4 related product websites/pages to identify content, UX, design, and conversion gaps

• Create an improved sitemap covering the main company site, product pages, contact/waitlist flows, provider pages, and legal/support links
• Define brand style direction, page hierarchy, reusable components, and content sections for a consistent product ecosystem

2. Main Expeditise Website Redesign

• Design and implement a modern homepage with clear messaging, product cards, interactive sections, calls to action, and polished visual hierarchy
• Improve navigation, footer, contact section, waitlist flow, and product discovery paths
• Create responsive layouts for desktop, tablet, and mobile with strong performance and accessibility practices

3. Product Website and Product Page Improvements

• Improve pages for ExpressReply, ExpressExplain, ExpCalendar, and any additional Expeditise product with clearer benefits, workflows, and calls to action
• Add product screenshots, mockups, icons, feature comparison sections, FAQ sections, and demo/preview areas
• Create reusable product-page templates so future Expeditise products can be launched quickly and consistently

4. Dynamic and Interactive Design Features

• Add interactive UI components such as animated hero sections, product carousels, feature tabs, hover states, comparison cards, scroll-based sections, and expandable FAQs
• Include product images and media galleries with optimized image loading and alt text
• Build smooth transitions while keeping the site fast, professional, and easy to maintain

5. Service Provider Connection Workflows

• Build forms and user flows for service providers to request collaboration, submit information, or connect with the Expeditise team
• Create product-specific lead capture and routing so inquiries are categorized by product, service type, and user role
• Connect forms to a backend database, email notification system, CRM-style sheet, or lightweight admin panel

6. Backend, CMS, and Admin Support

• Create API endpoints or serverless functions for contact forms, waitlists, provider onboarding, and product-interest submissions
• Develop a simple admin dashboard or database schema for reviewing leads, provider

submissions, product interest, and website analytics-ready events
• Implement basic security practices including validation, spam prevention, rate limits, and safe handling of submitted data

7. Testing, Deployment, and Documentation

• Test the site across browsers and devices and verify responsiveness, page speed, accessibility, and form reliability
• Prepare deployment instructions and environment configuration for the selected hosting platform
• Document the website architecture, component library, backend endpoints, database schema, admin workflow, and maintenance guidelines

Desired Background: Students should have experience or interest in the following areas:

• Frontend development using React, Next.js, Vue, Webflow, Framer, or similar modern web tools
• UI/UX design, responsive layouts, product storytelling, wireframing, and usability testing
• Backend development with Node.js, Python, serverless functions, REST APIs, databases, or Firebase/Supabase
• Image optimization, product media creation, animation libraries, and interactive web components
• Basic knowledge of security, form validation, analytics, SEO, accessibility, and deployment workflows
• Interest in startups, AI products, service marketplaces, and customer-facing product design

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: https://www.expeditise.com

Arizona State University, EASE Program - EASE App: A Digital Platform for Supporting Neurodivergent Student Success in Engineering

Proposer: Deana Delp deana.delp@asu.edu

Project Contact: Deana Delp deana.delp@asu.edu

Project Description: This project focuses on the development of a digital education platform, delivered through a mobile app (Android and iOS), to support neurodivergent engineering students participating in the EASE (Employment Assistance and Social Engagement) program at Arizona State University. The platform is designed to provide individualized, non-academic support that addresses common challenges such as time management, social interaction, and adapting to the less structured college environment. The project involves: (1) analyzing the data to guide app design, (2) comparing an initial mock-up with other designs, (3) developing an initial prototype, and (4) gathering user feedback through interviews and surveys to refine usability and accessibility to finalize the design.

Student Learning Goals: Through this project, the students will gain hands-on experience in user-centered design and interdisciplinary collaboration between the Fulton Schools of Engineering and the College of Health Solutions. The work involves analyzing qualitative and quantitative data, and applying findings to real-world software development using tools such as Figma, Android Studio, and Xcode. The students also develop skills in accessibility-focused design and learn to incorporate feedback from neurodivergent users.

Deliverables: Mobile app for iOS and Android systems

Desired Background: Previous mobile app development experience

Delivery Mode: In-person

Additional Materials: https://docs.google.com/presentation/d/1FHOULWqut-sBjdeRIlawZN-bQfKsTVke/edit?usp=sharing&ouid=108448559769684265512&rtpof=true&sd=true

General Dynamics Mission Systems - Edge-AI Intrusion Detection on a Space-Grade GPU Platform

Proposer: Glen Abousleman Glen.Abousleman@gd-ms.com

Project Contact: Jon Nethercutt Jon.Nethercutt@gd-ms.com

Project Description: This project develops an edge-AI intrusion-detection demonstration running entirely on a commercial NVIDIA Jetson Orin developer kit. Modern space and defense platforms increasingly place high-performance AI compute directly on the vehicle, where it must operate autonomously and detect threats without waiting on a ground link. The student team will build an on-board machine-learning application that continuously monitors system and interface telemetry (CPU/GPU load, memory, power, network traffic, sensor streams) and flags anomalous or intrusive behavior in real time.

The demonstration pairs the GPU with a companion security function: a reprogrammable GPU can be updated in months, whereas fielded cryptographic hardware takes years, so using the GPU as an adaptive intrusion detector in front of, or alongside, a security unit is a compelling architecture. This aligns with active interest from GDMS's cryptographic engineering group, which already collaborates with ASU.

The team will define a threat/anomaly model, generate and inject synthetic anomalies (for example simulated bit-flips, telemetry drift, or malicious network patterns), train a lightweight ML model that runs on the Orin, and measure detection accuracy, false-alarm rate, latency, and the compute overhead the detector adds to the platform. The result is a working, demonstrable capability plus a documented characterization of how well edge-AI intrusion detection performs on this class of hardware.

All work is performed on a commercial Orin developer kit using unclassified, general engineering methods, so there are no export-control constraints on the student team. GDMS will provide the hardware target and technical mentorship throughout both semesters.

Student Learning Goals: Students will gain hands-on experience with edge AI and machine learning (model training, quantization, and deployment on an embedded GPU), embedded Linux and hardware bring-up on the NVIDIA Jetson platform, real-time telemetry acquisition and sensor interfacing, cybersecurity and anomaly-detection concepts, data pipeline and software engineering, and formal systems/project management. The interdisciplinary structure mirrors a real engineering program: EE, CS/SwE, and Engineering Management students collaborate the way industry sub-teams do.

Deliverables: Project deliverables are phased through the two-semester sequence and include:

- Fall - Concept of operations, threat/anomaly model, and system requirements.*
- Fall - Orin dev-kit bring-up with a telemetry acquisition framework capturing

system and interface data.

- Fall - Baseline ML detection pipeline trained on synthetic/nominal data, with an initial performance report.
- Spring - Integrated on-board detector running in real time on the Orin, with a companion security/crypto interface demonstrated.
- Spring - Characterization report: detection accuracy, false-alarm rate, latency, and compute overhead.
- Spring - Final demonstration at the ASU Showcase, source code, and documentation package

Desired Background: Background information required to successfully execute the project includes:
Helpful (not required) background includes: Python and basic machine learning / data science; embedded Linux and C/C++; digital hardware and interface bring-up (EE); networking and cybersecurity fundamentals; and project management for the Engineering Management lead. Students do not need all of these - the mix across the team is what matters.

Delivery Mode: In-person

Required Agreements:

- ["US Citizenship requirement"]

MentaMorph Inc - Educator Dashboard & Classroom Management System

Proposer: Susan Meredith susan@mentamorph.com

Project Contact: Susan Meredith susan@mentamorph.com

Project Description: This project creates a web-based dashboard for educators and instructors to manage classrooms, track student progress, and monitor license usage. It integrates with backend systems to present real-time insights and may include integration with external platforms such as Google Classroom.

Student Learning Goals: Students will learn how to design user-centered applications that translate complex data into intuitive interfaces. They will also gain experience aligning technical systems with real-world user workflows.

Deliverables:
- Functional educator dashboard
- Student progress and grading tools
- Instructor license usage visibility
- Real-time or near-real-time data integration
- Integration prototype or research

Desired Background:
- Front-end development
- API integration
- UI/UX design principles

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

Appy.yo - eGift: Universal Digital Gift Card Platform

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: eGift needs to be built from scratch. The goal is to create a universal digital gift card platform that makes it easier for people to give, receive, and manage gift cards in one place. Gift cards can be difficult to track, limited by retailer, or inconvenient when people want flexible gifting options. eGift would explore a more universal and user-friendly way to manage digital gifting, potentially supporting group gifting, personalized messages, and flexible redemption options.

Student Learning Goals: Front-end development, back-end development, UI/UX design, payment systems interest, digital wallets, e-commerce, database design, API integrations, security/privacy awareness, JavaScript, React.

Deliverables: A fully functioning software solution built from scratch. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.

Desired Background: Front-end Developer 1–2
Back-end Developer 1–2
UI/UX Designer 1
Payments/E-commerce Research Support 1
QA/Product Support 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Appy.yo - ePoste: AI-Powered Content Planning Tool for Small Businesses

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: ePoste has already been deployed, but it needs stronger UI/UX improvements to make it easier, cleaner, and more intuitive for users. Small businesses often struggle to plan and create consistent social media content because they lack time, marketing experience, or access to affordable tools. ePoste is designed to help users generate, organize, and manage content ideas in a simple way. The current need is to improve the user experience, strengthen the interface, and support a fully functioning, deployable version that is polished for real users.

Student Learning Goals: UI/UX design, front-end development, product design, user testing, AI/content generation, deployment support, JavaScript, React, API integrations.

Deliverables: Students should review the current deployed product, identify UI/UX improvements, support front-end updates, test the user flow, and help move the product toward a polished, fully functioning version. Students should be able to attend bi-weekly check-ins and contribute approximately 5–10 hours per week, depending on the program expectations.

Desired Background: UI/UX Designer 1–2
Front-end Developer 1–2
AI/Content Tool Developer 1
Deployment/QA Support 1
Product Researcher 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Allgoodaccounting.ca (Sandria Goodall) - Excel to App: Financial Software Transformation

Proposer: Sandria Goodall allgoodcloud@gmail.com

Project Contact: Sandria Goodall Allgoodcloud@gmail.com

Project Description: Details
Allgoodaccounting.ca, a growing accounting firm, currently relies on an Excel-based tool to generate year-to-date financial reports. As the company expands, the limitations of Excel in handling larger datasets and providing an enhanced user experience have become apparent. The goal of this project is to transition this tool into a more scalable and efficient application. Students will design and develop a functional program that replicates the existing Excel tool's capabilities while improving usability and efficiency. This project provides an opportunity for students to apply their knowledge of software development, user interface design, and data management. By the end of the project, students will have created an application that streamlines the company's financial reporting process, supporting Allgoodaccounting.ca's continued growth and operational efficiency.

Student Learning Goals: Domain expertise and knowledge
Providing specialized knowledge in the project subject area, with industry context.

Skills, knowledge and expertise
Sharing knowledge in specific technical skills, techniques, methodologies required for the project.

Hands-on support
Direct involvement in project tasks, offering guidance, and demonstrating techniques.

Tools and/or resources
Providing access to necessary tools, software, and resources required for project completion.

Regular meetings
Scheduled check-ins to discuss progress, address challenges, and provide feedback.

Deliverables: The primary deliverable is a working application that replicates the functionality of the existing Excel tool, with improved scalability and user experience. Students will provide a user-friendly interface, ensuring ease of use for non-technical staff. Additionally, comprehensive documentation will be delivered, detailing the application's features, installation process, and user guide. The project will culminate in a presentation demonstrating the application's capabilities and benefits to Allgoodaccounting.ca

Desired Background: Senior-level 4th-year and graduate Computer Science students with an Accounting background. An accounting background is preferred for at least one of the students.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Feminine Intelligence Agency - FIA Cybersecurity Fair Platform — Security Hardening, Notifications, and Live Pilot Deployment

Proposer: Lauren Nignon lauren.nignon@gmail.com

Project Contact: Lauren Nignon lauren.nignon@gmail.com

Project Description: The FIA CyberFair platform is a working web application (ASP.NET Web Forms, C#, .NET Framework 4.7) that delivers peer-to-peer cybersecurity education, built to help women and other participants assess their cybersecurity risk, take recommended microcourses, and get matched with trained student "helpers." A previous capstone team built and deployed a complete MVP covering four user roles (Participant, Helper, University Admin, Super Admin), including account creation, a risk-scoring quiz, course recommendations, session enrollment with waitlists, helper certification tracking, gamified badges, and full audit logging. The application is live on a temporary host and fully documented.

This continuation project takes that MVP to a production-ready, real-world pilot. The new team will (1) harden the application's security to a standard appropriate for a cybersecurity product, (2) finalize deployment on the sponsor's permanent domain, (3) add an email notification system, (4) design and run a live pilot with real participants and helpers and iterate on what it reveals, and (5) in the second semester, migrate the highest-traffic data from flat XML files to a SQL Server database. The team inherits a well-documented codebase, a GitHub repository, complete technical and user guides, hosting credentials, a super-admin login, and direct access to the sponsor and the prior team's lead for onboarding.

Student Learning Goals: Students will take a real, deployed application from MVP to production. This is a professional refactoring and hardening challenge, not greenfield busywork. They will work inside a live production ASP.NET / C# codebase and: apply practical application security (password hashing with BCrypt/PBKDF2, XSS/CSRF mitigation, input sanitization against XPath injection, HTTPS enforcement) against documented OWASP-style risks; own real DevOps (domain configuration, SSL, automated data backup); build a new feature end to end (email notifications); gain modern database experience by migrating high-traffic data from XML to SQL Server; and design, execute, and analyze a structured user pilot, turning real user feedback and usage data into prioritized engineering work. The security, C#, DevOps, and data-architecture skills transfer directly to modern stacks and are exactly what employers screen for. Students will also practice professional sponsor collaboration through weekly check-ins.

Deliverables: A security-hardened build: hashed password storage migrated across all read/write/compare points, HTML-encoding of user-rendered input, anti-forgery tokens on sensitive forms, sanitized identifiers in queries, and enforced HTTPS.

Finalized production deployment on cyber.feminineintelligence.agency with SSL and a

documented, repeatable data-backup procedure.
A working email notification system covering key participant, helper, and admin events (enrollment, waitlist movement, room links, schedule changes, cancellations). Delivery via a standard email service (e.g., SendGrid, AWS SES, or SMTP — provider to be confirmed with the sponsor); the deliverable is the integration, not the provider choice.
A designed and executed pilot: a written test plan, recruited participants/helpers, a live run, and a results report with findings, usage metrics, and prioritized recommendations.
Bug fixes and improvements implemented in response to pilot feedback.
A short independent security review/checklist confirming the hardening work (so the team is not the sole judge of its own security fixes).
Second-semester database deliverable: migrate the highest-traffic data (enrollments, completions, audit log) from XML to SQL Server, with the remaining low-traffic files optionally left as XML. Pursued after the primary deliverables above are complete and stable.
Updated technical documentation, user guides, and a final handoff package.

Desired Background: Coursework or comfort in an object-oriented language (C# strongly preferred; Java or C++ acceptable with willingness to ramp on C#/.NET). Familiarity with web development fundamentals (HTML/CSS, HTTP, client-server model) and version control with Git. Interest in cybersecurity and application security is highly valued, as is any experience with databases (SQL), user testing, UX, or technical writing. No hardware or embedded experience is required.
Tech-stack note: the app runs on ASP.NET Web Forms and .NET Framework 4.7. This is established production technology rather than the newest framework, and that is the point — you will harden and ship a real application used by real people, and the security, C#, DevOps, and SQL skills you build transfer directly to modern stacks.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: www.feminineintelligence.agency

Feminine Intelligence Agency - FIA Social Media Insight Engine: Create Graph Database and Integrated Knowledge Pipeline

Proposer: Lauren Nignon lauren.nignon@gmail.com

Project Contact: Lauren Nignon lauren.nignon@gmail.com

Project Description: The Feminine Intelligence Agency (FIA) is a media and education organization that produces social content on relationship safety, manipulation awareness, and social discernment — helping people recognize coercive and manipulative behavioral patterns. FIA maintains an analytical framework (a large library of concepts such as behavioral patterns, warning signs, manipulation tactics, and vulnerability and trauma types) in Notion.

A prior ASU capstone team built and deployed a working prototype of the Insight Engine, an AI application that (1) uses a web-search agent to surface trending news and media topics, (2) matches a chosen trend against concepts in FIA's knowledge base, and (3) uses a generative AI model to produce social media posts (X/Twitter, Instagram, Facebook) in FIA's analytical voice, saved to a content library for review, comment, and approval. The system is deployed across Google Cloud Run (backend), Vercel (frontend), the Gemini API (generation), a Neon PostgreSQL database, and a Neo4j Aura graph database.

The prototype proves the concept, but the knowledge base at the heart of the system was not built correctly and is not maintainable. It needs to be rebuilt from scratch rather than patched, using a free or low-cost, always-available database that the owner can populate and update herself without writing code.

This project asks a new team to take over the deployed codebase and do two things. First, rebuild the knowledge base from the ground up: select the right free or low-cost, always-available architecture (for example the PostgreSQL database already in the stack with the pgvector extension for semantic search), design a clean schema, and build an ingestion path that loads FIA's framework in bulk from the existing Notion data with no hand-entry. Second, build a high-quality content pipeline — most likely in Python — that combines FIA's knowledge base with trending inputs (such as trending social media posts or newly published academic papers) and moves them through a systematic, semantic process to produce content that is genuinely original and high quality, in both long form and short form. This pipeline connects to FIA's existing Python article-generation code so the whole system works end to end: from a trending input plus knowledge-base concepts, to a drafted long-form article or short-form social post, to FIA's review and approval before it goes out. FIA will provide step-by-step guidance on raising output quality, and the core of the work is for the students to build this pipeline and keep iterating it until the content is consistently excellent. The team will also improve editing and make it easy to get approved content out to the platforms FIA uses — formatted and ready to copy, export, or schedule.

Student Learning Goals: Students will inherit, operate, and substantially improve a real, already-deployed full-stack AI product with a real client — experience much closer to industry work than a greenfield build. They will gain hands-on experience with modern

generative-AI application development, including large language model APIs, semantic data and embeddings, retrieval-augmented generation (grounding an AI model in a curated knowledge source), and the discipline of iterating a content pipeline against a high quality bar until its output is consistently excellent. They will make and defend real architectural decisions — choosing the database and schema and rebuilding the knowledge base from scratch, then designing a content pipeline that produces long- and short-form pieces and integrates with FIA's existing Python article code — and they will integrate third-party APIs (for example pulling in trending inputs and exporting formatted content), which is realistic engineering and integration work.

The team will work across the full stack (React/Next.js and TypeScript on the front end, Python services on the back end) and across cloud infrastructure (Google Cloud Run, Vercel, and managed databases), practicing deployment, monitoring, cost control, and secure credential handling. Because the client is non-technical, students will also practice translating requirements, communicating trade-offs, and designing an interface a non-engineer can operate independently — skills in requirements gathering, client communication, UX for non-technical users, documentation, and clean project handoff that are directly valued by employers.

Deliverables: By the end of the project the team is expected to deliver:

Knowledge-base architecture and schema (first milestone). A short written plan for rebuilding the knowledge base from scratch: the chosen free or low-cost, always-available database, the data schema, and how FIA's framework will be represented. Selection criteria include cost (must fit a free or low-cost tier), always-on availability (no auto-pausing), ease of population and updates by a non-technical user, and retrieval speed and accuracy. FIA will approve the plan before implementation begins.

Rebuilt knowledge base. A reliably hosted, free-or-low-cost knowledge base, built from scratch, that stays available and returns consistent results to the AI pipeline. A strong candidate is the PostgreSQL database already in the stack (Neon) with the pgvector extension for semantic retrieval, which keeps the system on a single free-tier database and removes the separate Neo4j service.

Owner-maintainable import pipeline. A bulk import / ingestion pipeline and an interface that lets the non-technical owner load and update the knowledge base herself without writing code or database syntax. Direct import from Notion is the goal; FIA will provide a clean export of the knowledge base (CSV or JSON) as a fallback if direct Notion API import proves unreliable given rate limits or data structure.

Content pipeline (long form and short form). A high-quality, semantic content pipeline — most likely in Python — that combines FIA's knowledge base with trending inputs (trending social media posts or newly published academic papers) and moves them through a systematic process to produce genuinely original, high-quality content in both long form (articles) and short form (social posts). Retrieval reliably reflects FIA's framework, and the pipeline integrates with FIA's existing Python article-generation code so the two work as one system. FIA will provide step-by-step guidance for raising quality, and the team iterates on the pipeline until the output is consistently excellent.

Content library and review workflow. A content library with a simple list view, edit capability, an approval toggle, and a scheduled publish date (scheduling uses each platform's native scheduling feature where available). This is intentionally lightweight, not a full editorial CMS with version history or role-based permissions.
Getting approved content out. Once a piece is approved, the system makes it easy to publish: content is formatted for each target platform (TikTok, Instagram, Facebook, YouTube, X) and can be copied, exported, or scheduled so FIA can post it. Fully automated direct posting is not required — several platforms restrict it anyway — though a team may add one-click direct publishing to a single platform as an optional stretch goal.
Stable, low-cost deployment. A documented deployment with monitoring that targets a free or near-zero recurring cost: the team keeps services within free tiers wherever possible, consolidates or removes paid services (such as the separate Neo4j instance), and flags any component that would require a paid plan for client approval before incurring it. Monthly reporting of any API and infrastructure costs (the main expected cost is Gemini API usage, measured by the prior team at well under a cent per generation). All platform ownership transferable to the client.
Documentation and handoff. Updated README and architecture docs, setup/onboarding instructions, a user guide written for the non-technical owner, and a final handoff package and demo.

Desired Background: Students should have solid full-stack web development skills: React/Next.js and TypeScript on the front end, and a back-end language (the current system uses Python). Familiarity with REST API integration and with relational databases (SQL) is important, and experience or interest in integrating third-party APIs is valuable. Strong candidates will also have interest or coursework in generative AI and large language models, retrieval-augmented generation or vector/semantic search, and cloud deployment (Google Cloud Platform and/or Vercel). Comfort reading, refactoring, and extending an existing codebase rather than starting from scratch is valuable, as is an interest in AI agents and in building interfaces for non-technical users. No hardware or embedded-systems experience is required.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: www.feminineintelligence.agency

Kona Kapital - Financial Planning Web App

Proposer: Diana Bristow diana.bristow@icloud.com

Project Contact: Diana Bristow diana.bristow@icloud.com

Project Description: KONA is a digital financial wellness platform designed to empower women to take control of their financial futures through personalized planning, education, and professional guidance.

The platform will utilize artificial intelligence to engage users in a conversational assessment process, gathering information about their financial goals, income, expenses, savings habits, risk tolerance, family situation, and long-term objectives. Based on this information, KONA will generate customized budgets, financial roadmaps, and actionable recommendations tailored to each user's unique circumstances.

In addition to automated planning tools, KONA will connect users with qualified financial professionals when specialized advice is needed. Through a referral network, users can be matched with advisors who possess expertise relevant to their specific needs, such as retirement planning, investing, insurance, debt management, entrepreneurship, estate planning, or major life transitions.

The platform will be supported by a comprehensive financial knowledge database containing budgeting methodologies, investment principles, tax considerations, financial planning frameworks, and educational resources. This database will serve as the foundation for AI-generated recommendations while promoting financial literacy and informed decision-making.

The goal of KONA is to reduce barriers to financial planning, increase financial confidence among women, and provide accessible, personalized guidance that bridges the gap between self-directed financial management and professional financial advice.

Student Learning Goals: Students participating in the KONA project will gain hands-on experience in the design, development, and implementation of an AI-powered financial planning platform. The project will expose students to a broad range of modern technologies, software platforms, and business applications commonly used in today's technology and financial services sectors.

Students will have the opportunity to work with web development platforms, cloud-based technologies, artificial intelligence tools, database design, and user experience development. The project will also provide exposure to cloud infrastructure solutions such as Amazon Web Services (AWS), software integration, data management, and automation tools.

A key component of the project will involve designing and building the underlying financial planning engine. Students will develop a structured database and spreadsheet framework

that stores financial planning assumptions, budgeting rules, investment parameters, and financial recommendations. User responses collected through the platform will be analyzed against this data to generate personalized budgets, financial plans, and educational recommendations.

Throughout the project, students will gain practical experience in:

* AI-assisted user interactions and decision support systems
* Website and web application development
* Database architecture and management
* Cloud computing and hosting environments
* Financial data organization and modeling
* Spreadsheet and calculation engine design
* User interface and user experience (UI/UX) development
* Software integration and workflow automation
* Testing, quality assurance, and project management

The project combines elements of financial technology (FinTech), artificial intelligence, data analytics, and software development, providing students with exposure to both technical and business-focused problem-solving. Students will also gain experience working with real-world requirements, stakeholder feedback, and iterative development processes while contributing to a platform designed to improve financial literacy and accessibility for women.

Deliverables: Expected Deliverables

The KONA Capstone project is expected to produce a functional prototype of an AI-powered financial planning platform designed to assist women in developing personalized financial plans and accessing professional financial guidance.

Key deliverables will include:

1. Functional Website Platform
   * Responsive website accessible from desktop and mobile devices.
   * User registration and account management functionality.
   * Secure collection and storage of user information.
2. AI-Powered Financial Planning Interface
   * Conversational questionnaire to gather financial information, goals, and priorities.
   * AI-assisted generation of personalized budgets, savings strategies, and financial planning recommendations.
   * Ability to adapt recommendations based on user responses and changing circumstances.
3. Financial Planning Database and Calculation Engine
   * Structured financial planning framework containing budgeting rules, financial assumptions, planning models, and educational content.

* Spreadsheet and/or database-driven calculation engine capable of generating customized financial plans.
* Administrative tools for updating assumptions, financial parameters, and planning recommendations.

4. Privacy, Compliance, and Legal Framework
* Privacy policy and user consent framework.
* Terms of service and website use agreements.
* Appropriate financial planning disclaimers and risk disclosures.
* Data security recommendations and compliance considerations for handling personal financial information.

5. Advisor Referral and Matching System
* Directory and matching functionality connecting users with qualified financial professionals based on their needs.
* Referral tracking capabilities.
* Administrative tools for managing advisor profiles and referral relationships.

6. Professional Review and Monetization Features
* Ability for users to request a professional review of their AI-generated financial plan.
* Payment processing framework for premium services.
* Optional subscription or fee-based service models for future expansion.

7. Educational Resource Centre
* Financial literacy content covering budgeting, debt management, investing, retirement planning, insurance, and major life events.
* AI-assisted guidance directing users to relevant educational materials.

8. Documentation and Deployment
* Technical documentation outlining system architecture and development processes.
* User documentation and administrative guides.
* Deployment recommendations utilizing cloud-based infrastructure such as Amazon Web Services (AWS).

The final deliverable will be a working prototype that demonstrates how artificial intelligence, financial planning methodologies, and professional advisor networks can be combined to provide accessible, personalized financial guidance while maintaining appropriate privacy, compliance, and consumer protection standards.

Desired Background: This project is suitable for senior undergraduate students, graduate students, or interdisciplinary teams with an interest in technology, business, finance, and entrepreneurship.

Preferred candidates include students in their fourth year of study or enrolled in Master's programs in areas such as:

* Computer Science
* Software Engineering
* Information Technology

* Data Science and Analytics
* Artificial Intelligence
* Business Administration
* Finance
* Financial Technology (FinTech)
* Entrepreneurship and Innovation
* User Experience (UX/UI) Design

Students do not need to possess extensive financial planning knowledge at the outset; however, an interest in financial services, personal finance, financial literacy, and emerging technologies would be beneficial.

The project will provide exposure to the development of a real-world startup concept, allowing students to gain experience in:

* Building a technology-based business solution
* Applying artificial intelligence to practical consumer applications
* Developing software and web-based platforms
* Understanding financial services and consumer financial planning
* Working with privacy, compliance, and regulatory considerations
* Product development, testing, and market validation
* Entrepreneurship and small business operations

Students will have the opportunity to collaborate directly with the project sponsor and industry professionals, gaining insight into both the technical and business challenges involved in launching and scaling a financial technology platform. The experience is particularly valuable for students interested in careers in software development, financial services, consulting, product management, startups, and innovation-driven organizations.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

Appy.yo - FlexyGig: AI-Supported Flexible Work Marketplace

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: FlexyGig is an app concept that needs further development and AI implementation. Many workers have skills that can apply across different types of jobs, but they may struggle to find flexible work that matches their location, availability, and experience. Employers also need easier ways to find reliable short-term or part-time workers. FlexyGig would connect workers and employers through profiles, job boards, filtering, messaging, and AI-supported matching. The current need is to build the app and implement AI features that improve job and worker matching.

Student Learning Goals: Mobile app development, front-end development, back-end development, UI/UX design, AI implementation, matching algorithms, marketplace platforms, search/filter functionality, messaging systems, JavaScript, React Native, database design.

Deliverables: A fully functioning app or MVP with AI-supported matching and marketplace functionality. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.

Desired Background: Mobile or Front-end Developer 1–2
Back-end Developer 1
UI/UX Designer 1
AI/Matching Developer 1
Product Researcher 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Flutter Energy - Flutter Energy: Smart Industrial Energy Management Platform

Proposer: Ehsan Daneshgar ehsan@launchpath.ca

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: As industries, transportation, and buildings continue to electrify, electricity has become one of the most strategic and rapidly growing operational challenges for businesses. Artificial Intelligence, industrial IoT, and energy storage are transforming how organizations manage and optimize energy, creating exciting opportunities to build the next generation of intelligent energy systems.

Flutter Energy is developing an AI-powered industrial energy optimization platform that helps commercial and industrial facilities reduce electricity costs, improve operational efficiency, and lower carbon emissions through intelligent energy management.

The platform combines artificial intelligence, real-time energy monitoring, battery energy storage, IoT devices, and enterprise system integration to optimize how facilities consume and manage electricity. Rather than simply forecasting energy demand or electricity prices, the platform analyzes operational data and recommends intelligent actions such as battery charging and discharging, energy load shifting, equipment scheduling, and automated power management.

Students will contribute to the next phase of the platform by designing and integrating a Battery Energy Storage System (BESS) with the existing cloud-based software platform, developing communication between embedded hardware and cloud services, implementing battery charging and monitoring capabilities, and supporting automatic switching between utility and battery power.

In addition, students will evaluate the existing software platform, identify opportunities to improve functionality, usability, performance, reliability, and workflow, and implement approved enhancements. The project may also explore integration with industrial IoT devices, smart energy meters, and enterprise systems to support intelligent decision-making and future autonomous energy optimization.

The final solution will demonstrate a scalable cyber-physical platform that integrates hardware, software, and AI to support intelligent energy management in commercial and industrial environments.

Student Learning Goals: Students participating in this project will have the opportunity to:

- Contribute to the development of a real-world industrial energy management platform.
- Evaluate and improve an existing commercial software and hardware solution.
- Design and integrate a Battery Energy Storage System (BESS) with a cloud-based energy platform.
- Develop embedded software for battery charging, monitoring, and intelligent power management.
- Implement communication between embedded hardware, IoT devices, and cloud-based

services.
- Explore AI-driven energy optimization, predictive analytics, and operational decision support.
- Gain experience with embedded systems, IoT communication, hardware/software integration, cloud computing, and system testing.
- Collaborate directly with an industry startup while following professional software engineering, systems engineering, and product development practices.

Deliverables: - Review, test, and evaluate the existing Flutter Energy platform.
- Document findings and recommend improvements to system functionality, usability, performance, and reliability.
- Design and integrate a functional Battery Energy Storage System (BESS).
- Implement battery charging, monitoring, health management, and automatic switching between battery and utility power.
- Develop reliable communication between embedded hardware, IoT devices, and the Flutter Energy cloud platform.
- Implement approved software and hardware enhancements that improve system capabilities.
- Perform end-to-end integration testing, validation, and troubleshooting.
- Deliver updated technical documentation, setup guides, wiring diagrams, APIs, and source code.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:

- Embedded Systems
- Electrical Engineering
- Computer Engineering
- IoT and Connected Devices
- Energy Storage Systems
- Power Electronics
- Artificial Intelligence / Machine Learning
- Python
- C/C++
- Cloud Computing
- REST APIs and IoT Communication Protocols (MQTT, Modbus, OPC-UA, etc.)
- Hardware/Software Integration
- Industrial Automation
- System Testing and Debugging
- UI/UX, Software Design, or Product Improvement

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

GM3 Solutions LLC - Forged Soft Skills AI — Engineering for Scale, Building for Technical Interviews

Proposer: Gilbert Mora gilmora@gm3solutions.net

Project Contact: Gilbert Mora gilmora@gm3solutions.net

Project Description: Forged Soft Skills AI is a working platform, not a concept. Today, real users can record themselves answering behavioral interview questions on camera, and the system analyzes their eye contact, pacing, filler words, tone, and emotion using computer vision and speech models, then delivers AI coaching through an integrated Claude-powered chat. The architecture already spans a Next.js frontend, a Python GPU-based sensing pipeline (MediaPipe, DeepFace, AssemblyAI), a PostgreSQL database, and live integrations with Anthropic's API. This next phase is not about adding more features. It is about taking a system that works in a demo and making it work at the scale. It's about expanding its scope to interview prep that goes beyond behavioral questions into technical communication. The team will tackle two concrete, production-grade engineering problems: (1) making the existing analysis pipeline faster, cheaper, and more reliable under real usage, and (2) designing and shipping a new module where users upload a resume and a job description, the system generates tailored technical interview questions, and gives feedback not just on what was said, but on how clearly and completely it was communicated. This is real infrastructure work on a system already running in production, sponsored directly by the platform's founder, with weekly engineering reviews and a documented architecture you will extend — not redesign from scratch.

Student Learning Goals: Students will work directly inside a production codebase with an existing architecture, existing users, and real operating constraints — the kind of environment most capstones can only simulate. The experience covers four areas that map directly to in-demand engineering skills: distributed systems and async processing (redesigning a GPU analysis pipeline to run reliably and quickly under concurrent load); cost-aware engineering (instrumenting a multi-vendor AI/cloud system — GPU compute, LLM APIs, storage — to track and control real operating cost, a skill rarely taught but constantly demanded in industry); applied AI/ML (building an LLM-driven question generation system from unstructured resume and job description input, and designing an evaluation approach for open-ended spoken answers where no fixed answer key exists); and engineering governance (operating under a real change-management process, including a structured change request and review system with a non-academic sponsor, mirroring how engineering decisions get made on a real team with a real stakeholder). Students will also directly own one of three concurrent workstreams — platform reliability and cost engineering, technical interview module (AI/data pipeline), or technical interview module (scoring and coaching) — giving every team member a defined area of ownership rather than a shared, undifferentiated task list.

Deliverables: Fall semester: a documented and completed ownership/access migration of all backing services to the sponsor's organization; a measurably faster and more cost-visible

analysis pipeline (defined latency target, per-session cost tracking, and a hard spending cap on GPU usage); a reduced and rebalanced interview session length with supporting question-rotation logic; and a working resume + job description upload flow that generates tailored technical interview questions and reuses the existing recording infrastructure to capture responses. Spring semester: full scoring and coaching for the technical interview module, applying the existing communication-analysis pipeline to technical answers and adding a new structural-completeness signal (clearly distinguished from technical correctness); a stable, documented platform ready for continued development by a future team; and a final architecture and operations report, in the same format as the current sponsor documentation, reflecting everything built and changed across both semesters.

Desired Background: This is an interdisciplinary project spanning full-stack software engineering, applied AI/ML, and data engineering, and it does not require any one student to have all three. Helpful backgrounds include: web application development (React/Next.js, REST APIs, relational databases); Python and basic computer vision or audio processing concepts (no deep ML research background required — the existing models are pre-built; the work is integration, reliability, and pipeline design); experience with or interest in LLM-based applications (prompt design, structured output, working with the Anthropic or OpenAI APIs); and an interest in production operations — thinking about cost, latency, and failure handling as first-class engineering concerns, not afterthoughts. No single student needs to check every box. A team with a mix of these strengths across its 5–6 members will cover the full scope well.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://docs.google.com/document/d/1D14WUH2hkBjrsFtbA-Ye0fAZ5oVMOAwg/edit?usp=sharing&ouid=107705574438781964844&rtpof=true&sd=true

FoxiPrint - FoxiPrint AI & Systems Internship

Proposer: Sally Kong info@foxiprint.com

Project Contact: Hannah Cohen hannah@foxiprint.com

Project Description: This internship is designed for students who want to work on real-world systems that directly impact business growth.
At FoxiPrint, you won't just be completing tasks—you'll be contributing to the development of AI-driven tools, automation systems, and a scalable digital platform that supports marketing, sales, and operations.

CORE FOCUS AREAS::

1. Website Development & Pre-Launch Optimization
* Work directly on refining the FoxiPrint website ahead of its public launch (target ~2 months)
* Improve user experience, product flow, and overall site performance
* Identify gaps, test features, and contribute ideas to strengthen the platform before launch
* Collaborate with the team to ensure the site is fully optimized, functional, and scalable at launch

👉 This is a unique opportunity to work on a platform that is actively being prepared for launch—your work will directly impact what goes live.

2. Search & Analytics Systems
* Implement and optimize:
* Google Search Console metrics
* Bing Webmaster Tools
Track:
- Traffic sources
- Keyword performance
- Conversion behavior

Build dashboards to visualize insights

3. AI Outreach Engine (High Impact)

Students will help design and build:

* A system that identifies local Arizona businesses
* AI-generated personalized outreach messages
* Automated sending + follow-up workflows
* Lead tracking and qualification logic

Goal: Turn outbound sales into a scalable system

4. Backend & Data Systems
* Improve or build internal systems (pricing tools, order flows, CRM logic)
* Work with APIs and databases to streamline operations
* Explore automation opportunities across the business

5. Cloud & Infrastructure
* Deploy and manage systems using AWS
* Follow best practices for version control (Git/GitHub)
* Build scalable and maintainable architecture

TOOLS & TECHNOLOGIES
* JavaScript / React / Node.js
* Python (for automation + AI workflows)
* APIs + database systems
* AWS / cloud services
* AI/LLM tools

Deliverables
* Improved website with measurable performance gains
* Functional AI outreach prototype or system
* Search + analytics dashboards
* Documentation of systems built and decisions made
* Final presentation of your work and impact

What We're Looking For
* Curiosity and willingness to learn
* Ability to think beyond assignments
* Strong communication and collaboration
* Deep Interest in AI, systems, and real-world problem solving

Student Learning Goals: Students will gain hands-on experience working on real-world systems that directly impact a growing business, including AI-driven development, web optimization, automation, and data systems.

Top-performing students who stand out as rising contributors will be considered for long-term roles with FoxiPrint, with the potential to transition from internship to full-time positions.

Deliverables: * Refinement and optimization of the FoxiPrint website in preparation for launch
* Implementation of user tracking and analytics systems (Google Search Console, Bing Webmaster Tools) to measure performance and user behavior

* Design and development of an in-house AI Outreach Engine from the ground up, including:
    - Identifying and sourcing local business leads (initial focus on the Arizona market, with plans to expand nationwide following successful development, testing, and optimization)
    - Building systems for personalized AI-generated outreach
    - Developing automated messaging and follow-up workflows
    - Creating internal logic for lead tracking and qualification

(Note: this will be built internally—not using pre-built or outsourced AI tools)
👉 This system is intended to become a core revenue driver for FoxiPrint

Clear documentation of systems built, including approach, challenges, and solutions
Final presentation outlining contributions, system impact, and recommendations for future development

Desired Background: * Students in Computer Science, Software Engineering, Data Science, or related fields
* Experience with JavaScript, React, Python, or similar technologies
* Passion for AI and automation, with a builder mindset—eager to design, test, and develop systems from scratch that create real-world impact
* Strong communication skills and willingness to ask questions
* Ability to work independently and collaboratively

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

General Dynamics Mission Systems - FPGA AI Engine-Based Network Intrusion Detection and Security Monitoring System

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Jamey Creary Jamey.Creary@gd-ms.com

Project Description: Design and implement a proof-of-concept intrusion detection system using the dedicated AI Engine array within an AMD/Xilinx Versal Adaptive SoC FPGA. The FPGA programmable logic handles packet parsing and feature extraction at line rate, while the AI Engine tiles run a trained ML model for real-time anomaly detection and traffic classification. The project draws comparison to fixed-function on-chip AI approaches such as Apple's Neural Engine and evaluates the trade-offs of programmable FPGA AI cores for embedded security applications.

Tasks:

- Research on-chip AI architectures (AMD Versal AI Engine, Apple Neural Engine) and their applicability to network security monitoring.
- Design a PL-based packet ingestion and feature extraction pipeline integrated with the AI Engine array via AXI-Stream and the NoC.
- Train, quantize, and deploy an ML anomaly detection model onto the Versal AT Engine tiles using the Vitis AT workflow.
- Develop a PS-based monitoring dashboard to display detection results, alerts, and system performance metrics.
- Benchmark inference latency, throughput, detection accuracy, and resource utilization against traditional PL-only approaches.

Goals:

- Demonstrate real-time AT Engine-based intrusion detection on live or simulated network traffic at 1G/10G line rate.
- Produce a comparative analysis of FPGA AT Engine inference versus fixed-function neural processors for embedded security.
- Document a repeatable workflow for training, quantizing, and deploying ML security models onto Versal hardware.
- Evaluate trade-offs between on-chip AT Engine inference and traditional PL-based packet inspection approaches.
- Deliver a functional proof-of-concept with measured performance benchmarks on the Versal development kit.

Student Learning Goals:

- FPGA architecture and design with emphasis on heterogeneous SoC platforms (Versal AT Engine, PL, PS).
- VHDL/Verilog for packet processing and C/C++ for AT Engine kernel development.
- Machine learning fundamentals including model training, quantization, and deployment with Vitis AT.
- Network security concepts including intrusion detection, anomaly detection, and

traffic classification.
· COTS FPGA board development with AMD Vivado and Vitis tools.

Deliverables: · Weekly status reports (metrics, meeting minutes, work breakdown, roadblocks) and bi-weekly hardware demonstrations.
· Trained and quantized ML model with documented training methodology and accuracy metrics.
· Functional proof-of-concept demonstrating AT Engine-based intrusion detection on the Versal development kit.
· Organized repository containing all RTL, AT Engine kernels, ML model artifacts, and PS application code.
· Final technical report comparing on-chip AT Engine inference with traditional FPGA-based and fixed-function approaches.

Desired Background: Background information required to successfully execute the project includes:
US Citizenship required.
Best fit degrees, emphasis areas and interests for participants:
· Computer Science or Computer (Systems) Engineering students
· Electrical Engineering with emphasis on digital design or signal processing.
· Cybersecurity or Data Science / Machine Learning.
Valuable skills and interests for participants:
· FPGA Design, VHDL/Verilog, and C/C++ development
· Familiarity with Python and ML frameworks (PyTorch, TensorFlow)
· Familiarity with AMD Vivado/Vitis and interest in network security

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- US Citizenship requirement
- Intellectual property agreement

Additional Materials: Supplies/Equipment Required: · Development PC and AMD Vivado/Vitis/Vitis AI tools provided by team members and General Dynamics. · AMD Versal AI Core Development Kit (VCK190 or equivalent), provided by General Dynamics. · Network traffic datasets for ML training (publicly available: CICIDS, NSL-KDD, or UNSW-NB15).

General Dynamics Mission Systems - FPGA Left/Right Domain Separation with Post-Implementation Isolation Verification

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Jamey Creary Jamey.Creary@gd-ms.com

Project Description: Implement a left/right domain separation architecture on a Xilinx/AMD FPGA and verify that isolation is maintained from RTL through the final bitstream. The design uses the AMD/Xilinx Isolation Design Flow (IDF) with Pblock constraints, fence registers, and trusted routing to enforce physical separation between two trust domains. The Vivado Isolation Verifier (VIV) confirms IDF rule compliance at the netlist level, and Enverite PV-Bit from Graf Research performs Bitstream Equivalence Checking (BEC) to prove those isolation properties are preserved in the bitstream.
Tasks:

- Design a left/right separated FPGA with two isolated Pblock regions, fence cells, and trusted inter-region boundary crossings per IDF rules.

- Run the Vivado Isolation Verifier (VIV) at floorplan and routed DCP stages to confirm IDF DRC compliance (IDF-VIV2-3, IDF-VIV2-5, IDF-VIV2-6).
- Perform bitstream-level verification using Enverite PV-Bit BEC to prove BEL, PIP, and resource configurations match the VIV-verified netlist.
- Develop RTL-level formal property checks to prove no unintended information flow between domains.
- Explore AI/ML techniques to automate analysis of VIV and PV-Bit reports to support accreditation evidence packages.

Goals:

- Demonstrate a working left/right separated FPGA design with two independent applications in physically isolated regions.
- Produce clean VIV DRC reports and a PV-Bit BEC attestation proving isolation from netlist through bitstream.
- Deliver a complete verification artifact package (RTL formal proofs, VIV DRC reports, PV-Bit BEC attestation).
- Develop an AI-assisted tool to consolidate separation evidence to support accreditation reports.
- Produce a recommendations document for applying this methodology to production FPGA designs.

Student Learning Goals:

- FPGA architecture, physical implementation constraints, and isolation region design.
- VHDL/Verilog development and Tcl scripting for Vivado automation and constraints.
- Hands-on experience with VIV and Enverite PV-Bit verification tools on COTS FPGA hardware.

· Formal verification concepts for proving RTL-level design isolation properties.
· Multi-level security concepts, Single Chip Crypto and security requirements.

Deliverables: · Weekly status reports (metrics, meeting minutes, work breakdown, roadblocks) and bi-weekly hardware demonstrations.
· Left/right separated FPGA design with Pblock constraints, fence registers, and trusted routes running on hardware.
· VIV DRC reports (floorplan and routed DCP) and Enverite PV-Bit BEC attestation report.
· AI-assisted verification tool for automated separation evidence generation.

· Final technical report with verification methodology, tool evaluations and security compliance recommendations.

Desired Background: Background information required to successfully execute the project includes:
US Citizenship required.
Best fit degrees, emphasis areas and interests for participants:
· Computer Science or Computer (Systems) Engineering students
· Electrical Engineering with emphasis on digital design.
· Cybersecurity.
Valuable skills and interests for participants:
· FPGA Design, VHDL/Verilog, and Tcl/Python scripting
· Familiarity with AMD/Xilinx Vivado and FPGA implementation flows
· Interest in security engineering, formal verification, and high-assurance systems

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: · Development PC and AMD Vivado tools provided by team members and General Dynamics. · Enverite PV-Bit license from Graf Research Corporation, provided by General Dynamics. · Xilinx/AMD FPGA Development Kit (e.g., KCU105 or ZCU102), provided by General Dynamics.

General Dynamics Mission Systems - FPGA Wire-Speed Cryptographic Protocol Inspection and Replay Engine

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Phil Amantia Phil.Amantia@gd-ms.com

Project Description: Design and implement an FPGA-based network security test instrument that performs real-time, wire-speed inspection of Gigabit Ethernet traffic targeting cryptographic protocol fields — including TLS handshakes, IPsec ESP headers, and MACsec security tags — with the ability to capture, selectively modify, and replay packets at line rate with deterministic, sub-microsecond latency. The platform serves

as a hardware-accelerated security testing tool for verifying the correctness and robustness of cryptographic protocol implementations. Software-based tools such as Scapy and tcpreplay are limited by OS scheduling jitter and CPU throughput, typically sustaining 100–300 Kpps with inspection; this FPGA approach processes every frame in a fully pipelined architecture at the full 1.488 Mpps Gigabit Ethernet line rate with zero packet loss and nanosecond-precision timing.

Tasks:

· Research wire-speed packet processing architectures and existing FPGA-based network security tools. Survey cryptographic protocol specifications (TLS 1.2/1.3, IPsec ESP, MACsec 802.1AE) to identify security-critical fields for inspection.

· Bring up the Gigabit Ethernet PHY interface (RGMII/GMII) using open-source MAC IP cores and verify basic frame reception and transmission.

· Design and implement a streaming packet processing pipeline with L2/L3/L4 header parsing, extracting Ethernet, IP, TCP/UDP, and protocol-specific fields at one word per clock cycle.

· Implement an application-layer identification stage capable of detecting and parsing TLS record headers, TLS handshake messages (ClientHello, ServerHello), IPsec ESP Security Parameter Index (SPI) and sequence numbers, and MACsec SecTAG fields.

· Design a configurable pattern match engine using BRAM-based rule tables supporting up to 64 concurrent match rules with wildcard and mask support, checked in parallel via CAM-style lookup.

· Implement per-rule action processing: PASS (forward unmodified), DROP (discard), CAPTURE (store
to DDR ring buffer with nanosecond timestamp), MODIFY (field overwrite with Ethernet FCS
recalculation), and INJECT (insert pre-loaded frame from replay buffer).

· Develop a DDR-backed packet capture ring buffer with 64-bit nanosecond timestamps and PCAP-format export capability.

· Build a host control plane application (Python over UART or Ethernet socket) for programming match rules, configuring actions, displaying live traffic statistics, and exporting captured packets.

· Execute security test scenarios against real protocol endpoints: TLS version downgrade attack, IPsec anti-replay window verification, and MACsec sequence number rollover testing.

Goals:

· Demonstrate zero-packet-loss inspection of Gigabit Ethernet traffic at full line rate (1.488 Mpps for minimum-size frames) with a measured pipeline latency under 100 nanoseconds.

· Successfully execute a TLS downgrade attack by modifying a ClientHello version field in-flight and observing the target server's response.

· Verify IPsec anti-replay window correctness by capturing and replaying ESP packets with controlled sequence numbers and documenting the receiver's accept/reject behavior.

· Produce a quantitative comparison of packet replay timing precision: FPGA engine (nanosecond-resolution, zero jitter) versus Scapy/tcpreplay (millisecond-resolution, OS-induced jitter).

· Deliver a functional proof-of-concept with live inline demonstration: FPGA sitting between two
endpoints, performing real-time protocol inspection and selective packet modification while
Wireshark confirms the results.

Student Learning Goals: · FPGA architecture and high-performance digital design using VHDL or SystemVerilog, including pipelined streaming architectures and clock-domain crossing.

· Gigabit Ethernet PHY/MAC interfacing (RGMII/GMII), packet framing, and CRC/FCS computation.

· Network protocol internals at the byte level: Ethernet, IPv4, TCP, UDP, TLS record and handshake protocols, IPsec ESP, and MACsec 802.1AE.

· Network security testing concepts including man-in-the-middle techniques, protocol conformance testing, replay attacks, and downgrade attacks.

· DDR memory controller integration for high-throughput packet buffering and timestamped capture.

· Python scripting for network tool development, PCAP file generation, and traffic analysis.

Deliverables: · Weekly status reports (metrics, meeting minutes, work breakdown, roadblocks) and biweekly hardware demonstrations.

· Fully pipelined RTL source code for the packet processing pipeline, pattern match engine, action engine, and capture/replay subsystems, organized in a version-controlled repository.

· Python host control plane application with rule editor, live traffic dashboard, and PCAP export functionality.

· Documented security test results: TLS downgrade, IPsec replay, and MACsec

sequence number rollover tests with packet captures and analysis.

· Performance benchmark report: measured throughput, pipeline latency, and timing precision compared against software-based alternatives.

· Final technical report documenting the architecture, protocol parsing methodology, test scenarios, experimental results, and lessons learned.

Desired Background: Background information required to successfully execute the project includes:

US Citizenship required.

Best fit degrees, emphasis areas and interests for participants:

· Computer Engineering or Electrical Engineering with emphasis on digital design and networking.

· Computer Science with networking or cybersecurity emphasis.

· Cybersecurity with interest in protocol analysis and penetration testing.

Valuable skills and interests for participants:

· FPGA design experience with VHDL, Verilog, or SystemVerilog.

· Understanding of network protocols (TCP/IP stack, Ethernet) at the packet level.

· Familiarity with network security tools (Wireshark, Scapy, tcpreplay).

· Familiarity with AMD Vivado design tools or Microchip Libero SoC.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: · Kintex-7 with dual GigE PHY or Artix-7 with GigE PHY or similar FPGA development board, provided by General Dynamics. Alternative: Microchip PolarFire Video Kit. · Development PCs for hosting applications provided by the Project Team. · Gigabit Ethernet switch and cables for test network setup provided by General Dynamics · AMD Vivado Design Suite (WebPACK or full license), obtained by the Project Team via Student licensing with General Dynamics support. · Linux workstation or VM for running TLS/IPsec endpoints (OpenSSL, strongSwan) as protocol test targets provided by the Project Team · Wireshark and Scapy (free, open source) for traffic analysis and comparison benchmarking.

General Dynamics Mission Systems - FPGA-Based Hardware-Accelerated Cryptographic Implementation Fuzzer

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Phil Amantia Phil.Amantia@gd-ms.com

Project Description: Design and implement an FPGA-based fuzzing platform that generates high-volume, structured test inputs and drives them into a target cryptographic module at hardware speed, monitoring for incorrect outputs, state corruption, timing anomalies, and implementation defects. The platform instantiates a known-good reference cryptographic core alongside the device under test (DUT), enabling cycle-

accurate output comparison at rates exceeding one million test vectors per second. Software-based fuzzing tools are limited to approximately 10K–100K tests per second when targeting hardware crypto implementations; this FPGA-native approach eliminates that bottleneck by communicating with the DUT over a direct bus interface, achieving 100–1000x throughput improvement for vulnerability discovery in cryptographic IP.

Tasks:

· Research hardware fuzzing methodologies and existing approaches to cryptographic implementation testing (NIST CAVP, differential fuzzing, fault injection).

· Design and implement a parameterized test vector generation engine with multiple mutation strategies: random generation (LFSR/xoshiro PRNG), targeted bit-flip mutation, boundary value injection, and control signal malformation.

· Instantiate a verified AES-128 reference core and a separate device-under-test (DUT) core with configurable fault injection points for validation of the fuzzing harness.

· Implement a cycle-accurate comparator and oracle engine that detects output mismatches, timing deviations, and FSM hang conditions between the reference and DUT.

· Design a DDR-backed circular trace buffer to log failing test vectors, DUT outputs, and timestamps for post-run forensic analysis.

· Develop a UART/USB control interface and Python host application for configuring mutation modes, monitoring test progress, and exporting failure logs.

· Extend the platform to support SHA-256 fuzzing to demonstrate generality beyond a single cipher.

· Benchmark FPGA fuzzer throughput against an equivalent software fuzzing approach (Python/C driving the same DUT over SPI) and characterize detection rates across mutation strategies.

Goals:

· Demonstrate detection of deliberately injected cryptographic implementation defects (skipped rounds, corrupted S-box entries, missing key schedule rotations) at a sustained rate exceeding 1 million test vectors per second.

· Produce a quantitative comparison of FPGA-based versus software-based fuzzing

throughput, showing 100x or greater speedup for hardware crypto IP testing.

· Validate the fuzzing platform against NIST Cryptographic Algorithm Validation Program (CAVP) known-answer test vectors to confirm the reference model and comparator correctness.

· Demonstrate platform generality by supporting at least two distinct cryptographic algorithms (AES-128 and SHA-256) with algorithm-specific mutation strategies.

· Deliver a functional proof-of-concept with a live demonstration: enable a DUT fault via switch input, observe the anomaly counter incrementing in real time.

Student Learning Goals: · FPGA architecture and digital design using VHDL or SystemVerilog, including parameterized and reusable IP core development.

· Cryptographic algorithm internals (AES round structure, SubBytes/ShiftRows/MixColumns/AddRoundKey, SHA-256 compression function) at the register-
transfer level.

· Hardware verification and test methodology, including self-checking testbenches, coverage analysis, and differential testing.

· Security testing concepts including fuzz testing strategies, mutation-based testing, and cryptographic implementation validation (NIST CAVP/CMVP frameworks).

· Embedded system integration: DDR memory controllers, UART interfaces, and host-FPGA communication protocols.

· Python scripting for hardware test automation, data visualization, and statistical analysis of test results.

Deliverables: · Weekly status reports (metrics, meeting minutes, work breakdown, roadblocks) and biweekly hardware demonstrations.

· Parameterized RTL source code for the fuzzing engine, mutation engine, comparator, trace logger, and target interface, organized in a version-controlled repository.

· Verified AES-128 and SHA-256 reference cores validated against NIST CAVP test vectors.

· Python host application for fuzzer configuration, real-time monitoring, and failure log analysis.

· Quantitative benchmark report comparing FPGA fuzzer throughput and detection rates against software-based fuzzing of the same DUT.

· Final technical report documenting the architecture, mutation strategies, experimental results, and lessons learned.

Desired Background: Background information required to successfully execute the project includes:
US Citizenship required.

Best fit degrees, emphasis areas and interests for participants:

· Electrical Engineering with emphasis on digital design or VLSI.
· Computer Engineering or Computer Science with hardware focus.
· Cybersecurity with interest in hardware security and cryptographic implementation testing.

Valuable skills and interests for participants:

· FPGA design experience with VHDL, Verilog, or SystemVerilog.
· Understanding of cryptographic algorithms (AES, SHA) at the block diagram or RTL level.
· Familiarity with AMD Vivado or Microchip Libero SoC design tools.
· Python programming for test automation and data analysis.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: · FPGA Development Board based on AMD/Xilinx Artix-7 or similar provided by General Dynamics. Alternative: Microchip PolarFire Evaluation Kit. · Development PCs for hosting applications provided by the Project Team. · AMD Vivado Design Suite (WebPACK or full license), obtained by the Project Team via Student licensing with General Dynamics support. · USB-UART cable for host communication and logic analyzer or ILA for debug. · NIST CAVP test vector files (publicly available from csrc.nist.gov).

FreshPick - FreshPick: Computer Vision Platform for Food Quality Inspection

Proposer: Ehsan Daneshgar ehsan@launchpath.ca

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: FreshPick is an AI-powered platform that helps food processors automate produce quality inspection using computer vision, machine learning, and modern cloud technologies. By combining intelligent image analysis with enterprise software, the platform enables businesses to improve inspection consistency, streamline quality control workflows, generate compliance reports, and gain actionable operational insights.
A strong foundation has already been developed, including AI-assisted inspection workflows, computer vision capabilities, dashboards, analytics, reporting, user management, ERP integration, and cloud-based backend services. This project focuses on advancing the platform into its next stage of development.
Students will begin by evaluating the existing platform through functional, technical, and usability testing to understand its current capabilities, identify opportunities for improvement, and recommend enhancements that strengthen the overall product. Once approved, students will implement those improvements while expanding the platform with new functionality aligned with the product roadmap.
Project work may include enhancing AI-assisted inspection workflows, improving computer vision pipelines, expanding analytics and reporting capabilities, optimizing platform performance and scalability, strengthening enterprise integrations, and refining the overall user experience to create a more robust, intelligent, and production-ready solution.

Student Learning Goals: - Work on an enterprise software platform for the food technology industry.
- Evaluate an existing product to identify technical, functional, and usability improvements.
- Design, propose, and implement meaningful product enhancements.
- Develop modern frontend and backend application features.
- Improve reporting, dashboards, analytics, and operational workflows.
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:

- Gain exposure to computer vision and AI-assisted inspection within a real-world software platform.
- Participate in software testing, validation, deployment, and technical documentation.
- Apply agile software engineering and collaborative product development practices.

Deliverables: - Review and evaluate the existing platform to identify opportunities for improvement.
- Recommend and implement approved enhancements to existing features, workflows, and overall system performance.
- Develop new functionality that expands the platform's capabilities.
- Improve inspection workflows and supporting computer vision functionality where appropriate.
- Enhance analytics, reporting, enterprise integrations, and platform scalability.
- Conduct functional, usability, and system testing to validate improvements.
- Deliver updated technical documentation, deployment guides, and source code.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:

- Full-stack software development
- React, TypeScript, or modern frontend frameworks
- Backend development (Node.js, REST APIs)
- Database design and cloud application development
- Artificial intelligence and machine learning
- Computer vision and image processing
- Data analytics and visualization
- UI/UX design
- Software testing, debugging, and system optimization

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

FreshPick - FreshPick: Computer Vision Platform for Food Quality Inspection

Proposer: Ehsan Daneshgar ehsan@launchpath.ca

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: FreshPick is an AI-powered platform that helps food processors automate produce quality inspection using computer vision, machine learning, and modern cloud technologies. By combining intelligent image analysis with enterprise software, the platform enables businesses to improve inspection consistency, streamline quality control workflows, generate compliance reports, and gain actionable operational insights.
A strong foundation has already been developed, including AI-assisted inspection workflows, computer vision capabilities, dashboards, analytics, reporting, user management, ERP integration, and cloud-based backend services. This project focuses on advancing the platform into its next stage of development.
Students will begin by evaluating the existing platform through functional, technical, and usability testing to understand its current capabilities, identify opportunities for improvement, and recommend enhancements that strengthen the overall product. Once approved, students will implement those improvements while expanding the platform with new functionality aligned with the product roadmap.
Project work may include enhancing AI-assisted inspection workflows, improving computer vision pipelines, expanding analytics and reporting capabilities, optimizing platform performance and scalability, strengthening enterprise integrations, and refining the overall user experience to create a more robust, intelligent, and production-ready solution.

Student Learning Goals: - Work on an enterprise software platform for the food technology industry.
- Evaluate an existing product to identify technical, functional, and usability improvements.
- Design, propose, and implement meaningful product enhancements.
- Develop modern frontend and backend application features.
- Improve reporting, dashboards, analytics, and operational workflows.
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:
Students participating in this project will have the opportunity to:

- Gain exposure to computer vision and AI-assisted inspection within a real-world software platform.
- Participate in software testing, validation, deployment, and technical documentation.
- Apply agile software engineering and collaborative product development practices.

Deliverables: - Review and evaluate the existing platform to identify opportunities for improvement.
- Recommend and implement approved enhancements to existing features, workflows, and overall system performance.
- Develop new functionality that expands the platform's capabilities.
- Improve inspection workflows and supporting computer vision functionality where appropriate.
- Enhance analytics, reporting, enterprise integrations, and platform scalability.
- Conduct functional, usability, and system testing to validate improvements.
- Deliver updated technical documentation, deployment guides, and source code.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:

- Full-stack software development
- React, TypeScript, or modern frontend frameworks
- Backend development (Node.js, REST APIs)
- Database design and cloud application development
- Artificial intelligence and machine learning
- Computer vision and image processing
- Data analytics and visualization
- UI/UX design
- Software testing, debugging, and system optimization

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

Kids At Risk Action - From Searches to Insight: Data Engineering a KARA Topic Discovery Platform

Proposer: Mike E. Tikkanen mike@invisiblechildren.org

Project Contact: Mike E. Tikkanen mike@invisiblechildren.org

Project Description: turn many hundreds of google searches for articles, studies, and statistics on multiple child abuse and child protection topics into searchable and readable information on KARA's online database

Student Learning Goals: Students would get a full-stack data/UX experience: designing a real data pipeline and search interface that turns raw Google results into usable child-welfare knowledge for KARA.

Data engineering and tooling
Design and implement workflows to capture, clean, and organize results from hundreds of topic-specific searches (titles, sources, tags, brief descriptions) into structured tables or documents.

Learn to build a basic "data pipeline": defining schemas, handling duplicates and low-quality links, and maintaining data quality so the KARA database stays reliable over time.

Information architecture and search
Practice information architecture by deciding how to group topics, subtopics, tags, and metadata so different audiences (advocates, students, reporters) can actually find what they need.

Implement and tune search and filtering (by topic, geography, year, source type, etc.), and evaluate how well different designs help users discover relevant child-abuse and protection information.

UX design and usability
Conduct light user research with KARA and test users to understand how they search now, where they get stuck, and what an intuitive, readable results page looks like for non-technical users.

Apply UX principles (layout, labeling, visual hierarchy, accessibility) to turn dense lists of links into readable, scannable summaries with clear signals of source quality and relevance.

Ethics, curation, and critical thinking
Develop judgment about source credibility and bias when curating articles and studies on sensitive topics like child abuse, learning to prioritize rigorous and responsible information.

Engage with questions of trauma-informed presentation, avoiding sensationalism while still

surfacing hard truths, and writing safe, neutral topic labels and descriptions.

Teamwork and professional skills
Work as a multidisciplinary team (data, software, UX, content) using version control, tickets, and documentation to coordinate an evolving, multi-semester system.

Produce professional-grade documentation and handoff materials so future student teams and KARA staff can extend the database and search tools without starting from zero.

Deliverables: The main deliverable would be a working, searchable KARA knowledge index plus the data and documentation needed to grow it over time.

Product and data deliverables
Searchable online database (or integrated KARA module) containing hundreds of curated links to articles, studies, and statistics, each with consistent metadata (title, source, date, topic tags, brief summary).

Defined data model and populated dataset (CSV/JSON or database tables) describing how topics, subtopics, tags, and sources are structured so future teams can keep ingesting new material.

Search and UX deliverables
Functional search and filter interface that lets users query by keyword and refine by topic, geography, date range, source type, or other key facets, returning readable, well-organized results.

UX package with sample wireframes/screens, design rationale, and accessibility decisions (e.g., labeling, color/contrast, mobile layout) tailored to KARA's audiences.

Curation, quality, and ethics
Written curation guidelines covering: which sources qualify, how to handle duplicates, how to summarize neutrally, and how to name sensitive topics in a trauma-informed way.

A short quality-check report showing sampling of entries reviewed for accuracy, broken links, and tagging consistency, plus a plan for ongoing maintenance.

Technical and handoff materials
Version-controlled repository (or Airtable/DB schema export) with setup instructions, scripts/notebooks for loading/updating data, and notes on any automation used for collecting search results.

Handoff brief (2–3 pages or slide deck) summarizing what was built, current coverage (topics, number of links), known gaps, and concrete recommendations for the next semester's team.

Desired Background: The project works best with a multidisciplinary team combining data, software, UX, and content/subject-matter strength.

Data and software background
Students in computer science, software engineering, data science, informatics, or information systems who can design schemas, write scripts/notebooks, and build/maintain a basic search interface or database-backed web tool.

Comfort with version control, working with CSV/JSON, basic APIs or web tools for collecting/cleaning results, and structuring a simple "data pipeline" from raw search output into a clean table.

UX, information architecture, and content
Students with UX, HCI, or information-architecture interests (e.g., Human Systems Engineering, Graphic Information Technology – UX) to design navigation, tagging structures, and readable result layouts.

Strong writing and critical-reading skills to summarize articles and studies neutrally, label topics clearly, and help define trauma-informed naming and curation guidelines.

Social issue interest and collaboration
Work with KARA and if possible, at least one student with interest or coursework related to social work, psychology, criminology, public policy, or child welfare who is motivated by child-abuse/child-protection issues and can help interpret content.

Team members ready to interact with a community partner, gather feedback, and document decisions so future semesters can extend the KARA knowledge index without starting over.

Delivery Mode: In-person

Additional Materials: https://invisiblechildren.org/2025/10/21/sad-stories-about-child-abuse-in-america-today/

SkipCourse LLC - Frontend Team (SkipCourse Portfolio, WBL, & Transfer UI)

Proposer: Karl Ernsberger karl@skipcourse.com

Project Contact: Karl Ernsberger karl@skipcourse.com

Project Description: SkipCourse is an enrollment-readiness platform that converts workplace learning into structured, competency-aligned documentation for registrars. The SCV1 repository contains the React-based frontend that students, faculty, and registrars use to upload artifacts, view competency maps, track progress, and review audit-ready portfolios.

This Capstone team will modernize, expand, and stabilize the SCV1 frontend. Students will work on UI/UX improvements, component refactoring, accessibility upgrades, new feature development, and integration with backend APIs. The team will help evolve SCV1 into a production-ready, scalable interface used by real institutions.

Student Learning Goals: Students will gain hands-on experience building a real edtech product used by schools and colleges. They will work with:

React, TypeScript, and modern component architecture

Firebase Storage & Firestore integration patterns

API consumption and state management

UI/UX design for education workflows

Accessibility and responsive design standards

Real sprint cycles, GitHub issue tracking, and PR workflows

Collaboration with a backend Capstone team delivering new APIs

Students will learn how to build maintainable, scalable frontend systems in a real startup environment.

Deliverables: The frontend team will deliver:

Refactored React components for portfolio viewing, evidence mapping, and course progress dashboards

Improved file upload workflows, including error handling for Firebase Storage permission issues

New UI for competency alignment, evidence chips, and standards visualization

Registrar-facing dashboards for reviewing audit-ready portfolios

Student progress visualization tools, including course completion, standards mastery, and project evidence

Accessibility improvements (ARIA, keyboard navigation, contrast compliance)

Responsive design updates for mobile and tablet users

Integration with new backend endpoints created by the backend Capstone team

Documentation for future SkipCourse developers

Desired Background: Ideal students will have experience with:

React (required)

TypeScript or JavaScript

CSS frameworks (Tailwind, MUI, or similar)

REST API consumption

Firebase familiarity (optional but helpful)

GitHub workflows and pull requests

Basic UI/UX design principles

No prior edtech experience required — just strong frontend fundamentals and willingness to collaborate.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials:
https://docs.google.com/document/d/1IMhJ9k5kwgO1vt6X2lA97bedUpk-bjN7nw5c5rQ4SyU/edit?usp=sharing

MentaMorph Inc - Game Experience & Unity Development

Proposer: Susan Meredith susan@mentamorph.com

Project Contact: Susan Meredith susan@mentamorph.com

Project Description: This project enhances the MentaMorph game experience within Unity, focusing on usability, engagement, and integration with backend systems. It includes tutorials, UI/UX improvements, avatar systems, and data tracking integration.

Student Learning Goals: Students will develop skills in game development and interactive system design, while learning how front-end environments connect to backend services for data collection and personalization.

Deliverables:
- Improved onboarding/tutorial system
- Enhanced Unity UI/UX
- Avatar or player identity features
- Data integration hooks
- Polished gameplay experience

Desired Background:
- Unity and C#
- Game design fundamentals
- Basic API understanding

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

Etherea - Global Social Impact App & Edtech Platform

Proposer: Celina Mankarios celina.mankarios@gmail.com

Project Contact: Celina Mankarios celina.mankarios@gmail.com

Project Description: About Etherea: Etherea is a global social enterprise building an innovative app that uses AI, AR, psychology and emerging research to help youth change their lives (with 6 personal development themes: Mental Health, Physical Health, Financial Freedom, Spreading Kindness, Personal Growth and Relationship-building) and global issues (with 9 world issue themes including Poverty & World Hunger, Gender Equality, LGBTQ+ Rights, Climate Change & the Environment, Animal Welfare, Racial Justice, Disability Rights, Children's Rights, and Current Affairs). Users can do theme-based impactful missions for points, explore animated/AR educational worlds, join team challenges for mega prizes, attend events with world leaders, find relevant resources (books, podcasts, videos and movies), shop ethical/eco-friendly small businesses run by marginalized communities and more!

It's a one-stop hub for resources, education, activism and measurable impact with content and features created in collaboration with professors from top universities like Harvard, Cornell, and the University of Toronto, UN diplomats, theme-based experts, and over 100 of the most inspiring/impressive youth activists from over 80 countries around the world, including Forbes 30 Under 30, Diana Award Winners, BBC 100 Women, Vogue 21 Under 21, TIME 100 honorees, high-level policy advisors and huge non-profit founders helping hundreds within their community.

Etherea will be accessible to youth through the app stores and is also being implemented in schools globally, with pilot programs being launched in the Democratic Republic of Congo, Thailand, the Bahamas, Canada, Ghana and South Africa.

About the Project: Students would work with our product managers, design team and Development & Testing Subcommittee to implement improved/updated features of the Etherea app using React Native, in preparation for MVP launch by year-end. This would include fixing any minor issues in the code, updating our educational worlds with more corrected content, and adding paid features such as our Premium Subscription, Social Circles with ads, Events, and Small Business Brand Shopping sections.

Student Learning Goals: Students will be able to learn:
- Using React Native for app development and how to launch a mobile app on IOS and Android
- About world issues with the content and platform being shaped by leading global humanitarian figures
- Lean Startup Methodology/Agile Development- constantly testing and iterating to refine a product with product-market fit across sprints

- Working with diverse teams including designers, testing subcommittee leaders, product managers, content creators and other partners.

Deliverables: Transfer figma designs into functioning IOS code in React Native to launch the MVP app features by the end of November and work on updated features following MVP launch. Features may include: a habit-building home page, an impact calculator, a page with educational gamified worlds and customizable land, a resources section, brands shopping section, mega challenges and events sections and/or in-app social media feature.

Desired Background: Background experience with IOS/Android app development, React Native, cybersecurity and/or Firebase. If possible, any students with UI/UX experience too would also be great.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

IEEE Industry Engagement Committee - IEEE Impact++ Platform: Leveraging Data Mining, Analytics, and AI for Automated Impact Assessment and Quantification

Proposer: Lina Karam karam@asu.edu

Project Contact: Matthew Scott mscott@ieee.org

Project Description: The work for this project contributes to the design and development of a new IEEE Impact++ platform, an initial version of which was designed and proposed by the Industry Impact Working
Group and the IEEE Impact++ SubWorking Group of the IEEE Industry Engagement (IEC) committee. Members of the team of students working on this project are expected to contribute to
font-end and backend design and development, algorithms for data mining, data analytics, AI/machine learning, metrics for automated assessment, documentations and presentations, testing and validation, scaling the current initial prototype into a deployable end-to-end system that has a front-end, back-end and offline data mining service and algorithms and software for measuring impact.

Student Learning Goals: In addition to enhancing their software engineering, algorithm development and implementation in relevant areas such as font-end and backend design and development, algorithms for data mining, data analytics, AI/machine learning, metrics for automated assessment, the students will learn about real-world software engineering and what it takes to transform a system from a prototype level to a real-world full-scale deployable product. The students will also be collaborating with software engineers and computer scientists from various industries and will learn about the importance of effective oral and written communication, project management and team collaborations.

Deliverables: Deployable end-to-end system that has a front-end, back-end and offline data mining service and algorithms and software for measuring impact.
Documentation
Fully commented code
Testing documentation
Validation
Final report including measuring performance.

Desired Background: CS students with very good coding, analytical, and communication skills and with a high degree of reliability and who can seriously commit to this project and make things happen.
Experiences with the following would be a plus:
(1) Backend development experiences including RESTful APIs, integrating AI services and APIs, and so on;
(2) Frontend development experiences such as web interfaces, interactive pages, and so on;

(3) Supporting data processing and AI feature integration.
Please see also attached file for additional desired qualifications and project requirements.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: NDA and IP Agreement:
https://drive.google.com/file/d/1gigblsLiIIJw7Ko7FW5hca9p5axZaq0a/view?usp=drive_link

Qualcomm Inc. - Improved 3D modeling in Digital Twin

Proposer: Emma Lacey elacey@qti.qualcomm.com

Project Contact: Lokesh Jain lokejain@qti.qualcomm.com

Project Description: Large scale outdoor 3D models provided by map vendors do not use semantic information to refine shapes of objects. For example, foliage is often represented as coarse polyhedrons. The project will use semantic understanding of the object and refine its surfaces to reflect common knowledge. Refining the existing mesh models will be done by applying targeted geometric operations.

Student Learning Goals: Students get to work on latest Computer Vision segmentation models and large scale 3D models.

Deliverables: Final working prototype

Desired Background: Need to have: Python
Nice to have: Any experience in computer vision machine learning models

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Qualcomm Inc. - Improved 3D modeling in Digital Twin

Proposer: Emma Lacey elacey@qti.qualcomm.com

Project Contact: Lokesh Jain lokejain@qti.qualcomm.com

Project Description: Large scale outdoor 3D models provided by map vendors do not use semantic information to refine shapes of objects. For example, foliage is often represented as coarse polyhedrons. The project will use semantic understanding of the object and refine its surfaces to reflect common knowledge. Refining the existing mesh models will be done by applying targeted geometric operations.

Student Learning Goals: Students get to work on latest Computer Vision segmentation models and large scale 3D models.

Deliverables: Final working model

Desired Background: Need to have: Python
Nice to have: Any experience in computer vision machine learning models

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

InvenTrade - InvenTrade: Industrial Inventory Management and Marketplace Platform

Proposer: Ehsan Daneshgar ehsan@launchpath.ca

Project Contact: Maryum Mazhar maryum@launchpath.ca

Project Description: InvenTrade is a cloud-based industrial inventory management and marketplace platform that helps manufacturers, contractors, distributors, and industrial businesses manage inventory while discovering and redistributing surplus, discontinued, and hard-to-find industrial parts.

The platform combines inventory management, intelligent search, regional product discovery, and marketplace functionality to help businesses reduce procurement delays, recover value from idle inventory, lower operational costs, and support a more sustainable circular economy. By connecting businesses with locally available industrial inventory, InvenTrade enables faster sourcing while reducing waste and dependence on long supplier lead times.

An existing platform has already been developed, including secure authentication, inventory management, image uploads, product listings, search capabilities, recommendation features, and AWS cloud infrastructure. This project focuses on advancing the platform into its next stage of development by evaluating the existing system, improving core workflows, and expanding functionality to create a more scalable and production-ready marketplace.

Student Learning Goals: Students participating in this project will have the opportunity to:
- Contribute to the development of an AI-powered industrial inventory management and discovery platform.
- Design and implement scalable full-stack features using modern web technologies.
- Develop and enhance AI-powered search, recommendation, and intelligent inventory matching capabilities.
- Build and improve inventory management workflows, marketplace functionality, and user experience.
- Work with AWS cloud infrastructure, databases, authentication, REST APIs, and system integrations.
- Participate in product planning, testing, debugging, performance optimization, and usability improvements.
- Gain hands-on experience collaborating with an early-stage startup using agile software development practices

Deliverables: - Review and evaluate the existing InvenTrade platform, architecture, and user experience.
- Document findings and recommend improvements to functionality, usability, scalability, and performance.

- Design and implement approved enhancements to the Inventory Management System and industrial marketplace.
- Develop and improve AI-powered search, intelligent product matching, and recommendation capabilities.
- Enhance inventory management, buyer, supplier, and procurement workflows.
- Improve cloud infrastructure, API integrations, database performance, and platform reliability.
- Conduct functional, usability, and performance testing, including bug fixes and quality assurance.
- Deliver technical documentation, implementation guides, deployment documentation, and production-ready source code.

Desired Background: Students with experience or interest in one or more of the following areas are encouraged to apply:

- React, TypeScript, or modern JavaScript frameworks
- Java and Spring Boot
- REST API development
- PostgreSQL or relational databases
- AWS cloud services
- Full-stack web application development
- AI or recommendation systems
- UI/UX design and usability evaluation
- Software testing, debugging, and performance optimization

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

Qualaces Inc - Jobque.ai Core AI Backend Team

Proposer: Aakash Patel aakash@qualaces.com

Project Contact: Aakash Patel Aakash@qualaces.com

Project Description: Join Jobque.ai and work on building an AI-powered career technology platform focused on transforming how people discover jobs, manage applications, and navigate their career journey. This project involves developing intelligent systems that leverage AI, automation, and modern cloud technologies to create personalized user experiences and streamline job-seeking workflows. jobque.ai is live. Jobque Chrome extension is live. Please check it out and see if you like to build a backend for an AI agent.

Student Learning Goals: Students and team members may contribute to areas such as AI integration, backend development, recommendation systems, workflow automation, analytics, and scalable cloud infrastructure built in AWS. The project provides hands-on experience building real-world products used by actual users while collaborating in a startup-style environment focused on innovation and rapid execution.

Deliverables: **Expected Deliverables**

1. Design and develop features for AI-powered web or mobile applications.
2. Build and integrate frontend and/or backend components.
3. Implement APIs and AI-enabled workflows.
4. Contribute to testing, debugging, and performance optimization.
5. Participate in sprint planning, code reviews, and team collaboration.
6. Deliver a working feature, prototype, or production-ready component by project completion.

Desired Background: **Desired Background**

1. Background in Computer Science, Software Engineering, Information Technology, or related fields.
2. Familiarity with JavaScript/TypeScript and modern web or mobile frameworks (React, React Native, etc.).
3. Basic understanding of APIs, databases, and cloud technologies.
4. Interest in AI, machine learning, and emerging technologies.
5. Strong problem-solving skills and willingness to learn.
6. Prior project experience, hackathons, app development, or published applications is a plus.

Delivery Mode: In-person

Required Agreements:

- US Citizenship requirement
- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: jobque.ai & https://chromewebstore.google.com/detail/jobque-agent/kbbefdejaeljoemfoegfilpjlfolejgk

Joyify My Day - Joyify My Day: Building a Private AI-Powered GPS for Lingering Thoughts

Proposer: Tina Capotorto tcapotorto@gmail.com

Project Contact: Tina Capotorto tcapotorto@gmail.com

Project Description: Joyify My Day is developing a personalized app for mid-career professionals dealing with office drama. Anyone who's lived it knows the meeting ends, but the thoughts don't. Joyify My Day has created proprietary Thought-Maps that work like a GPS for lingering thoughts. Users start with the thought they're already having, and the app matches them with a Thought-Map that guides them one believable thought at a time toward greater clarity.

A working AI-powered MVP has already been developed, including the front-end and back-end source code and technical handover documentation. The current MVP dynamically matches a user's input to a Thought-Map and guides them through a sequence of thoughts. It currently supports one office-drama scenario and does not include user accounts, persistent user data, or progress tracking.

Student Learning Goals: Students will have the opportunity to work with an existing AI-powered MVP (v1.0), identify technical requirements, recommend alternative solutions, implement new functionality, and prepare the application for early-user validation and market entry.

The project provides hands-on learning across several areas:

- Full-stack application development: Work with and extend an existing React front end and FastAPI/Python back end. And evaluate any alternative options to v1.0.
- AI integration and matching: Understand how the existing application uses OpenAI to match natural-language user input with proprietary Thought-Maps, then expand the application to additional Thought-Maps. Evaluate any alternative options to current v1.0
- UX and data visualization: Apply updated brand and UX requirements and design a visual progress tracker that turns stored user progress into a useful front-end experience.
- Authentication and data architecture: Design and implement secure user accounts and determine what information needs to be stored to support progress tracking while minimizing the collection of private user information.
- Privacy and cybersecurity: Evaluate the application's security architecture, identify vulnerabilities, apply appropriate security controls, and explore ways to separate identifiable account information from private user input.
- Privacy-preserving analytics: Research and recommend an analytics approach that allows the company to understand user behaviour, such as app usage, progression, completion and drop-off, without giving the company access to the private thoughts users enter.
- Cloud infrastructure and DevOps: Evaluate deployment and hosting options for the next stage of the MVP. Students can compare managed Docker-compatible platforms and self-managed options, including containers, VPS/cloud servers, Docker, Docker Compose, hosting architecture, HTTPS, environment variables, monitoring and deployment

workflows, and recommend an approach based on security, scalability, maintainability and cost.

- Technical decision-making: Students will research alternatives, explain trade-offs, make recommendations, and document why their selected architecture is appropriate for an early-stage company looking to expand and grow usage.

Deliverables: A working MVP that Joyify My Day can put in front of early users to validate the product experience, gather structured insights and prepare for the Founding Members pilot.

* Updated MVP: A functioning version of the existing application incorporating JoyifyMyDay's current branding, content, calculations and user experience requirements.
* Expanded Thought-Map functionality: Integration of additional proprietary Thought-Maps and office-drama scenarios into the existing AI-powered matching experience. The current MVP supports one scenario, so this project will extend the existing functionality.
* Secure user accounts: Registration, login and account functionality that allows users to securely return to the application and track their progress over time.
* Progress tracker: A user-facing chart or visualization that allows users to see their progress and patterns across sessions.
* Privacy-preserving analytics: Implementation of an appropriate analytics solution that provides insight into how users navigate and engage with the application, including completion and drop-off points, without providing access to the private thoughts users enter.
* Security and privacy enhancements: Review and strengthening of the application's security architecture, including authentication, access controls, secure data handling and data minimization. The current MVP is publicly accessible, has no authentication and does not persist user data, so these additions will require new security and privacy considerations.
* Hosting and deployment recommendation: Research and comparison of appropriate infrastructure options, including containers, VPS/cloud hosting, Docker and managed Docker-compatible platforms. Students will recommend an approach based on security, cost, scalability, maintenance and the needs of an early-stage product.
* Deployment: Deploy the enhanced MVP using the selected architecture and configure the environment required for early-user testing.
* Technical documentation and handover: Updated documentation covering the application architecture, deployment process, third-party services, environment configuration, security considerations and instructions for maintaining and extending the application.

Desired Background: Students should be curious problem-solvers who are interested in taking an existing AI-powered MVP, evaluating the technical decisions already made, researching alternatives, and collaboratively developing a secure, scalable product for real-world user validation and eventual market entry.

Front-End Development - 2 students

Experience or interest in:

- React and JavaScript
- Responsive web application development
- UI/UX implementation and accessibility
- Data visualization and development of a user progress tracker
- Translating established brand guidelines into a polished digital experience
- Integrating front-end functionality with APIs and authenticated user accounts

Back-End Development - 2 students
Experience or interest in:

- Python and FastAPI or similar back-end frameworks
- REST APIs and front-end/back-end integration
- Databases and data architecture
- User authentication and account management
- OpenAI or other AI/API integrations
- Cloud deployment, Docker and containerized applications
- Exploring managed hosting, containers, VPS/cloud servers and deployment architecture
- Analytics implementation and privacy-conscious data collection

Cybersecurity - 1 student
Experience or interest in:

- Web application security
- Secure authentication and access controls
- Privacy-by-design and data minimization
- Secure API and credential management
- Threat modelling and vulnerability assessment
- Encryption and secure storage practices
- Reviewing third-party services and how user information moves through the application
- Developing security recommendations appropriate for an application handling highly private user-entered information

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement
- Non-disclosure agreement

Additional Materials: https://joyifymyday.com/
https://www.linkedin.com/company/joyify-my-day/?viewAsMember=true
https://www.linkedin.com/in/tinacapotorto/

DigiClips - LANGUAGE TRANSLATION PROJECT

Proposer: Bob Shapiro bobshapiro40@gmail.com

Project Contact: Henry Bremers hbremers@gmail.com

Project Description: LANGUAGE TRANSLATION PROJECT

https://www.digiclipsinc.com/digiview/DigiClips_Language_Translator.htm

Language translation needs to be done to generate media results reports worldwide.
Plus Language Translation for the frontends for General Public Subscribers and be part of the options page Information and Administration. Also for the backend.

Need to first use Spanish for presentation then through ADMINISTRATION
14D. Language translation be able to add on more language
https://www.digiclipsinc.com/digiview/DigiClips_Language_Translator.htm

Was worked on by ASU Team 6 and ASU Team 9

It needs a way to present this project in the cloud on AWS Amazon Lightsail on www.digiclipsproject.com .

Student Learning Goals: We have weekly meetings and are available 24/7 with the students. A real world work experience and work with other capstone teams and have DIGICLIPS DISCORD MEETINGS Link: https://discord.gg/eNp7bjdjUK.

Deliverables: With in about 30 days before the end of semester and put up on AWS Amazon Lightsail on www.digiclipsproject.com/languagetranslation

Desired Background: Software to be familiar with would be Ubuntu-Mate 22.04, C, nodejs, react, angular 20 https://angular.dev/ typescript, mysql relational database, putting up on AWS Amazon Lightsail.
We have students work remotely on our local system through digi-frontend for development testing debugging then put up on DigiClips Github and for final deliverable and presentation put up front end on AWS Amazon Lightsail www.digiclipsproject.com

Delivery Mode: In-person

Additional Materials:
https://docs.google.com/document/d/17WlsZPaOyOgE4diyGexDlsz7_DeR6d6UBUCS5YCKrvQ/edit?usp=sharing

Leeg Inc. - Leeg × ASU Capstone Partnership: Career OS & Workforce Intelligence

Proposer: Shannon Burch shannon@leegworks.com

Project Contact: Shannon Burch shannon@leegworks.com

Project Description: Who Is Leeg
We are changing the way individuals manage their careers and how they interact with their employer. Leeg is building the Career Operating System for the AI era — a platform that puts career ownership back where it belongs: with the individual.

We're a fast-paced, female-founded startup building deeply human technology. Our product is intimate and personalized, and so is the way we work. We're unapologetically human-centric — with each other, with our users, and with the partners we build alongside. We move with velocity, we love to have fun, and we learn, learn, learn. Always.

Our team is intentionally built around a model where people do many things at once — we have a strong fractional culture, where talented humans contribute across multiple roles and projects. It keeps us sharp, curious, and close to the work. If you join us, you'll be treated like one of the team — not an intern, not a contractor, not a student doing an assignment.
What We're Building
Two connected products, one mission.

The Leeg App — a Career OS owned entirely by the individual. Your career, your data, your control. The Leeg App is a personalized career operating system that helps individuals chart their growth, build skills, and navigate their professional journey. Career intelligence specific to you builds over time — the more you use the app, the smarter and more personalized it becomes. Your career becomes something you manage daily, 2-5 minutes at a time. And no individual user data is ever shared with their organization. Privacy isn't a feature — it's the foundation.

The Workforce Intelligence Portal — aggregated insight for organizations. On the other side of the platform, employers gain access to a Workforce Intelligence Portal where layers of behavioral data, labour market signals, and original research combine into intelligence about their organization's AI-era skill readiness. Organizations see the forest; individuals own their trees.

Together, these tools redefine the relationship between employee and employer — built on trust, transparency, and shared progress.

Student Learning Goals: What You'll Be Doing
This project requires Frontend Engineering, Backend Engineering, and Project Management

talent working alongside Leeg's Technology and Product team. If you thrive in an autonomous environment that's always building and always releasing, this is the project for you.

Individually, you'll own a specific part of the build during each sprint. Collectively, your work will significantly move our products and the user experience forward. Always be shipping is in our DNA, and you'll feel that from week one.

We're building something that doesn't exist today and reshaping the 100-year-old institution of how we work. Over the course of two semesters, you'll encounter real-world scenarios and develop skills that complement what you're learning in the classroom. Our goal is to grow your confidence as you prepare to enter the workforce.

A note on fit. Velocity, curiosity, and ownership matter to us. If you like ambiguity, ask questions before being asked, and get energy from shipping things people actually use — you'll thrive here. If you're looking for a tightly defined ticket queue and a slow, predictable rhythm, this probably isn't your project, and that's okay. We want this to be a great fit for you as much as for us.
How We'll Support You
We work on a steady cadence: two huddles per week to check progress, work through challenges, and communicate next steps — and a sprint release that keeps us shipping.

That said, nothing waits for a huddle. Slack is a powerful tool on our team, where we ensure communication is available to everyone helping develop the Leeg suite of tools with as little friction as possible.

You'll have direct access to Leeg's product, technology, and growth leadership throughout the project — coaching, unblocking, and celebrating wins right alongside you. Because we run with a fractional model, you'll also be working next to experienced operators who bring perspective from across the industry — exposure that's hard to find in a traditional internship.

Deliverables: Project Outcomes
By the end of the two-semester engagement, the ASU team will have made tangible contributions to a live, market-facing product. Specifically, you can expect to:

Ship real features to real users. Your code and product decisions will be part of the Leeg suite used by actual individuals and organizations — not a sandbox or class exercise.
Own a meaningful piece of the build. Each team member will have a clearly defined area of responsibility within the Leeg App, the Workforce Intelligence Portal, or the supporting infrastructure.
Experience the full product lifecycle. From sprint planning through release, you'll see how features move from idea to user — including the messy, real-world parts that don't make it

into textbooks.
Build a portfolio-ready body of work. Walk away with concrete, demonstrable contributions you can point to in interviews — features shipped, problems solved, decisions made.
Develop professional fluency. Sprint cadence, async communication, working across functions, navigating ambiguity, and contributing in a startup environment are all skills you'll practice every week.
Grow your confidence entering the workforce. Our north star for this partnership is that each student leaves more prepared, more confident, and more clear-eyed about the kind of work and team they want to join after graduation.
Why This Matters
AI has the power to change the way we work. Leeg is making it easy for individuals and organizations to chart their course as the world changes around all of us.

Desired Background: Desired Student Background
(It is not expected that all students have all desired experience or knowledge, however collectively the project goes better if most experiences are included) Leeg is building the future of career management learning and growing is foundational to our culture.
Across All Roles
Comfortable with ambiguity and self-directed work; able to make progress without waiting to be told what to do next
Strong written communication, especially async (Slack-native)
Curious about AI, the future of work, or career development as a problem space
Excited by startup pace and a "ship early, ship often" culture
Open to feedback, fast iteration, and changing direction when the user or data tells us to
Frontend Engineer
Working proficiency in React (or similar modern framework) and TypeScript
Comfort with responsive design, component libraries, and modern CSS approaches (Tailwind, CSS-in-JS, or similar)
Familiarity with REST APIs and consuming backend services
Bonus: experience with mobile-first design, accessibility, or design system work
Bonus: an eye for UX and the ability to collaborate with designers
Backend Engineer
Working proficiency in at least one modern backend language/stack (Node.js, Python, or similar)
Familiarity with relational databases (Postgres a plus — we run on Supabase)
Comfort building and documenting REST APIs
Understanding of authentication, data privacy, and basic security practices (especially important given Leeg's privacy-first stance)
Coursework or hands-on exposure to Python
Basic SQL skills — comfortable writing queries, joining tables, and exploring data
Curiosity about AI, machine learning, or data science — coursework, side projects, or independent learning all count
Willingness to learn data pipelines, AI APIs (Anthropic, OpenAI), and modern data tools on

the job
Bonus: exposure to AI/LLM APIs, vector databases, or data pipelines
Bonus: experience with Supabase, serverless architectures, or cloud platforms

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Outer Snail Productions, LLC - Legalese - User Profile and Notification Agent

Proposer: Eric Trant eric@outersnail.com

Project Contact: Eric Trant eric@outersnail.com

Project Description: This project is to work with an existing application and website to expand user profile features and include a daily workflow to provide users notifications based on keywords and topics they have selected. It will also include working with the chief architect to improve the consistency and reliability of large language model prompt results.

Student Learning Goals: Browser-based application development. Application of LLM and prompt engineering.

Deliverables: An updated version of the existing Legalese application.

Desired Background: Web development, prompt engineering, automated workflow implementation & agentic application.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Objectways Technologies - Liceum Crowdsourcing Application Enhancement

Proposer: A.K. Sako ak.sako@objectways.com

Project Contact: A.K. Sako ak.sako@objectways.com

Project Description: Liceum is an existing crowdsourcing application where individuals can find exciting AI projects and companies can discover qualified AI talents. There are many enhancement opportunities that students can contribute to.

Student Learning Goals: Students will gain hands-on experience with frontend technologies such as Next.js, React, Typescript, backend technologies such as Node.js, Nest.js, RDBMS, etc. Also, they will be exposed to software development lifecycle along with source code management, CI/CD, and deployment.

Deliverables: At least 5 functional features reviewed and tested.

Desired Background: Javascript, typescript, React.js, Node.js. Next.js and Nest.js experience are plus.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://liceum.ai/

Exposure Creative LLC - Managing a website and adding features

Proposer: Sophia DuBois sophiakdubois@gmail.com

Project Contact: Sophia DuBois sophiakdubois@gmail.com

Project Description: Last years seniors helped code a website for me. It helps showcase creative freelancers work. This includes: photographers, videographers, makeup artists, and hair stylists. You can check out the landing page if you're interested and if you choose this project then I can show you the code they've completed of the website that will be shown once it's launched! https://www.theexposurecreative.com/

Student Learning Goals: The Students will learn how to navigate pre-existing code and add to it. We will not only focus on creating code that works but they will learn how to collaborate with others in order to finish by a deadline and create aesthetics according to the brief given

Deliverables: Monitor and deliver the sign up sheets of the creatives. Fine tune the pre-existing website. Create add ons to the website and make sure the system doesn't crash.

Desired Background: The system we use is GitHub so as long as you're familiar with that coding then you should be good! It's ok if you're beginner at it! Everyone in the group will work together

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://www.theexposurecreative.com/

Youbound LLC - Margyn MarketBridge: A Retailer-Ready Decision Pilot Layer

Proposer: Jeff Moe jeff@youbound.app

Project Contact: Jeff Alan Moe jeff@youbound.app

Project Description: ASU Margyn MarketBridge takes the reusable Margyn decision-rehearsal engines being developed through Seattle University and WSU Vancouver and turns them into a retailer-ready pilot layer. Students will build a demo-grade prototype using synthetic, sample, or public product/retail data that accepts a shopping scenario (via barcode, URL, seller, or manual entry), displays relevant retail signals, and evaluates it through the Margyn five-call structure — Proceed, Verify, Substitute, Pause, or Ask Human — with a plain-language explanation, human override capability, and an event log. Proprietary Margyn decision logic stays sponsor-side; students build to a clean, documented interface, making the system portable to a future retailer partnership

Student Learning Goals: Students will gain full-stack web development experience, practical API/interface design (building to a bounded external contract rather than owning the full system), data modeling for retail/product signals, and human-in-the-loop UX design — translating an abstract decision architecture into a usable, demo able product. This mirrors real product engineering work in retail tech, trust & safety, and fintech

Deliverables: 1) Product/shopping-scenario intake (barcode, URL, seller, manual entry); 2) retail signal display using sample data; 3) retailer-style configuration layer; 4) five-call decision output (Proceed/Verify/Substitute/Pause/Ask Human) via a documented interface to the decision logic; 5) plain-language explanation screen; 6) human override with logging; 7) persistent event log; 8) API/interface documentation; 9) a retailer-facing demo script

Desired Background: Web development (frontend + backend), REST API design, basic database/data modeling. Helpful but not required: interest in e-commerce/retail systems or human-in-the-loop product design

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

Maricopa County Attorney's Office - Maricopa County Attorney's Office

Proposer: Lieutenant John M. Dorsey dorseyj@mcao.maricopa.gov

Project Contact: 602-849-4555 dorseyj@mcao.maricopa.gov

Project Description: The Maricopa County Attorney's Office employs approximately 1,000 people with about 250 of them being lawyers who prosecute the majority of felony cases that occur in Maricopa County.

We have approximately 60 sworn detectives who support these prosecutors by following up on cases for them and closing gaps or issues through further investigation as required. The cases are police reports submitted for prosecution from Valley police agencies; i.e. Phoenix Police Department, Tempe Police Department, Scottsdale Police Department, the Sheriff's Office, etctera,

Our detectives who support the prosecutors write police reports of various sorts to include supplements to criminal cases for the mentioned police departments. We also author original police reports and other similar documents related to the uniquenesss of our duties.

Currently the office does not have an automated/comprehensive report writing/records management system. We write our reports in a Word document and save the documents on a shared-drive and track the reports on an Excel Spreadsheet. We write from 50-100 reports a year and these reports could vary from one page to several pages. We also do background investigations on people we are considering for employment and other similar reports and if we added all of the reports together we complete about 300 reports a year.

We need a comprehensive records management system that modernizes our current system so we can author, catalog, and store these documents. The system needs to be secure, self-populating, automatically numbers reports and is not difficult to maintain and update. We want to create the most state-of-the-art-system possible.

So , far we've done two Capstone's on this project with ASU, and I'd say we are about a 1/3 of the way completed. The students have done a FANTASTIC job, but due to the size of the project it takes a while to complete, but so far things are going great and we do not want to interrupt this excellent work and trajectory.

Therefore we are requesting another Capstone project and will probably need one more after this one and we will have a comprehensive records management system at the Maricopa County Attorney's Office that could be a model system for other prosecutorial agencies, thanks to Arizona State University!

Student Learning Goals: The students will have an opportunity to work in one of the biggest offices in the country. The goal of the office is to prosecute felony-criminal cases of individuals committting crimes in our communities with the ultimate goal of making our community as safe as possible. The students will be contributing to this goal by making our Investigations Division more effective and efficient in supporting prosecutors by developing a comprehensive report writing system that supports our detectives so they can better support our attorneys.

The students will have an opportunity to work in and to lead teams in brining this conception to reality. They will be able to speak in front of groups of people and explain technology to them and hone their communication and team building skills. They will also get an opportunity to see how government works and interact with the law enforcement side of the Maricopa County Attorney's Office, They will also get a glimpse into how our Valley police departments work and how the criminal justice process works.

The Maricopa County Attorney's Office employees are very friendly, and the past students have enjoyed their interactions with the office and I believe future students will also have a positive experience. The work they will be participating in is very important and noble as they will be significantly contributing to the effectiveness and efficiency of a very important office that effects crime and the quality of life in our community.

Deliverables: A comprehensive records managements system for the approximately 60 detectives of the Maricopa County Attorney's Office.

Desired Background: Prefferred, strong background in writing software and building systems from the ground up, or creating, originating systems.

Delivery Mode: In-person

Additional Materials: N/A

SDH Systems - MenuIQ: Menu Price Intelligence for Independent Restaurants

Proposer: Vishwak Thatikonda vishwakt@gmail.com

Project Contact: Vishwak Thatikonda vishwakt@gmail.com

Project Description: Independent restaurant owners operate on margins of three to five percent, yet almost none of them price strategically. Most set prices by gut feel or by glancing at the place down the street. The result is money left on the table on dishes customers would happily pay more for, and lost covers on dishes priced above what the local market will bear. Enterprise revenue management software exists, but it is built and priced for chains. The independent owner has nothing.

MenuIQ gives that owner a data-driven view of their menu. The system ingests menus (the owner's own plus local competitors) from PDFs and photographs, extracts structured item data, and builds a comparable-dish index for a local market. The hard technical problem sits right here: deciding that "Rigatoni alla Bolognese" at one restaurant and "House Pasta, Meat Sauce" at another are the same dish for pricing purposes. This is entity resolution over noisy, inconsistent, human-written text, and everything the product does depends on getting it right. Students will build it as a real machine learning system with a hand-labeled benchmark and measured precision and recall, rather than assuming a language model can eyeball it.

On top of that foundation, owners enter recipe costs to compute true item margin, and the system combines margin with local price positioning to produce ranked, explainable pricing recommendations grounded in the established menu engineering framework (Kasavana and Smith's classification of items as stars, plowhorses, puzzles, and dogs) rather than in model intuition.

The second semester tackles the question every owner actually asks: did the change work? Naively correlating a price change with next week's revenue is meaningless, because seasonality, weather, staffing, and local events swamp the signal. Students will instead build a proper difference-in-differences framework that uses unchanged items on the same menu as a control group, producing price elasticity estimates with honest confidence intervals and clearly flagging when there is not enough data to say anything at all. Learning to build a system that says "I don't know yet" is a large part of the educational value.

A conversational interface lets owners ask questions in plain language ("which appetizers should I raise?", "am I competitive on burgers?"). Deliberately, the language model handles language, and all numerical comparison and statistical work runs through deterministic, testable code paths. Students will learn where a language model belongs in a system and, just as importantly, where it does not.

The project is validated with real restaurant owners in the Phoenix metro area, not with fictional sample data.

Semester 1 delivers a working competitive positioning product: menu ingestion, the entity resolution engine with a measured benchmark, a local menu corpus, recipe costing, margin calculation, and a positioning dashboard. This is a complete, demonstrable system on its own.

Semester 2 adds measurement and intelligence: the price change tracker, the difference-in-differences impact framework, elasticity estimation, recommendation ranking, the conversational interface, and validation with real owners.

Student Learning Goals: Students will gain hands-on experience with:

- Document understanding and extraction: building a robust ingestion pipeline that turns PDFs and phone photographs of real menus into structured records (item name, section, description, price, modifiers), including handling multi-column layouts, decorative typography, and poor-quality images.
- Entity resolution and record linkage: the core technical challenge, matching semantically equivalent dishes across restaurants using embeddings, blocking strategies for efficient candidate generation, and a trained matching model, evaluated against a hand-labeled ground truth set with reported precision, recall, and F1.
- Relational data modeling: designing a PostgreSQL schema for restaurants, menus, versioned menu items, recipe costs, competitor observations, price change events, and revenue reports, with temporal correctness so that historical analysis reflects what was true at the time.
- Causal inference and applied statistics: difference-in-differences design, control group selection, price elasticity estimation, confidence intervals, and the discipline of quantifying uncertainty instead of asserting conclusions. Very few undergraduates encounter this, and it transfers to any data-driven role.
- Applied LLM engineering with appropriate boundaries: using the Claude API for extraction, natural language querying, and explanation generation, while routing all arithmetic and statistical reasoning to deterministic code. Students will learn structured output design, grounding responses in retrieved database records, and evaluating model reliability.
- Domain modeling from real literature: implementing menu engineering theory from hospitality research rather than inventing heuristics, and understanding restaurant unit economics well enough to build software professionals would trust.
- Full-stack product engineering: REST API design, authentication and multi-tenancy, dashboard and data visualization, and a workflow simple enough for a busy owner to use on a phone between shifts.
- Working with real users: onboarding real restaurant owners, observing where the product fails in practice, and iterating on feedback from people who do not care about the technology.

Professional practice: version control, code review, CI, testing, and biweekly design reviews with the sponsor acting as a technical stakeholder.

Deliverables: End of Semester 1

Menu ingestion pipeline accepting PDF and image uploads, extracting structured item data with measured accuracy on a held-out set of real menus
Local menu corpus covering a defined geographic area, assembled from owner uploads and permitted public sources
Dish entity resolution engine with a hand-labeled evaluation benchmark and published precision, recall, and F1 scores
PostgreSQL schema covering restaurants, menus, versioned items, categories, recipe costs, competitor observations, price changes, and revenue reports
Recipe costing module producing true per-item food cost and contribution margin
Competitive positioning dashboard showing per-item and per-category price placement against comparable local offerings
Semester 1 design document and demonstration

End of Semester 2

Price change logger capturing before and after prices with timestamps and owner notes
Difference-in-differences impact measurement framework using unchanged menu items as controls
Price elasticity estimates with confidence intervals, including explicit reporting when data is insufficient for a conclusion
Recommendation engine ranking pricing opportunities by expected impact, with plain-language explanations of the reasoning
Conversational query interface answering natural language pricing questions grounded in the restaurant's own data
Validation study with at least two real restaurant partners, documenting what worked, what failed, and what they would not use
Deployed application, technical documentation, test suite, and final presentation
Open-source repository with setup instructions and architecture documentation

Desired Background: Required across the team (not required of every student):

Proficiency in Python and JavaScript or TypeScript
Coursework or practical experience with relational databases and SQL
Familiarity with Git and collaborative development workflows
Willingness to talk directly with restaurant owners, who are the real users and the real source of truth

Valuable for specific roles:

Machine learning or NLP coursework, particularly anything touching embeddings, text similarity, or classification, for the entity resolution work
Statistics coursework, especially regression or experimental design, for the impact measurement work. A student from statistics, economics, or business analytics would be an excellent fourth team member.
Computer vision or document processing experience for the ingestion pipeline
Frontend and data visualization skills for the dashboard and owner-facing interface
Any experience with cloud deployment (AWS, GCP, or similar)

Delivery Mode: In-person

Additional Materials: N/A

Allgoodaccounting.ca (Sandria Goodall) - Mobile Delivery App Development

Proposer: Sandria Goodall allgoodcloud@gmail.com

Project Contact: Sandria Goodall allgoodcloud@gmail.com

Project Description: This project involves developing a mobile application that facilitates a pick-up and drop-off delivery service. The app will allow users to sign up, input pick-up times, and notify local drivers of items ready for collection. Drivers will also sign up, providing car details and their preferred geographic working area.

The app will include a payment-processing feature that calculates costs based on kilometres travelled. The project will enable learners to apply their knowledge of mobile app development, user interface design, and payment integration. The tasks are designed to be completed by a team with a shared skill set.

Student Learning Goals: Core tasks:

- Design the user interface for both user and driver sign-up processes

- Develop the notification system for pick-up requests

- Implement the payment processing feature based on distance

- Test the app for functionality and user experience

Deliverables: Deliverables:

- Mobile application

- Development process documentation

- Testing results report

- Recommendations for future improvements

Desired Background: Computer Science

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

- Intellectual property agreement

Additional Materials: N/A

ModelScan - ModelScan: An Automated Detection Framework for Unsafe Deserialization in ML Model Artifacts

Proposer: Anish Subash asubash@asu.edu

Project Description: Machine learning workflows depend on loading serialized artifacts (model checkpoints, tokenizers, configs) from sources like the Hugging Face Hub and PyPI. Many common loaders (pickle.load, torch.load, joblib.load, and framework-specific equivalents) can execute arbitrary code embedded in the file itself, but no widely-adopted tool exists to systematically scan a repository, package, or model file and flag these risks before a user loads them. This project builds ModelScan, an open-source static/dynamic analysis tool that scans ML artifacts and codebases for unsafe deserialization patterns, classifies findings by risk level, and generates actionable reports. The tool will support scanning individual model files, GitHub repositories, and PyPI packages, and will ship with a rule engine that can be extended as new unsafe patterns are identified. Over the course of the project the team will also run the tool against a real corpus of public ML repositories to validate detection accuracy and produce a dataset of findings, giving the tool a concrete evaluation benchmark.

Student Learning Goals: Students will design and build a full security tool from the ground up: static analysis/taint-tracking in Python, AST parsing, rule-engine architecture, and CLI/reporting tooling. They'll learn software engineering practices for a security product (testing, versioning, documentation, CI), plus the underlying vulnerability research skills (understanding serialization internals, exploit primitives, risk classification) needed to design accurate detection rules. Students will also get exposure to the ML supply-chain security space and, if findings warrant, the professional disclosure process for any real vulnerabilities the tool surfaces during evaluation.

Deliverables: Semester 1: Core scanning engine (AST-based static analysis for Python serialization sinks), CLI tool, initial rule set covering pickle/torch/joblib sinks, test suite, and a benchmark corpus of known-vulnerable/known-safe files for validation.
Semester 2: Expanded rule engine (additional frameworks/formats, e.g. ONNX, safetensors edge cases), GitHub/PyPI repository scanning support, HTML/JSON report generation, evaluation run against a real corpus of public repositories with a published findings dataset, and final documentation/release.
Milestones will be tracked incrementally (monthly feature targets, GitHub commit history, sprint-style check-ins) so progress is visible and demoable at every stage rather than concentrated at the end.

Desired Background: Comfort with Python, interest in static analysis/security tooling, coursework or CTF experience in vulnerability concepts is a plus but not required. (At least one other teammate with an orange belt on pwn.college

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

EcoClaim Solutions - Multi-Peril Resilience ROI Modeling Platform for Insurance Claims and Sustainable Reconstruction

Proposer: Tom Herman therman@ecoclaim.ca

Project Contact: Tom Herman therman@ecoclaim.ca

Project Description: We are developing an advanced resilience ROI calculator for the insurance and construction industries that combines climate risk modeling, building science, construction materials data, technical product specifications, environmental impact metrics, and cost data to evaluate long-term resilience investment decisions.

The system is designed to determine whether higher-performance building products and repair strategies, such as impact-resistant roofing, upgraded insulation, or moisture-resistant materials, provide a measurable financial and environmental return over the expected lifespan of a building.

The project involves integrating multiple complex datasets, including:

* Regional climate and peril risk data
* Construction material performance and durability data
* Embodied carbon and emissions factors
* Building service life modeling
* Insurance loss and replacement cost scenarios
* Product pricing and lifecycle cost analysis

The research component focuses on developing and validating methodologies for predicting long-term resilience outcomes under uncertain environmental conditions, including competing hazards, material degradation, regional climate variability, and evolving building standards.

This internship is well suited for students interested in AI, climate technology, software engineering, actuarial modeling, sustainability analytics, construction technology, or data science.

Student Learning Goals: * Work on a real-world climate technology and insurance innovation project involving AI, sustainability analytics, and resilience modeling
* Gain experience integrating large datasets related to climate risk, construction materials, building performance, embodied carbon, and insurance claims
* Participate in research and experimentation involving predictive modeling, lifecycle analysis, and resilience ROI calculations
* Contribute to development of AI-assisted tools for environmental impact analysis and decision support
* Learn about software architecture, data validation, technical uncertainty analysis, and quality assurance methodologies used in advanced R&D environments

* Collaborate with a multidisciplinary team spanning sustainability, software engineering, insurance, and construction domains
* Develop experience documenting experimental approaches, technical findings, and analytical methodologies aligned with Canadian SR&ED research practices

Deliverables: Expected Deliverables *

* Web-based resilience ROI calculator prototype
* Built using AI-assisted coding tools, APIs, and EcoClaim's aggregated datasets
* Inputs for climate/peril risk, building characteristics, material options, product specifications, cost, lifespan, and insurance time horizon
* ROI outputs comparing standard LKQ repair options against higher-resilience alternatives
* Sensitivity analysis showing how results change based on assumptions such as hazard probability, product lifespan, cost premium, avoided loss, and discount rate
* QA framework to validate calculations, data consistency, edge cases, and report outputs
* Documentation of data sources, assumptions, calculation methodology, limitations, and testing results
* Optional dashboard or report output summarizing financial, resilience, and carbon impacts

Desired Background: Desired Background *

* Student or recent graduate in computer science, engineering, data science, environmental studies, actuarial science, construction management, or related field
* Interest in AI, climate technology, sustainability, insurance, or construction innovation
* Experience with Python, JavaScript, SQL, data analysis, or statistical modeling is an asset
* Familiarity with APIs, databases, spreadsheets, or GIS/climate datasets is beneficial
* Strong analytical and problem-solving skills
* Ability to work with complex and sometimes ambiguous technical requirements
* Interest in applied research and experimentation within a startup environment

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: Previous Project / Background EcoClaim has already developed multiple software platforms related to insurance claims sustainability, embodied carbon measurement, climate resilience, and construction material analysis. Existing systems include TRAX and ReBuild Better (RBB), which process construction estimates, environmental product declaration (EPD) data, waste diversion information, emissions factors, and resilience-related material comparisons. The company has completed proof-of-concept work involving: * Building service life and durability modeling * Climate and peril

risk scoring * Embodied carbon calculations * Material substitution analysis * AI-assisted extraction of construction estimate data * Lifecycle and resilience calculations using Factor Method approaches (EN 15686) The project also builds on prior SR&ED-supported research involving AI orchestration, data consistency validation, environmental calculation methodologies, and multi-factor resilience modeling. Resources Available to the Student * Existing EcoClaim software platforms and APIs * Access to aggregated climate, construction material, EPD, and insurance-related datasets * Existing AI-assisted extraction and reporting tools * Prior proof-of-concept methodologies and technical documentation * Internal QA and testing frameworks * Access to AI coding tools and LLM-based development workflows * Mentorship from EcoClaim leadership and technical contributors involved in software engineering, sustainability analytics, insurance workflows, and resilience modeling * Existing cloud infrastructure, databases, and reporting systems supporting rapid prototyping and experimentation

MHP COnsulting LLC My Hockey Pro - My Hockey Pro Digital Card Creator Platform

Proposer: Chris Myhro myhro@phoenix1.io

Project Contact: Chris Myhro myhro@phoenix1.io

Project Description: My Hockey Pro has already built the first version of Phoenix1 Cards, a digital hockey card creation platform, and a second version is planned for release shortly. This capstone project will focus on scaling, refining, and productizing the existing platform so it can better support players, teams, coaches, families, scouts, and potential hockey industry partners. Rather than building a card creator from scratch, students will work with an existing product and help improve its architecture, user experience, data workflows, performance, reliability, and readiness for future integrations.

A major goal of the project is to prepare Phoenix1 Cards for a potential partnership and data integration with Elite Prospects, subject to approved and properly licensed API access. Students may design and prototype an Elite Prospects integration workflow that allows player search, player matching, profile verification, stat import, team/league data mapping, and auto-population of card fields where permitted. The project should avoid unauthorized scraping and instead focus on API-based integration, data governance, caching, error handling, attribution requirements, and compliance with partner data usage rules.

The student team will evaluate the current Phoenix1 Cards application, identify technical debt and scaling needs, improve the product workflow, strengthen the template/card rendering system, and build features that make the platform easier to operate as a commercial sports technology product. Potential work areas include account management, admin tools, template management, player profile linking, batch card generation, roster imports, image/PDF export improvements, analytics, QA/testing, deployment improvements, and documentation. Stretch goals may include QR codes linking to verified player profiles, team-branded card collections, social sharing, subscription-ready feature controls, and a technical proof-of-concept for Elite Prospects data integration.

Student Learning Goals: Students will gain experience working on an existing real-world sports technology product rather than a purely theoretical prototype. They will learn how to review an existing codebase, identify product and technical gaps, prioritize improvements, work through sponsor feedback, and deliver production-minded software over the course of the capstone. The project includes full-stack development, UI/UX refinement, API integration planning, data modeling, image/file processing, performance improvement, testing, deployment, documentation, and product management.

Students will also gain experience designing a responsible third-party data integration. A key learning opportunity will be planning how Phoenix1 Cards could integrate with Elite Prospects through approved/licensed API access, including player search, data matching, field mapping, caching, rate-limit handling, partner attribution, and data quality checks. This

gives students exposure to real-world issues that professional software teams face when connecting a product to an external data provider.

The sponsor will provide access to the current Phoenix1 Cards product, product requirements, brand assets, sample card templates, anonymized player/team data, and feedback throughout the project. The sponsor will meet with the team approximately every two weeks and will be available to answer reasonable project questions between meetings. Students will finish the project with experience contributing to a real product that has a clear market, real users, and potential hockey industry partnership value.

Deliverables: Expected deliverables include:

1. Current product assessment: Review the existing Phoenix1 Cards application and document current architecture, user workflows, technical debt, scaling risks, and recommended improvements.

2. Product roadmap and requirements document: Define the highest-priority features and improvements for scaling Phoenix1 Cards, including MVP requirements, stretch goals, user stories, acceptance criteria, and milestone plan.

3. UI/UX improvements: Improve the card creation workflow, saved card experience, admin/template management flow, and overall usability for players, parents, teams, and internal My Hockey Pro administrators.

4. Scalable card/template system: Improve or extend the template system so cards can support multiple layouts, team branding, player photos, logos, stat fields, QR codes, and export formats.

5. Elite Prospects integration plan and proof-of-concept: Design and, if approved API access is available, prototype a licensed API-based integration for player search, player matching, stat import, field mapping, and profile linking. If API access is not available during the project, students should deliver a documented integration design using mock data and clearly defined API assumptions.

6. Data and roster workflows: Support improved workflows for entering, importing, validating, and managing player/team data, potentially including CSV roster import, bulk card generation, and duplicate player matching.

7. Performance and reliability improvements: Identify and improve bottlenecks related to rendering cards, uploading images, saving projects, exporting images/PDFs, and loading templates.

8. Admin/product operations tools: Build or improve sponsor-facing tools for managing card templates, brand assets, sample cards, user content, product settings, and data

mappings.

9. Testing and quality assurance: Provide manual test cases and, where practical, automated tests for key workflows such as login, card creation, image upload, template selection, export, roster import, and API/data integration.

10. Deployment and handoff package: Provide a working demo or staging deployment, final code repository, setup guide, technical documentation, data model documentation, API/integration notes, testing notes, and final presentation/demo.**

Desired Background: Students should have interest or experience in full-stack web application development, software engineering, UI/UX design, API integrations, databases, cloud deployment, and product scaling. Helpful skills include JavaScript/TypeScript, React or a similar front-end framework, Node.js or Python, REST APIs, database design, Git/GitHub, testing, and cloud services.

Additional helpful experience includes image rendering, Canvas/SVG, PDF generation, file storage, authentication, admin dashboards, data import/export, caching, analytics, CI/CD, Figma, Firebase, Supabase, AWS, or similar tools. Interest in hockey, sports technology, digital collectibles, player profiles, scouting, or fan engagement would be beneficial but is not required.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement
- Non-disclosure agreement

Additional Materials: phoenix1.io Sponsor will provide access to the existing Phoenix1 Cards product, current product requirements, brand assets, sample card templates, anonymized player/team data, sample roster files, example card outputs, test accounts, and sponsor feedback. Sponsor may also provide access to source code, staging environments, product analytics, and third-party API documentation or credentials when appropriate and approved. Any Elite Prospects integration work will use approved/licensed API access or mock data if API access is not available during the project. Phoenix1 Cards / My Hockey Pro: https://phoenix1.io Elite Prospects: https://www.eliteprospects.com Elite Prospects API information: https://www.eliteprospects.com/api. “Students will help scale an existing hockey technology product and prepare it for professional-grade data integration with Elite Prospects.”

Nextiva - Nextiva Sales Operations, Analytics & AI Enablement

Proposer: Srimaan Edara srimaan.edara@nextiva.com

Project Contact: Srimaan Edara srimaan.edara@nextiva.com

Project Description: The Nextiva Sales team works with customers, partners, and internal teams to help grow the business. As the sales organization grows, several manual tasks are taking up more time, including answering common questions, creating sales reports, reviewing performance data, and preparing for Quarterly Business Reviews (QBRs).

In this project, students will identify opportunities to use AI, automation, and analytics to make these processes faster and easier. They may build tools that help answer common questions, create repeatable dashboards and reports, summarize sales data, identify important trends, and support QBR preparation.

The project will remain broad so students can adjust the scope based on the most valuable opportunities they discover. A stretch goal may include designing or prototyping an improved workflow for tracking and registering new sales opportunities.

Student Learning Goals: tudents will gain hands-on experience working with a real sales organization at a SaaS company. They will meet with members of the Nextiva Sales team to understand current challenges and turn business needs into practical technical solutions.

Students will have opportunities to work with:

Artificial intelligence and generative AI
Workflow automation
Data analysis and visualization
Sales and CRM data
Dashboard and report development
Process improvement and documentation

They will also practice interviewing stakeholders, gathering requirements, building prototypes, testing solutions, and presenting recommendations to a business audience. No previous sales experience is required.

Deliverables: An AI-powered or automated workflow that helps the Sales team respond to common questions or complete repetitive tasks.
A standardized analytics and reporting package that helps sales teams understand performance, pipeline, activity, and other key trends.
A QBR toolkit that may include templates, AI-generated summaries, dashboards, or automated data collection.
Documentation explaining how the tools work and recommendations for future improvements.

Stretch goal: A proposed or prototyped workflow for tracking and registering new sales opportunities

Desired Background: Students should have an interest in technology, business operations, artificial intelligence, data analytics, or process improvement.

Experience with tools such as Excel, SQL, Power BI, Tableau, Python, Salesforce, Power Automate, Zapier, or similar platforms is helpful but not required. Strong communication, problem-solving, and teamwork skills are also valuable.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://www.nextiva.com/

Banner Health, Banner University Medical Center Phoenix - Novel Application for Cardiac Arrest Prevention and Risk Statification

Proposer: Jordan Johnson jordan.johnson6@bannerhealth.com

Project Contact: Jordan Johnson jordan.johnson6@bannerhealth.com

Project Description: Every year in the United States, over 300,000 patients in our hospitals suffer from cardiac arrest! For 5-25% of these patients, their doctors were unable to properly adjust hospital care to prevent cardiac arrest because they didn't know their patients were at risk. We need better tools to measure, calculate, and stratify this risk.

The Banner University Medical Center Phoenix Cardiac Electrophysiology team is sponsoring a new EKG analysis application project that will finally shed light on which patients are at the highest risk. We hope to partner with a team of bright ASU computer science students to design an app that will allow emergency department and hospital physicians to quickly take a picture of a patient's EKG and return key risk measurements that go way beyond the antiquated EKG interpretation algorithms we still use today. This project will be a great mix of design patterns, image processing, composition over inheritance, application and user interface design, as well as HIPAA compliance and cybersecurity.

The first phase of this project will be to design a HIPAA-compliant application for hypothesis testing with the entire Banner Health patient population database. This first app will allow the cardiac electrophysiology and internal medicine teams to retrospectively digitize and analyze EKG waves of patients who have suffered from cardiac arrest or Torsades de Pointes. The student design team would work closely with cardiac electrophysiologists to construct the novel application. Medical residents and students in the Banner Health system will be able to harvest large amounts of data and EKGs for testing. There will be an IRB-approved study conducted with the beta application after application validation, and the student design team will be included on scientific publications based on the findings.

Student Learning Goals: As listed in the project description, this project will be a great mix of design patterns, image processing, composition over inheritance, application and user interface design, as well as HIPAA compliance and cybersecurity. Medical field applications and research have many hoops to jump through, but this project will provide valuable experience navigating those requirements. Students will have the opportunity to work with a multidisciplinary team of medical learners, physicians, engineers, and hospital system informatics professionals to design a useful tool. There will also be an opportunity to explore and dive into medical physiology with great physicians and teachers leading the team.

Deliverables: Beta application that accomplishes the following:
- Allows the user to capture an image of EKGs on paper or on a computer screen
- De-identifies the EKG
- Digitizes each EKG wave
- Identifies many wave attributes of the EKG, particularly focusing on the T-wave (wave start/stop point, peak amplitude, AUC, activation angle, etc.)
- Outputs EKG metrics to a HIPAA-compliant dataset for retrospective studies on cardiac arrest
- Based on initial retrospective studies with the beta app, output a calculated risk score for cardiac arrest

Desired Background: Python, imaging and signal processing, application design, user interface design

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

Pixel Pirate Studio - Off-Road Champion Mobile Game Backend Update

Proposer: Obi Onyejekwe obi.onyejekwe@gmail.com

Project Contact: Obi Onyejekwe obi@pixelpiratestudio.com

Project Description: Off-Road Champion – Backend & Security Intern Project
Project Goal

Students will improve Off-Road Champion's server features and security by building/maintaining backend APIs, hardening authentication and betting integrity, adding monitoring, and strengthening fraud prevention, while documenting everything for the core engineering team.

Intern Tracks

• Backend Features Intern (APIs, game services, payouts support)

• Security Intern (auth, abuse prevention, encryption, threat modeling)

• Backend QA Intern (tests, load testing, bug reproduction, verification)

• React Admin Tools Intern (internal dashboard to manage users, matches, payouts, flags)

Phase 1: Onboarding & System Map (Week 1)

Tasks

• Diagram current architecture: clients → API → DB → payments/crypto provider → logs

• Inventory all endpoints and sensitive flows (login, wallet, match results, payouts)

• Identify "trust boundaries" (what comes from device vs server)

Deliverables

• Backend architecture diagram

• Endpoint inventory + data classification (public vs sensitive)

• "Top 10 risks" list (initial threat model)

Phase 2: Backend Feature Updates (Weeks 2–4)

Core Features Students Can Own

• Match Service

• Create match, join match, lock bet, start match, submit results (server-validated), finalize

• Wallet / Ledger Service

• Transaction ledger entries for bets, fees, refunds, payouts (auditable, immutable-style records)

• Tournament / Leaderboard Service

• Server-side computation and caching, anti-spam rate limits

• Player Profile Service

• Device/session tracking, country/region flags, ban status, KYC-ready fields if needed later

Deliverables

• PRs for new/updated endpoints

• DB migrations (if needed) + rollback plan

• API docs (OpenAPI/Swagger or structured README)

Phase 3: Security Hardening (Weeks 3–6)

Security Requirements Students Implement

• Authentication

• Secure session handling (JWT rotation/short TTL), refresh tokens, device binding

• Role-based access control (RBAC) for admin actions

• Betting Integrity

• Server is source of truth for: bet amount, pot, fees, winner, payout approval

• Idempotency keys for payouts and critical updates (prevents double-charging/double-

pay)

• API Protection

• Rate limiting + bot detection basics

• Input validation + schema validation on every endpoint

• Audit logs for all sensitive actions (payouts, bans, match finalization)

• Data Protection

• TLS enforced, secrets management, encryption at rest where applicable

• PII minimization and access controls

• Abuse/Fraud Rules

• Flag suspicious patterns: repeated matchups, abnormal win streaks, VPN/device churn, rapid bet escalations

Deliverables

• Threat model doc + mitigations shipped

• Security checklist + “definition of done” for sensitive endpoints

• Admin-only audit log viewer (if you have a React admin panel)

Phase 4: Reliability, Testing, and Observability (Weeks 5–8)

• Tasks

• Automated Tests

• Unit tests for services

• Integration tests for match lifecycle + payout flows

• Load Testing

• Simulate concurrent matches and payouts (k6/JMeter/etc.)

• Monitoring

• Structured logs, request IDs, error dashboards

• Alerts for payout failures, spike in declines, abnormal latency

• Deliverables

• Test suite + coverage report

• Load test results and recommended capacity targets

• Monitoring/alert runbook (“If X happens, do Y”)

Final Student Deliverables (Short List)

• Backend architecture + endpoint documentation

• 2–5 shipped backend feature improvements (match/wallet/tournament/profile)

• Security upgrades (auth hardening, rate limiting, audit logs, idempotency)

• Automated tests + load test report

• Admin tooling updates (React dashboard) for moderation/auditing

• Demo Day presentation + portfolio-ready case study

Student Learning Goals: In this backend-focused Off-Road Champion internship, students gain hands-on experience working on a live competitive gaming server where real cash and crypto transactions are involved, learning how to build and secure production-level APIs, implement server-authoritative match logic, and protect sensitive betting flows from fraud and abuse. They deepen their understanding of authentication, rate limiting, data validation, encryption, audit logging, and threat modeling while contributing measurable improvements to backend performance and reliability. Through structured sprints, code reviews, testing, and load simulations, students experience how scalable systems are designed, monitored, and hardened in real-world environments, leaving the program with practical backend engineering skills, security awareness, and portfolio-ready

documentation that demonstrates impact on a revenue-generating platform.

Deliverables: 1. Backend Feature Updates

2–5 production-ready API improvements (match lifecycle, wallet, leaderboard, or player profile services)

Updated endpoint documentation (OpenAPI/README format)

Database updates or migrations (if required) with rollback plan

2. Security Enhancements

Threat model document outlining key risks and mitigations

Authentication improvements (JWT handling, session controls, RBAC)

Rate limiting and input validation implemented on critical endpoints

Idempotency protection for payouts and sensitive transactions

Audit logging for match finalization and payouts

3. Betting Integrity Improvements

Server-authoritative match result validation

Secure pot calculation and fee deduction logic

Fraud detection rules (flagging abnormal patterns)

Suspicious activity monitoring proposal

4. Testing & Reliability

Unit and integration test coverage for new features

Load test report simulating concurrent matches and payouts

Error handling improvements and structured logging updates

5. Monitoring & Admin Tools

Backend performance report (latency, uptime metrics)

Alerts or monitoring improvements for payout failures

React-based admin dashboard updates (user flags, match audits, transaction review tools)

6. Final Presentation

Demo of deployed improvements

Security checklist summary

Measurable KPI impact (latency reduction, reduced fraud risk, improved stability)

Portfolio-ready case study documentation

Desired Background: • Computer Science, Software Engineering, Cybersecurity, or related field

• Experience with React and basic backend/API development (Node.js, Next.js, or similar)

• Understanding of REST APIs, authentication (JWT), and database fundamentals

• A full understanding of React

• Familiarity with basic security principles (encryption, rate limiting, input validation)

• Interest in gaming systems, fintech, or secure transaction platforms

• Strong problem-solving skills and ability to work in agile sprint environments

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Pixel Pirate Studio - Off-Road Champion Mobile Game Update

Proposer: Obi Onyejekwe obi.onyejekwe@gmail.com

Project Contact: Obi Onyejekwe obi@pixelpiratestudio.com

Project Description: Project Title: Off-Road Champion Feature & Chimoney Integration Update
Platform: Unity (iOS & Android)

Backend: Cloud server with geolocation, leaderboard, and token economy

3rd Party Integration: Chimoney SDK

Interns will work hands-on, improving Off-Road Champion, a competitive racing game featuring real cash and crypto prize tournaments. The project focuses on upgrading UI/UX, optimizing graphics and performance, strengthening the betting system's security, and improving download speed and overall gameplay stability. Students will collaborate in weekly sprints, ship measurable improvements to a live product, and gain real-world experience in game development, product optimization, and secure digital systems.

PHASE 1: UI/UX Updates (Weeks 1–3)
1. Lobby & Race Flow Redesign
2. UX Improvements
3. Micro-Interactions

PHASE 2: Testing & Gameplay QA (Weeks 2–5)
1. Gameplay Fairness Testing
2. Betting System Testing
3. Stress Testing

PHASE 3: Graphics Optimization (Weeks 3–6)
1. Asset Optimization
2. Lighting Optimization
3. FPS Testing

PHASE 4: Download Speed Improvements (Weeks 4–6)
1. App Size Reduction
2. First Launch Optimization

PHASE 5: Betting System Security Improvements (Weeks 5–8)
1. Anti-Cheat Improvements
2. Encryption
3. Identity & Account Security
4. Risk Monitoring

Student Learning Goals: The Off-Road Champion internship is designed to give students a real-world, startup-level product development experience where they work hands-on across UI/UX design, gameplay testing, graphics optimization, and betting system security. Interns don't just observe, they ship measurable improvements inside a live game that includes real cash and crypto competition, learning how performance optimization, anti-cheat systems, download efficiency, and user psychology directly impact retention and revenue. They collaborate in weekly sprints, present progress like a professional studio team, analyze performance metrics, and understand how design decisions affect fairness, security, and global scalability. By the end of the program, students walk away with portfolio-ready deliverables, documented impact metrics, cross-functional teamwork experience, and a deep understanding of how to build and scale a competitive, skill-based gaming product in today's market.

Deliverables: 1. UI/UX Deliverables

• Updated Figma design files (organized and developer-ready)

• Improved race flow wireframes (Open App → Enter Race)

• Redesigned wallet and betting screens

• Updated design system (buttons, typography, states, modals)

• Interactive prototype walkthrough

• Before/After UX comparison presentation

• Usability test summary (5–10 test participants minimum)

2. Testing & QA Deliverables

• Structured bug report log (severity, reproduction steps, screenshots)

• Gameplay balance analysis report

• Betting system validation checklist (pot calculation, fees, payouts)

• Stress test documentation (latency, disconnect handling)

• Exploit discovery and prevention recommendations

• Weekly QA summary reports

3. Graphics & Performance Deliverables

• Optimized asset package (reduced poly counts & texture sizes)

• Performance benchmark comparison (Before vs After FPS)

• Mobile device test matrix (low, mid, high-tier devices)

• Lighting & shader optimization report

• Final optimized build ready for staging

• Documentation of compression techniques used

4. Download & Load Speed Deliverables

• App size reduction analysis report

• Asset audit (unused files removed)

• First-launch time improvement metrics

• CDN or asset bundle restructuring proposal

• Install completion rate improvement analysis

5. Betting System Security Deliverables

• Threat model documentation

• Anti-cheat improvement recommendations

• Server-side validation documentation

• Encrypted data flow diagram

• Fraud detection logic proposal

• Risk assessment matrix

6. Data & Analytics Deliverables

• Retention analysis (Day 1, Day 7 trends if available)

• Player behavior breakdown

• Funnel drop-off analysis

• Revenue impact projection from improvements

• Dashboard mockup for tracking performance metrics

7. Professional Development Deliverables

• Weekly sprint presentations

• Demo Day final presentation

• Portfolio-ready case study

• GitHub or documentation contribution

• Written reflection on learning outcomes

Desired Background: 1. Game Development / Programming

• Computer Science, Game Development, Software Engineering

• Experience with Unity

• Basic knowledge of C#, C++, or JavaScript

• Understanding of multiplayer systems or server-client logic (preferred)

2. UI/UX Design

• Graphic Design, Product Design, Interaction Design

• Experience with Figma or Adobe XD

• Understanding of mobile app flows and usability principles

• Interest in gaming interfaces and user psychology

3. QA / Data Analytics

• Computer Science, Information Systems, Data Science

• Experience with testing frameworks or structured bug reporting

• Ability to analyze player behavior and retention metrics

• Strong attention to detail

4. 3D Art / Technical Art

• 3D Animation, Game Art, Digital Media

• Experience with Blender, Maya, or similar tools

• Understanding of polygon optimization and texture compression

• Knowledge of lighting and performance optimization for mobile

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Kids At Risk Action - Offline / low-bandwidth resource access

Proposer: Mike E. Tikkanen mike@invisiblechildren.org

Project Contact: Mike E. Tikkanen mike@invisiblechildren.org

Project Description:
Create an offline-capable "digital resource kit" (web or mobile app plus local cache) so front-line workers and families can access KARA content where connectivity is poor, similar to EPICS digital library projects.

Engineering work: local storage, sync logic, compression, and a simple, robust UI.

Student Learning Goals: Students on the offline / low-bandwidth resource access project would learn how to design and build technology that works under real-world constraints, not just ideal campus conditions. They would gain experience analyzing user needs for front-line workers and families with spotty or expensive connectivity, then translating those needs into technical requirements (offline storage, sync logic, simple UI). Learners would practice implementing and testing local caching, data compression, and synchronization strategies, and they would confront tradeoffs among performance, security, and usability. Throughout, they would also develop skills in human-centered design and ethical, trauma-informed thinking—iterating with KARA on content structure, safety warnings, and interface choices to be sure the tool is not only functional but also respectful and accessible to stressed, time-pressed users.

Deliverables: Working prototype application (web, Android, or cross-platform) that lets users browse a curated subset of KARA resources offline, then syncs updates when a connection is available.

Local data package and sync logic: a defined content bundle (e.g., JSON/SQLite) with documented structure, plus code that handles download, storage, conflict handling, and basic encryption where appropriate.

User interface and UX documentation: wireframes or screenshots, navigation map, and a brief rationale explaining design choices for low-bandwidth, trauma-informed, and accessibility needs.

Technical documentation: setup instructions, architecture diagram, and notes on dependencies so KARA or future student teams can maintain and extend the system.

Short evaluation summary: results from limited user or stakeholder testing (even small-N), capturing performance in poor-connectivity scenarios and key feedback for the next iteration.

Desired Background: For the offline / low-bandwidth access project, the ideal student background would include:

Software / systems skills

Coursework in computer science, software engineering, or related fields.

Experience with at least one of: web app development (HTML/CSS/JS, a front-end framework), Android/mobile development, or cross-platform frameworks (e.g., React Native, Flutter).

Familiarity with local storage (IndexedDB, SQLite, device file systems) and basic client–server or API concepts.

Data and architecture basics

Comfort working with structured data formats (JSON, CSV, simple databases).

Intro experience designing simple data models and understanding sync/versioning tradeoffs.

Human-centered / social impact interest

Genuine interest in child welfare, public health, or social-impact tech, with sensitivity to trauma-related content.

Some exposure to user-centered design or usability testing is a plus.

A strong team could pair at least one more back-end-oriented student (sync, storage, security) with one more front-end / UX-oriented student (interface, accessibility, user flows).

Delivery Mode: In-person

Additional Materials: https://invisiblechildren.org/ , https://youtu.be/1_dtQLZwL28

GreyBox Project - Offline Connectivity and Content Delivery for Education and Health Support on Android Devices (Project UNI)

Proposer: Valentin Kravtchenko valentin.kravtchenko@grey-box.ca

Project Contact: Zahleija Fontes-Janusz zahleija.fontes-janusz@grey-box.ca

Project Description: UNI is an open-source, low-cost device and Android application that creates a local Wi Fi learning hub for students and teachers, even in places without reliable or safe internet access. With a simple upload flow, schools, NGOs, or instructors can make slides, PDFs, videos, databases, and even copies of entire websites available offline for free to any learner connected to UNI.

The platform is designed as an offline first "local internet" for education: users connect to the UNI Wi Fi network from standard Android devices and access curated learning content. Ongoing work is extending UNI with deeper integrations for course platforms (e.g., Moodle), collaborative file tools (e.g., Nextcloud), and AI powered offline search so future deployments can support richer classroom workflows without external connectivity.

Maturity level:
Level 7 – Prototype ready for demonstration in an appropriate operational environment (on a 1–9 technology readiness scale). UNI is our most mature project, with pilots in 12 countries and field demonstrations already conducted.

Social impact:
UNI directly supports Sustainable Development Goals 4 (Quality Education), 9 (Industry, Innovation and Infrastructure), and 17 (Partnerships for the Goals).

Student Learning Goals:

- Build and ship offline-first features used by real users (education + field settings), not toy demos.
- Work across a "full stack": Android host environment + backend services + web UI, with real constraints (devices, connectivity, power, UX).
- Practice applied security + privacy-by-design for local networks and shared devices.
- Learn instrumentation/analytics for offline usage + performance, turning logs into impact signals.
- Operate in an agile delivery loop: scoped epic → PRs → testing → demoable release notes.

Deliverables: IT Needs:
We need to design and implement the following capabilities (students may prioritize and select from these; the project uses Android as the host platform, with PHP and Java for the

backend and HTML, JavaScript, and CSS for the frontend):

- Add user management functions (role-based access levels and privileges).
- Improve the analytics dashboard (usage, impact analysis, and reporting functions).
- Add ethernet and 4G connectivity while hotspot and captive portal are activated.
- Improve Moodle integration to allow multiple users to connect concurrently.
- Improve Moodle performance and load times on UNI, and explore safe upgrade paths to newer Moodle versions with stability testing alongside other UNI features.
- Add multilingual support for both the web interface and the Android application.
- Expand support for unrooted Android mobile devices, including compatibility validation for Android 11–16.
- Improve the web interface's UX/UI for low-connectivity and offline-first usage.
- Improve Nextcloud integration to support more file types, more stable offline live collaboration, and better runtime performance on UNI hardware.
- Enhance admin and classroom workflows in Nextcloud (tighten admin-only permissions, improve logging, and complete stress/deployment testing on UNI devices).
- Improve the AI Kiwix Wikipedia search experience (optimize the embedding pipeline, evaluate more Java-friendly or higher-quality models, and reduce the need to scan every entry).
- Expand AI search to support more concurrent users and potentially additional ZIM-based content beyond Wikipedia.
- Refactor the UNI codebase to improve scalability, maintainability, and system design while preserving existing functionality.

Desired Background:

- Solid fundamentals in CS/CSE software engineering (clean code, debugging, working from existing codebases).
- Comfortable with web basics (HTML/CSS/JavaScript) and backend concepts (APIs, authentication, data storage).
- Familiarity with Android and/or Java or willingness to ramp up quickly (capstone teams can split roles).
- Working knowledge of Git (branches, pull requests, reviews) and writing usable technical documentation.
- Bonus: basic networking concepts (Wi-Fi, captive portals, local services) and security fundamentals.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

GreyBox Project - Offline Connectivity and Content Delivery for Education and Health Support on Android Devices (Project UNI)

Proposer: Valentin Kravtchenko valentin.kravtchenko@grey-box.ca

Project Contact: Zahleija Fontes-Janusz zahleija.fontes-janusz@grey-box.ca

Project Description: UNI is an open-source, low-cost device and Android application that creates a local Wi Fi learning hub for students and teachers, even in places without reliable or safe internet access. With a simple upload flow, schools, NGOs, or instructors can make slides, PDFs, videos, databases, and even copies of entire websites available offline for free to any learner connected to UNI.

The platform is designed as an offline first "local internet" for education: users connect to the UNI Wi Fi network from standard Android devices and access curated learning content. Ongoing work is extending UNI with deeper integrations for course platforms (e.g., Moodle), collaborative file tools (e.g., Nextcloud), and AI powered offline search so future deployments can support richer classroom workflows without external connectivity.

Maturity level:
Level 7 – Prototype ready for demonstration in an appropriate operational environment (on a 1–9 technology readiness scale). UNI is our most mature project, with pilots in 12 countries and field demonstrations already conducted.

Social impact:
UNI directly supports Sustainable Development Goals 4 (Quality Education), 9 (Industry, Innovation and Infrastructure), and 17 (Partnerships for the Goals).

Student Learning Goals:

- Build and ship offline-first features used by real users (education + field settings), not toy demos.
- Work across a "full stack": Android host environment + backend services + web UI, with real constraints (devices, connectivity, power, UX).
- Practice applied security + privacy-by-design for local networks and shared devices.
- Learn instrumentation/analytics for offline usage + performance, turning logs into impact signals.
- Operate in an agile delivery loop: scoped epic → PRs → testing → demoable release notes.

Deliverables: IT Needs:
We need to design and implement the following capabilities (students may prioritize and

select from these; the project uses Android as the host platform, with PHP and Java for the backend and HTML, JavaScript, and CSS for the frontend):

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- Improve the AI Kiwix Wikipedia search experience (optimize the embedding pipeline, evaluate more Java-friendly or higher-quality models, and reduce the need to scan every entry).
- Expand AI search to support more concurrent users and potentially additional ZIM-based content beyond Wikipedia.
- Refactor the UNI codebase to improve scalability, maintainability, and system design while preserving existing functionality.

Desired Background: • Solid fundamentals in CS/CSE software engineering (clean code, debugging,
working from existing codebases).

- Comfortable with web basics (HTML/CSS/JavaScript) and backend concepts (APIs, authentication, data storage).
- Familiarity with Android and/or Java or willingness to ramp up quickly (capstone teams can split roles).
- Working knowledge of Git (branches, pull requests, reviews) and writing usable technical documentation.
- Bonus: basic networking concepts (Wi-Fi, captive portals, local services) and security fundamentals.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: N/A

Pixel Pirate Studio - OpenMap Software Update

Proposer: Obi Onyejekwe obi.onyejekwe@gmail.com

Project Contact: Obi Onyejekwe obi@pixelpiratestudio.com

Project Description: Project Overview

Interns will work directly on OpenMap, a data-driven mapping platform, helping improve performance, expand datasets, and build a responsive mobile version of the application. Students will play a hands-on role in testing the live platform, optimizing speed and scalability, integrating AI-powered features, and implementing automated data collection workflows.

Key Responsibilities

• Test and debug the OpenMap web platform

• Optimize front-end and backend performance

• Add and validate new public and structured datasets

• Implement AI features and AI bots for data analysis and user interaction

• Build and refine a responsive mobile version of the app

• Develop and manage data scraping pipelines

Student Learning Goals: Interns will gain real-world experience building and scaling a live SaaS data platform. They will work across the full stack, front-end React development, backend architecture, AI integration, and data engineering, while learning performance optimization techniques and production-level deployment workflows. Students will leave with practical experience in AI implementation, automation, and building scalable, map-based web applications.

Deliverables: * Performance optimization report and implemented improvements
* Working mobile-responsive version of OpenMap
* AI-powered feature or bot integration
* New validated datasets integrated into the platform
* Documented data scraping workflow

Desired Background: * Strong knowledge of React
* Backend development experience (Node.js, Python, or similar)
* Experience implementing AI models or AI bots
* Knowledge of data scraping and API integrations

* Familiarity with database systems and cloud deployment
* Interest in data platforms, mapping tools, or civic tech

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

Appy.yo - OpenSheet: AI-Powered Software Tool for Spreadsheet-Based Workflows

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: Many small businesses rely on spreadsheets to manage daily operations, but spreadsheets can become difficult to maintain, scale, and turn into useful workflows. OpenSheet is intended to help users transform spreadsheet-based work into a more organized software experience. The project needs software development support to create a functioning computer-based tool that can organize information, support workflows, and potentially use AI to help users understand and manage spreadsheet data. The goal is to make spreadsheet-heavy work easier, smarter, and more scalable.

Student Learning Goals: Software development, front-end development, back-end development, UI/UX design, spreadsheet logic, AI implementation, database design, workflow automation, Python, JavaScript, React, API integrations.

Deliverables: A fully functioning software solution or MVP. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.

Desired Background: Software development, front-end development, back-end development, UI/UX design, spreadsheet logic, AI implementation, database design, workflow automation, Python, JavaScript, React, API integrations.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Parents Challenge - Parent App 2.0

Proposer: Deborah Hendrix dhendrix@parentschallenge.org

Project Contact: Deborah Hendrix dhendrix@parentschallenge.org

Project Description: Parents Challenge Parent App 2.0 builds upon the successful foundation established during the 2025–2026 school year. While Version 1.0 introduced families to a centralized platform for program communication and resources, Version 2.0 will significantly expand the app's functionality to create a more seamless and efficient experience for both families and staff.

Student Learning Goals: This project provides Arizona State University students with a unique, real-world opportunity to apply classroom knowledge to a meaningful community initiative. Working alongside Parents Challenge, students will engage in the full software development lifecycle, including user research, application design, programming, testing, integration, and implementation. Through this experiential learning opportunity, students strengthen their technical, project management, communication, and problem-solving skills while producing a technology solution that advances educational access and parent engagement in the community.

Deliverables: By Fall 2027, Parents Challenge will launch Parent App 2.0, providing families with a comprehensive, user-friendly platform that simplifies their participation in the program. The enhanced app will enable parents to:

Access scholarship and funding information in one centralized location.
Complete required forms and submit documentation electronically.
Register for parent sessions and events with ease.
Receive real-time notifications, reminders, and program updates.
View program requirements, important deadlines, and educational resources.
Experience a streamlined, mobile-friendly platform that reduces paperwork and improves communication.

This enhanced functionality will improve operational efficiency, increase data accuracy, and allow Parents Challenge to better serve a growing number of families while maintaining a lean administrative structure.

Desired Background: We are seeking students with backgrounds in computer science, software development, coding, and user interface/user experience (UI/UX) design to develop and enhance the Parents Challenge Parent App. In addition, students with project management experience are needed to coordinate project activities, manage timelines, facilitate communication, and ensure project milestones and deliverables are successfully achieved. Together, this multidisciplinary team will create a high-quality, user-centered application that improves the experience for Parents Challenge families and staff.

Delivery Mode: In-person

Additional Materials: Will be provided as part of orientation - will include documentation developed by last year's team.

Appy.yo - Physio App: AI-Guided Exercise and Injury Management Platform

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: The Physio App needs UI/UX support and AI functionality, especially around animated exercise guidance. Many people are given stretching or strengthening exercises but struggle to remember how to do them correctly, stay consistent, or understand proper movement. Physiotherapists also need better ways to assign routines, explain exercises clearly, and track patient progress. This app would allow users to receive guided routines, view animated exercise demonstrations, and follow personalized injury management plans. The current focus is to improve the user experience and explore AI-generated or AI-supported exercise animations.

Student Learning Goals: UI/UX design, mobile app development, AI implementation, animation, exercise visualization, front-end development, back-end development, health app design, database design, JavaScript, React Native, Python.

Deliverables: A fully functioning app or MVP with improved UI/UX and AI-supported exercise animation. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.Students should help improve the user experience, design clear exercise flows, support AI or animation features, and test the experience from routine creation to exercise completion. Students should attend bi-weekly check-ins and contribute approximately 5–10 hours per week.

Desired Background: UI/UX Designer 1
Mobile or Front-end Developer 1–2
Back-end Developer 1
AI/Animation Developer 1
Health Research Support 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Expeditise LLC - Pinluma.com: Machine Learning Interactive Art Experiences and Collaborative Art Platform

Proposer: Ali Sarabi asarabi1@asu.edu

Project Description: Pinluma is an early-stage startup in the ASU Venture Devil program building a collaborative art platform where an audience creates artwork together from their own phones. A shared canvas is displayed on a screen, projection, or web page; anyone can join instantly from a phone browser with no app to install, and every participant contributes to a single evolving piece. The current website already hosts three early works. These pieces prove the core interaction model, but they rely on simple touch input and hand-written visual rules.

This capstone project has two connected phases. In the first phase, the team will design and build a new generation of interactive works driven by machine learning and phone sensors. This includes real-time image processing of the phone camera feed converted into art, using techniques such as style transfer, person and object segmentation, edge and depth extraction, hand and body pose tracking, and live palette extraction from the scene in front of the user. It also includes using the gyroscope, accelerometer, and device orientation so that tilting, aiming, shaking, and steering the phone moves, deforms, and paints objects on the shared canvas. Models should run on-device in the browser wherever possible, so the experience stays real-time, private, and inexpensive to host, with server-side inference used only where a model is too heavy for a phone.

In the second phase, the team will build the backend that turns individual works into a product. Today every piece is custom-built and custom-hosted, and each new artwork has to reinvent the same plumbing. The goal is a platform that runs the artwork on the artist's behalf, so an artist can bring a piece, set it up, and publish it to a live session while Pinluma handles everything around it. This includes hosting and running the piece, accounts and roles, session and room management with QR-code joining, real-time synchronization between phones, configurable parameters and presets, media and session storage, replay and export of a finished piece, personal and public galleries, share links, embeddable widgets, moderation controls for public events, and participation analytics. The artist keeps full creative and technical control of the artwork itself; what gets simplified is publishing it, running it in front of an audience, and letting participants keep and share what they created.

Students are strongly encouraged to use agentic AI throughout their own development work: coding agents to build and refactor features, AI-assisted debugging and code review, generated art assets and reference material, and automated test writing. Strong pieces produced during the semester will be published on pinluma.com and credited to the student team, giving students a public, demonstrable portfolio of interactive work.

Student Learning Goals: Students will gain hands-on experience building real-time, ML-driven interactive experiences that run on ordinary phones and are used by real audiences, combining applied machine learning, creative coding, mobile web engineering, and product backend development in a startup setting.

• Running machine learning models in real time inside mobile browsers using TensorFlow.js, MediaPipe, ONNX Runtime Web, and WebGPU/WebGL, under strict latency, memory, and battery constraints
• Applying real-time computer vision to a live camera feed: segmentation, pose and hand tracking, depth and edge extraction, style transfer, and colour analysis
• Turning phone sensors (gyroscope, accelerometer, device orientation, touch, microphone) into expressive artistic input
• Creative coding and generative graphics: shaders, particle systems, physics, and high-frame-rate Canvas/WebGL rendering
• Designing multi-user real-time systems with WebSockets or WebRTC, including session management, state synchronization, latency handling, and scaling to many simultaneous phones
• Building product backends: authentication, database schema design, media storage, sharing and embedding, moderation, and an artist-facing dashboard
• Applying agentic AI and LLM APIs to creative tooling, including prompt-to-parameter generation, variation, and assisted composition
• Working in an agile startup workflow with weekly demos, live playtests with real participants, and iterative feedback

Deliverables: 1. Concept Design and Technical Plan
• Review the existing Pinluma works, document the current interaction model and its limitations, and design three to four new ML and sensor-driven pieces with sketches and interaction storyboards
• Benchmark candidate on-device models on real phones (frame rate, model size, battery impact, iOS vs. Android) and select the technical stack

2. Real-Time Image Processing Artworks
• Live phone camera capture in the browser, with permission handling, orientation, resolution and frame-rate control, and graceful fallbacks
• Real-time models applied to the live feed (style transfer, segmentation, edge and depth maps, hand and body landmarks, dominant-colour extraction), running on-device wherever possible
• Conversion of model output into art: generative brush strokes, particles, evolving geometry, or shader effects driven by what the camera sees

3. Sensor Interaction and Shared Canvas
• Gyroscope, accelerometer, and device-orientation input (including the iOS permission flow), with smoothing, calibration, and gestures such as tilt, aim, flick, and shake

• A single reusable input layer combining camera, sensor, touch, and audio for use across every piece
• Real-time synchronization between many phones and the shared display, with QR-code joining, reconnection, and load testing at dozens of simultaneous participants

4. Artist Platform and Backend (Phase 2)
• Accounts, roles, sessions, and permissions, plus a hosted runtime and documented piece API so an artist can register a piece and have the platform serve, run, and publish it
• An artist dashboard to configure a piece and its presets, preview it, open or schedule sessions, and review past ones
• Media and session storage, recording and replay, export of a finished piece, and moderation controls for public events

5. Sharing, Galleries, and Agentic AI Tooling
• Personal and public galleries, share links, social preview cards, embeddable widgets, and participation analytics for artists
• An agent that turns an artist's description of a piece into a working parameter set, palette, and behaviour configuration, with variation tools and cost guardrails

6. Testing, Deployment, and Documentation
• Cross-device testing on a range of iOS and Android phones and browsers, measuring frame rate, input latency, and battery usage
• Deployment pipeline, environment configuration, and monitoring for the real-time services
• Documentation of the architecture, input layer, model choices, API and database schema, artist workflow, and a guide for building a new piece

Desired Background: Students should have experience or interest in the following areas:

• Frontend and creative coding with JavaScript/TypeScript and React or Next.js, plus Canvas or WebGL work with three.js, p5.js, PixiJS, or custom shaders
• Machine learning and computer vision, especially running models in the browser with TensorFlow.js, MediaPipe, ONNX Runtime Web, or WebGPU
• Mobile web development, including device sensor APIs, permission flows, and performance tuning on real phones
• Backend development with Node.js or Python, real-time services using WebSockets or WebRTC, database design, and cloud media storage
• Knowledge of agentic AI, LLM APIs, or generative models is highly recommended
• A strong creative and visual sense; interest in generative art, motion design, games, music visualization, or interactive installations is a major plus
• Interest in startups and in shipping public-facing work that real audiences use

Students who bring their own artistic ideas are encouraged to propose and build them. Deep prior machine learning experience is not required; curiosity, creativity, and

willingness to learn matter more.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: https://pinluma.com

Expeditise LLC - Pinluma: Collaborative Art Platform, Hosted Runtime, and Artist Publishing Backend

Proposer: Ali Sarabi asarabi1@asu.edu

Project Description: Pinluma is an early-stage startup in the ASU Venture Devil program building a collaborative art platform where an audience creates a single artwork together from their own phones. A shared canvas is displayed on a screen, projection, or web page, anyone joins instantly from a phone browser with no app to install, and every participant contributes to the same evolving piece. Three works are already live at pinluma.com: Luminae, Sillage, and Chroma.
Every one of those pieces is custom-built and custom-hosted, and each new artwork reinvents the same plumbing: sessions, joining, synchronization between phones, storage, and sharing. This capstone team will build the platform underneath: a hosted runtime that runs an artist's piece on the artist's behalf, so publishing a collaborative artwork becomes a normal product workflow instead of a bespoke engineering project. The artist keeps full creative and technical control of the artwork itself; the platform takes over everything around it.

The centre of the project is the piece API: a documented, versioned interface describing how an artwork registers with the platform, receives its configuration and its stream of participant input, and reports state back. Everything else follows from that contract. The platform serves and runs registered pieces, manages sessions and rooms with QR-code joining, synchronizes state in real time between many phones and the shared display, stores media, records and replays sessions, exports finished artworks, and presents them through galleries, share links, and embeds. Migrating the three existing works onto the platform is the proof that the API is good enough.

The hard engineering problem is real time at scale under bad conditions. A live session may involve dozens or hundreds of phones of wildly varying age and speed, on venue wifi or cellular, with people joining late, reconnecting, and leaving throughout. The collective artwork has to stay coherent and visually stable regardless. Live events also raise the operational stakes: a session that dies during a show cannot be retried, so monitoring, graceful degradation, moderation controls, and an emergency stop are engineering requirements rather than extras.

If the companion Pinluma interactive-artwork project runs in the same semester, that team is the first customer of this platform and the piece API is the interface between the two, which makes API design and documentation part of the real work rather than an exercise. Students are also strongly encouraged to use agentic AI in their own development workflow, including coding agents, AI-assisted review, and generated assets, while remaining accountable for everything they ship.

Student Learning Goals: Students will gain hands-on experience building a real-time, multi-tenant platform that other developers build on top of and that real audiences use live, combining distributed systems, API design, media engineering, and product backend work in a startup setting.

• Designing real-time multi-user systems with WebSockets or WebRTC, including presence, state synchronization, late join, reconnection, and latency compensation
• Scaling and load-testing a live service to hundreds of concurrent clients on unreliable networks, with graceful degradation rather than failure
• Designing and versioning an API and SDK that third-party developers build against, and documenting it well enough that someone else can ship with it
• Safely running third-party code: sandboxing, resource limits, and isolation so one artwork cannot destabilize the platform
• Multi-tenant platform design with accounts, roles, workspaces, and permissions
• Media engineering: storage, session recording and replay, and rendering a finished artwork to image or video
• Trust and safety engineering, including moderation tooling, reporting, and controls for public live events
• Observability and operations for a service that fails in public: monitoring, alerting, runbooks, and incident response
• Using agentic AI as a working development tool while staying fully accountable for the code that ships

Deliverables: 1. Platform Foundation and Piece API
• A documented, versioned piece API and SDK defining how an artwork registers, receives configuration and participant input, and reports state, with a reference implementation an artist can copy
• A hosted runtime that serves and runs a registered piece, with sandboxing and resource limits so a misbehaving artwork cannot take down a session or the platform
• Accounts, roles (artist, participant, administrator), per-artist workspaces, and permissions

2. Sessions, Real-Time Synchronization, and Scale
• Session and room lifecycle: create, open, join by code or QR, spectator and display views, close, and archive
• Real-time state synchronization between many phones and the shared display, handling late join, reconnection, dropped clients, and uneven device performance
• Load testing to hundreds of concurrent participants, with documented limits, graceful degradation, and resilience to weak venue networks

3. Artist Dashboard and Publishing
• Tools to register or upload a piece, configure its parameters and presets such as palette, physics, audio, and duration, preview it privately, and publish it
• Session management for artists: schedule or open a session, monitor it while it runs, and

review past ones
• Migration of the three existing works (Luminae, Sillage, Chroma) onto the platform as proof that the piece API is sufficient for real artworks

4. Media, Galleries, and Sharing
• Session recording and replay, plus export of a finished artwork as an image or video
• Personal and public galleries where participants find what they made, see collaborators credited, and browse past sessions
• Share links, social preview cards, embeddable widgets for artist websites, and a public works page driven by the backend instead of hard-coded pages

5. Moderation, Safety, and Live-Event Operations
• Moderation controls for public sessions: reporting, participant blocking, content filtering, session locks, and an emergency stop
• Operational tooling for live events, including a session health view, alerting, and a documented runbook for running a show
• Participation analytics per session and per piece, covering joins, duration, contributions, and device breakdown

6. Testing, Deployment, and Documentation
• Automated tests including multi-client integration tests that simulate many simultaneous participants
• Deployment, secrets management, media storage, and monitoring for the real-time production services
• Documentation of the architecture, piece API reference, artist guide, and live-event runbook

Desired Background: Students should have experience or interest in the following areas:
• Backend development with Node.js, Python, or Go, including REST APIs, databases, and cloud storage
• Real-time services using WebSockets or WebRTC, and interest in distributed systems problems such as state synchronization, presence, and latency
• Frontend development with React or Next.js for the artist dashboard and public gallery pages
• API and SDK design, and the discipline to document an interface other developers depend on
• Media processing and video export with tools such as FFmpeg is a plus
• Deployment, monitoring, and operations for services that need to stay up during a live event
• Knowledge of agentic AI, LLM APIs, or generative models is highly recommended
• Interest in startups, creative technology, live events, and platforms that other people build on

This project pairs with the Pinluma interactive-artwork project. Teams that want to build the artworks themselves should look at that proposal; this one is for students who would rather build the platform that carries them.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: https://pinluma.com

MentaMorph Inc - Platform, Identity & Payment/Licensing System (AWS Backbone)

Proposer: Susan Meredith susan@mentamorph.com

Project Contact: Susan Meredith susan@mentamorph.com

Project Description: This project develops the core platform infrastructure for MentaMorph, including user identity, organizational access, and integrated payment and licensing systems. The system must support certified instructors and organizations in purchasing access through bulk licenses (e.g., blocks of 10, 100, or 1,000 users) or affiliate-style enrollment models. It will include domain-based access (schools, companies), role-based permissions (admin, instructor, student), and flexible onboarding methods such as email invitations and privacy-preserving synthetic IDs. The project may also explore integration with platforms such as GoHighLevel for payment flows and customer management.

Student Learning Goals: Students will design a real-world SaaS platform that integrates authentication, authorization, and monetization. They will learn how business models (licensing, affiliate systems) directly shape technical architecture, while developing skills in cloud infrastructure, APIs, and secure system design.

Deliverables: • AWS-based authentication and user management system
- Role-based access control (admin, instructor, student)
- Payment integration and workflows
- Licensing system (block, seat, or hybrid model)
- Admin/instructor dashboard for managing users and licenses
- Architecture and business logic documentation

Desired Background: • Backend development (Node.js, Python, or similar)
- Familiarity with AWS and APIs
- Understanding of authentication systems
- Bonus: payment systems or SaaS experience

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

ASU/NASA Psyche Mission - Predicting Solar Wind Conditions with AI

Proposer: Cassie Bowman cbowman5@asu.edu

Project Contact: Cassie Bowman cbowman5@asu.edu

Project Description: This capstone project will explore how solar wind prediction tools could be enhanced with the help of AI methods. Solar wind prediction is at the heart of many technological applications, from ground based electrical system to satellites and deep space missions. NASA and the US government have been investing considerable resources into computational models that take in solar images and predict the plasma and electromagnetic disturbances that can be emitted by the Sun, affecting planetary magnetic fields, planetary atmospheres and surfaces, spacecraft, and astronauts. While these models have been in use and constant development in the past few decades, many aspects of space missions require considerably higher resolution and faster models. The NASA Psyche mission, exploring the asteroid 16 Psyche, is an example of such a mission, where its goal of detecting the asteroid's magnetic field may require knowledge of the surrounding solar wind magnetic field. However, the size of the asteroid and its fast rotation (4.2 hours) means that current state of the art solar wind models cannot provide the needed time and spatial resolution. This project will explore the intersection between current state of the art numerical modeling techniques and how they may be enhanced and refined using machine learning and artificial intelligence. It would involve using existing models and real spacecraft data to design and train new models, and compare their performance to test data. This exciting possibility may prove useful to the NASA Psyche mission as well as many more space weather applications, and would position learners at the cutting edge of current predictive physical models.

Student Learning Goals: Learners would be introduced to the natural phenomena related to the Sun, solar wind, and eruptions. Emphasis would be given to impacts on planets, atmospheres, and technological systems. Learners would be introduced to the types of spacecraft data measuring plasmas and electromagnetic disturbances in space. Learners will gain hands-on experience with existing predictive models, and learn about the underlying principles, capabilities, and limitations. Learners would be guided in training AI models on existing data, both from models and from space measurements. Teams could explore and chose between different methods.

This is an exciting opportunity to test your problem solving and creativity! You will become part of a larger community of students working on a range of projects with Psyche and have the opportunity to meet members of the Psyche mission team. We look forward to having you become part of the Psyche team through this unique capstone project.

Deliverables: Teams would produce code and documentation (e.g., Jupyter notebooks). Teams would create a write-up of results and their analysis of their model's performance. Teams would create a short presentation summarizing their results.

Desired Background: Curiosity and willingness to learn new topics and tackle unsolved problems.
Familiarity with python, MATLAB, or similar coding environments (easy plotting, interactive memory management).
Understanding of the general principles of neural networks and other machine learning methods is preferred.
Background in electromagnetism is a plus, but students would be introduced to the needed physics principles.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: https://psyche.ssl.berkeley.edu/get-involved/capstone-projects/

PreSwing Golf, LLC - PreSwing Coach: Connecting Teaching Professionals to Their Students

Proposer: Scott Blanford scott@preswinggolf.com

Project Contact: Scott Blanford scott@preswinggolf.com

Project Description: PreSwing is a voice first golf coaching platform launching in 2026. An iOS app and Apple Watch companion deliver short audio cues in the seconds before a shot, organized into thirty skills such as Driver, Wedges, Bunker Play, and Putting, alongside GPS distances across more than 45,000 courses and shot tracking. A clip on hardware device called Genius follows, giving golfers the same voice guidance without a phone or watch. The company is an early stage startup in Scottsdale.

PreSwing today serves the golfer alone. This project builds the missing half of the relationship: a connection between a teaching professional and the students they coach. A pro should be able to shape what a specific student hears and practices, and a student should leave every lesson with concrete work to do before the next one.

The team will not be handed a specification. The first phase is discovery. The team interviews teaching professionals and their students to determine what this feature should be, and validates or rejects the hypotheses below before writing production code. A well argued case for killing one of these is a successful outcome.

Hypotheses to test:

Per student tip customization, so a pro can adjust the audio cues a specific student hears inside each skill.

Video and personal audio, so a pro can record into a skill and add their voice to the student's My Coach section. This matters because a growing share of instruction now happens over video calls.

Assigned drills with student logged results. We believe this is the strongest. A pro assigns a drill during a lesson, such as a greenside bunker drill measured by how many of ten balls finish inside ten feet, and the student logs results during solo practice. The team should investigate whether the app's GPS layer can measure results automatically, and whether AI can turn a student's practice history into a pre lesson summary for the pro.

Invite code connection, so a pro connects to a student through a code the pro issues. No public directory of professionals is in scope.

The team may build a web dashboard for the pro, a pro mode inside the iOS app, or both. Discovery decides.

Lesson scheduling is a stretch goal to be specified during discovery and built only if the coaching work lands early.

Work happens in a fork of the production repository. Code that proves out gets merged by the company afterward.

Student Learning Goals: Students practice the full arc of product development rather than implementing a fixed backlog. The project opens with real user research: structured interviews with practicing PGA teaching professionals and their students, synthesis of findings, and a written specification the team defends to the sponsor before building.

AI is part of how this company works and will be part of how the team works. Students are expected to use AI coding tools such as Claude Code or Cursor, and to use AI to draft and refine product requirements documents, on the condition that they can explain and defend every line they ship. Learning where these tools accelerate real work and where they quietly produce plausible garbage is one of the more valuable things a graduating engineer can carry into a first job.

On the engineering side the team works inside a shipping commercial codebase. That means extending existing Swift and SwiftUI code, integrating against a live REST API with its authentication and entitlement model, respecting branch discipline and code review, and writing work a professional team will merge into a paid product. The team also confronts the constraints of an app used outdoors, one handed, in sunlight.

The sponsor is the CEO, who spent twenty five years building enterprise software platforms before founding this company. Reviews are substantive technical reviews. Students get direct access to domain experts on the coaching side, including a former PGA Teacher of the Year.

Deliverables: First semester:

Discovery report covering interviews with a minimum of eight teaching professionals and eight of their students, with findings and a clear recommendation on which hypotheses to pursue and which to abandon.

Product requirements document and interaction design for the recommended feature set, approved by the sponsor before implementation.

Technical design covering data model, API surface, platform choice for the pro facing tool, and integration plan with the existing backend.

Working prototype of the core assignment loop, rough is fine.

Second semester:

Functioning implementation in a fork of the PreSwing repository, covering the invite code connection, per student content customization, and the drill assignment and result logging loop.

Automated tests on the core flows at a coverage level agreed with the sponsor.

Developer documentation sufficient for the company's engineers to take ownership, including setup, architecture, and known limitations.

Usability validation with at least three real teaching professionals, documented.

Written specification for lesson scheduling, whether or not it is built.

Final presentation, demonstration, and an engineering handoff session.

Desired Background: Golfers, caddies, and anyone who has taken a lesson and then failed to remember a word of it on the first tee are strongly encouraged to apply. Passion for the sport is not required, but it is a real advantage on this project. The team is designing for a player standing over a shot, and students who understand that moment will ask better questions in interviews and make better product calls. A background in any competitive sport with coaching and structured practice translates well.

Useful across the team: Swift and SwiftUI, or React and TypeScript if the team builds a web dashboard. Experience consuming REST APIs, working with relational data models, and using Git in a branch and pull request workflow. Comfort with AI assisted development tools, and the judgment to review what they produce.

At least one student comfortable with iOS development and Xcode, since the student facing side lives in an existing SwiftUI app.

At least one student genuinely interested in user research and interviewing. This project begins with talking to people, and a team that treats that as an obstacle will build the wrong thing.

Students who want to see how an early stage company makes product decisions under commercial pressure will get more from this than students who want a fixed specification.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement
- Non-disclosure agreement

Studio XE - Production Release Engineering for a Cross-Platform Mobile Application

Proposer: David Lutz david.lutz@communityhdp.com

Project Contact: Rohith Reddy Byreddy rohith@snackiebeans.com

Project Description: Studio XE's mobile application is built with React Native and Expo and covers 19 screens — authentication, two-factor verification, inbox and messaging, tasks, referrals, documents, accounts, data tables, and an assistant interface. The application runs and demonstrates well. It is also, by any professional standard, not shippable: no automated tests, no crash or error reporting, no release pipeline, and no presence in either app store. Everything between “this runs on a developer's phone” and “a customer's chief financial officer installs it from the App Store” is missing.
The student team builds that missing half. Term one produces the release and verification infrastructure a shippable mobile product requires — crash reporting, a signed build-and-submit pipeline for both platforms, an automated smoke test across all 19 screens, a device-validated OS support matrix — and gets the application accepted into Apple TestFlight and Google Play internal testing (a minimal build is submitted within the first six weeks so store-review latency is discovered early). Term two addresses the two deficiencies that would otherwise undermine it: an offline cache with staleness indicators and optimistic writes (the app currently assumes constant connectivity), and a shared component definition consumed by both web and mobile so the two clients stop drifting apart. The sponsor provides developer accounts, an Expo Application Services subscription, and physical test devices; all work occurs against a synthetic dataset with no access to production systems or protected health information.

Student Learning Goals: Students learn release engineering — the discipline that separates running code from a shipped product, and that coursework almost never covers: crash and error reporting with a triage runbook, signed builds from a single command, store submission and review, staged rollout, device-matrix validation on physical hardware, and a measured crash-free session rate target (99.5%+ over the final 30 days).
The second term adds distributed-client architecture: offline caching with explicit staleness semantics, optimistic writes with documented conflict resolution, and a cross-client parity contract shared by web and mobile. For a graduating student this produces an unusually strong portfolio artifact: a released application, in public app stores, with their work in it.

Deliverables: • Crash and error reporting live across all build configurations, with a triage runbook.
- An automated build-and-submit pipeline producing signed builds for iOS and Android from a single command.
- The application accepted into Apple TestFlight and Google Play internal testing.
- An automated smoke test covering all 19 screens, running in continuous integration, plus a documented, device-validated OS support matrix.
- An offline cache with staleness indicators, optimistic writes with documented

conflict handling, and a shared component definition consumed by both web and mobile for at least five components.

- A crash-free session rate of 99.5 percent or better across the final 30 days, and a final live demonstration on physical devices.

Desired Background: • Mobile: two members with React Native or strong React experience (Expo, Expo Router, TypeScript).

- Release engineering: one member with interest in CI/CD and build pipelines (EAS Build and Submit).
- Sponsor provides Apple/Google developer accounts, EAS subscription, and physical test devices before term start.
- Out of scope: server API changes, new feature screens, any work requiring production data.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

ProfessorDM Technologies Inc. - ProfessorDM Version 1 MVP: AI-Powered Social Media Question Monetization and Knowledge Platform

Proposer: Kristin Topping info@professorDM.ca

Project Contact: Kristin Topping info@professorDM.ca

Project Description: ProfessorDM Technologies Inc. is building an AI-enabled platform that allows expert content creators to turn questions received through social media into paid, high-quality responses and reusable knowledge assets.

For this capstone, the student team will build Version 1 of the ProfessorDM Minimum Viable Product (MVP), using the Sweetlife Flora Instagram account as the initial live test environment. Sweetlife Flora has an established audience of more than 165,000 followers, giving the team the opportunity to develop against a genuine creator use case and support real-world beta testing.

The MVP will connect to Instagram and process incoming direct messages through an AI-assisted workflow. The system will identify which messages are genuine expert questions, distinguish them from social messages and spam, determine whether more information is required, and generate clarification prompts when necessary. Once a question is ready for the expert, the system will search the creator's existing knowledge base using semantic search and generate a draft response for the creator to review, personalize and approve.

The creator will work through a secure, web-based dashboard that includes a unified question inbox, answer-review tools, safety and moderation workflows, follow-up questions, transaction status, and access to the creator's growing Knowledge Database. Approved answers will be stored as searchable knowledge so that expertise developed through one-to-one interactions becomes a reusable digital asset.

The project will also include a consumer-facing searchable knowledge experience, allowing followers to search approved creator knowledge, browse relevant answers, and submit personalized questions. Payment functionality will be integrated using an appropriate test or production-ready payment workflow, subject to project and university requirements.

AI is used throughout the system for:

- message and intent classification;
- question clarity assessment and clarification;
- semantic knowledge retrieval;
- AI-assisted response drafting;
- image-context processing where appropriate;
- safety screening and moderation;

- knowledge indexing and retrieval.

A defining feature of ProfessorDM is its human-in-the-loop architecture. The system does not replace the expert. AI organizes, retrieves, drafts and screens information, while the creator remains responsible for reviewing and approving the final response.

The goal of the capstone is to deliver a working, beta-capable Version 1 MVP, not simply a demonstration or disconnected prototype. The completed system should be sufficiently functional to support controlled beta trials using Sweetlife Flora's Instagram audience. Results from those trials will be used to identify technical, usability and AI-performance improvements required before broader commercial release.

This project gives fourth-year Computer Science and Computer Systems Engineering students the opportunity to build a real AI-enabled SaaS product involving full-stack development, social API integration, applied AI, semantic search, databases, cloud infrastructure, cybersecurity, payments, moderation and human-centred system design within one integrated capstone project.

Student Learning Goals: This capstone will give students the opportunity to design and build a real, beta-capable AI-enabled SaaS product for an operating startup, rather than a classroom-only demonstration.

Students will work across the full software development lifecycle: requirements analysis, system architecture, implementation, integration, testing, deployment, documentation, and beta evaluation. The project is intentionally multidisciplinary and will require the team to combine several areas of senior-level Computer Science and Computer Systems Engineering into one integrated system.

Students will gain practical experience in:

1. Applied AI and machine learning integration, including intent classification, question clarity assessment, semantic search, retrieval-augmented generation, AI-assisted drafting, multimodal input, and safety filtering.
2. Full-stack software development, including a web-based creator dashboard, consumer-facing knowledge experience, backend APIs, authentication, databases, and cloud deployment.
3. Social-platform integration, including working with the Meta/Instagram API, webhooks, message routing, attachments, permissions, and third-party platform constraints.
4. Database and search architecture, including relational data models, embeddings, vector search, knowledge indexing, and retrieval of relevant prior answers.
5. Human-in-the-loop AI design, where AI supports the expert but does not replace them. Students will need to design workflows that require creator review, editing, approval, and escalation.
6. AI safety and moderation, including the detection and routing of potentially harmful

questions or responses and the design of moderation workflows.
7. Secure systems engineering, including authentication, access control, privacy, secure API integration, logging, error handling, and protection of user-generated content.
8. Payment and transaction integration, using a suitable test or production-oriented payment workflow as project constraints permit.
9. Software testing and validation, including unit, integration, usability, AI-performance, and end-to-end workflow testing.
10. Real-world beta testing, using Sweetlife Flora as the initial creator environment and evaluating the MVP with actual or recruited users to identify technical and usability improvements before broader release.

The project will also expose students to the realities of building software for a startup environment. They will need to make architecture trade-offs, manage scope, respond to changing requirements, document technical decisions, work with external APIs, and distinguish between what is theoretically possible and what is reliable enough for an MVP.

By the end of the capstone, students should have experience building a complex system in which AI, web applications, APIs, cloud services, data infrastructure, security, and user experience all have to work together. The final product is intended to become ProfessorDM Version 1 and to inform the company's next stage of commercial development.

Deliverables: The student team will deliver a working, beta-capable Version 1 MVP of ProfessorDM, supported by the technical documentation and testing needed for ProfessorDM Technologies Inc. to evaluate the system and prepare for broader commercial development.

Expected deliverables include:

1. Live Instagram integration for the Sweetlife Flora account, including inbound direct-message handling, supported attachments, webhook processing, and outbound clarification and answer delivery.
2. AI classification pipeline that identifies expert questions, messages requiring clarification, social messages, spam or irrelevant content, and safety-sensitive questions.
3. Question clarification workflow that identifies missing information, generates clarification prompts, receives revised questions, and reassesses whether they are ready for expert review.
4. Semantic Knowledge Database containing creator-approved Q&A content, embeddings, metadata, and semantic search capability.
5. AI-assisted response generation that uses the consumer's question, clarification details, relevant Knowledge Database content, and available image context to prepare a draft response.
6. Human-in-the-loop answer workflow allowing the creator to review, edit, replace, approve, or decline AI-generated drafts before anything is sent to the consumer.
7. Safety and moderation system for screening incoming questions, AI-generated drafts,

and creator-edited responses, including appropriate flagging and moderation queues.
8. Web-based Creator Dashboard with Inbox, Sent, and Moderation views; question status; source information; attachments; AI draft review; follow-up handling; Knowledge Database access; and relevant transaction information.
9. Consumer-facing searchable knowledge experience where users can search approved creator knowledge, browse relevant answers, and submit personalized questions.
10. Payment/transaction workflow, using Stripe test mode or another agreed implementation, supporting question payment, payment confirmation, transaction status, and refund or beta-test workflows.
11. Follow-up question capability that maintains context with the original paid question and supports creator-reviewed AI assistance for follow-up responses.
12. Secure authentication and data handling for creator and consumer users, including appropriate access controls, logging, error handling, and protection of user-generated content.
13. Cloud deployment of the MVP in a stable test environment suitable for controlled beta use.
14. Beta testing support using Sweetlife Flora's Instagram audience, including test scenarios, issue tracking, usability feedback, AI-performance observations, and defect identification.
15. Technical documentation, including system architecture, database design, API documentation, deployment instructions, AI configuration, security considerations, known limitations, and maintenance guidance.
16. Final beta findings and MVP recommendations identifying defects, workflow issues, AI limitations, usability improvements, and recommended changes before wider commercial release.
17. Complete source-code repository and handover package so ProfessorDM Technologies Inc. can continue development after the capstone.

The key outcome is not simply a prototype presentation. The expected result is an integrated Version 1 system that can be operated, tested, and evaluated with real users to generate evidence for the next stage of ProfessorDM development.

Desired Background: The preferred team is five fourth-year Computer Science / Computer Systems Engineering students plus one Engineering Management student serving as Project Manager.

The five technical students should collectively cover the following areas. Individual students do not need to have experience in every area, but the team should have a strong mix of these skills.

1. Full-Stack Web Development Student: Strong background in frontend and backend development, preferably with technologies such as JavaScript/TypeScript, React or similar frameworks, Node.js or Python, REST APIs, authentication, and responsive web application

design. This student would contribute heavily to the Creator Dashboard and consumer-facing knowledge experience.
2. AI / Machine Learning Student: Interest or experience in applied AI, natural language processing, large language model APIs, prompt engineering, embeddings, semantic search, retrieval-augmented generation, classification, multimodal AI, and AI safety. This student would focus on the ProfessorDM AI pipeline, including question classification, clarification, knowledge retrieval, response drafting, and moderation.
3. Backend / Systems Integration Student: Strong software engineering and systems background with experience in APIs, webhooks, asynchronous processing, third-party integrations, and data flow between services. This student would be particularly important for the Meta/Instagram integration, message routing, Stripe integration, and coordination between the AI, database, and application layers.
4. Database / Cloud / DevOps Student: Experience or strong interest in relational databases, PostgreSQL, vector search or pgvector, cloud services, deployment, CI/CD, monitoring, logging, containerization, and scalable application infrastructure. This student would help design and deploy the data, search, and cloud environment required to support beta testing.
5. Security / Testing / Software Quality Student: Background or interest in cybersecurity, secure authentication, access control, privacy, API security, testing, software quality assurance, fault handling, and system reliability. This student would help ensure the MVP is suitable for controlled real-world beta use and would contribute significantly to integration testing, security review, moderation workflows, and beta defect tracking.

Project Manager - Engineering Management Student Requested

ProfessorDM Technologies Inc. is specifically requesting that the sixth team member be an Engineering Management student who will serve as the Project Manager.
The Engineering Management student would not be expected to perform the same level of software development as the five CS/CSE students. Instead, this role would focus on coordinating the technical work and keeping the project aligned with the Version 1 MVP objectives.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://drive.google.com/drive/folders/16QM-FACeScJBKj3ITSaY8wxvTrGb-5PY?usp=share_link

Grey-Box - Project Mesh

Proposer: Valentin Kravtchenko valentin.kravtchenko@grey-box.ca

Project Contact: Valentin Kravtchenko valentin.kravtchenko@grey-box.ca

Project Description: Project Mesh is an open-source application that enables Android devices to connect and send files and messages over long distances without requiring internet connectivity. As the app has grown, its codebase has outpaced its original design. This project modernizes that foundation: continuing a refactor already underway, the team will migrate the app's legacy screens onto a modern, testable MVVM architecture, build the unit test suite the project has never had, and surface the hidden bugs that a more testable structure brings to light

Student Learning Goals: Modern Android Architecture - Learn Google's official app architecture guidance hands-on: unidirectional data flow, a single-state-holder MVVM pattern, Kotlin Flows, and lifecycle-aware state collection in Jetpack Compose

Refactoring Legacy Code - Practice safely decomposing a large class into testable, layered components through small, behavior-preserving, incrementally-verified changes

State Management & Repository Pattern - Build screens around one immutable UI state object and a repository as the single source of truth, replacing ad-hoc state scattered across the UI

Testing & Bug Hunting - Write JVM unit tests with fakes and mocking (kotlinx-coroutines-test, MockK) that pin down real behavior and surface hidden bugs. This is a great way to learn a codebase deeply, even without diving straight into the refactoring

Collaborative Software Engineering - Work in pairs and as a team using Git version control, code reviews, Architecture Decision Records (ADRs), agile practices, and technical documentation

Portfolio Development - Create tangible, demonstrable work that showcases modern Android architecture skills highly relevant to industry roles

Deliverables: - Adopt the new MVVM architecture across Project Mesh's legacy screens.

- Build a robust unit test suite for each migrated screen.

- Discover and investigate bugs — both long-standing and newly introduced.

- Design documentation to ensure future work can easily adhere to the new architecture.

Desired Background: - Basic programming experience (any language)
- Interest in software architecture, clean code, and writing maintainable systems

- Experience with any object-oriented or typed language (Java, C#, TypeScript, etc.) (Nice to have)
- Experience with Kotlin, Android development, Jetpack Compose, unit testing, or Git would all be huge pluses, but aren't expected.

Delivery Mode: In-person

Required Agreements:

- ["Non-disclosure agreement"]

Additional Materials: N/A

Manakiki LLC (Protocol AI) - Protocol AI: Clinical Trial Management & Data Analytics Web Platform

Proposer: Nand Modi nmodi@manakiki.net

Project Contact: Nand Janakbhai Modi nmodi@manakiki.net

Project Description: Protocol AI is an early-stage clinical research technology product developed by Manakiki LLC. We are building a web-based Clinical Trial Management System (CTMS) and a complementary clinical data analytics module that help research teams manage study protocols, track participant data, and visualize trial metrics. The capstone team will design and develop core web application features (front-end user interfaces and back-end services), implement secure data handling that meets healthcare compliance standards (HIPAA, ICH-GCP), build interactive analytics/visualization dashboards, and integrate workflow automation. The platform will provide students hands-on experience building a real-world, compliance-aware healthcare SaaS product.

Student Learning Goals: Students will gain practical, end-to-end experience building a production-oriented web application: requirements gathering with a startup sponsor, full-stack web development, database and API design, secure handling of sensitive (healthcare) data, data visualization/analytics dashboards, agile development, and version control via GitHub. They will also learn how regulatory and compliance considerations (HIPAA, ICH-GCP, CDISC data standards) shape software design in the healthcare domain.

Deliverables: 1) A functional web application module for the CTMS (front-end + back-end) with documented features; 2) An analytics/visualization dashboard for clinical trial metrics; 3) Secure, well-structured data model and API documentation; 4) Source code with documentation in a shared repository (GitHub); 5) A final demo, technical documentation, and handoff materials. NDA and IP assignment to the sponsor are expected.

Desired Background: Web/full-stack development (JavaScript/TypeScript, React or similar front-end frameworks, REST APIs, and relational or NoSQL databases). Familiarity with cloud deployment, secure authentication, and data visualization libraries is helpful. Interest in healthcare technology and data privacy/compliance is a plus but not required.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Surgestreams ProXimite - ProXimite: AI-Powered Supply Chain Intelligence Platform for the U.S. Space Industry

Proposer: Asim Robinson asim.robinson@surgestreams.com

Project Contact: Asim Robinson asim.robinson@surgestreams.com

Project Description: ProXimite is a live, working supply chain intelligence platform purpose-built for the U.S. space and aerospace industry. The platform connects aerospace buyers (satellite manufacturers, launch providers, defense primes) with verified domestic suppliers of space-grade components — from rad-hard microelectronics and cryogenic valves to composite pressure vessels and optical inter-satellite links.

The space economy still relies on spreadsheets, cold calls, and personal networks. Over 75% of qualified small and mid-sized manufacturers are invisible to prime contractors. It currently takes 4–6 months to vet and onboard a single new supplier. ProXimite solves this with AI-driven supplier discovery, automated certification verification, real-time supply chain risk monitoring across 12 critical chokepoints, and built-in ITAR/EAR export control compliance.

The platform is already in active development with a functioning backend (FastAPI/Python), PostgreSQL database with pgvector for semantic search, AI agent layer (CrewAI with Claude/OpenAI), a React/TypeScript frontend, and a comprehensive HTML-based admin dashboard with live supplier data, RFQ workflows, and compliance tracking. The company is a member of the Commercial Space Federation's Space Supply Chain Council (S2C2) and has 150+ suppliers on its waitlist.

The student team will work directly with the founder on shipping production features, improving the AI matching engine, building out the compliance automation layer, and enhancing the marketplace UX — all on a real product with real industry users.

Student Learning Goals: Students on this project will gain hands-on experience in:

* Full-stack development on a production application, working with FastAPI (Python), React/TypeScript, PostgreSQL, and Docker in a real deployment environment, not a classroom exercise
* Applied AI/ML engineering, including building and tuning NLP-based certification parsing, vector embedding search (pgvector), semantic supplier matching, and multi-agent AI systems using CrewAI and large language models such as Claude and OpenAI
* Regulated industry software development, learning how ITAR, EAR, NIST 800-171, and SOC 2 compliance requirements shape software architecture, data handling, and access control decisions
* Aerospace domain knowledge, including understanding the space supply chain, critical component chokepoints, certification standards such as AS9100, Nadcap, and QPL, and how

procurement works in a high-consequence industry
* Startup product development through customer discovery using Mom Test methodology, iterating on real user feedback, shipping features weekly, and making trade-offs between speed and quality
* API design and systems integration, including building RESTful APIs, integrating with payment service providers such as Stripe, implementing role-based access control, and designing for environment separation between commercial and controlled data
* Professional collaboration, working with an industry founder who has operated at SpaceX, NASA, Amazon, and the U.S. Navy, while engaging with real aerospace supply chain stakeholders

Deliverables: 1. AI Matching Engine Enhancements — Improve the semantic supplier matching system by refining vector embeddings for supplier capability profiles, implementing RFQ-to-supplier matching with ranked results, and adding confidence scoring

2. Automated Certification Verification Module — Build an NLP pipeline that ingests supplier certification documents such as AS9100, ITAR registration, and Nadcap, extracts key fields including expiry date, scope, and classification, and flags expirations or compliance gaps

3. Supply Chain Risk Dashboard — Develop a real-time risk monitoring dashboard that tracks lead times, capacity constraints, and supplier health signals across the 12 critical chokepoint categories, including rad-hard electronics, COPVs, and cryogenic valves

4. Buyer/Supplier Role-Based UX Improvements — Enhance the marketplace frontend with proper role-gated views, streamlined RFQ submission workflows, quote comparison tools, and order tracking capabilities

5. Denied-Party Screening Integration — Implement automated screening of all platform users and transaction counterparties against the BIS Entity List, OFAC SDN List, and DDTC Debarred Parties databases

6. API Documentation and Testing Suite — Create comprehensive API documentation using OpenAPI/Swagger, build integration tests for all critical workflows, and improve the CI/CD pipeline

7. Final Technical Report and Demo — Deliver a written technical report documenting architecture decisions, implementation details, and recommendations for future development, along with a live demo of all completed features

Desired Background: * Proficiency in Python, preferably with FastAPI or Flask experience, and/or TypeScript/React

* Experience with PostgreSQL or other relational databases and SQL
* Familiarity with REST API development and modern web application architecture
* Comfort with Git, version control workflows, and collaborative development

Strongly Preferred (at least 1 team member):

* Experience with AI/ML concepts such as embeddings, vector search, NLP, or working with LLM APIs including OpenAI and Anthropic/Claude
* Experience with Docker and containerized development environments
* Coursework or interest in cybersecurity, compliance, or data governance

Nice to Have:

* Interest in the space/aerospace industry, defense technology, or supply chain management
* Exposure to SQLAlchemy/Alembic for Python ORM and database migrations
* Familiarity with Tailwind CSS, component libraries such as Radix UI/Shadcn, or frontend design systems
* Experience with CI/CD pipelines, automated testing, or DevOps practices

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Qualaces Inc - Qualaces Core AI Frontend Team

Proposer: Aakash Patel aakash@qualaces.com

Project Contact: Aakash Patel Aakash@qualaces.com

Project Description: Join the Qualaces Core AI Frontend Team and work on building modern AI-powered web and mobile experiences used in real-world applications. This project focuses on developing intuitive, scalable, and high-performance frontend systems that integrate with AI-driven platforms and services. Students may contribute to products such as Jobque.ai, gaining hands-on experience building production-grade user interfaces, integrating APIs, and developing next-generation AI experiences in a fast-paced startup environment. Excited to solve job seekers' problems?

Student Learning Goals: Overall, be responsible for updating and maintaining the frontend of jobque.ai & chrome extension: https://chromewebstore.google.com/detail/jobque-agent/kbbefdejaeljoemfoegfilpjlfolejgk?pli=1

1. Gain hands-on experience building production-grade AI-powered web and mobile applications
2. Work in a real startup-style engineering environment with modern frontend technologies
3. Learn how to integrate frontend systems with AI platforms, APIs, and cloud services
4. Collaborate with developers, designers, QA engineers, and product teams on real features
5. Improve practical skills in React, React Native, UI/UX, debugging, testing, and deployment workflows
6. Contribute to real products such as Jobque.ai that are actively being developed and used Receive mentorship, technical guidance, code reviews, and exposure to industry best practices
7. We use AWS tech stack.

Deliverables: 1. Build and enhance frontend features for AI-powered applications
2. Develop responsive and user-friendly web/mobile interfaces
3. Integrate APIs and backend services into frontend applications
4. Participate in testing, debugging, code reviews, and deployments
5. Contribute production-quality code and technical documentation

Desired Background:
1. Basic knowledge of frontend development using React, React Native, JavaScript, or TypeScript
2. Familiarity with APIs, Git/GitHub, and modern web/mobile development workflows
3. Passion for UI/UX, problem-solving, and building real-world applications
4. Interest in AI-powered products and modern frontend technologies
5. Someone who has built or published projects/apps, or is highly motivated to do so
6. Willingness to learn and contribute in a fast-paced startup environment

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- US Citizenship requirement
- Intellectual property agreement

Additional Materials: https://jobque.ai/ & https://chromewebstore.google.com/detail/jobque-agent/kbbefdejaeljoemfoegfilpjlfolejgk?pli=1

Qualaces Inc - Qualaces core ai Quality Assurance team

Proposer: Aakash Patel aakash@qualaces.com

Project Contact: Aakash Patel Aakash@qualaces.com

Project Description: The Qualaces Core AI Quality Assurance Team develops AI-powered solutions that transform how software quality is planned, executed, and measured. Students will work on real-world projects involving test automation, AI-assisted test generation, quality analytics, and defect prediction. Team members will gain hands-on experience with modern QA practices, software testing tools, cloud technologies, and artificial intelligence. The project focuses on improving software quality, release confidence, and engineering productivity through innovation. Participants will collaborate with experienced QA professionals while contributing to products used by real customers and organizations.

Student Learning Goals: ### Student Learning Experience

* Gain hands-on experience in manual testing, test automation, and AI-assisted quality assurance practices used in modern software development.
* Learn how to design, execute, and maintain test cases for web, mobile, API, and cloud-based applications.
* Work with industry-standard tools such as Jira, TestRail, GitHub, automation frameworks, and AI-powered testing solutions.
* Develop skills in defect analysis, root cause investigation, quality metrics, and release management processes.
* Collaborate with experienced QA professionals and software engineers while contributing to real-world projects used by customers.

Deliverables: ### Expected Deliverables

* Create and execute comprehensive test cases for web, mobile, API, and AI-powered applications.
* Identify, document, and track software defects while providing detailed reproduction steps and impact analysis.
* Contribute to automated test scripts and regression test suites to improve testing efficiency and coverage.
* Produce quality metrics, test execution reports, and release readiness assessments for stakeholders.
* Collaborate with engineering teams to improve product quality and successfully support software releases.

Desired Background: ### Desired Background

* Currently pursuing or recently completed a degree/diploma in Computer Science, Software Engineering, Information Technology, or a related field.
* Basic understanding of software testing concepts, web/mobile applications, and software development lifecycles.
* Someone who loves to break things, is naturally curious about how systems work, and enjoys finding issues before customers do.
* Passionate about Quality Assurance, software quality, and delivering exceptional user experiences through thorough testing and attention to detail.
* Strong analytical, problem-solving, and communication skills with an interest in quality assurance and emerging AI technologies.

Delivery Mode: In-person

Required Agreements:

- US Citizenship requirement
- Non-disclosure agreement

Additional Materials: https://apps.apple.com/kz/developer/qualaces-inc/id1855369279 & https://play.google.com/store/apps/developer?id=Qualaces+Inc & jobque.ai & https://chromewebstore.google.com/detail/jobque-agent/kbbefdejaeljoemfoegfilpjlfolejgk

Qualaces Inc - Qualaces Full stack team

Proposer: Aakash Patel aakash@qualaces.com

Project Contact: Aakash Patel Aakash@qualaces.com

Project Description: Join the Qualaces Full Stack Team and work on building modern, scalable web and mobile applications used in real-world environments. This project focuses on developing end-to-end solutions, from intuitive frontend experiences to robust backend systems and APIs. Team members will gain hands-on experience building production-grade applications while collaborating in a fast-paced environment using modern technologies and development practices. You could be working in-house or on a client project

Student Learning Goals: ### **Student Learning Experience**

1. Gain hands-on experience building real-world full stack applications used in production environments.
2. Learn modern frontend and backend development practices, architecture, and system design.
3. Work with APIs, databases, cloud technologies, and deployment workflows.
4. Experience agile development through sprint planning, collaboration, and code reviews.
5. Improve problem-solving, debugging, and software engineering best practices.
6. Collaborate with experienced engineers and gain exposure to startup and industry workflows.

Deliverables: 1. Design and develop frontend and backend application features.
2. Build and integrate APIs and database-driven functionality.
3. Implement responsive web and mobile user experiences.
4. Contribute to testing, debugging, and performance optimization.
5. Participate in sprint planning, code reviews, and team collaboration.
6. Deliver a working feature, prototype, or production-ready component by project completion.

Desired Background: 1. Background in Computer Science, Software Engineering, Information Technology, or related fields.
2. Familiarity with frontend technologies (HTML, CSS, JavaScript/TypeScript, React) and backend concepts.
3. Basic understanding of APIs, databases, and cloud technologies.
4. Experience with mobile development or modern frameworks is a plus.
5. Strong problem-solving skills and willingness to learn.
6. Prior project work, hackathons, internships, or app development experience is a plus.

Delivery Mode: In-person

Required Agreements:

- US Citizenship requirement
- Intellectual property agreement
- Non-disclosure agreement

Additional Materials: qualaces.com | jobque.ai

Qualaces Inc - Qualaces Mobile development Team

Proposer: Aakash Patel aakash@qualaces.com

Project Contact: Aakash Patel Aakash@qualaces.com

Project Description: We are serious about building useful apps. We aim to get 100K users in each app by end of 2027. In this project, you will continue hands-on mobile development for apps already published on the Apple App Store and Google Play Store. You may also build a new app from scratch and help publish it to the app stores.

App Store: https://apps.apple.com/kz/developer/qualaces-inc/id1855369279

Play Store: https://play.google.com/store/apps/developer?id=Qualaces+Inc

Student Learning Goals: This is a highly coding-intensive Capstone project designed for students who genuinely enjoy building software and solving real-world problems. You will gain hands-on experience working on production mobile applications that are published on the Apple App Store and Google Play Store.

Students will collaborate in a fast-paced startup-style environment, contribute to real features, and learn practical software engineering skills used in the industry. Clear guidance, mentorship, vision, and technical direction will be provided throughout the project to support your growth and success.

Deliverables: 1. Contribute production-quality code to mobile applications
2. Develop and test app features independently and collaboratively
3. Participate in code reviews, debugging, and deployments
4. Maintain proper documentation and Git workflows
5. Potentially publish or support publishing apps to App Store and Play Store

Desired Background: 1. Basic knowledge of mobile app development (React Native, iOS, or Android)
2. Familiarity with Git/GitHub and software development workflows
3. Passion for coding, problem-solving, and building real-world applications
4. Someone who has published apps to the App Store or Play Store, or is highly motivated and passionate about doing so
5. Willingness to learn and work in a fast-paced startup environment

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: https://apps.apple.com/kz/developer/qualaces-inc/id1855369279 & https://play.google.com/store/apps/developer?id=Qualaces+Inc

U-Haul International - Rapid Simulation Agent (RSA)

Proposer: Brian Rutherford brian_rutherford@uhaul.com

Project Description: Create a system that will be used by U-Haul Development teams for quickly provisioning test data and cloud resources. Create a web portal, an accompanying MCP server, and the middleware services that support them.

This system will provide tooling for use in manual/automated testing scenarios, including:

- Create test rental contracts
- Create test customer accounts
- Create test equipment inventory
- Create test team member or partner user accounts
- Create test locations
- Simulate Rate Limiting scenarios for a given operation
- Provision, Update, and Destroy Azure resources in a sandbox environment

Team members will use the portal for ad-hoc requests and/or the MCP server for agentic work. The web portal will include UI selections to specify the desired output from any operation alongside a "chat" experience where users can request what they need similar to interacting with the MCP server in an agentic prompt.

Student Learning Goals: Students will gain experience working with systems that serve millions of customers each year while also having the chance to build something entirely new for use by other developers.

Deliverables:

- Web portal hosted in Azure
- MCP server hosted in Azure
- Middleware services hosted in Azure
- Skills files for Agentic tools
- C# Source Code for all projects (.NET Framework 11+)

Desired Background: Students will have an understanding of basic web development, middleware development, and AI tooling frameworks.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement
- Non-disclosure agreement

MentaMorph Inc - Real-World Market Simulation Integration

Proposer: Susan Meredith susan@mentamorph.com

Project Contact: Susan Meredith susan@mentamorph.com

Project Description: This project integrates historical and/or live financial market data into MentaMorph scenarios, enabling simulations of real-world economic events such as booms and crashes.

Student Learning Goals: Students will work at the intersection of software, data, and finance, learning how to model complex real-world systems in an interactive environment.

Deliverables: • Market data integration (historical or live)

- Real-world scenario simulations
- Data pipelines or APIs
- Modeling documentation

Desired Background: • API and data handling experience

- Interest in financial systems
- Backend or data programming

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://assets.cdn.filesafe.space/KgbobB9K3SHlswdmMom9/media/6a21f180f607d4002b8293e7.pdf and more will be provided after NDAs

Emirates Airlines - Robotics Spatial Mapping Create a rapid

Proposer: Mohammad Khoori mohammad.khoori@emirates.com

Project Contact: Mohammad Khoori

Project Description: Large operational facilities such as warehouses, aircraft hangars and cargo terminals are constantly
changing. Racks are relocated, staging areas are reconfigured, equipment is moved and temporary
obstacles appear. Robots and digital-twin systems need accurate spatial information, but traditional
survey and modelling methods can be slow, specialist-dependent and difficult to repeat.

Student Learning Goals: Gain hands on experience on real challenges large organization are solving and have an impact on them

Deliverables: Full version of solution to be developed and handed over

Desired Background: Mix of understanding of Computer science and hardware integrations and data scenarios

Delivery Mode: In-person

Required Agreements:

- ["Non-disclosure agreement"
- "Intellectual property agreement"]

Additional Materials:
https://drive.google.com/file/d/1IirEpgvIR2P7Kd5uFUKrBd1HhLN8NBfA/view

Appy.yo - Rollodex: Smart Contact Management and App Scanning Platform

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: Rollodex needs UI/UX improvements, deployment support, and stronger app scanning functionality. Many small businesses and teams manage contacts, contractors, vendors, and professional relationships across disconnected spreadsheets, emails, notes, and forms. Rollodex is designed to help users collect, organize, search, and manage contact information in one place. The current focus is to improve the user experience, support deployment, and build scanning functionality that can help users capture contact information more easily.

Student Learning Goals: UI/UX design, front-end development, back-end development, deployment support, database design, search/filter functionality, AI-assisted organization, scanning technology, OCR or image-to-data extraction, JavaScript, React, Python, API integrations.

Deliverables: A fully functioning and deployable software solution with improved UI/UX and scanning functionality. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.

Desired Background: UI/UX Designer 1
Front-end Developer 1–2
Back-end Developer 1
AI/Scanning Developer 1
Deployment/QA Support 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

California Polytechnic State University - ROVER2 - Multimodal Robot Control Platform

Proposer: Javier Gonzalez-Sanchez javiergs@calpoly.edu

Project Contact: Javier Gonzalez-Sanchez javiergs@calpoly.edu

Project Description: This project explores remote interaction with robotic devices through native desktop (Java, C/C++), web (JavaScript), and virtual reality (Unity/C#) interfaces connected through a shared communication infrastructure. The goal is to provide responsive, synchronized control and real-time monitoring across platforms while addressing challenges in distributed robotic systems such as communication latency, state synchronization, fault handling, and software quality.

The project builds on components developed by a previous student team, including prototypes for desktop, web, communication infrastructure (MQTT and ZeroMQ), and Meta Quest VR. These components will be reviewed, tested with physical robotic arms (UR e-Series and UFACTORY), and reused or redesigned as appropriate. The platform will also be extended to integrate a TurtleBot 4 mobile robot, building on an existing prototype that requires further development. The resulting system will provide a reliable and extensible foundation for multimodal robot interaction and future capabilities such as digital twins, additional sensing, and cloud-supported robotic services.

Student Learning Goals: Students will gain hands-on experience in key software engineering areas, including software maintenance and evolution, program comprehension, legacy code modernization, refactoring, system integration, software testing, and quality assurance. Rather than starting from scratch, they will evaluate an existing codebase, understand its architecture, identify technical debt and limitations, and make informed decisions about what to reuse, refactor, or redesign.

Students will develop skills in object-oriented software development, distributed systems, real-time communication, hardware-software integration, and cross-platform development. They will integrate components developed in multiple programming languages—including Java, C/C++, C#, and JavaScript—into a single distributed system spanning desktop, web, VR, communication, and robotic components. Remotely controlling and monitoring physical robots will expose them to practical challenges including latency, state synchronization, communication failures, safety constraints, and differences between expected and actual system behavior.

The project will apply and reinforce Agile software development through iterative development, backlog and issue management, Git-based workflows, pull requests and code reviews, continuous integration, automated and integration testing, documentation, and regular demonstrations. Team members will specialize in different areas while sharing responsibility for system integration, software quality, and end-to-end testing.

Deliverables: * Review of the inherited codebase, architecture, functionality, technical debt, and recommendations for reuse, refactoring, or redesign.

* Working system supporting remote control and real-time monitoring of physical robotic arms through desktop, web, and Meta Quest VR interfaces.

* Extension and completion of the existing TurtleBot 4 prototype for integration with the shared ROVER2 communication infrastructure and interfaces.

* Reliable MQTT/ZeroMQ-based communication, state synchronization, error handling, and recovery mechanisms across system components.

* New, refactored, and documented code, with identified technical debt addressed where appropriate.

* Unit, integration, and system-level tests covering critical functionality and communication among components.

* Performance evaluation results addressing communication latency, state synchronization, fault handling, and remote operation with physical robots.

* Organized multi-language codebase with Git history, issues, pull requests, code reviews, and release/version information.

* README files, architecture and interface documentation, build/setup instructions, and troubleshooting guidance sufficient for a future team to continue development.

* End-to-end demonstration of ROVER2 operating with physical robotic devices through multiple interfaces.

Desired Background: * Students should have a solid foundation in object-oriented programming and software development, with proficiency in at least one of the primary languages used in the project: Java, C/C++, C#, or JavaScript. A team with complementary backgrounds is encouraged; not every student is expected to know every technology. For example, a student with web development experience working alongside a student with experience or interest in Unity and C# could provide an excellent combination of skills.

* They should be comfortable reading an existing codebase, debugging unfamiliar software, and working with modular software architectures.

* Familiarity with Git and GitHub is expected, including branches, commits, merges, and pull requests.

* Students should have a basic understanding of networking and distributed

communication concepts, such as client-server architectures, APIs, sockets, and publish-subscribe messaging.

* Helpful experience, but not required, includes Meta Quest development, web development, robotics or IoT, automated testing and continuous integration, and cloud or container-based services. An interest in working across software and physical robotic systems is particularly valuable.

Delivery Mode: In-person

Required Agreements:

- ["None"]

Kids At Risk Action - Safe reporting / referral decision support tool

Proposer: Mike E. Tikkanen mike@invisiblechildren.org

Project Contact: Mike E. Tikkanen mike@invisiblechildren.org

Project Description: Design a tool that helps non-experts understand when and how to report suspected abuse, what to expect, and what local resources exist—without collecting sensitive data.

Student Learning Goals: Students would experience a multi-semester, real-client design studio where they define, build, test, and refine an interactive tool in partnership with KARA.

Project-based, real-world design
Apply classroom knowledge (coding, UX, data modeling) to a concrete community project, moving through full cycles of defining requirements, prototyping, testing, and iterating.

Work with a "customer" (KARA) to understand needs, gather feedback, and refine the storytelling tool so it is accurate, usable, and sustainable in practice.

Teamwork and professional skills
Collaborate on a multidisciplinary team (software, UX, data, social-issue focused members), coordinating tasks, timelines, and integration across semesters.

Practice communication through design reviews, presentations, and documentation, building confidence in explaining technical and ethical choices to non-technical stakeholders.

Human-centered and reflective learning
Engage in human-centered design for a sensitive social issue, learning to incorporate stakeholder perspectives and ethical considerations into technical decisions.

Reflect on how engineering skills can advance social responsibility, strengthening attitudes toward community engagement and appreciation of diverse perspectives in problem-solving.

Deliverables: Expected deliverables for the interactive storytelling tool function would center on a working system plus the artifacts needed for handoff and growth over multiple semesters.

Product deliverables
Interactive web experience

Deployed or deployable prototype that lets users step through one or more CPS/foster-care pathways (e.g., report → investigation → placement → permanency) with clickable choices, timelines, and explanations.

Basic analytics or logging to understand how users move through stories (what they click, where they drop off) so future teams can improve impact.

Story and data structures

A documented "story schema" describing events, decision points, outcomes, and metadata (age, placement type, time in care), with at least a small library of KARA-vetted scenarios.

Configuration files or simple admin interface that allows future teams (or KARA) to add/edit stories without rewriting core code.

Design and ethics deliverables
UX and accessibility package

Wireframes, user flows, screen mockups, and rationale for accessibility and trauma-informed choices (language, content warnings, options to skip/exit, color and icon decisions).

Results from at least one round of user or stakeholder testing (methods, participants, key findings, and prioritized improvements).

Ethical and content guidelines

Short guideline document on privacy, anonymization, and respectful representation of children and families, to guide future content additions.

Clear statements of scope and disclaimers so users understand this is an educational storytelling tool, not legal advice or an official CPS interface.

Technical and continuity deliverables
Code and technical documentation

Version-controlled repository with readme, architecture diagram, setup instructions, and notes on dependencies and deployment steps.

Developer-oriented instructions for adding new story branches, connecting to external data

sources later, and integrating with other KARA tools if those exist.

Handoff and next-steps brief

5–10 slide or 2–3 page summary outlining what was done, what remains, known limitations, and recommended next phases for subsequent EPICS or capstone teams.

Desired Background: For this KARA interactive storytelling tool, the ideal is a multidisciplinary EPICS-style team with strengths in software, UX, and human-centered social impact work.

Technical majors and skills
Computer science / software engineering / informatics students to build the web app, manage data structures for story paths, and handle integrations or basic analytics.

Students from related technical majors (computing, information technology, systems engineering) comfortable with version control, APIs, and front-end frameworks.

UX, design, and human-centered roles
Students with user experience or design focus (e.g., Graphic Information Technology – UX, Human Systems Engineering – UX) to lead research, information architecture, visual design, accessibility, and trauma-informed interaction patterns.

Comfort with human-centered design methods such as interviews, journey mapping, prototyping, and iterative usability testing with a community partner.

Social issue and communication strength
Work with KARA personnel or at least one team member interested in or minoring in social work, psychology, criminology, or related social sciences, who can help interpret CPS/foster-care processes and keep the narratives respectful and accurate.

Students with solid written and oral communication skills to present to KARA and EPICS faculty, document decisions, and create clear content and disclaimers for the storytelling tool.

Delivery Mode: In-person

Objectways Technologies - SafeMod - Content Moderation Application

Proposer: A.K. Sako ak.sako@objectways.com

Project Contact: A.K. Sako ak.sako@objectways.com

Project Description: SafeMode is an existing content moderation application where customers can submit text, image, audio and video contents to detect possible harmful materials. While major functionalities are in place, there are some enhancement opportunities that students can contribute to.

Student Learning Goals: Students will gain hands-on experience in frontend technologies such as Typescript and backend technologies such as Python. Also, they will be exposed to different LLMs used for content moderation.

Deliverables: At least 5 functional features reviewed and tested.

Desired Background: Javascript, Typescript, Python, and Git.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://safemod.ai/

Qualaces Inc - Sales, Marketing & Community Growth Specialist & Content creator

Proposer: Aakash Patel aakash@qualaces.com

Project Contact: Aakash Patel Aakash@qualaces.com

Project Description: We are looking for an AI-savvy Sales, Marketing & Community Growth Specialist to help grow the user base, brand awareness, and online presence of our software products.

Our portfolio includes:

Four mobile applications available on both the Apple App Store and Google Play Store
JobQue.ai – an AI-powered job alert platform
A Chrome Extension that complements our job search ecosystem

The goal is to build engaged communities around our products, increase downloads and active users, grow our social media presence, and establish a strong brand across multiple digital channels.

Student Learning Goals: ### Student Learning Experience

This position provides a unique opportunity to gain hands-on experience working directly with startup founders and contributing to the growth of real-world technology products used by customers.

Students will have the opportunity to:

* Learn how early-stage startups acquire and retain users.
* Gain practical experience in digital marketing, growth marketing, and community building.
* Use industry-leading AI tools to create content, automate workflows, and improve marketing efficiency.
* Develop skills in social media strategy, content creation, audience engagement, and brand building.
* Learn App Store Optimization (ASO) and techniques for increasing app downloads and user engagement.
* Participate in the development and execution of marketing campaigns across multiple channels.
* Analyze marketing and product performance metrics to make data-driven decisions.
* Build professional communication skills through user engagement, partnership outreach, and community management.
* Gain exposure to SaaS products, mobile applications, browser extensions, and AI-powered technologies.
* Contribute ideas directly to product growth strategies and see the impact of their work in

real time.
* Build a portfolio of marketing campaigns, content assets, community initiatives, and growth projects that can be showcased to future employers.
* Experience working in a fast-paced entrepreneurial environment where creativity, experimentation, and innovation are encouraged.

By the end of the placement, students will have developed practical skills in marketing, growth strategy, AI-assisted content creation, analytics, and community management while helping scale multiple technology products.

Deliverables: Lead Generation & Follow-Up, Community Building, Content Creation & AI-Powered Marketing, Growth & User Acquisition, Social Media Management, Sales Support

Desired Background: ### Desired Background

* Degree or diploma in Marketing, Business, Communications, Journalism, Digital Media, Public Relations, or a related field.
* Experience in digital marketing, social media management, community management, content creation, growth marketing, or SaaS marketing.
* Demonstrated ability to grow social media audiences, online communities, or consumer applications.
* Experience working with startups, technology companies, mobile apps, SaaS products, or AI-powered platforms.
* Familiarity with AI tools such as ChatGPT, Claude, Gemini, Canva AI, Midjourney, Perplexity, or similar productivity tools.
* Understanding of App Store Optimization (ASO), Search Engine Optimization (SEO), email marketing, and content marketing best practices.
* Experience using marketing and analytics platforms such as Google Analytics, LinkedIn Analytics, Meta Business Suite, HubSpot, Mailchimp, or similar tools.
* Strong written and verbal communication skills with the ability to create engaging content for multiple audiences.
* Self-starter who can work independently, manage multiple projects, and execute growth initiatives with minimal supervision.
* Experience building and managing communities on LinkedIn, Reddit, Discord, Facebook Groups, Slack, or similar platforms is considered a strong asset.
* Knowledge of job search, career development, recruitment technology, mobile applications, or productivity software is an asset.
* Portfolio showcasing social media growth, content creation, community building, or successful marketing campaigns is preferred.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- US Citizenship requirement

Additional Materials: https://apps.apple.com/kz/developer/qualaces-inc/id1855369279 & https://play.google.com/store/apps/developer?id=Qualaces+Inc & jobque.ai & https://chromewebstore.google.com/detail/jobque-agent/kbbefdejaeljoemfoegfilpjlfolejgk

School of Computing and Augmented Intelligence - Arizona State University - SCAI - ASU Capstone Showcase

Proposer: Steven Osburn sdosburn@asu.edu

Project Description: The ASU Capstone Showcase is a web platform for judging and showcasing student projects across different ASU events. Last semester we continued building on that foundation, and now we're entering Phase three by building upon that work through integrating new features to accommodate other ASU programs' needs pertaining to the capstone showcase event and expanding the functionalities it supports. You'll be working with the Capstone II team and reporting to Professor Osburn for progress.

Student Learning Goals: Full stack web development, database configuration, technical documentation, and unit testing.

Deliverables: A fully functional and tested Phase three release with new features and documentation for future teams.

Desired Background: Web development, databases, version control (Git), debugging, and unit testing.

Delivery Mode: In-person

Additional Materials: https://showcase.asucapstone.com/

ASU - SCAI - SCAI - ASU Capstone Ticketing Application

Proposer: Satya Parupudi parupudi@asu.edu

Project Description: Centralized ticketing system that allows for efficient communication, tracking, and management of capstone project issues between students and instructors.

Student Learning Goals: Web development, database configuration, process management

Deliverables: Continuing next phase of the project.

Desired Background: Frontend development, Backend development

Delivery Mode: In-person

Additional Materials: N/A

zevari ai - Scaling an AI Sales Agent Platform: Multi-Tenant Microservices, Public APIs, and Autonomous Agent Workflows

Proposer: John Peslar john@launchpadfast.com

Project Description: Zevari is a live, revenue-generating AI sales platform used by real customers every day. It runs autonomous agents that research prospects, write personalized outreach, manage multi-channel campaigns across LinkedIn and email, and route replies back into a CRM. The product currently serves roughly 150 paying customers and is targeting 1,000 within the next 6 to 12 months. That growth target is the whole project: the architecture that comfortably serves 150 tenants will not survive 1,000 without real engineering work.

The capstone team will own a scoped slice of that scaling effort end to end. Work is split across three tracks, and the team will touch all three rather than being siloed:

1. Backend and microservices. Decompose parts of the current service into independently deployable services with clear contracts. Design the job queue and worker model that runs agent tasks asynchronously, add retry and idempotency semantics, and build the observability needed to debug a system that is mostly running without a human watching it.

2. APIs, public and internal. Zevari exposes a public API to customers and a larger set of internal APIs consumed by our own agents and dashboards. The team will design and implement new endpoints, version them properly, add authentication and per-tenant rate limiting, write the OpenAPI specification, and build the developer documentation a third-party

3. AI agents and frontend. Build one or more new agents in the platform (for example: a lead-scoring agent, a reply-classification agent, or a research agent that assembles a prospect brief from public sources), including prompt design, tool definitions, evaluation harness, and cost and latency controls. Then build the frontend surface where a customer configures, monitors, and audits that agent.

The exact scope is set with the team in week one and locked into a written statement of work, so the semester ends with something shipped rather than something half-started. Candidate scopes are sized so a team of three can complete a working, tested, demoable system in one semester.

This is not a sandbox or a throwaway prototype. Work is done against a real codebase, in a real staging environment, with real production constraints. Code that meets our review bar has a genuine path to production, and students will be told explicitly which parts of their work shipped.

Student Learning Goals: Students will learn what it takes to scale a production SaaS system by an order of magnitude, which is a problem you cannot practice in coursework because it only appears when real customers are already depending on the software.

Concretely, the team will gain hands-on experience with:

- Distributed systems design: service boundaries, asynchronous job processing, queueing, idempotency, retries, and graceful degradation when a downstream provider fails.
- Production API engineering: REST design, authentication and authorization, multi-tenant data isolation, rate limiting, versioning, OpenAPI specs, and writing docs for an external developer audience.
- Applied AI engineering, beyond calling an API: prompt and tool design, structured output, model fallback and cost control, latency budgets, and how to build an evaluation harness so you can prove an agent got better instead of guessing.
- Modern frontend work: building interfaces on top of asynchronous, long-running, sometimes-failing backend operations, which is a much harder UI problem than CRUD. reading an unfamiliar production codebase, writing a design doc before writing code, code review, testing, CI, staged rollout, and instrumenting your own work so you can see it running.

Process and mentorship. The team runs one-week sprints. There is a scheduled 1-hour working session with the sponsor every two weeks at minimum, and a shared channel with a same-business-day response commitment for questions in between.
Students present their design decisions and defend the tradeoffs, the same as any engineer on the team. Code is reviewed by the sponsor with written feedback, not just approved.

Career value. Students finish with a demonstrable, non-trivial system they built inside a real commercial product, a public API and documentation with their name on the design, and a concrete story about scaling and AI agent infrastructure that is directly relevant to the roles they will be interviewing for. The sponsor will provide detailed references and LinkedIn recommendations for students who deliver.

Deliverables: 1. Architecture and design document covering the target service decomposition, data model changes for multi-tenant scale, API contracts, and the specific scaling bottlenecks the work addresses.

2. Working backend services, merged and running in a staging environment, with the asynchronous job/worker pipeline that executes agent tasks, including retry, idempotency, and failure handling.

3. Public API surface: implemented and versioned endpoints, authentication, per-tenant rate limiting, an OpenAPI/Swagger specification, and developer-facing documentation with runnable examples.

4. Internal API endpoints and the client libraries or interfaces our own agents and dashboards use to consume them.

5. At least one new AI agent, deployed to staging, including prompt and tool definitions, structured output handling, model fallback behavior, and cost/latency instrumentation.

6. Agent evaluation harness: a repeatable test set and scoring method that answers "is this agent actually performing correctly" without manual spot checking.

7. Frontend interface for configuring, running, and auditing the new agent, matching the existing product design system.

8. Automated test suite (unit plus integration) covering the new services and endpoints, wired into CI.

9. Load or scale test results demonstrating behavior at target tenant volume, with identified bottlenecks and recommendations.

10. Operational handoff: README, runbook, deployment notes, and a recorded demo, such that our team can run and extend the work after the semester ends.

11. Final presentation and demo to the sponsor.

All code is delivered in our repository with clean commit history and passing CI.

Desired Background: Required across the team as a whole:
- Proficiency in at least one modern backend language. Our stack is primarily Python and TypeScript/Node, so comfort with either is the strongest fit.
- Solid understanding of REST API design and HTTP fundamentals.
- Working knowledge of relational databases and SQL (we use PostgreSQL).
- Git-based collaboration, branches, and code review.

Strongly preferred (nice to have on at least one team member each):
- React or a similar modern frontend framework, plus TypeScript.
- Docker and general familiarity with deploying a service somewhere other than localhost.
- Experience calling an LLM API (OpenAI, Anthropic, or similar), including prompting and structured output.
- Exposure to message queues, background workers, or event-driven architecture.
- CI/CD, automated testing, or basic observability tooling.

Not required, and we will teach it:
- Prior experience with AI agent architecture, multi-tenant SaaS design, or production scaling. Curiosity and willingness to read an unfamiliar codebase matter more than a checklist of technologies.

Mindset we are looking for: students who want to see their code run in front of real paying customers, who will ask questions early instead of stalling for a week, and who are comfortable with a scope that is real and therefore occasionally messy.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://docs.zevari.ai

Northern Watch Services Inc. - Scan a Watch

Proposer: Thomas Sandrin Thomas@NorthernWatchServices.com

Project Description: See a watch you like? Snap or upload a photo to discover the model, learn more, and locate nearby authorized retailers.

Student Learning Goals: By the end of the internship, the student should be able to:

- Understand how AI + product + data come together
- Build a real-world feature from concept to launch
- Work with APIs, structured data, and UX flows
- Think like a product manager + builder

Deliverables: End of first or 2nd semester

Desired Background: Currently pursuing a degree in:

- Computer Science
- Software Engineering
- Data Science
- Artificial Intelligence
- Business / Product Management (with strong tech interest)

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: WatchDNA.com

NA - Scrappy

Proposer: Paul Mann paulneumann13@gmail.com

Project Contact: Paul Mann paulneumann13@gmail.com

Project Description: Scrappy is a native (iOS or Android) and a web app. Assisted by AI it helps neighbors connect with other neighbors with extra craft and art supplies. Got some extra googly eyes or pipe cleaners? Instead of treking to the thrift store, connect with a neighbor down the street who could use them! This subscription based model will unlock additional benefits to users that want to organize their inventory and supplies, get tips on where to find local recycling centers, how to safely dispose of materials, and so much more. Scrappy is also the corgi mascot! A helpful (and definitely playful) doggo, who is knowledgable companion.

Student Learning Goals: Organize a pre-existing Copilot build, collect and develop graphics and animations with sounds, test the app, and launch it.

Deliverables: A finished app that can launch into the market.

Desired Background: App development, graphics, animations, etc. Familiarity with Copilot and Loveable.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials:
https://copilot.microsoft.com/conversations/join/A8iy4NRXJxBfQra4M5YWj

Feminine Intelligence Agency - Script to Scorecard: An AI Pipeline That Profiles Movie Characters by Behavior

Proposer: Lauren Nignon lauren.nignon@gmail.com

Project Contact: Lauren Nignon lauren.nignon@gmail.com

Project Description: The Feminine Intelligence Agency (FIA) is an education platform that teaches people to recognize manipulation and coercive control. FIA has a program (Python, using LLMs) that reads a movie script and produces an evidence-based scorecard for each character, scoring behavior trait by trait, citing the character's actual lines, and matching the character to one of FIA's 17 behavioral categories. FIA also has an existing web app (SvelteKit and TypeScript, backed by Postgres).

This project has two connected parts. First, students continue and harden the Python analysis program so it reliably turns scripts into scored characters and scales to a library of many films. Second, they build a public wiki in the web app that publishes those characters and lets readers and expert reviewers comment on each analysis and confirm or challenge whether it is accurate, so the results become trustworthy rather than unreviewed model output. The analysis program is the back-office engine that produces the data; the wiki is the public product that displays and validates it. The validated characters, sorted across FIA's 17 categories, also give FIA a labeled dataset it can later use as a test set for its diagnostic chatbots.

Two features are deliberately out of scope for this project and left for later: a public “ask about the movies” question-and-answer tool over the character database, and a layer for visitors to attach their own personal experiences (which will need its own consent, de-identification, and any required IRB work before it is built).

What FIA Provides (so the team can start without stalling)

Access to the existing repository (the Python program and the SvelteKit app), plus a short architecture and database overview and a working local setup.

Movie scripts or dialogue transcripts with usage rights confirmed, and a named list of about 5 to 8 characters per film (each with at least roughly 10 lines of dialogue) to score.

A target size for the library, for example about 20 films and 100 characters.

Model API keys and a usage budget [dollar cap: FIA to specify], with a sample of scored output.

A committed scorecard-review capacity: at least two named reviewers [FIA to name] with a weekly quota, an expected turnaround, and a way to resolve reviewer disagreement, plus a named contact [FIA to name] for weekly check-ins.

Ownership: the character library and labeled dataset are FIA's; code ownership and licensing follow ASU's capstone IP agreement, to be confirmed with the program.

Student Learning Goals: Students extend a real, running system end to end and gain experience in applied AI and NLP (hardening an LLM-based analysis pipeline, verifying its

output against source text), full-stack web development (SvelteKit/TypeScript, Postgres), data modeling, measuring classification accuracy against an expert-reviewed set, and building a moderated, public-facing product. They finish with a working pipeline, a labeled dataset, and a deployed wiki as concrete portfolio pieces.

Deliverables: Phase 1 — The analysis pipeline (continue and harden the Python program). Make the existing Python scoring program run reliably and return structured output (character name, scores across all traits, and cited quotes per trait), including an automated check that verifies each cited quote against the source script by fuzzy string matching (for example, a Levenshtein-ratio or embedding-similarity threshold around 0.85, tuned with FIA; anything below it is flagged for human review) so the model cannot fabricate quotes. Prove it on at least 5 films, validated with FIA, producing at least 25 scored characters. Validated means an FIA domain expert reviews each scorecard and accepts it or gives corrections; as an initial target set with FIA, roughly 90% of cited quotes pass the trace check and about 85% of trait scores agree with the expert (agreement counted within one point), with stable results across repeated runs (the same script reproduces its scores within one point). After a go/no-go decision on these first 25, scale across at least 20 films to build a library of at least 100 scored characters spanning all 17 categories (the target can flex if validation is slower than the agreed weekly quota, as long as all 17 categories stay represented).
Export the library as JSON (character name, category label, trait scores, and quote citations), which becomes FIA's labeled dataset and future chatbot test set.
Phase 2 — The wiki (publish and validate in the web app).
Public character pages that display each scorecard, a short analysis, and quotes, accessible and responsive.
User accounts and a commenting/review layer where the public and expert reviewers comment on characters and confirm or challenge whether the analysis is accurate. Roles are defined (visitor, reviewer, admin); reviewer comments are always visible; visitor comments appear immediately but are subject to moderation and are auto-hidden once flagged by a set number of users pending review; an admin dashboard shows a triage queue sorted by flag count and is where FIA marks an analysis as validated; and moderation removes comments that name or accuse real people. The build includes standard account security, comment retention and deletion, and a target moderation-response window [FIA to set]. Automated tests (for example, Vitest or Playwright) cover the permission and moderation rules.
Out of scope (future phases): a public “ask about the movies” natural-language Q&A over the database; a personal-experience contribution layer (which needs its own consent, de-identification, moderation, and any required IRB work).
Supporting throughout: automated tests, a simple CI/CD setup (for example, GitHub Actions with a Vercel or Render deploy), documentation, and a deployed demo. With six students, the team can split into a pipeline group and a full-stack group, each with a named lead. The wiki can be built in parallel with pipeline scaling but goes public only once the character library is populated, so the launch is never an empty site. Phase 1 stands on its own, so a

reliable pipeline and labeled dataset are valuable even if the wiki runs long.

Desired Background: Web development with a modern framework and TypeScript (the app is in SvelteKit; React transfers well) and relational databases such as Postgres (required).
Python for the existing scoring pipeline, and any experience with LLM APIs or data analysis (a plus).
Comfort reading and extending an existing codebase.
Non-CS/CSE students are welcome and useful: psychology or statistics students can help review the analyses and judge accuracy, and design students can help with the public experience. FIA provides all domain knowledge.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: www.feminineintelligence.agency

Selfmadee LLC - Selfamdee - AI-Powered E-Commerce Platform

Proposer: Jaron Lodge jaron@scorenn.com

Project Description: Selfmadee is an AI-powered operating system that helps e-commerce brands manage functions such as inventory, supply chain, sales, accounting, cash flow, and marketing within one connected platform. Many e-commerce businesses currently store operational data across inconsistent spreadsheets, sales channels, accounting systems, supplier documents, and disconnected software tools.

Student Learning Goals: Students will gain experience designing and developing a real-world, full-stack AI application in collaboration with an industry sponsor. They will work with semi-structured business data, spreadsheet processing, schema matching, data normalization, confidence scoring, and human-in-the-loop AI workflows.

Students will also gain practical experience with frontend and backend development, relational databases, API design, cloud deployment, automated testing, system security, and technical documentation. The team will compare deterministic, AI-based, and hybrid approaches and evaluate them based on accuracy, reliability, cost, latency, and usability.

Deliverables: The student team will contribute to the design, development, testing, and improvement of the Selfmadee platform, an AI-powered operating system for e-commerce businesses. The exact feature set will be finalized during project kickoff based on platform priorities, student interests, technical feasibility, and the capabilities of the assigned team.

Desired Background: Students should collectively have experience or interest in:

Full-stack web development.
React, TypeScript, JavaScript, Python, or similar technologies.
Backend APIs, SQL, and relational databases.
AI, machine learning, large language models, or workflow automation.
Cloud services, serverless functions, and application deployment.
Third-party platform integrations and data synchronization.
Automated testing, debugging, security, and performance improvement.
UI/UX design and dashboard development.
Git, GitHub, and collaborative software development.
E-commerce, inventory, accounting, supply chain, or business operations.

Experience with React, Supabase/PostgreSQL, AWS, Shopify, Amazon, QuickBooks, or AI APIs would be helpful but is not required.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

- Intellectual property agreement

Appy.yo - Shopful: AI-Powered Local Clothing and Store Discovery App

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: Shopful's overall concept and look have been built out, but the project needs stronger UI/UX and AI implementation. Consumers often see clothing online or have a style in mind but do not know where to find similar items nearby. Small and local retailers also need better ways to be discovered by shoppers who are looking for specific products. Shopful's main focus is to use AI to help users search for clothing and identify nearby stores that may carry similar styles or items.

Student Learning Goals: AI implementation, UI/UX design, front-end development, back-end development, location-based search, product matching, image search, API integrations, mobile app development, JavaScript, React.

Deliverables: Students should help improve the existing design, define the AI-powered search experience, support clothing and store matching logic, and help develop or improve the working product. Students should attend bi-weekly check-ins and contribute approximately 5–10 hours per week.

Desired Background: AI/Product Matching Developer 1–2
UI/UX Designer 1
Front-end or Mobile Developer 1–2
Back-end Developer 1
Research and Development 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

SkillArea.com - SkillArea Completion

Proposer: Luke Gara info.SkillArea@gmail.com

Project Contact: Luke Gara info.SkillArea@gmail.com

Project Description: Finishing the SkillArea Platform : Store & Learning

Student Learning Goals: - Web Development
- Project Management

Deliverables: SkillArea Store and SkillArea Learning Sub-Platforms Completed Completed

Desired Background: Web Development

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: SkillArea.com

SkillArea.com - SkillArea Completion

Proposer: Luke Gara info.SkillArea@gmail.com

Project Contact: Luke Gara info.SkillArea@gmail.com

Project Description: Finishing the SkillArea Ecosystem Platform : SkillArea.com
Platform Current Completion Level : 65%

Student Learning Goals: - IT Project Execution Experience
- Project Management Experience
- Collaboration Experience

Deliverables: A finished marketable platform

Desired Background: Full Stack Web Developers

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: SkillArea.com

SkipCourse LLC - SkipCourse Backend Architecture, API Development & Cloud Run Services

Proposer: Karl Ernsberger karl@skipcourse.com

Project Contact: Karl Ernsberger karl@skipcourse.com

Project Description: SkipCourse's backend powers the competency-mapping engine, evidence verification workflows, student progress tracking, and portfolio generation. The skipcourse-backend-cloud-run repository contains the Node.js/TypeScript backend deployed on Google Cloud Run, integrated with Firestore, Firebase Storage, and Neo4j Graph-RAG assessment workers.

This Capstone team will expand backend capabilities, improve reliability, build new APIs for the frontend team, and strengthen the platform's data architecture. Students will work on real production services supporting active pilot schools.

Student Learning Goals: Students will gain experience in:

Cloud-native backend development (Google Cloud Run)

Node.js + TypeScript service architecture

Firestore data modeling and querying

Firebase Storage file handling and permissions

Neo4j graph database integration

API design, documentation, and versioning

Authentication & authorization patterns

Real DevOps workflows, CI/CD, and containerization

Collaboration with a frontend Capstone team consuming their APIs

This project provides deep exposure to modern backend engineering in a real startup environment.

Deliverables: The backend team will deliver:

New REST API endpoints for competency mapping, evidence logs, and portfolio generation

Improved Firestore data models for standards, skills, topics, and course progress

Fixes for Firebase Storage upload permission issues (“missing or insufficient permissions”)

Graph-RAG assessment worker improvements, including better precision/recall in competency inference

Registrar-facing backend workflows for audit-ready portfolio assembly

Student progress computation services, including mastery scoring and standards tracking

Containerized Cloud Run deployments with CI/CD pipelines

Logging, monitoring, and error-handling improvements

Backend documentation for future SkipCourse developers

Desired Background: Ideal students will have experience with:

Node.js + TypeScript

REST API development

Firestore or NoSQL databases

Google Cloud Platform (Cloud Run, IAM, logging)

Docker or containerization

Authentication/authorization concepts

Neo4j or graph databases (optional but valuable)

GitHub workflows and PR collaboration

Students should be comfortable working with real production systems and collaborating with a frontend team.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials:
https://docs.google.com/document/d/1IMhJ9k5kwgO1vt6X2lA97bedUpk-bjN7nw5c5rQ4SyU/edit?usp=sharing

grey-box.ca - Smart Hotspot Access Control & Usage Monitoring on Android Devices (Project CyberCafe)

Proposer: Valentin Kravtchenko valentin.kravtchenko@grey-box.ca

Project Contact: Stephane Bisson stephane.bisson@grey-box.ca

Project Description: Project Cybercafe is an open-source software that runs on an Android device and enables “smart hotspot” functionality to share internet connectivity with other Wi-Fi devices. Because connectivity is often expensive, limited, or restricted in the regions we target, Cybercafe includes controls that let operators manage and monitor how the hotspot is used for internet access.

Maturity level: Level 3: Analytical and experimental critical function and/or proof of concept (TRL scale, out of 9)
Social impact :SDG 9: Industry, innovation, and infrastructure, SDG 11: Sustainable cities and communities, SDG 16: Peace and justice (strong institutions)

Student Learning Goals: - Build a real connectivity product: turning an Android device into a controlled internet-sharing hotspot for constrained environments.
- Improve a mobile-first web interface and onboarding/setup UX so non-technical users can deploy it.
- Implement user management and access rules (accounts, privileges, quota/bandwidth awareness) with clear status feedback.
- Bridge Android’s Linux layer, Bash services, and a PHP web UI, learning how to wire and harden cross-layer systems.
- Improve reliability and observability (error handling, basic logging, stability under intermittent networks).

Deliverables: IT needs: We need to develop the following functions (your choice; the project uses Java, PHP, and Bash):
- Improve the UX of the mobile-first website
- Implement website functionality, bridging the gap between the Linux system underpinning Android and the PHP-based website
- Improve robustness of the interconnection between the website/PHP front-end with the Bash/Linux back-end
- Add user management
- Implement a method for alerting those who use the system of account status (bandwidth use, for example)
- UX/UI improvement to make initial setup of the software easier/possible

Desired Background: - Strong software engineering fundamentals and comfort working in existing codebases (debugging, reading legacy code, shipping incremental fixes).
- Ability to work in PHP + web basics (HTML/CSS/JS) and/or Linux/Bash; Java familiarity is a plus.

- Basic understanding of networking concepts (Wi-Fi/hotspot behavior, bandwidth constraints) and security hygiene.
- Comfortable with Git and writing clear documentation (setup steps, troubleshooting notes).
- Bonus: experience improving mobile-first UX, onboarding flows, or admin dashboards.

Delivery Mode: In-person

Required Agreements:

- ["None"]

Additional Materials: https://www.grey-box.ca/project-cybercafe/

Qualcomm Inc. - Smart Recycling System

Proposer: Emma Lacey elacey@qti.qualcomm.com

Project Contact: Jing Lin jinglin@qti.qualcomm.com

Project Description: A camera captures waste items and sends images to an AI model running on the MPU for classification (e.g., plastic, metal, glass). Based on the prediction and confidence, the system communicates to the MCU, which controls servos or actuators to physically sort the item into the correct bin.

Student Learning Goals: This project gives students hands-on experience with modern edge AI systems, combining computer vision, embedded programming, and real-world hardware integration. It strengthens practical skills like AI model usage (classification), Arduino programming, system design, and debugging, while also teaching how to build a complete end-to-end intelligent system (camera → AI → actuation).

Deliverables: Final working prototype published to GitHub with supporting documentation

Desired Background: Need to have: Python basics, Arduino C/C++ basics, Basic ML classification workflow, Hardware wiring, Debugging mindset
Nice to have: Computer vision concepts, Model tuning, UI/dashboard (App Lab web UI), Mechanical design

Delivery Mode: In-person

Surgestreams ProXimite - Surgestreams Idea-to-Launch Framework (ITLF) — AI-Driven Startup Launch Automation Platform

Proposer: Asim Robinson asim.robinson@surgestreams.com

Project Contact: Asim Robinson asim.robinson@surgestreams.com

Project Description: The Surgestreams Idea-to-Launch Framework is a production-grade, hybrid automation platform that helps startups go from a working MVP to a fully launched product. The platform combines a Java/Spring Boot 3 API backend with n8n visual workflow automation, enabling both engineers and non-technical users to orchestrate complex launch activities.

Student Learning Goals: oday the platform supports:
- **Submission Portal** — Founders/interns submit MVP links, pitch decks, and startup details; submissions go through an approval pipeline that triggers automated workflows.
- **AI-Powered Content Generation** — OpenAI GPT-4 generates platform-specific launch copy for LinkedIn, Twitter/X, Reddit, email campaigns, and video scripts.
- **Social Media Automation** — Scheduled, multi-platform social posting via API integrations (LinkedIn, Twitter, Reddit).
- **Investor Outreach** — Automated personalized email campaigns, HubSpot CRM syncing, SendSpark video outreach, and meeting-booking tracking.
- **Community Building** — Discord announcements, Slack notifications, and new-member onboarding flows.
- **Analytics Dashboard** — Real-time web dashboard with engagement trends, conversion funnels, top-post rankings, and campaign performance metrics.
- **MCP (Model Context Protocol) Integration** — An AI sidecar that validates and provisions n8n workflows programmatically, including a circuit breaker, rate-limit handling, and node-level schema validation.

The ongoing development team will extend the platform with new integrations, improve the AI pipeline, harden the deployment infrastructure, and build out the front-end experience.

Team members will gain hands-on experience in:

- **Full-Stack Java Development** — Spring Boot 3, Spring Data JPA, Spring Security, REST API design, and Lombok.
- **AI/LLM Integration** — Prompt engineering, OpenAI API usage, and content generation pipelines using GPT-4.
- **Workflow Automation** — Designing and provisioning n8n visual workflows via JSON-RPC / MCP protocol.
- **Third-Party API Integration** — OAuth flows and API clients for LinkedIn, Twitter, Reddit, HubSpot, Discord, Slack, Stripe, SendSpark, and more.
- **DevOps & Containerization** — Docker, Docker Compose, multi-service orchestration

(Java API + n8n + PostgreSQL + MCP sidecar).
- **Production Engineering Patterns** — Circuit breakers, exponential-backoff retry logic, rate limiting, fault-tolerant service communication.
- **Testing & Quality** — JUnit 5, AssertJ, parameterized tests, integration tests, and achieving high test coverage on complex adapters.
- **Agile Collaboration** — Working in an existing codebase with established conventions, contributing via feature branches and pull requests.

Deliverables: 1. **Front-End Overhaul** — Build a modern React/Next.js (or Thymeleaf-enhanced) submission portal and analytics dashboard to replace or extend the current server-rendered views.
2. **New Integration Clients** — Implement at least 3 additional platform integrations (e.g., Product Hunt auto-posting, Stripe payment tracking, Calendly meeting sync) following the existing client pattern.
3. **Enhanced AI Pipeline** — Add multi-model support (e.g., Anthropic Claude alongside OpenAI), A/B content experiments, and tone/style customization per platform.
4. **Workflow Catalog Expansion** — Create 5+ new n8n workflow templates for use cases like onboarding drip campaigns, feedback loops, and post-launch metrics collection; ensure all pass MCP validation.
5. **CI/CD Pipeline** — Set up GitHub Actions for automated build, test, Docker image publishing, and staging deployment.
6. **Comprehensive Test Suite** — Achieve ≥ 85% line coverage across all modules with unit, integration, and end-to-end tests.
7. **Technical Documentation** — Architecture decision records (ADRs), API reference (Swagger/OpenAPI), deployment runbook, and onboarding guide for future contributors.
8. **Security Hardening** — Implement OAuth 2.0 / JWT-based authentication for all API endpoints, secret management via Vault or AWS Secrets Manager, and role-based access control.

Desired Background: | Area | Skill Level | Details |
|---|---|---|
| **Java / Spring Boot** | Intermediate–Advanced | Spring Boot 3, REST APIs, JPA/Hibernate, dependency injection |
| **SQL / PostgreSQL** | Intermediate | Schema design, queries, migrations |
| **Docker / DevOps** | Beginner–Intermediate | Docker Compose, container debugging, CI/CD basics |
| **JavaScript / TypeScript** | Intermediate | For front-end work and n8n workflow customization |
| **React or Next.js** | Beginner–Intermediate | For building the submission portal and dashboard UI |
| **AI / LLM APIs** | Beginner–Intermediate | Prompt engineering, OpenAI or Anthropic API usage |

Git / GitHub	Intermediate	Feature branches, pull requests, code review workflow
API Integration	Beginner–Intermediate	Experience consuming REST APIs, OAuth flows
Testing	Beginner–Intermediate	JUnit 5, Mockito, or similar test frameworks

Bonus (not required):
- Experience with n8n or similar workflow automation tools (Zapier, Make)
- Familiarity with MCP (Model Context Protocol) or JSON-RPC
- Interest in startup ecosystems, growth hacking, or product launches

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Independent Contractor - Switch-Accessible Word-Based Writing Tool for Totally Blind Students with Complex Learning Needs

Proposer: Bruce McClanahan mcclanahan.bruce@gmail.com

Project Contact: Bruce McClanahan mcclanahan.bruce@gmail.com

Project Description: A switch-accessible word-based writing tool would be developed for totally blind students with complex learning needs. This app would also be accessible for CVI students in Phase I, totally blind students with moderate cognitive disabilities, and the general special education population. (Clicker Sentence Sets and PixWriter are examples of Word Based Writing Tools, however they are only accessible for the general special education population.) (CVI students in Phase I access everything through the auditory channel.)

Student Learning Goals: Develop a switch-accessible word based writing tool. A review of the literature will reveal that roughly 50% - 65% of blind and low vision students have at least one additional disabilities. A review of the literature will reveal that no software is developed for this population group. Why this is could be a very complex discussion. Societal changes over the last 40 years have greatly increased the percentage of blind and low vision students with additional disabilities.

Deliverables: Develop a switch-based word-based writing tool for the populations listed. Develop and implement a marketing program for this software. The software could be cross-platform, but needs to be developed for the iPad. Most special education teachers use the iPad for their students with complex learning needs.
The software that would be developed would remain the intellectual property of either the students or if the ASU Computer Science department.

Desired Background: Knowledge of assistive technology

Delivery Mode: In-person

Additional Materials: https://www.wssb.wa.gov/services/statewide-technology

Objectways Technologies - Tensoract Data Annotation Templates

Proposer: A.K. Sako ak.sako@objectways.com

Project Contact: A.K. Sako ak.sako@objectways.com

Project Description: Develop data annotation templates to train AI models. The types of input data include text, image, audio, and video.

Student Learning Goals: Mainly writing production grade frontend code using Javascript. Some templates require backend APIs where students can learn developing them. Also, students will experience software development lifecycle along with source code management, CI/CD, and deployment.

Deliverables: At least 5 functional production grade templates reviewed and tested.

Desired Background: Javascript, HTML, and Git.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Anton Intelligence - The Cost of What's Next: Helping health plans forecast financial exposure ahead of a drug's FDA approval.

Proposer: Ryan Kerr ryan.kerr@antonintelligence.com

Project Contact: Ryan Kerr ryan.kerr@antonintelligence.com

Project Description: A single infused therapy can cost a health plan seven figures for one patient, and roughly twenty more are heading toward an FDA decision within eighteen months. When one lands, a plan has to decide quickly whether to cover it, on what terms, and at what price. The plans that decide well are the ones that saw it coming.

Almost everything needed to see it coming is already public: regulatory filings, trial registrations, investor disclosures, and federal pricing files. It is also nearly unusable. Formats do not match, sources are revised without notice, and nothing connects a number to where it came from. Today the work is done by an analyst, by hand, in a spreadsheet.

You will build the application that replaces that.

It tracks physician-administered drugs approaching FDA action across three categories: cell therapy, gene therapy, and traditional medical-benefit biologics. There are two halves to it. A pipeline captures the public data, translates every source into one consistent structure, stores it, records where each value came from, and watches for sources that change or go stale. An application turns that into what a health plan needs: what is coming, what it would cost them specifically, how confident we are, and where each number came from.
That changes real decisions. Formulary design. Provider contracting. Where a therapy gets administered. Whether a plan is ready when the first claim arrives or caught out by it. Done well, it helps keep coverage available and member costs lower.

The technical approach is yours to propose. We are not handing you a stack. You will evaluate the options, weigh the trade-offs, and bring us a recommendation. Our CTO and VP of Engineering will review it with you and approve the approach before you start building, the same way we sign off on architecture internally. Expect a real conversation and hard questions rather than a rubber stamp. Two things we will be looking for: that it will hold up in production, because we intend to run this, and that it is affordable to operate. You will report what a refresh cycle costs.

What we provide. Accounts on our own AI subscriptions, including Claude Code and the other agentic coding tools our engineers use every day. A funded environment and any API budget the project needs. Direct access to our CTO, our VP of Engineering, our data engineering team, and our clinical pharmacy team for the questions no engineer can answer alone. Optional desk space at our office, a short walk from campus. Real code review, on the same terms we review each other.

The agreement. This project requires a confidentiality and intellectual property agreement, similar to what members of our own team sign. It will not stop you from presenting this project at the ASU showcase, describing your work in interviews, or listing it on your resume.

Student Learning Goals: Technical. Data engineering against sources you do not control and cannot fix. Data modeling that has to hold up under many differently shaped inputs. Provenance and lineage, meaning a system where every value knows where it came from, which is increasingly required in regulated industries and almost never taught. Append-only design and as-of queries, including the subtle failure where a system uses information that did not yet exist. An architecture you choose, defend to senior engineers, and are accountable for on cost. And agentic AI tooling applied to a real codebase, with people who can show you where it helps and where it does not.

Professional. You will present options and trade-offs to a CTO and a VP of Engineering and be asked hard questions. You will work somewhere tests and documentation are part of the deliverable. And you will practice reporting honestly on what your own system fails to catch.

Domain. You will finish understanding how the most expensive drugs in the world get priced, covered and paid for in the United States. That makes you immediately useful to health plans, health systems, pharmacy benefit managers and health technology companies. No healthcare background is expected; our clinical team will teach you the parts that matter.

How we work with you. A recurring one-hour meeting, weekly to begin with and every other week after that, or weekly throughout if your team prefers. Video call as the primary format, so scheduling stays easy around your class load. In-person working sessions at our Tempe office, including kickoff and the showcase. Questions answered within one business day. And a knowledge transfer session with the pharmacist who built the current version of this analysis by hand, so you understand what to capture and which sources can be trusted.

Deliverables: • A recommended technical approach, with trade-offs, reviewed by our technical leadership

- The data model and a decision on where the data lives, documented well enough for someone else to use
- Automated capture running unattended, plus a working path for sources that have no interface
- Monitoring that surfaces possible developments for review rather than acting on them unverified
- Every number in the system records where it came from, what document, what date, and when it was retrieved

- Monitoring that tells you when a source has changed, stopped updating, or started returning something unexpected
- The impact model that translates national data into a specific plan's expected exposure
- The application interface, and a consistent client-ready output generated rather than assembled by hand
- An evaluation with real numbers, including what the system misses and what it costs to run
- Documentation and a written handoff

Desired Background: Four lanes of work, covered between the team rather than by each person.

- Data engineering: building things that run on a schedule and keep running
- Database and data modeling: designing the structure many different sources translate into
- API development: exposing that data cleanly to whatever consumes it
- User interface: turning it into something a non-technical decision maker can act on quickly

We are not screening for a particular stack, since the approach is not decided yet. Bring real experience in whichever tools you know well and be ready to make the case for them.
No healthcare background needed. Our clinical team will teach you the domain.

Delivery Mode: In-person

Required Agreements:

- ["Non-disclosure agreement"
- "Intellectual property agreement"]

Additional Materials: N/A

Being Love Incorporated DBA The Good Goodbye - The Good Goodbye

Proposer: Kimberlee Klein and Christina McInnis thegoodgoodbye8@gmail.com

Project Contact: Kimberlee Klein eelrebmik@gmail.com

Project Description: The Good Goodbye is a North American end-of-life planning, education, marketplace and community platform designed to help individuals and families navigate end-of-life planning, death, grief and the many decisions that follow. The platform combines educational resources, service providers, products, events and AI-guided support to help users find relevant information and resources at the right time. Our long-term goal is to make end-of-life planning less overwhelming and more accessible while respecting individual, cultural, religious and family preferences. The Good Goodbye is currently developing its platform through student partnerships and is expanding an existing MVP built by (Phase 1) a previous BCIT student team and (Phase 2) a WGU student team.

Student Learning Goals: The Good Goodbye is seeking an ASU student team to continue development of an existing web platform created by 2 previous student teams. The current platform includes user accounts, a service-provider marketplace, educational resources, events, administrative functionality and the underlying database structure. The WGU Students took on creating the first AI user journey foundations guiding someone through a set of questions for when someone has just passed away. (example. did a doctor pronounce the person dead? Where did the death take place?)
ASU Students will be working on Phase 3 - Expanding upon the AI User Journeys and AI Experience created in Phase 2. The ASU Student will review the codebase and continue this phase focusing on developing and improving an AI-guided user experience that helps people navigate end-of-life planning, advance planning, caregiving, death and bereavement. Users will be able to select a journey based on their situation and interact with an AI Guide that asks questions, understands their responses and provides personalized next steps.

The project will also include development of AI-assisted planning tools for areas such as estate-preparation information, wills and planning worksheets, advance planning, funeral wishes, digital assets and other end-of-life planning documents. These tools are intended to help users organize information for subsequent review with appropriate professionals rather than replace professional legal or financial advice.

We also would like to implement - PHASE 4 if time permits – MONETIZATION: Provider subscriptions,Payment processing, Provider billing dashboard, Subscription management
Students will work with the existing codebase and database rather than rebuilding existing functionality. The goal is a functional, maintainable MVP that can be expanded by future student teams. *****We can decide together what the scope of this cohort can take on. *****

Deliverables: - We are in process of completing the User Journey 1 now -Someone Close To Me Has Died and we will continue with User journey 2 (I'm Planning for myself), and User

Journey 3 (I'm planning for someone close to me).the foundations for the AI integration has already been set up.
- We would like to create an Estate Planning Studio where these documents are filled out with the client:
Last Will (draft)
Executor worksheet
Beneficiary worksheet
Asset inventory
Liability inventory
Family information
Guardian planning
Funeral wishes
Letter of Wishes
Digital asset inventory
---If time permits: Monetization----

Desired Background: Computer Science, Software Engineering, or related background. Experience with React, TypeScript, Git/GitHub, databases (Supabase/PostgreSQL), and web application development. Comfortable contributing to an existing codebase in a collaborative team environment.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: www.thegoodgoodbye.com

Northern Watch Services Inc. - TimeGem.Press | The Global Watch & Jewelry News Platform

Proposer: Thomas J. Sandrin, MBA Thomas@NorthernWatchServices.com

Project Contact: Thomas Sandrin Thomas@NorthernWatchServices.com

Project Description: TimeGem.Press is a new digital media platform designed to become the global newswire and press distribution hub for the watch and jewelry industry.

The platform will be built primarily on Shopify and will integrate Artificial Intelligence (AI) and RSS aggregation technology to automatically collect, organize, and publish industry news from brand press lounges, media publications, and official announcements.

The goal is to create a centralized platform where:

- Watch and jewelry brands can publish and distribute press releases
- Journalists and media professionals can discover verified industry news
- Retailers and collectors can stay informed about product launches and market developments

In addition to aggregating news from brand press lounges and RSS feeds, the platform will offer a paid press release distribution service, allowing brands to upload their official announcements and media assets directly to the platform.

The project will also explore the development of a mobile application for iOS and Android, enabling users to access industry news, receive alerts about new press releases, and explore brand announcements in real time.

Students participating in this project will contribute to the development of both the web platform and the mobile applications, helping bring a new global media platform to life within the watch and jewelry industry.

Student Learning Goals: This project offers students the opportunity to work on a real-world technology and media platform, gaining practical experience in modern digital product development.

Students will gain hands-on experience in:

E-commerce Platform Development

- Building and customizing a Shopify-based platform
- Creating custom Shopify sections and Liquid templates
- Integrating external APIs and data feeds

AI and Data Integration

- Implementing AI tools for content categorization and summarization
- Integrating RSS feeds and automated content aggregation
- Structuring large datasets of articles and press releases

Mobile App Development

- Designing and developing mobile applications for iOS and Android
- Connecting the mobile apps to the platform's content database
- Building push notifications for news alerts

User Experience and Product Design

- Designing intuitive interfaces for news discovery
- Structuring media content and article layouts
- Creating responsive mobile and desktop experiences

Media and Platform Strategy

- Understanding how digital news platforms operate
- Learning about content distribution systems
- Exploring monetization strategies for media platforms

Students will work on a live industry platform, contributing directly to a project intended to serve a global audience in the watch and jewelry sector.

Deliverables: Students participating in the project will contribute to the development of the following components:

1. Shopify Platform Development

- Creation of the core TimeGem.Press Shopify website
- Development of custom templates and sections
- Integration of RSS feeds for automated news aggregation
- Creation of a press release submission system for brands
- Database structure for brands, articles, and press releases

2. AI Content Processing

- Implementation of AI tools to summarize and categorize articles
- Automated tagging by brand, category, and topic
- Filtering and organizing aggregated news content

3. Mobile Application Development

- Development of a TimeGem.Press mobile app for iOS
- Development of a TimeGem.Press mobile app for Android
- Integration with the platform's content database
- Push notification system for breaking news and press releases

4. User Interface and Experience

- Design of the platform's homepage and news feed structure
- Creation of article pages and brand press rooms
- Responsive design optimized for desktop, tablet, and mobile

5. Documentation and Testing

- Technical documentation of the platform architecture
- User testing and feedback analysis
- Recommendations for future platform improvements

Desired Background: e are looking for students with interest or experience in the following areas:

Web Development

- Shopify / Liquid
- HTML, CSS, JavaScript
- API integrations

Mobile Development

- iOS development (Swift)
- Android development (Kotlin or Java)
- Cross-platform frameworks such as Flutter or React Native

Artificial Intelligence / Data

- Natural language processing
- Content classification
- Data aggregation and automation

UX/UI Design

- Digital product design
- Responsive interface design
- Media platform user experience

Interest in Media or Luxury Industries (Optional)

Students interested in digital media, journalism, watches, luxury products, or brand communication may find this project particularly engaging.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: NorthernWatchServices.com

Allgoodaccounting.ca (Sandria Goodall) - Times up Mobile App Development

Proposer: Sandria Goodall allgoodcloud@gmail.com

Project Contact: Sandria Goodall allgoodcloud@gmail.com

Project Description: This project involves developing a mobile app called Time's Up, which helps users efficiently manage their daily tasks based on personal preferences. The app will utilize a formula to schedule both fixed and variable tasks, optimizing the user's time management.

Users will input fixed tasks that block their schedule, such as work and rest, while variable tasks will be grouped by similarity and urgency. The app will dynamically rearrange the schedule if any changes occur, ensuring optimal time utilization.

Core tasks:

- Develop the app's user interface and user experience design

- Implement the scheduling algorithm based on user inputs

- Test the app for functionality and usability

- Gather user feedback for improvements

- Document the development process and code

Student Learning Goals: Industry expertise & knowledge
Technical skills and methodologies
Hands-on support
Tools and/or resources
Regular meetings

Deliverables: Deliverables:

- Functional mobile app

- User interface design files

- Development and testing documentation

- User feedback report

Desired Background: Computer Science

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

Feminine Intelligence Agency - Toxic Workplace Simulator: Taking an AI-Graded Roleplay Trainer from MVP to Deployed Product

Proposer: Lauren Nignon lauren.nignon@gmail.com

Project Contact: Lauren Nignon lauren.nignon@gmail.com

Project Description: The Feminine Intelligence Agency (FIA) builds tools that help people recognize and respond to manipulation and bad-faith behavior, a skill we call social discernment. Our flagship tool, the Toxic Workplace Simulator, is a browser-based roleplay game where players take secret roles in a difficult workplace scenario (for example a manipulative manager, a bystander, or the target of harassment), act the scene out in real time by voice, and then receive AI-generated scores and feedback on how they handled it. Players also earn points for correctly guessing each other's hidden roles. The goal is a safe place to rehearse hard workplace conversations before having them for real.

This year's capstone team delivered a strong MVP: a multiplayer web app (React and TypeScript on Supabase) with lobby creation and join-by-code, live voice via WebRTC, browser-based speech-to-text, an OpenAI-powered grading engine, a scenario editor with import/export, and the beginnings of a solo mode. The system is documented and has been handed off to the sponsor.

The MVP works but is not yet ready for real users, and this year's team was candid about the gaps. The team's first task, before any forward progress, is to compile a complete, systematic inventory of every remaining bug and known issue. Using a structured triage process the sponsor provides, they will produce a single organized database of defects to fix, so the year starts from a clear, shared picture of the codebase rather than guesswork, and so priorities are set by evidence. From there, the team turns this prototype into a deployed, reliable product along four connected tracks: (1) ship it, meaning deploy to a live domain, add real accounts, and close the security gaps the current team flagged; (2) fix the core reliability problem by replacing the browser speech recognition (which works well only in Chromium and sometimes fails to hear players at all) with a server-side speech-to-text pipeline that works across browsers; (3) build a solo, AI-driven practice mode so one person can rehearse against AI-played characters without gathering a group; and (4) make the feedback genuinely useful, so scores are consistent and, more importantly, the app tells players specifically what they did well and how to respond better next time. The deliverables below are split into a required core and optional stretch work so the team has a clear definition of success, beginning with that bug inventory.

Student Learning Goals: Students own a real, sponsor-backed product end to end and ship it to production. Along the way they gain hands-on experience with a modern full-stack TypeScript/React codebase; real-time systems (WebRTC voice and Supabase realtime); integrating third-party AI services (server-side speech-to-text such as Whisper or Deepgram, and LLM grading via the OpenAI API); prompt engineering plus building an evaluation set to measure grading quality; authentication, web-security hardening, and

vulnerability remediation; and cloud deployment and DevOps (Vercel/Netlify/Cloudflare, domains, secrets and environment management, and API cost monitoring). Just as valuable, they practice the real-world skill of inheriting and extending a documented codebase with real technical debt, designing for an emotionally sensitive subject, handling user data responsibly, and working directly with a sponsor to turn honest MVP limitations into a shipped product. Students seeking an "AI plus real-time media plus full-stack" portfolio piece will find all three here, capped by a live, demoable product.

The team also dogfoods the product. Rather than only meeting each week, they play the game together every week across all three scenario families (Toxic Workplace, Cult Jam, and Dysfunctional Family), using their own experience as players to find what is broken, what is boring, and where better game incentives would make it more engaging and more instructive. This makes the project a strong fit for anyone interested in psychology or sociology, acting and improvisation, or game design and player dynamics, not only backend engineering.

Deliverables: Deliverables are tiered. Core items are the definition of a successful year. Stretch items are pursued if the team moves quickly. Each core item lists how "done" is judged.

Core (required):

Complete bug and issue inventory (first deliverable). Using the sponsor's systematic triage process, audit the existing MVP and produce a single organized database of every remaining bug and known issue, each with severity, area, and reproduction notes. This becomes the team's prioritized worklist for the year. Done when: the sponsor and team agree the inventory is complete and triaged, and it is the reference used to sequence the tracks below.

Deployed application on a live domain over HTTPS, with production secrets configured and a documented deploy-and-rollback process. Done when: a committee member can reach the live URL and the team can redeploy or roll back from documentation.

Cross-browser reliable transcription. Replace the browser Speech Recognition API with a server-side speech-to-text pipeline. Done when: roleplay speech is transcribed on current desktop Chrome, Edge, Safari, and Firefox, and on a labeled test set of recorded sessions the pipeline captures the large majority of spoken turns (target agreed with sponsor, e.g. ≥90% of utterances) with no total "heard no one" failures. Note: because grading runs on the full transcript after the scene, this pipeline does not need sub-second latency; only the optional live-caption display does, which removes the main real-time-latency risk.

User authentication and accounts with saved session history so players can track progress over time. Done when: a user can register, log in, complete a session, and see it in their history on next login.

Security pass. Audit and fix the vulnerabilities the current team flagged, specifically unsanitized user-supplied links (cross-site-scripting risk) and user image/asset uploads (malicious-file risk), plus a dependency-vulnerability scan. Done when: identified issues are remediated and documented in a short security report.

Solo, AI-driven practice mode: at least one fully working scenario where a single user rehearses against AI-played characters and receives feedback. Done when: a user completes

one solo scenario end to end and receives a score and written feedback.
Improved grading and feedback, consistent and coaching-style, grounded in FIA's social-discernment framework (a sponsor priority). Done when: a documented rubric exists; on a labeled evaluation set of at least 15–20 sessions the grader agrees with human raters within one point on the target proportion of items (e.g. ~80%); and pilot users rate the written feedback as helpful at 4/5 or better on average.
API cost monitoring and usage throttling for the LLM and speech services, so runaway usage cannot silently burn the sponsor's budget. Done when: per-session and monthly usage are visible and a configurable cap or throttle is in place.
Data-privacy and consent handling for voice and transcripts (see PART B). Done when: users see a recording notice and consent before capture, retention and deletion behave as specified, and an in-product notice states that AI feedback is educational practice, not an HR, legal, or clinical assessment.
Updated documentation and clean handoff: engineering handbook, deployment guide, and a refreshed "what's next" list.
Stretch (if time allows): saved-progress analytics and trend views; multiple solo scenarios with branching AI behavior; broader scenario library growth beyond the core set; mobile-responsive polish; bulk scenario import/export improvements; live on-screen captions during roleplay.

Desired Background: Solid programming fundamentals and comfort (or eagerness to get comfortable) with JavaScript/TypeScript and React. Helpful but not required: Node.js, relational databases and backends (Postgres / Supabase or similar), REST and realtime APIs, and any exposure to AI/LLM APIs, prompt engineering, WebRTC/audio, or authentication and web security. Above all, students should be self-directed, comfortable picking up an existing codebase, and interested in the mission of helping people handle harassment and toxic dynamics at work. Team members from outside CS/CSE are welcome and would strengthen scenario writing, feedback design, and the user experience. Because the team dogfoods the game every week, this project also suits students drawn to psychology or sociology, acting and improvisation, or game design and player dynamics, who will help shape the incentives that make it more engaging and more instructive.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials: www.feminineintelligence.agency

ProsGrow AI, Inc. - Unified Serverless LLM API & Enterprise Control Platform

Proposer: Elaine Liu elaine@prosgrow.ai

Project Contact: Elaine Liu elaine@prosgrow.ai

Project Description: ProsGrow AI is building a next-generation, OpenRouter-style unified LLM API gateway that empowers developers to access any Large Language Model (OpenAI, Anthropic, and various open-source models) through a single unified API and a single API key. While providing a seamless, usage-based serverless API for developers, the platform also features a robust Enterprise Control Plane that guarantees Zero Data Retention (ZDR), multi-tenant capacity isolation, and strict organization budget enforcement.

The capstone team will build critical production features on top of our existing robust architecture. Specifically, the team will focus on three core areas: 1) Integrating real-world payment gateways (Stripe), 2) Implementing and testing Tier-1 multi-provider failover routing for LLM endpoints, and 3) Developing comprehensive developer-facing API documentation and technical engineering blogs to showcase our zero-data-retention and capacity-isolation capabilities.

Student Learning Goals: Students will gain hands-on, invaluable experience building and scaling a modern, production-ready AI model platform. Throughout this project, they will learn:

1. AI Model Platform & Inference Engineering: Deep dive into the architecture of an LLM API gateway, learning how model inference works, how to manage prompt caching, and how to implement auto-failover mechanisms for high availability across different AI providers.
2. Azure Cloud Technologies: Deploying and managing scalable services using Microsoft Azure cloud infrastructure, including virtual machines, networking, and managed database services.
3. Full-Stack Development: Working extensively with Python on the backend (API design, routing, performance optimization) and JavaScript (HTML/CSS/Vanilla JS) on the frontend for the management console.
4. Enterprise Software Standards: Working with Python, PostgreSQL, Redis, Docker, and rigorous CI/CD test-driven development workflows to ensure the reliability of a high-concurrency system.
5. Advanced Model Optimization (Phase 2): Once the core platform is fully operational, students may have the unique opportunity to participate in integrating and optimizing state-of-the-art AI models developed by university professors.

Deliverables: 1. End-to-End Production Deployment: Successfully deploy, configure, and maintain the OpenRouter-style serverless API platform in a production environment, ensuring all components (Gateway, Console, Ledger) run smoothly end-to-end.

2. Provider Integration Testing: Build and execute end-to-end tests for various external LLM providers to guarantee accurate token counting and system stability.

3. Comprehensive API Documentation: Write and publish all developer-facing API documentation, providing clear quickstarts, endpoint references, and integration guides for users to easily adopt our platform.

4. Technical Engineering Blogs: Write and publish technical blog posts detailing the platform's architecture, such as how we handle capacity isolation, zero-data-retention, or high-concurrency routing, to showcase our engineering capabilities to enterprise clients.

Desired Background: 1. Hands-on Full-Stack Capabilities: Proven ability to execute tasks independently across both frontend (JavaScript/HTML/CSS) and backend (Python/FastAPI). We need doers who can write clean, production-ready code.

2. System Stability & Operations: Strong focus on reliability. Candidates must be capable of deploying, monitoring, and ensuring the platform runs stably end-to-end without constant supervision.

3. Scalability Mindset: Experience or strong interest in building high-concurrency systems. The team will be responsible for ensuring the architecture can scale to support a large volume of users and enterprise API traffic.

4. Tech Stack: Familiarity with Python, SQL databases (PostgreSQL), Docker, and Git is required. Experience with Microsoft Azure and API gateway architecture is a major plus.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Pixel Pirate Studio - UNOMi Software Update

Proposer: Obi Onyejekwe obi.onyejekwe@gmail.com

Project Contact: Obi Onyejekwe obi@pixelpiratestudio.com

Project Description: UNOMi Plugin Update Internship – Project Breakdown
Interns will support ongoing feature updates and testing for the UNOMi 3D animation plugin across Maya, Blender, Unreal Engine, and Cinema 4D. The team will focus on testing performance, stability, and cross-platform compatibility while helping implement and validate new features such as auto facial expression generation and text-to-animation workflows. Students will assist with debugging, documenting issues, optimizing scripts, and collaborating with engineers to improve automation accuracy and real-time performance inside production pipelines.
Key Responsibilities
* Test plugin functionality across Maya, Blender, Unreal Engine, and Cinema 4D
* Assist in implementing and validating auto facial expression features
* Support development and testing of text-to-animation tools
* Debug scripts and identify performance bottlenecks
* Document bugs, workflow improvements, and feature updates
* Provide feedback from a creative and gameplay perspective
Required Skills
* Python
* Node.js / JavaScript
* C++
* Working knowledge of Unreal Engine
* Familiarity with Maya, Blender, or Cinema 4D
* Basic understanding of 3D modeling and animation principles
Bonus
* Interest in gaming, real-time engines, or interactive storytelling

Student Learning Goals: Students will gain hands-on experience working on a real-world AI-powered 3D animation plugin used across Maya, Blender, Unreal Engine, and Cinema 4D. They will learn how production-grade plugins are built, tested, and optimized across multiple 3D platforms while strengthening their skills in Python, C++, and JavaScript within live pipelines. Interns will deepen their understanding of facial animation systems, procedural automation, and text-to-animation workflows, while also improving debugging, performance optimization, and cross-platform compatibility skills.
Beyond technical growth, students will experience collaborative software development, agile workflows, version control practices, and structured QA testing. They will also gain insight into how AI tools integrate into creative pipelines, preparing them for careers in gaming, animation, virtual production, and real-time 3D development.

Deliverables: * Cross-platform testing reports for Maya, Blender, Unreal Engine, and Cinema 4D
* Bug logs with clear reproduction steps and suggested fixes
* Performance testing documentation (load times, processing speed, stability benchmarks)
* Implemented and tested updates for auto facial expression features
* Implemented and tested updates for text-to-animation functionality
* Script improvements or optimizations in Python, C++, or JavaScript
* Compatibility verification across different software versions
* Technical documentation outlining new features and workflow improvements
* End-of-term presentation or demo showcasing testing results and feature enhancements

Desired Background: * Students pursuing Computer Science, Game Development, Animation, Technical Art, or a related field
* Experience with Python, C++, and/or Node.js / JavaScript
* Familiarity with Unreal Engine and at least one 3D platform (Maya, Blender, or Cinema 4D)
* Basic understanding of 3D modeling, rigging, and animation workflows
* Experience working with plugins, scripting, or tool development (academic or personal projects)
* Interest in AI-driven tools, automation, or real-time animation systems
* Passion for gaming, interactive media, or virtual production
* Strong problem-solving skills and attention to detail

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: N/A

The Sturge-Weber Foundation - Using Artificial Intelligence to Enhance Operations, Engagement, and Fundraising for the Sturge-Weber Foundation

Proposer: Karen Ball kball@sturge-weber.org

Project Contact: Julia Terrell jterrell@sturge-weber.org

Project Description: The Sturge-Weber Foundation (SWF) is a global nonprofit dedicated to improving the lives of individuals affected by Sturge-Weber syndrome (SWS), a rare and complex neurological condition. With a small team and a wide range of programs—including patient support, Clinical Care Networks (CCNs), scientific meetings, and fundraising initiatives—SWF must operate efficiently while continuing to grow its impact and financial sustainability.

This capstone project explores how artificial intelligence (AI) can be applied across SWF's existing systems, including its website and CRM (Auxiliary), to improve operations, strengthen stakeholder engagement, and increase fundraising effectiveness.

The goal is to identify, design, and prototype AI-driven tools that streamline internal workflows, enhance communication, and support revenue growth through smarter donor, sponsor, and grant strategies.

Student Learning Goals: Key areas of focus include:

Website Optimization & Intelligent Navigation:

Evaluate the SWF website to identify opportunities for AI integration, such as smart search, personalized content recommendations, and chatbot support to help users quickly find relevant resources, events, and ways to get involved or donate.

CRM (Auxiliary) Intelligence & Automation:

Leverage AI to enhance how SWF uses its CRM by automating workflows, segmenting audiences, identifying high-potential donors or sponsors, and improving outreach to patients, families, and partners.

Clinical Care Network (CCN) Coordination:

Design tools to help track, manage, and optimize the 25+ Clinical Care Network sites, including scheduling coordination, communication tracking, and identifying gaps in geographic or specialty coverage.

Scientific Meeting & Event Management:

Develop AI-assisted solutions to support planning and execution of conferences, roundtables, and educational events. This may include automated scheduling, attendee matching, agenda optimization, and post-event analytics.

Fundraising & Donor Insights:

Apply AI to analyze donor behavior, predict giving patterns, recommend personalized outreach strategies, and identify new donor segments to improve retention and growth.

Sponsorship Strategy & Corporate Engagement:

Build tools or models that help identify and prioritize corporate partners (e.g., pharma, biotech, medtech), track sponsorship pipelines, and tailor proposals based on company

interests and past engagement.
Grant Writing & Prospect Identification:
Explore AI tools that assist in identifying relevant grant opportunities, drafting proposals, summarizing program impact, and aligning SWF initiatives with funder priorities to increase success rates.
Scheduling & Resource Management Tools:
Create or recommend systems that streamline scheduling across programs—such as patient sessions, expert roundtables, and internal coordination—reducing manual workload and improving visibility.
AI-Powered Communication Tools:
Explore tools that assist in drafting emails, summarizing meetings, generating reports, and maintaining consistent messaging across stakeholders.
Ethical, Practical, and Scalable Implementation:
Ensure all solutions are feasible for a small nonprofit team, with a focus on ease of use, cost-effectiveness, data privacy, and long-term sustainability.

Deliverables: The project will deliver a strategic roadmap and/or working prototypes demonstrating how AI can be integrated into SWF's current operations to improve efficiency, strengthen relationships, and increase funding opportunities. Solutions should be practical, scalable, and designed to support immediate implementation.
By leveraging AI across both operations and revenue generation, this project aims to help SWF expand its reach, better serve its community, and build a more sustainable future for individuals living with Sturge-Weber syndrome.

Desired Background: Computer science

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Varsity Media Foundation - Varsity Media Foundation - Sports Livestream Platform & Digital Media Management System

Proposer: Vince D'Aliesio info@varsitysportsshow.com

Project Contact: Vince D'Aliesio info@varsitysportsshow.com

Project Description: Develop, maintain, and enhance Varsity Media Foundation's sports livestream platform and digital media ecosystem. This project focuses on supporting live sports broadcasting through a web-based platform that integrates livestreaming, subscription management, content delivery, scoreboard overlays, and media administration tools. Students will work on improving existing systems while developing new features that increase usability, scalability, and long-term sustainability for a nonprofit sports media organization.
The platform currently includes website infrastructure, livestream integration, user authentication, subscription services, scoreboard overlays tools on OBS, and media management workflows. Future teams will continue enhancing and integrating these systems into a more complete production-ready platform.

Student Learning Goals: Students will gain real-world software engineering experience by working directly with a nonprofit client and an active production environment.
Students will develop skills in:
Full-stack web development
Frontend UI/UX design
Cloud deployment and hosting
Authentication and user management
Subscription and paywall systems
Livestream technology and media workflows
System integration and API development
Agile software development
Client communication and project management
Technical documentation and product handoff

Deliverables: Website enhancements and feature improvements
Livestream management and broadcasting workflows
Subscription and membership management features
Administrative dashboards and content management tools
Integration between frontend, backend, storage, and media services
Improvements to OBS Scoreboard Overlay Systems with different Sports
Automated media archiving and replay management
Production deployment and operational documentation
Sponsor handoff and system maintenance documentation

Desired Background: Experience with the following is beneficial:
JavaScript / TypeScript
React / Next.js
HTML / CSS
REST APIs
Git / GitHub
Cloud hosting platforms
Database systems
UI/UX design
Interest in sports media, livestream production, or digital content platforms is helpful but not required.

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement
- Non-disclosure agreement

Additional Materials: https://varsitysportsshow.com/

Pixels to Players - Video Game - Live Playtesting Feedback

Proposer: sam nulf samnulf@gmail.com

Project Contact: Sam Nulf samnulf@gmail.com

Project Description: Overview

Pixels to Players is a SaaS platform designed for player research for game developers. The platform streamlines playtesting, allowing developers to find players, validate gameplay mechanics, and optimize user experiences before launching their games. This presents a significant business opportunity in a rapidly expanding gaming industry where player feedback is critical for a game's success.

Game Showcases are lively events where developers share their work, often a game demo, to gather feedback and generate interest in their product. We would like to provide a feature to quickly collect first impressions from players during live events.

Student Learning Goals: Students will connect directly with us for mentorship throughout the project.

We will support students:

//with weekly check ins and meaningful feedback to ensure success

//by providing resources for product onboarding, software tools, repository management

//mentorship and access to industry experts for consultation

//co-created tasks that align with individual strengths

Deliverables: The successful team will:

//prototype workflow in Figma using UI style features, proper workflow and UX design principles (candidates will be given access to Figma Team Project)

//author customer User Story that addresses game testing sessions (templates provided) // User Stories written and submitted via Google Docs or Word

//identify appropriate candidates to A/B Test workflows for authentic customer feedback (user testing)

//write and upload source code features to the GitHub repository under an MIT License

//follow developer workflow as outlined in our Project Readme (eg. QA branching, Push/Pull Requests)

Desired Background: Candidates should be familiar with Vue, Javascript, HTML and Firebase.

Repository is Github.

A technical document is available with additional information including links: to access the source code, to access the database and RESTful API documentation.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: pixelstoplayers.com

Viking Roots - Viking Roots – Mobile App Development

Proposer: Diana Bristow diana.bristow@icloud.com

Project Contact: Diana Bristow diana.bristow@icloud.com

Project Description: This project focuses on designing and building the mobile experience for an AI-powered social platform centred around photos, identity, and connection.

The existing platform allows users to upload and tag photos, identify people across images, and explore how they are connected. The mobile app will focus on making these core interactions simple, fast, and intuitive for everyday use.

Rather than replicating the web platform, the goal is to create a mobile-first experience that feels natural and easy to use, with a strong focus on usability and performance.

Students will design and build a functional mobile application, with flexibility in the technical approach depending on the team's strengths. This may include native or cross-platform development.

Student Learning Goals: Students will work on a real product with an existing backend and active development roadmap.

They will gain experience designing mobile-first user experiences, working with APIs, and building something intended for real users rather than demonstration.

The project will run in a collaborative, team-based environment with regular check-ins and clear direction. Students will have opportunities to take ownership of features, contribute to product decisions, and present ideas, similar to a real development team.

Deliverables: By the end of the project, the team will deliver a functional mobile application that supports the core platform experience and is ready for continued development.

The app should include a clean and intuitive user experience for uploading and tagging photos, navigating connections, and interacting with the platform in a mobile context. Integration with existing backend services will be required.

All development should be production-ready, clearly structured, and supported by practical documentation so the app can be maintained and expanded beyond the project.

Desired Background: By the end of the project, the team will deliver a functional mobile application that supports the core platform experience and is ready for continued development.

The app should include a clean and intuitive user experience for uploading and tagging

photos, navigating connections, and interacting with the platform in a mobile context. Integration with existing backend services will be required.

All development should be production-ready, clearly structured, and supported by practical documentation so the app can be maintained and expanded beyond the project.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Machinavision inc - Virtual Reality, 3D Visualization, 3D Modeling, User Interface (UI) Design

Proposer: Cedricaymeric Niango niangocedricaymeric@estimarmine.com

Project Contact: niangocedricaymeric@estimarmine.com
niangocedricaymeric@estimarmine.com

Project Description: Main Objective

Develop a desktop-based geological resource estimation software combining 3D modelling, real-time 3D visualisation, and an embedded AR module, with an intuitive UI to improve geological interpretation, validation, and collaboration.

Goals

AR : Provide immersive desktop AR interaction with geological models, including clipping, annotations, and layer control.
3D Modelling : Implement implicit (FastRBF) and explicit tools for lithology, domains, and resource shells from drillhole data.
3D Visualisation : Deliver high-performance OpenGL/VTK rendering with dynamic sectioning, colour mapping, and legend editing.
UI Design : Create an intuitive desktop interface for streamlined workflows, legend management, and AR tool integration.

Student Learning Goals: Students working on Estimarmine+ will gain practical experience developing advanced engineering software for the mining and geoscience industry. They will apply programming, mathematics, and data analysis skills to solve real-world industrial problems in a collaborative, multidisciplinary environment.

Through this experience, students will learn software development practices, scientific computing, and 3D data visualization while working with professional tools and real datasets. They will also develop key professional skills such as problem-solving, teamwork, communication, and project management.

By the end of the program, students will be able to design and implement technical solutions, contribute to production-quality software, and understand how engineering software is developed and used in industry.

Deliverables: AR Module: Immersive desktop AR with clipping, annotations, and layer control.
3D Modelling : FastRBF implicit modelling, explicit boundary tools, and block model integration.
3D Visualisation : OpenGL/VTK rendering, dynamic sectioning, colour mapping, and legend editing.

UI Design : Streamlined interface with legend management, measurements, and export tools.

Desired Background: Computer Science
Software Engineering
Computer Engineering
Data Science
Applied Mathematics

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement

Additional Materials:
https://www.linkedin.com/company/104760925/admin/dashboard/

Northern Watch Services Inc. - Virtual Try ON

Proposer: Thomas J. Sandrin thomas@northernwatchservices.com

Project Contact: Thomas thomas@northernwatchservices.com

Project Description: This project will focus on enhancing and expanding WatchDNA's existing AI-enhanced Augmented Reality (AR) Virtual Try-On technology and transforming it into a scalable public Shopify application.

The current WatchDNA technology enables customers to virtually try on watches using a mobile or desktop device. The goal of this project is to significantly improve the existing solution while expanding the technology beyond watches to support a broader range of jewellery and fashion accessories.

Students will work with our existing platform to improve the accuracy, realism, performance, and overall user experience of the AR technology, including body-part detection, product positioning, sizing, scaling, movement, and realistic product placement.

A key area of development will be adapting the technology to recognize different parts of the body and accurately position different product categories, including:

* Watches and bracelets – wrist detection and placement
* Rings – hand and finger detection
* Necklaces and pendants – neck and upper-body detection
* Earrings – face and ear detection
* Other jewellery and accessories – exploring additional applications of the technology

Another major objective will be to develop the technology into a public Shopify app that brands, jewellers, and authorized retailers can install and integrate directly into their own Shopify e-commerce stores.

The application should allow merchants to connect their products to the AR experience, configure virtual try-on options, and provide customers with an easy way to experience products virtually directly from a product page.

The long-term objective is to transform WatchDNA's current watch-focused AR solution into a scalable multi-category Virtual Try-On platform that can support watches, jewellery, and accessories across multiple brands and retailers.

The project will provide students with hands-on experience in Artificial Intelligence, Augmented Reality, computer vision, body and object detection, e-commerce, Shopify app development, UX/UI, and real-world product development.

Student Learning Goals: Students will gain hands-on experience working on a real-world technology platform using a modern development stack that includes Vercel, Railway, GitHub, Shopify, AI, and Augmented Reality technologies.

They will learn how to work collaboratively within an existing codebase, use GitHub for version control and team development, deploy and test front-end applications through Vercel, and manage back-end services, databases, and application infrastructure using Railway.

Students will also gain practical experience improving AR and computer vision functionality, connecting front-end and back-end systems, integrating Shopify APIs, and developing features for a public Shopify application.

The project will expose students to the full product development lifecycle, including planning, coding, testing, debugging, deployment, user experience improvements, documentation, and preparing a technology solution for real-world commercial use.

By the end of the project, students will have practical experience building and improving a scalable application that combines AI, AR, e-commerce, cloud infrastructure, and collaborative software development.

Deliverables: Expected Deliverables – End of Semester

By the end of the semester, students are expected to deliver:

* An improved and more accurate version of WatchDNA's existing AR Watch Virtual Try-On technology.
* A working multi-category AR prototype supporting watches, rings, bracelets, necklaces, earrings, and other jewellery/accessories.
* A functional public Shopify app prototype allowing brands and retailers to integrate the Virtual Try-On experience into their Shopify stores.
* A basic merchant dashboard for connecting and managing products within the AR experience.
* Integration of the platform using Vercel, Railway, GitHub, and Shopify.
* Testing and optimization across mobile and desktop devices.
* A functional end-to-end demonstration using real watch and jewellery products.
* Complete technical documentation and recommendations for future development, scalability, and preparation for a potential Shopify App Store launch.

Desired Background: Students with an interest or background in Software Engineering, Computer Science, AI, Augmented Reality, Computer Vision, or Web/Application Development would be ideal for this project.

Experience or interest in the following areas would be beneficial:

* JavaScript / TypeScript, React, and modern web development
* AI, Computer Vision, and AR technologies
* Shopify development and API integrations
* GitHub and collaborative software development
* Vercel and Railway deployment and cloud infrastructure
* Front-end and back-end application development
* UX/UI and responsive mobile-first design
* APIs, databases, and third-party integrations

Previous experience with every technology is not required. We are looking for students who are curious, technically motivated, comfortable learning new technologies, and interested in contributing to the development of a real-world AI/AR e-commerce solution.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: https://watchdna.com/products/aerowatch-1942-a-91108-aa02-m

General Dynamics Mission Systems - Vision-Guided AI Agent for Autonomous GUI Testing on Edge Hardware

Proposer: Glen Abousleman glen.abousleman@gd-ms.com

Project Contact: Jason Landoll Jason.Landoll@gd-ms.com

Project Description: Modern software testing tools almost always require installing an agent, driver, or automation framework directly on the system under test. This is impossible or prohibited for many high-value targets: secure and air-gapped systems, BIOS/UEFI firmware menus, embedded and legacy platforms, and certification environments where the software configuration must not be altered. This project asks students to design and build an external, non-invasive automated testing rig that interacts with a target computer exactly as a human operator would — by watching the screen and manipulating a keyboard and mouse — with zero software installed on the target.

The system is a closed-loop feedback platform built from low-cost, portable hardware. A target PC's HDMI output is captured into NVIDIA edge hardware (the "eyes and brain"), which runs a computer-vision and AI/ML perception stack to understand the on-screen state. A Python-based agent coordinator decides the next action required to advance a test procedure, and issues that action to a dedicated microcontroller-based USB HID emulator (the "hands") that injects real keyboard and mouse events into the target's USB port. Because the rig only ever sees pixels and sends raw HID inputs, the target cannot tell it apart from a human operator — making the platform ideal for regression testing, certification support, and lab environments.

A central technical theme is decoupling localization from identification in the perception layer. Rather than relying on a single general-purpose model to both find and choose UI elements (which produces unreliable click coordinates), students will train a custom object-detection model (using a self-hosted, on-premises labeling workflow) to reliably locate generic interactive elements — buttons, text fields, menus, checkboxes — and pair it with OCR for text and a language model for semantic decision-making ("which of these is the Login button?"). This hybrid approach yields pixel-accurate actions while remaining flexible across different applications.

Operators interact with the platform through a dashboard / control layer — a web-based user interface that lets a user select which test cases or suites to run, launch and abort executions, monitor live status and step-by-step progress, watch the captured screen with detected elements highlighted in real time, and view, filter, and export pass/fail reports. The rig is engineered headless-first: the test engine is fully operable through a clean programmatic API, and the dashboard is a thin client on top of it. This keeps concerns cleanly separated, guarantees the system can still run unattended in automated pipelines, and makes the UI a replaceable, low-risk layer that greatly improves both day-to-day

debugging and the final demonstration.

Student Learning Goals: Students will gain hands-on, portfolio-quality experience across a broad, in-demand skill set:

- Applied Computer Vision & Machine Learning: Building an image dataset, training and evaluating a custom object detector, OCR integration, and deploying/optimizing models on edge hardware.
- Edge AI & Embedded Compute: Working with the NVIDIA edge hardware — model deployment, performance/latency benchmarking, and resource-constrained optimization.
- Embedded Systems & Hardware Interfacing: Microcontroller programming (Arduino Pro Micro / RP2040), USB HID protocol internals
- AI Agent & LLM Application Design: Prompt/tool design, structured output (JSON schema design), deterministic-first control flow with LLM-based fallback, and building reproducible, auditable agent behavior.
- Full-Stack & UI Development: Designing and building a web dashboard (e.g., Streamlit/Gradio or FastAPI + JavaScript), REST/WebSocket API design, real-time status streaming, and automated report generation/export.
- Data Labeling & MLOps Workflow: Standing up and operating a self-hosted labeling pipeline (i.e. CVAT), authoring an annotation specification, and managing dataset versioning and QA.

Deliverables: Project deliverables are phased through the two-semester sequence and include:

- Weekly verbal or written progress reports on work completed and problems encountered
- Trade study report with technology recommendations
- Preliminary Design Review
- Testing and validation report
- Final implementation of the system
- Comprehensive project documentation and user manual
- Project presentation and demonstration

Desired Background: Background information required to successfully execute the project includes:

Programming: Strong Python skills (required by at least one team member); general software engineering and Git familiarity.

Machine Learning / Computer Vision: Coursework or project experience in ML or CV is highly desirable (image datasets, object detection, model training/evaluation).

Embedded / Hardware: Familiarity with microcontrollers (Arduino or similar) and basic electronics; interest in USB/HID and hardware interfacing.

Linux: Comfort with the Linux command line, Docker/containers, and SSH (the Jetson and CVAT run on Linux).
Web / Full-Stack (helpful): Experience with a web framework (Streamlit/Gradio, or FastAPI + JavaScript/React) and REST/WebSocket APIs for the dashboard/control layer.
Helpful but not required: Experience with NVIDIA Jetson or other edge devices, TensorRT, LLM/AI-agent development, or software test/QA methodology.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement
- US Citizenship requirement

Additional Materials: All material will be provided by General Dynamics. Item Purpose Est. QtyEst. Unit Cost Est. Total NVIDIA Jetson Orin Nano Developer Kit Edge compute — perception & agent brain 1 $500 $500 microSD card (128 GB+) / NVMe SSD (256 GB+) Jetson OS & dataset storage 1 $80 $80 HDMI capture card Capture target PC video into Jetson 1 $60 $60 Arduino Pro Micro (ATmega32u4) or Raspberry Pi Pico (RP2040) USB HID keyboard/mouse emulator 2 $15 $30 USB cables, HDMI cables, adapters (USB-C, OTG) Interconnects — — $50 Breadboard, jumper wires, misc. electronics Prototyping the HID interface — — $30 Enclosure / mounting hardware (optional) Portable, packaged rig 1 $40 $40 Estimated Total ~$790 Software note: The core software stack — including the dashboard/control layer — is open-source and free (CVAT, OpenCV, Ultralytics YOLO, open-source VLMs/LLMs, Python,Streamlit/Gradio or FastAPI). Self-hosted CVAT, local model inference, and the web UI incur no licensing cost; cloud AI usage is limited to the offline authoring phase to control cost and preserve the zero-connectivity runtime.

NWS Agency - WatchDNA | Map Fetcher Update & Launch

Proposer: Thomas J. Sandrin, MBA Thomas@NWSAgency.com

Project Contact: Thomas J. Sandrin, MBA Thomas@NWSAgency.com

Project Description: Following up and finalizing a project utilizing AI, Vercel, Shopify, and Railway to complete the development and official launch of the new WatchDNA Store Locator. This includes deploying the beta version, integrating AI-powered map fetching and location services, optimizing performance and user experience, and ensuring seamless synchronization with the WatchDNA platform.

Beta Version: https://watchdna.com/pages/store-locator

Student Learning Goals: This project provided hands-on experience in applying AI to solve real-world business challenges while strengthening my technical skills in Shopify, Vercel, Railway, APIs, geolocation services, and full-stack deployment. It enhanced my understanding of integrating multiple cloud platforms, managing beta releases, optimizing user experience, and collaborating with AI-assisted development tools to deliver a production-ready solution. The experience reinforced agile project management, problem-solving, and the practical application of emerging technologies to create a scalable, user-focused product.

Deliverables: * Successfully launch the beta version of the new WatchDNA Store Locator.
* Develop and deploy an AI-powered map fetching and geolocation system.
* Integrate the platform with Shopify, Vercel, and Railway for a secure, scalable, and high-performance deployment.
* Connect the WatchDNA store database with Google Maps to automatically synchronize store information, including photos, website links, business details, opening hours, ratings, and location data.
* Improve the accuracy and completeness of retailer profiles through automated data synchronization.
* Optimize the user experience across desktop and mobile devices, including search, filtering, and map performance.
* Test, debug, and refine the platform based on user feedback and performance analytics.
* Prepare technical documentation covering the system architecture, AI implementation, database integration, deployment process, and future enhancements.
* Deliver a final presentation demonstrating the platform's functionality, business impact, and roadmap for expanding the Store Locator beyond the beta version.

Desired Background: We are looking for a student with an interest in Artificial Intelligence, Full-Stack Web Development, and Cloud Technologies. Experience with the following is preferred but not required:

* AI tools and API integrations
* JavaScript/TypeScript, React, or Next.js

* Shopify development and APIs
* Vercel and Railway deployment
* Google Maps Platform and geolocation services
* Databases (PostgreSQL, MySQL, or similar)
* Git/GitHub version control
* REST APIs and third-party integrations
* Strong analytical and problem-solving skills
* Ability to work independently while collaborating in an agile environment

A passion for innovation, user experience, and applying AI to solve real-world business challenges is highly valued.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Additional Materials: https://watchdna.com/pages/store-locator

Appy.yo - WeVibe: AI-Powered Dating and Compatibility Recommendation App

Proposer: Nadia Hosseinzadeh nadiahosse@gmail.com

Project Contact: Nadia Hosseinzadeh Hello@appyyo.com

Project Description: WeVibe requires AI features to support profile scanning, recommendations, and matching algorithms. Many dating apps focus heavily on photos and quick swiping, which can create shallow or frustrating user experiences. WeVibe is designed to create more thoughtful matches by analyzing profiles, preferences, values, and compatibility signals. The current need is to develop AI-powered profile recommendations and matching logic that can support a more meaningful dating experience.

Student Learning Goals: AI implementation, recommendation systems, algorithm design, UI/UX design, front-end development, back-end development, database design, mobile app development, psychology/personality framework research, JavaScript, React Native, Python.

Deliverables: A fully functioning software solution with AI profile scanning, recommendation logic, and matching algorithms. The final format can be discussed, but ideally the solution should support web, iOS, and Android experiences, either through a responsive web app, mobile app, or cross-platform build.

Desired Background: AI/Matching Algorithm Developer 1–2
UI/UX Designer 1
Front-end or Mobile Developer 1
Back-end Developer 1
Research and Development 1

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement

Rue Productions - WordWall + Rue File Splicer + Rue Funky

Proposer: Rubén Möller rueproductions@gmail.com

Project Contact: Rubén Möller rueproductions@gmail.com

Project Description: 1. Overview
Four related but independent modules that support experimental animation and production workflows at Rue Productions:

WordWall – Extend an existing JavaScript word-animation library with accurate video/audio word-level synchronization and improved motion effects.
Rue File Splicer – Fix and complete a missing or broken Python function inside an image-to-image file extraction app.
Rue Funky – Repair a working automation bot that triggers website commands to generate animation output. Issues exist in the Java–Python bridge and interface/execution layer.
Rue Match – Complete, stabilize, and refine a Qt-based desktop application for interactive image segmentation, facial-feature matching, multi-scene timeline management, and segment export.

All projects use GitHub. Required workflow: GitHub Desktop (no exceptions).

2. Project 1 – WordWall (Primary Focus)
Goal
Add robust video synchronization and modern word-timing to an existing JavaScript library that turns text or spoken audio into animated, scattered/moving word displays.
Current Capabilities

Words can be colored and “popped” into the background.
Basic animation and layout control exist in JavaScript.

Required New Functionality

Parse spoken sentences (especially from recorded interviews) into individual words with accurate start/end timing.
Synchronize animated words to video/audio so the spoken word appears large in the foreground, then shrinks and moves backward into the scene.
Support time-code driven display and emphasis effects (e.g. shake).
Overcome current JavaScript limitations that prevent cleanly advancing word data in the timeline.
Explore a clean bridge between the working JavaScript library and Python where it improves new effects.

First-Week Expectations

Clone the WordWall repository.
Work on a clean personal branch (to be created).
Read the README thoroughly.
Spend the first week exploring the package and preparing questions for the next meeting.

3. Project 2 – Rue File Splicer
Goal
Fix and complete a missing or broken Python function inside an image-to-image file extraction app.
Scope

Pure coding task.
The function is required for the app's pipeline.
Light HTML familiarity is useful but secondary.
Dedicated branch will be created once the new developer is onboarded.

4. Project 3 – Rue Funky (Automation Bot Repair)
Goal
Repair and stabilize a bot that triggers commands on a website to automate animation output generation.
Current Status

Core functionality works.
Interface and execution issues exist.
Problems span both the Java and Python sides and the bridge between them.

Scope

Diagnose and fix Python interface / execution problems.
Resolve issues in the Java–Python bridge so the bot reliably triggers website commands and produces correct animation output.
Keep existing working functionality intact.
Deliver clean, maintainable code that makes the tool dependable for production use.

5. Project 4 – Rue Match
Goal
Complete, stabilize, and refine Rue Match, a Qt-based desktop application for interactive image segmentation, facial-feature template matching, multi-scene timeline handling, and segment export.
What the application does

Rue Match lets a user load a source image (or sequence of scenes), segment it into regions using Watershed, K-Means, or SAM-based methods, then interactively match those segments to predefined facial-feature templates ("alphas"). The tool supports:

Click-to-select / click-to-remove contours
Highlight-all and candidate ranking modes
Per-feature parameter persistence
A timeline bar for navigating multiple scenes/frames
Assignment tracking and export of cropped segments + CSV / text reports
Project save/load (.rum files)

Current Status
Significant new functionality exists on feature branches (main-sanji, sanjith) that is not yet in main, including:

Full GUI layer (gui/ folder: canvas widgets, timeline, view, controller)
Watershed and K-Means segmentation backends
Contour removal (rem_at_point)
Extensive interactive matching, highlighting, and export logic

The application needs consolidation, bug-fixing, interface polish, and reliable integration so it can be used dependably in production.
Scope

Merge and stabilize the new GUI and backend code into a clean, working state.
Resolve any remaining interface, execution, or cross-component issues.
Ensure segmentation, matching, timeline navigation, click modes, and export all function reliably.
Improve code clarity, error handling, and maintainability where needed.
Preserve existing working behaviour while making the tool production-ready.

Student Learning Goals: 6. Student Learning Experience
This placement offers substantial hands-on learning across real production tools used in experimental animation and media workflows. The student will gain practical experience in the following areas:

Cross-language development — Working with JavaScript, Python, and Java in the same overall ecosystem, including diagnosing and repairing issues that span language boundaries (especially the Java–Python bridge in Rue Funky).
Media & timing systems — Implementing accurate word-level audio/video synchronization, time-code handling, and kinetic typography effects inside an existing animation library (WordWall).
Interactive desktop application development — Building and refining a full Qt GUI with custom widgets (clickable canvas, timeline bar), complex state management, and real-time

image processing feedback (Rue Match).
Computer vision & segmentation — Working with contour extraction, watershed, K-Means, and feature-matching pipelines.
Debugging and code repair — Diagnosing broken or incomplete functions, interface problems, and execution failures in live codebases rather than building from scratch.
Version control discipline — Mandatory use of GitHub Desktop with clean branching, commits, and collaboration practices on active repositories.
Production-oriented problem solving — Understanding how small technical components (file extraction, website command automation, word animation, feature matching) serve larger creative pipelines in animation and film production.
Independent ownership — Taking full responsibility for discrete modules, preparing technical questions, and driving progress through weekly check-ins.
Communication in a creative-technical environment — Collaborating directly with an experimental filmmaker/animator, translating creative requirements into working code, and iterating based on real production needs.

The student will leave with concrete, portfolio-ready contributions to working tools, stronger debugging skills across multiple languages, and direct experience supporting professional animation workflows.

Deliverables: 7. Expected Deliverables

Word Wall

Working video/audio word-level synchronisation.
Words that appear large in the foreground on speech, then shrink and recede into the scene.
Reliable time-code support and emphasis effects (e.g. shake).
Export of the final animation to common video formats (via codec) as well as sequential image frames with a separate synchronized audio track.
Clean, documented code committed to a personal branch.
Short technical notes or README updates explaining how the new features work.

Rue File Splicer

Fully functional Python function that correctly performs the required image-to-image file extraction.
Integration into the existing app pipeline without breaking other features.
Clear commits and brief documentation of the fix.

Rue Funky

Stable bot that reliably triggers website commands and produces correct animation output.
Resolved Java–Python bridge and interface/execution issues.
Existing working functionality preserved.

Clean, maintainable code with notes on what was fixed and how to run the tool.

Rue Match

Stable, consolidated application with the new GUI and segmentation backends properly integrated.
Reliable segmentation, interactive matching, timeline navigation, click modes, and export.
Clean, maintainable code with clear structure between backend scripts and GUI layers.
Brief documentation of major components and how to run/build the tool.

Overall

All work delivered via GitHub Desktop on assigned personal branches.
Progress demonstrated through regular commits and weekly check-in discussions.
Code that is readable, maintainable, and ready for continued use in production workflows.

Desired Background: 8. Desired Background

Solid working knowledge of Python (required).
Comfort reading and modifying existing JavaScript codebases.
Experience or strong interest in Java and/or Java–Python interoperability.
Familiarity with Qt / PyQt / PySide (or willingness to learn quickly) is a strong plus for Rue Match.
Experience with OpenCV, image segmentation, or computer-vision pipelines is highly desirable.
Familiarity with Git and version control (GitHub Desktop will be required).
Interest in media, animation, audio/video timing, or browser automation is highly desirable.
Ability to diagnose and repair incomplete or broken code rather than only building new features from scratch.
Self-directed work style with the ability to explore unfamiliar codebases, ask focused questions, and iterate based on feedback.
Clear written and verbal communication skills for weekly check-ins and technical discussions.

Prior professional experience is not required; demonstrated coding ability, curiosity, and reliability are more important.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: www.rueproductions.com

Nextiva - XBert AI: Intelligent Voice Agent Platform

Proposer: Srimaan Sri Edara srimaan.edara@nextiva.com

Project Contact: Srimaan Edara srimaan.edara@nextiva.com

Project Description: XBert AI is Nextiva's AI-powered voice agent designed to communicate naturally with customers, understand their intent, answer questions, complete routine tasks, and intelligently transfer complex requests to human employees. The platform can learn from a company's website, business information, FAQs, policies, workflows, and customer context to provide accurate and personalized responses.

The capstone team will explore, design, and develop capabilities that improve XBert's intelligence, reliability, integrations, analytics, and overall customer experience. Potential areas may include conversational workflows, knowledge retrieval, intent recognition, call routing, appointment scheduling, lead qualification, customer support automation, CRM integrations, human handoffs, testing frameworks, performance monitoring, and administrative tools.

The project will intentionally maintain a broad scope because XBert AI is an evolving product. Specific priorities and technical deliverables may change throughout the academic year based on product development, customer needs, and emerging voice-AI technologies. Nextiva's current XBert platform supports voice, SMS, and web chat, learns from business information, performs tasks such as appointment booking and request routing, and provides transcripts and analytics.

Student Learning Goals: Students will gain hands-on experience developing a production-oriented AI application within a real enterprise software environment. They will work with technologies and concepts such as conversational AI, large language models, speech recognition, text-to-speech, retrieval-augmented generation, APIs, cloud services, data pipelines, prompt engineering, system integrations, and modern web development.

Students will also learn how to translate product requirements into technical solutions, evaluate AI responses, measure latency and accuracy, design safe fallback behavior, protect sensitive information, and build reliable human handoff experiences. The team will collaborate with Nextiva stakeholders through regular meetings, demonstrations, technical reviews, and iterative feedback.

Because the product and AI ecosystem will continue evolving, students will experience an agile development process in which requirements may be refined as research, prototypes, and user testing reveal new opportunities.

Deliverables: The final deliverables will be determined collaboratively based on the highest-priority XBert initiatives during the project. Expected outputs may include:

A functional prototype or production-ready enhancement for XBert AI
One or more voice-agent workflows or business use cases
Integration with websites, knowledge sources, APIs, CRM systems, calendars, or Nextiva services
Testing and evaluation tools for conversational accuracy, task completion, latency, safety, and handoff quality
A dashboard or reporting capability for reviewing calls, transcripts, outcomes, and performance
Technical documentation covering system architecture, setup, testing, and future development
A final demonstration and presentation of the completed solution

The team is not expected to deliver every item listed above. Final deliverables will be selected and refined with Nextiva as XBert's product roadmap develops.

Desired Background: Students should have a foundation in computer science or software engineering and be comfortable learning unfamiliar technologies. Helpful experience includes Python, JavaScript or TypeScript, REST APIs, databases, cloud platforms, Git, web application development, machine learning, natural-language processing, or large language models.

Prior experience building voice agents is not required. Strong problem-solving, communication, documentation, teamwork, and willingness to experiment with rapidly changing AI technologies are highly valued.

Delivery Mode: In-person

Required Agreements:

- Non-disclosure agreement
- Intellectual property agreement

Additional Materials: https://www.nextiva.com/products/xbert

📚FUN·da·Mental 🧠✨ Wellness - 📲💫🧠 | FUN·da·Mental Breaks MVP | Brain Buddy-Guided Mental Wellness App

Proposer: Deesil Danny Silva movement@fundamental.fans

Project Contact: Deesil Danny Silva movement@fundamental.fans

Project Description: FUN·da·Mental is a common-sense wellness platform that helps mentally overstimulated students, young adults, and everyday people build a healthier relationship with their mind. This capstone project focuses on a software MVP of FUN·da·Mental Breaks: a playful, Brain Buddy-guided app that helps users take real breaks from constant input and reconnect with breath, body, senses, and real life. 😀📱🪡

This project is centered on building a meaningful pilot-ready MVP, not a clinical therapy product.

Core MVP areas may include:
- 🌟 simple onboarding + first-time user flow
- ⏱️ a “how much time do you have?” break selector
- 😀 guided break experiences built around stillness, movement, or sensory reset
- 🤖 Brain Buddy support prompts that make the experience feel welcoming and easy to follow
- 🪡 lightweight reflection, check-in, or progress tracking
- 🧰 content architecture that allows future break types to be added cleanly

Why this project is exciting:
- it sits at the intersection of software, UX, humane technology, and mental wellness
- it gives students a chance to build something purpose-driven and highly relevant to modern life
- it is designed as life-elevating tech that supports presence instead of competing for more attention

This project already has real brand direction and supporting materials behind it, including a website, booklet direction, survey learning loop, and Brain Buddy experience layer, so students would be building from a strong early foundation rather than starting from zero. ✨

Student Learning Goals: Students will work on a real early-stage product at the intersection of software development, user experience, ethical product thinking, and startup execution. They will gain hands-on experience translating a human-centered mission into a functional digital product that people can actually use. 🪡💻✨

Students can expect to learn through:

- 🤖 product scoping and requirement definition
- 🎨 user-centered UX thinking for a low-overstimulation experience
- ⚙️ frontend, backend, and systems collaboration
- 🐞 API / data-flow thinking for app features and content logic
- 🧪 testing, iteration, feedback loops, and sprint-based development
- 💬 presenting technical decisions clearly to a real sponsor / founder
- 💫 building in a startup-style environment where scope, clarity, and communication matter

Why this is a strong learning opportunity:
- students get exposure to meaningful product work, not just an isolated coding assignment
- the project invites both technical problem-solving and thoughtful product judgment
- the team will have room to shape architecture, UX decisions, and implementation strategy
- the work can become strong portfolio material because it combines software, impact, and real-world constraints

Sponsor involvement / collaboration plan:
- 📍 remote kickoff meeting: 60–90 minutes
- 🔁 weekly check-ins & meetings aligned with course cadence
- ⚡ responsive async communication and feedback within 1 business day
- 📊 shared demos, sprint reviews, and clear product-direction feedback throughout the project

This is a strong fit for students who want to build technology that is thoughtful, relevant, and designed to help humans, not just capture more attention. 🌍

Deliverables: Semester 1 deliverables
- 🧭 refined project brief, goals, use cases, and user stories
- 🗺️ product architecture and technical implementation plan
- 🎨 UX flows / wireframes for the core user journey
- 🛠️ repository setup, development workflow, and foundational system structure
- 🔌 backend / data model planning for content, users, and progress logic
- 🚀 working alpha of the core flow, such as:
  - onboarding
  - check-in
  - break selection
  - at least one guided break experience
- 📽️ sprint demos and progress documentation

Semester 2 deliverables
- 📱 integrated MVP with polished core user flows

- 🤖 Brain Buddy-guided support layer across key moments in the app
- 🪞 reflection, check-in, and/or lightweight progress tracking
- 🧩 content structure or admin/config logic for expanding break types over time
- 🧪 QA / testing plan and bug-fix rounds
- ☁️ deployment guidance or pilot-ready release prep
- 📚 final technical documentation, presentation, and handoff materials

Success for this project
- build a strong educational MVP that demonstrates real user value
- create a thoughtful technical foundation for future iteration
- deliver a product direction that is credible, usable, and meaningful

A production launch is not required. The goal is a strong, well-scoped MVP that reflects quality learning, quality engineering, and real product momentum. ✨

Desired Background: Minimum background
- 💻 strong software engineering fundamentals
- 🧱 experience building with a modern frontend stack
- 🔌 comfort working with APIs, data flow, and app logic
- 🌿 familiarity with Git / version control and collaborative development
- 🧪 willingness to test, iterate, and work in sprints
- 🎯 interest in product thinking and user experience

Helpful / strong plus skills
- ⚛️ React, React Native, Next.js, or comparable frontend frameworks
- 🟦 TypeScript / JavaScript
- 🌐 Node.js or similar backend experience
- 🗃️ Firebase, Supabase, PostgreSQL, or comparable data tools
- 📱 mobile app architecture or responsive web-app development
- 📊 analytics / event tracking
- 🤖 conversational UX, AI-assisted features, or prompt-aware interface thinking
- ♿ accessibility-minded and human-centered design instincts

Project fit note
This project is software-first. It does not require embedded systems or hardware knowledge.
Students who enjoy thoughtful product work, meaningful UX, and real-world app building will likely be a strong fit. 🌟

Delivery Mode: In-person

Required Agreements:

- Intellectual property agreement
- Non-disclosure agreement

Additional Materials: https://www.fundamental.fans/ | https://drive.google.com/file/d/1KqH5PvrE6OrhSb0EWBYIePxWSNW9LGcr/view?usp=drive_link | https://drive.google.com/file/d/1mawECWjjTwF5OgqgzZ5ur5uykPfiyBWM/view?usp=drive_link | https://forms.gle/oc5GeveZEVo5Dt8EA | Note: if a Summer 2026 slot opens due to dropouts, we'd love to be considered and can slot in quickly.